

attributes backend basex bibtex bitset  
bitset blobs boolean buffers builders bytemap  
caches callback callbacks catcodes characters  
chemistry commands containers context  
converters coroutine ctxrunner debug dir  
directives document documentdata drivers  
effects environment experiments figures file  
fio floats font fonts functions globaldata  
globals graphics gzip helper images  
interactions interfaces io job kdtree kpse  
language languages layouts lfs libraries lmx  
loaders logs lpdf lpeg lua luac luatex lxml  
math mathematics md5 metapost mime moduledata  
modules mp mplib nanojpeg noads node nodes  
number objects optional os package  
pagebuilders parametersets pdf pdfdecode pdfengine  
physics pngdecode posit potrace publications  
qrcodegen regimes resolvers sandbox scripts  
serial set sha2 sio socket sorters sparse  
statistics status storage streams string  
structures table tex texconditionals texconfig  
texconstants texifs texio texpd texpstex  
texmodes texsystemmodes thirddata token tokens  
tracers trackers typesetters updaters url  
userdata utf8 utilities vector visualizers  
xcomplex xdecimal xinterval xmath xml xzip  
zbuffer zip zlib

# ConTeXt LMTX

# global (alias: \_G)

|                           |          |                     |            |
|---------------------------|----------|---------------------|------------|
| <b>CONTEXT</b>            | number   | <b>dimen</b>        | function   |
| <b>LUAFORMAT</b>          | number   | <b>dir</b>          | table      |
| <b>LUA_MAJORVERSION</b>   | number   | <b>directives</b>   | table      |
| <b>LUA_MINORVERSION</b>   | number   | <b>document</b>     | table      |
| <b>LUAENGINE</b>          | string   | <b>documentdata</b> | table      |
| <b>LUAFORMATID</b>        | number   | <b>dofile</b>       | function   |
| <b>LUAEXFUNCTIONALITY</b> | number   | <b>drivers</b>      | table      |
| <b>LUAEXVERBOSE</b>       | string   | <b>effects</b>      | table      |
| <b>LUAEXVERSION</b>       | number   | <b>environment</b>  | table      |
| <b>LUAVERSION</b>         | number   | <b>error</b>        | function   |
| <b>MP</b>                 | function | <b>experiments</b>  | table      |
| <b>SAVEDREQUIRE</b>       | function | <b>figures</b>      | table      |
| <b>aes</b>                | table    | <b>file</b>         | table      |
| <b>arg</b>                | table    | <b>fio</b>          | table      |
| <b>assert</b>             | function | <b>floats</b>       | table      |
| <b>attributes</b>         | table    | <b>font</b>         | table      |
| <b>backends</b>           | table    | <b>fonts</b>        | table      |
| <b>basexx</b>             | table    | <b>functions</b>    | table      |
| <b>bibtex</b>             | table    | <b>getmetatable</b> | function   |
| <b>bit32</b>              | table    | <b>globaldata</b>   | table      |
| <b>bitset</b>             | table    | <b>globals</b>      | table      |
| <b>blobs</b>              | table    | <b>graphics</b>     | table      |
| <b>boolean</b>            | table    | <b>gzip</b>         | table      |
| <b>buffers</b>            | table    | <b>helper</b>       | table      |
| <b>builders</b>           | table    | <b>images</b>       | table      |
| <b>bytemap</b>            | table    | <b>inspect</b>      | function   |
| <b>caches</b>             | table    | <b>interactions</b> | table      |
| <b>callback</b>           | table    | <b>interfaces</b>   | table      |
| <b>callbacks</b>          | table    | <b>io</b>           | table      |
| <b>catcodes</b>           | table    | <b>ipairs</b>       | function   |
| <b>characterkern</b>      | function | <b>job</b>          | table      |
| <b>characters</b>         | table    | <b>kdtree</b>       | table      |
| <b>chemistry</b>          | table    | <b>kpse</b>         | table      |
| <b>collectgarbage</b>     | function | <b>language</b>     | table      |
| <b>commands</b>           | table    | <b>languages</b>    | table      |
| <b>containers</b>         | table    | <b>layouts</b>      | table      |
| <b>context</b>            | function | <b>lfs</b>          | table      |
| <b>converters</b>         | table    | <b>libraries</b>    | table      |
| <b>coroutine</b>          | table    | <b>lmx</b>          | table      |
| <b>ctxrunner</b>          | table    | <b>load</b>         | function   |
| <b>debug</b>              | table    | <b>loaders</b>      | data table |

|                        |          |                          |            |
|------------------------|----------|--------------------------|------------|
| <b>loadfile</b>        | function | <b>rawget</b>            | function   |
| <b>loadstring</b>      | function | <b>rawlen</b>            | function   |
| <b>logs</b>            | table    | <b>rawset</b>            | function   |
| <b>lpdf</b>            | table    | <b>regimes</b>           | table      |
| <b>lpeg</b>            | table    | <b>require</b>           | function   |
| <b>lua</b>             | table    | <b>resolvers</b>         | table      |
| <b>luac</b>            | table    | <b>sandbox</b>           | table      |
| <b>luatex</b>          | table    | <b>scripts</b>           | table      |
| <b>lxml</b>            | table    | <b>select</b>            | function   |
| <b>math</b>            | table    | <b>serial</b>            | table      |
| <b>mathematics</b>     | table    | <b>set</b>               | table      |
| <b>md5</b>             | table    | <b>setinspector</b>      | function   |
| <b>metapost</b>        | table    | <b>setmathparameters</b> | function   |
| <b>mime</b>            | table    | <b>setmetatable</b>      | function   |
| <b>moduledata</b>      | table    | <b>sha2</b>              | table      |
| <b>modules</b>         | table    | <b>sio</b>               | table      |
| <b>mp</b>              | function | <b>socket</b>            | table      |
| <b>mplib</b>           | table    | <b>sorters</b>           | table      |
| <b>nanjpeg</b>         | table    | <b>sparse</b>            | table      |
| <b>next</b>            | function | <b>statistics</b>        | table      |
| <b>noads</b>           | table    | <b>status</b>            | table      |
| <b>node</b>            | table    | <b>storage</b>           | table      |
| <b>nodes</b>           | table    | <b>streams</b>           | table      |
| <b>number</b>          | table    | <b>string</b>            | table      |
| <b>objects</b>         | table    | <b>structures</b>        | table      |
| <b>optional</b>        | table    | <b>table</b>             | table      |
| <b>optionalrequire</b> | function | <b>tex</b>               | table      |
| <b>os</b>              | table    | <b>texconditionals</b>   | data table |
| <b>package</b>         | table    | <b>texconfig</b>         | table      |
| <b>pagebuilders</b>    | table    | <b>texconstants</b>      | data table |
| <b>pairs</b>           | function | <b>texifs</b>            | data table |
| <b>parametersets</b>   | table    | <b>texio</b>             | table      |
| <b>pcall</b>           | function | <b>texisdefined</b>      | data table |
| <b>pdf</b>             | table    | <b>texmodes</b>          | data table |
| <b>pdfdecode</b>       | table    | <b>texsystemmodes</b>    | data table |
| <b>pdfx</b>            | table    | <b>thirddata</b>         | table      |
| <b>physics</b>         | table    | <b>toboollean</b>        | function   |
| <b>pngdecode</b>       | table    | <b>todecimal</b>         | function   |
| <b>posit</b>           | table    | <b>token</b>             | table      |
| <b>potrace</b>         | table    | <b>tokens</b>            | table      |
| <b>print</b>           | function | <b>tonumber</b>          | function   |
| <b>publications</b>    | table    | <b>tostring</b>          | function   |
| <b>qrcodegen</b>       | table    | <b>tracers</b>           | table      |
| <b>rawequal</b>        | function | <b>trackers</b>          | table      |

|                    |          |                  |          |
|--------------------|----------|------------------|----------|
| <b>type</b>        | function | <b>warn</b>      | function |
| <b>typesetters</b> | table    | <b>xcomplex</b>  | table    |
| <b>unpack</b>      | function | <b>xdecimal</b>  | table    |
| <b>updaters</b>    | table    | <b>xinterval</b> | table    |
| <b>url</b>         | table    | <b>xmath</b>     | table    |
| <b>userdata</b>    | table    | <b>xml</b>       | table    |
| <b>utf</b>         | table    | <b>xpcall</b>    | function |
| <b>utf8</b>        | table    | <b>xzip</b>      | table    |
| <b>utilities</b>   | table    | <b>zbuffer</b>   | table    |
| <b>vector</b>      | table    | <b>zip</b>       | table    |
| <b>visualizers</b> | table    | <b>zlib</b>      | table    |

## aes (extra lua)

**decode** function  
**encode** function

**random** function

# attributes (context)

## alternates table

- N attribute
- D data
- F enable
- F handler
- F register
- D registered
- D values

## applyoverloads function

## checkvalues function

## colorintents table

- N attribute
- D data
- F enable
- F handler
- F register
- D registered

## colors table

- N attribute
- F cmyk
- F cmyktogray
- F cmyktorgb
- F colorcomponents
- D counts
- B couple
- F currentmodel
- F currentnamedmodel
- D data
- N default
- F defineintermediatecolor
- F definemultitonecolor
- F defineprocesscolor
- F defineprocesscolordirect
- F definesimplegray
- F definespotcolor
- F definetransparency
- F enable
- F filter
- N forced
- F forcedmodel
- F forcesupport

- F formatcolor
- F formatgray
- F formatluminance
- F formatluminanceratio
- F getpagecolormodel
- F gray
- F graytohsb
- F handler
- T helpers
  - F complement
  - F inbetween
  - F justone
- F hsvtorgay
- F hsvtorgb
- F hwbtorgb
- F iccprofile
- F isblack
- F labtolch
- F labtorgb
- F labtoxyz
- F lchtolab
- F lchtorgb
- N main
- S model
- D models
- F mpcolor
- F mpnamedcolor
- F mpoptions
- F mprgbcolor
- F namedcolorattributes
- N none
- F popset
- F processcolorcomponents
- F pushset
- F register
- D registered
- F registerspotcolor
- F rgb
- F rgbbytes
- F rgbtocmyk
- F rgbtorgay

|                          |            |                       |            |
|--------------------------|------------|-----------------------|------------|
| F rgbtohsv               |            | F handler             |            |
| F rgbtoxyz               |            | F register            |            |
| N selector               |            | D registered          |            |
| F setlist                |            | F set                 |            |
| F setmodel               |            | <b>numbers</b>        | data table |
| F setpagecolormodel      |            | <b>ofnode</b>         | function   |
| D sets                   |            | <b>private</b>        | function   |
| F spec                   |            | <b>public</b>         | function   |
| F spot                   |            | <b>registervalue</b>  | function   |
| F spotcolorname          |            | <b>restore</b>        | function   |
| F spotcolorparent        |            | <b>save</b>           | function   |
| F spotcolorvalue         |            | <b>setcleaner</b>     | function   |
| B supported              |            | <b>showcurrent</b>    | function   |
| F toattributes           |            | <b>states</b>         | table      |
| F transparencycomponents |            | F disabletriggering   |            |
| B triggering             |            | F enabletriggering    |            |
| F usecolors              |            | F finalize            |            |
| D valid                  |            | F initialize          |            |
| F value                  |            | F process             |            |
| D values                 |            | F selective           |            |
| B weightgray             |            | F simple              |            |
| F xyztolab               |            | F stacked             |            |
| F xyztorgb               |            | F stacker             |            |
| <b>counts</b>            | data table | <b>system</b>         | function   |
| <b>define</b>            | function   | <b>tooverloads</b>    | function   |
| <b>effects</b>           | table      | <b>transparencies</b> | table      |
| N attribute              |            | N attribute           |            |
| D data                   |            | D data                |            |
| F enable                 |            | F enable              |            |
| F handler                |            | F forcesupport        |            |
| F register               |            | F handler             |            |
| D registered             |            | T names               |            |
| D values                 |            | N color               |            |
| <b>getvalue</b>          | function   | N colorburn           |            |
| <b>getvalues</b>         | function   | N colordodge          |            |
| <b>handlers</b>          | table      | N darken              |            |
| <b>hasvalues</b>         | function   | N difference          |            |
| <b>list</b>              | data table | N exclusion           |            |
| <b>markasoverload</b>    | function   | N hardlight           |            |
| <b>names</b>             | data table | N hue                 |            |
| <b>negatives</b>         | table      | N lighten             |            |
| N attribute              |            | N luminosity          |            |
| D data                   |            | N multiply            |            |
| F enable                 |            | N none                |            |

|                     |            |                          |
|---------------------|------------|--------------------------|
| N normal            |            | F definedlayoutcomponent |
| N overlay           |            | F enable                 |
| N saturation        |            | F forcesupport           |
| N screen            |            | F handler                |
| N softlight         |            | B hasorder               |
| N none              |            | D listwise               |
| F register          |            | F register               |
| D registered        |            | D registered             |
| B supported         |            | F resolve_begin          |
| F toattribute       |            | F resolve_end            |
| B triggering        |            | F resolve_reset          |
| F value             |            | F resolve_step           |
| D values            |            | D scopes                 |
| <b>unsetvalue</b>   | number     | F setfeatures            |
| <b>values</b>       | data table | F setup                  |
| <b>viewerlayers</b> | table      | F start                  |
| N attribute         |            | F stop                   |
| F cleanup           |            | B supported              |
| D data              |            | D values                 |
| F define            |            |                          |



# backends (context)

**closeout** function

**codeinjections** table

F registerboxresource

F registerfont

F restoreboxresource

**current** string

**defaults** table

T codeinjections

F prerollreference

F setautoprefix

F vfstartcolor

F vfstopcolor

T nodeinjections

F cmykcolor

F destination

F graycolor

F injectknockout

F injectoverprint

F reference

F rgbcolor

F spotcolor

F transparency

T registrations

F transparency

D tables

**getcallbackstate** function

**included** table

B context

B date

B id

B metadata

B pdf

**initialize** function

**latelua** function

**nodeinjections** table

**nothing** function

**openout** function

**registered** table

T pdf

T codeinjections

F addbookmarks

F analyzenormalsymbol

F analyzesymbol

F attachmentid

F bitmap

F clonefield

F copyimage

F defaultprofile

F definefield

F definefieldset

F defineviewerlayer

F discardpages

F embedfile

F embedimage

F enabletags

F exportformdata

F fieldsurrounding

F flushmergelayer

F getbookmarks

F getboxresourcebox

F getdefaultfieldvalue

F getfieldcategory

F getfieldset

F getformatoption

F getinfo

F getlinkstructureparent

F getreferencestructureobject

F getrefobj

F insertrenderingwindow

F jp2

F jpg

F maptag

F mergebookmarks

F mergecomments

F mergefields

F mergereferences

F mergerenditions

F mergeviewerlayers

F newimage

F png

F prerollreference

F presetsymbol

F presetsymbollist  
 F processrendering  
 F registerdocumentcloseaction  
 F registerdocumentopenaction  
 F registeredsymbol  
 F registerfiguremask  
 F registerfigurerange  
 F registerpagecloseaction  
 F registerpageopenaction  
 F registerpattern  
 F registersymbol  
 F ruleactionbox  
 F ruleactiondraw  
 F ruleactionfill  
 F ruleactionmp  
 F ruleactionstroke  
 F saveboxresource  
 F scanimage  
 F setattachmentsupport  
 F setautoprefix  
 F setencryption  
 F setfieldcalculationset  
 F setfigurecolorspace  
 F setfiguremask  
 F setformat  
 F setformsmethod  
 F setjobcomment  
 F setpagetransition  
 F settaggingsupport  
 F setupviewerlayer  
 F setuserdata  
 F setxmpfile  
 F startactualtext  
 F startalternativetext  
 F startlayer  
 F startunicodetoactualtext  
 F startunicodetoactualtextdirect  
 F stopactualtext  
 F stopalternativetext  
 F stoplayer  
 F stopunicodetoactualtext  
 F stopunicodetoactualtextdirect  
 F supportedformats  
 F unicodetoactualtext

F useviewerlayer  
 F validcompactor  
 F validfield  
 F validfieldcategory  
 F validfieldset  
 F vfstartcolor  
 F vfstopcolor  
 F wrapimage  
 S name  
 T nodeinjections  
 F addtags  
 F annotation  
 F attachfile  
 F changestackedlayer  
 F cmykcolor  
 F comment  
 F destination  
 F effect  
 F getboxresourcedimensions  
 F graycolor  
 F injectbitmap  
 F insertu3d  
 F knockout  
 F negative  
 F overprint  
 F positive  
 F reference  
 F rgbcolor  
 F spotcolor  
 F startalternate  
 F startlayer  
 F startstackedlayer  
 F stopalternate  
 F stoplayer  
 F stopstackedlayer  
 F transparency  
 F typesetfield  
 F useboxresource  
 T registrations  
 F cmykindexcolor  
 F cmykspotcolor  
 F getspotcolorreference  
 F grayindexcolor  
 F grayspotcolor

F `rgbindexcolor`  
F `rgbspotcolor`  
F `spotcolorname`  
F `transparency`  
T `tables`

|                      |          |
|----------------------|----------|
| <b>registrations</b> | table    |
| <b>setencryption</b> | function |
| <b>tables</b>        | table    |
| <b>writeout</b>      | function |

## basexx (extra lua)

**decode16**    function

**decode64**    function

**decode85**    function

**decodeLZW**   function

**decodeRL**    function

**encode16**    function

**encode64**    function

**encode85**    function

**encodeLZW**   function

**encodeRL**    function

## bit32 (context)

**arshift** function  
**band** function  
**bnot** function  
**bor** function  
**btest** function  
**bxor** function

**extract** function  
**lrotate** function  
**lshift** function  
**replace** function  
**rrotate** function  
**rshift** function

## bitset (context)

|                 |          |
|-----------------|----------|
| <b>asstring</b> | function |
| <b>get</b>      | function |
| <b>new</b>      | function |
| <b>reset</b>    | function |

|                 |          |
|-----------------|----------|
| <b>set</b>      | function |
| <b>totable</b>  | function |
| <b>traverse</b> | function |
| <b>wipe</b>     | function |

## blobs (context)

|                   |          |
|-------------------|----------|
| <b>append</b>     | function |
| <b>dimensions</b> | function |
| <b>dispose</b>    | function |
| <b>new</b>        | function |
| <b>pack</b>       | function |
| <b>strdp</b>      | function |
| <b>strhd</b>      | function |

|                |          |
|----------------|----------|
| <b>strht</b>   | function |
| <b>strwd</b>   | function |
| <b>tohpack</b> | function |
| <b>tonodes</b> | function |
| <b>tovpack</b> | function |
| <b>write</b>   | function |

## boolean (context)

**tocontext** function

**tonumber** function



## buffers (context)

|                       |          |
|-----------------------|----------|
| <b>append</b>         | function |
| <b>assign</b>         | function |
| <b>collectcontent</b> | function |
| <b>empty</b>          | function |
| <b>erase</b>          | function |
| <b>exists</b>         | function |
| <b>get</b>            | function |
| <b>getcontent</b>     | function |
| <b>getctxlua</b>      | function |
| <b>getlines</b>       | function |
| <b>getmkvi</b>        | function |
| <b>gettex</b>         | function |
| <b>loadcontent</b>    | function |

|                        |          |
|------------------------|----------|
| <b>parallel</b>        | table    |
| F define               |          |
| F hassomecontent       |          |
| F next                 |          |
| F place                |          |
| F reset                |          |
| F save                 |          |
| <b>prepend</b>         | function |
| <b>raw</b>             | function |
| <b>run</b>             | function |
| <b>samplefile</b>      | function |
| <b>splitsamplefile</b> | function |
| <b>undent</b>          | function |

# builders (context)

|                        |          |                                         |                                   |
|------------------------|----------|-----------------------------------------|-----------------------------------|
| <b>checkparcontext</b> | function | F start                                 |                                   |
| <b>checksidefloat</b>  | function | F stop                                  |                                   |
| <b>hpack</b>           | table    | F tag                                   |                                   |
| F qualityactions       |          |                                         | <b>parpassidentifier</b> function |
| F report               |          |                                         | <b>passnames</b> data table       |
| F show                 |          |                                         | <b>profiling</b> table            |
| <b>kernel</b>          | table    | F add                                   |                                   |
| F collapsing           |          | F adddrivers                            |                                   |
| F hyphenating          |          | F boxlinehandler                        |                                   |
| F kerning              |          | F distance                              |                                   |
| F ligaturing           |          | F fixedprofile                          |                                   |
| F mlisttohlist         |          | F get                                   |                                   |
| <b>paragraphs</b>      | table    | F getrivers                             |                                   |
| T constructors         |          | F list                                  |                                   |
| N attribute            |          | F pagehandler                           |                                   |
| F define               |          | F pagelinehandler                       |                                   |
| F disable              |          | F profilebox                            |                                   |
| F enable               |          | F profilelines                          |                                   |
| F handler              |          | F set                                   |                                   |
| T methods              |          | F setlines                              |                                   |
| F basic                |          | <b>registerpaspasscallback</b> function |                                   |
| F default              |          | <b>vpack</b> table                      |                                   |
| F hpack                |          | F qualityactions                        |                                   |
| F oneline              |          | F report                                |                                   |
| D names                |          | <b>vspacing</b> table                   |                                   |
| D numbers              |          | D categories                            |                                   |
| F set                  |          | F checkstrutdepth                       |                                   |
| F start                |          | F collapsevbox                          |                                   |
| F stop                 |          | T data                                  |                                   |
| T expansion            |          | D map                                   |                                   |
| F trace                |          | D skip                                  |                                   |
| F kepttogether         |          | D snapmethods                           |                                   |
| F registertogether     |          | F definesnapmethod                      |                                   |
| T solutions            |          | B fixed                                 |                                   |
| T splitters            |          | F flushcontributions                    |                                   |
| F define               |          | F forcepageflush                        |                                   |
| F optimize             |          | F forcestrutdepth                       |                                   |
| T registered           |          | F getnofpreviouslines                   |                                   |
| F reset                |          | F inject                                |                                   |
| F set                  |          | F injectdisable                         |                                   |
| F split                |          | F injectnowhite                         |                                   |

F injectpenalty

F injectskip

F interceptsamepagecontent

F pagehandler

F pageoverflow

F popatsame

F pushatsame

F setmap

F setskip

F snapbox

F synchronizepage

F tocategories

F tocategory

F vboxhandler

## byte**map** (context)

|                     |          |
|---------------------|----------|
| <b>bounds</b>       | function |
| <b>clip</b>         | function |
| <b>copy</b>         | function |
| <b>dimensions</b>   | function |
| <b>domain</b>       | function |
| <b>downgrade</b>    | function |
| <b>downsample</b>   | function |
| <b>fillgray</b>     | function |
| <b>fillrgb</b>      | function |
| <b>filter</b>       | function |
| <b>getbyte</b>      | function |
| <b>getbytes</b>     | function |
| <b>getluminance</b> | function |
| <b>getvalue</b>     | function |
| <b>hasbytegray</b>  | function |
| <b>hasbyterange</b> | function |

|                   |          |
|-------------------|----------|
| <b>hasbytergb</b> | function |
| <b>new</b>        | function |
| <b>newdomain</b>  | function |
| <b>overlay</b>    | function |
| <b>process</b>    | function |
| <b>reduce</b>     | function |
| <b>reset</b>      | function |
| <b>setgray</b>    | function |
| <b>setgrayadd</b> | function |
| <b>setgraymin</b> | function |
| <b>setrgb</b>     | function |
| <b>slicegray</b>  | function |
| <b>slicerange</b> | function |
| <b>slicergb</b>   | function |
| <b>totable</b>    | function |

## cache (context)

|                             |          |                             |          |
|-----------------------------|----------|-----------------------------|----------|
| <b>base</b>                 | string   | <b>loadcontent</b>          | function |
| <b>collapsecontent</b>      | function | <b>loaddata</b>             | function |
| <b>configfiles</b>          | function | <b>more</b>                 | string   |
| <b>contentstate</b>         | function | <b>savecontent</b>          | function |
| <b>defaults</b>             | table    | <b>savedata</b>             | function |
| <b>getfirstreadablefile</b> | function | <b>setfirstwritablefile</b> | function |
| <b>getreadablepaths</b>     | function | <b>setluanames</b>          | function |
| <b>getwritablepath</b>      | function | <b>treehash</b>             | function |
| <b>hashed</b>               | function | <b>usedpaths</b>            | function |
| <b>is_writable</b>          | function |                             |          |

## callback (basic tex)

|                          |          |                   |                     |
|--------------------------|----------|-------------------|---------------------|
| <b>find</b>              | function | <b>known</b>      | function            |
| <b>getcontracts</b>      | function | <b>list</b>       | function            |
| <b>getcontractvalues</b> | function | <b>names</b>      | function            |
| <b>getindex</b>          | function | <b>register</b>   | overloaded function |
| <b>getoptions</b>        | function | <b>setoptions</b> | function            |
| <b>getoptionvalues</b>   | function | <b>setstate</b>   | function            |
| <b>getstate</b>          | function | <b>testonly</b>   | function            |
| <b>getstatevalues</b>    | function | <b>usage</b>      | function            |

# callbacks (context)

|                       |          |                           |
|-----------------------|----------|---------------------------|
| <b>freeze</b>         | function | N linebreak_check         |
| <b>functions</b>      | table    | N linebreak_quality       |
| F start_page_number   |          | N local_box               |
| F stop_page_number    |          | N make_extensible         |
| <b>getcomment</b>     | function | N math_rule               |
| <b>getstates</b>      | function | N missing_character       |
| <b>known</b>          | function | N mlist_to_hlist          |
| <b>list</b>           | table    | N open_data_file          |
| N alignment           |          | N packed_vbox             |
| N append_adjust       |          | N page_boundary           |
| N append_line         |          | N paragraph_context       |
| N append_migrate      |          | N paragraph_pass          |
| N append_to_vlist     |          | N post_linebreak          |
| N balance_boundary    |          | N pre_dump                |
| N balance_check       |          | N pre_linebreak           |
| N balance_insert      |          | N pre_output              |
| N begin_paragraph     |          | N process_character       |
| N buildpage           |          | N process_jobname         |
| N define_font         |          | N quality_font            |
| N find_format_file    |          | N register_extensible     |
| N find_log_file       |          | N show_build              |
| N get_attribute       |          | N show_error_message      |
| N get_math_dictionary |          | N show_loners             |
| N get_noad_class      |          | N show_lua_call           |
| N glyph_run           |          | N show_vsplit             |
| N handle_overload     |          | N show_warning_message    |
| N handle_uinsert      |          | N show_whatsit            |
| N handle_uleader      |          | N start_file              |
| N hpack               |          | N start_run               |
| N hpack_quality       |          | N stop_file               |
| N hyphenate           |          | N stop_run                |
| N insert_boundary     |          | N tail_append             |
| N insert_check_split  |          | N test_only               |
| N insert_distance     |          | N trace_memory            |
| N insert_par          |          | N vpack                   |
| N intercept_lua_error |          | N vpack_quality           |
| N intercept_tex_error |          | N wrapup_run              |
| N italic_correction   |          | <b>names</b> table        |
| N kerning             |          | <b>optimizer</b> function |
| N ligaturing          |          | <b>optionbits</b> table   |
| N linebreak           |          | <b>optioncodes</b> table  |

|                   |          |               |
|-------------------|----------|---------------|
| N direct          |          | N frozen      |
| N trace           |          | N fundamental |
| <b>pop</b>        | function | N private     |
| <b>push</b>       | function | N selective   |
| <b>register</b>   | function | N set         |
| <b>report</b>     | function | N touched     |
| <b>statebits</b>  | table    | N tracing     |
| <b>statecodes</b> | table    |               |
| N disabled        |          |               |



## catcodes (context)

**names** data table

**numbers** data table

**register** function

# characters (context)

|                          |            |                       |            |
|--------------------------|------------|-----------------------|------------|
| <b>addcharacternames</b> | function   | T after_postscript    |            |
| <b>adobename</b>         | function   | T after_subscript     |            |
| <b>bidirectional</b>     | data table | T anudatta            |            |
| <b>blockrange</b>        | function   | T before_half         |            |
| <b>blocks</b>            | data table | T before_main         |            |
| <b>can_have_space</b>    | data table | T before_postscript   |            |
| <b>casemap</b>           | table      | T before_subscript    |            |
| <b>categories</b>        | data table | T below_mark          |            |
| <b>category</b>          | function   | T consonant           |            |
| <b>categorytags</b>      | data table | T dependent_vowel     |            |
| <b>classifiers</b>       | table      | T halant              |            |
| <b>collapsed</b>         | data table | T independent_vowel   |            |
| <b>combined</b>          | data table | T nukta               |            |
| <b>contextname</b>       | function   | T post_mark           |            |
| <b>csletters</b>         | data table | T pre_mark            |            |
| <b>data</b>              | data table | T ra                  |            |
| <b>decomposed</b>        | data table | T stress_tone_mark    |            |
| <b>description</b>       | function   | T twopart_mark        |            |
| <b>descriptions</b>      | data table | T vowel_modifier      |            |
| <b>detailtags</b>        | data table | <b>initialize</b>     | function   |
| <b>directions</b>        | table      | <b>is_character</b>   | data table |
| <b>emoji</b>             | table      | <b>is_command</b>     | data table |
| F compact                |            | <b>is_hyphenator</b>  | data table |
| F known                  |            | <b>is_letter</b>      | data table |
| F resolve                |            | <b>is_mark</b>        | data table |
| <b>emojized</b>          | function   | <b>is_nothing</b>     | data table |
| <b>encodes</b>           | data table | <b>is_punctuation</b> | data table |
| <b>entities</b>          | data table | <b>is_spacing</b>     | data table |
| <b>fallbacks</b>         | data table | <b>is_symbol</b>      | data table |
| <b>filters</b>           | data table | <b>lcchars</b>        | data table |
| <b>fractions</b>         | table      | <b>lccode</b>         | function   |
| <b>fromnumber</b>        | function   | <b>lccodes</b>        | data table |
| <b>fschars</b>           | data table | <b>lettered</b>       | function   |
| <b>fscodes</b>           | data table | <b>lhash</b>          | data table |
| <b>getrange</b>          | function   | <b>linebreaks</b>     | data table |
| <b>graphemes</b>         | data table | <b>lower</b>          | function   |
| <b>hangul</b>            | data table | <b>mathlists</b>      | data table |
| <b>indicgroups</b>       | table      | <b>mirrors</b>        | table      |
| T above_mark             |            | <b>otfscripts</b>     | data table |
| T after_half             |            | <b>packdata</b>       | table      |
| T after_main             |            | N luasize             |            |

|                              |            |                     |            |
|------------------------------|------------|---------------------|------------|
| N newsize                    |            | <b>subscripts</b>   | table      |
| N oldsize                    |            | <b>superscripts</b> | table      |
| N version                    |            | <b>synonyms</b>     | data table |
| <b>ranges</b>                | data table | <b>tex</b>          | table      |
| <b>registerentities</b>      | function   | D accentmapping     |            |
| <b>remap_hangul_syllable</b> | function   | D commandmapping    |            |
| <b>scripthash</b>            | table      | F defineaccents     |            |
| <b>setactivecatcodes</b>     | function   | F safechar          |            |
| <b>setlettercatcodes</b>     | function   | F strtoutf          |            |
| <b>setothercatcodes</b>      | function   | F strtoutfpattern   |            |
| <b>shape</b>                 | function   | F toutf             |            |
| <b>shaped</b>                | function   | F toutfpattern      |            |
| <b>shash</b>                 | data table | <b>textclasses</b>  | table      |
| <b>shchars</b>               | data table | <b>tonumber</b>     | function   |
| <b>shcodes</b>               | data table | <b>ucchars</b>      | data table |
| <b>showstring</b>            | function   | <b>uccode</b>       | function   |
| <b>specialchars</b>          | data table | <b>uccodes</b>      | data table |
| <b>specials</b>              | data table | <b>uhash</b>        | data table |
| <b>splits</b>                | table      | <b>unicodechar</b>  | function   |
| T char                       |            | <b>upper</b>        | function   |
| T compat                     |            | <b>verbosemarks</b> | data table |

# chemistry (context)

|                       |            |            |
|-----------------------|------------|------------|
| <b>component</b>      | function   | S line     |
| <b>define</b>         | function   | S lrn      |
| <b>definitions</b>    | data table | S lrt      |
| <b>format</b>         | string     | S mark     |
| <b>inlinechemical</b> | function   | S mir      |
| <b>instance</b>       | string     | S mov      |
| <b>method</b>         | string     | S move     |
| <b>molecule</b>       | function   | S number   |
| <b>moleculeparser</b> | userdata   | S oe       |
| <b>start</b>          | function   | S off      |
| <b>stop</b>           | function   | S r        |
| <b>syntax</b>         | table      | S rb       |
| T alkyl               |            | S rbd      |
| S category            |            | S rd       |
| T keys                |            | S rdb      |
| S ad                  |            | S rdd      |
| S adj                 |            | S rh       |
| S arrow               |            | S rm       |
| S au                  |            | S rn       |
| S b                   |            | S rot      |
| S bb                  |            | S rrn      |
| S bd                  |            | S rrt      |
| S bw                  |            | S rt       |
| S c                   |            | S rz       |
| S cc                  |            | S sb       |
| S ccd                 |            | S sd       |
| S cd                  |            | S sr       |
| S cz                  |            | S sub      |
| S dash                |            | S tb       |
| S db                  |            | S z        |
| S diff                |            | S zln      |
| S dir                 |            | S zlt      |
| S dr                  |            | S zn       |
| S ed                  |            | S zrn      |
| S ep                  |            | S zrt      |
| S es                  |            | S zt       |
| S et                  |            | S keyword  |
| S hb                  |            | N max      |
| S label               |            | T boat     |
| S ldb                 |            | S category |
| S ldd                 |            | T keys     |

|          |                |
|----------|----------------|
| S adj    | S rt           |
| S arrow  | S rz           |
| S b      | S sb           |
| S bb     | S sr           |
| S c      | S sub          |
| S cc     | S z            |
| S ccd    | S zln          |
| S cd     | S zlt          |
| S dash   | S zn           |
| S diff   | S zrn          |
| S eb     | S zrt          |
| S label  | S zt           |
| S line   | S keyword      |
| S lr     | N max          |
| S lrb    | T carbon       |
| S lrbd   | S category     |
| S lrd    | S keyword      |
| S lrh    | N max          |
| S lrn    | T chair        |
| S lrt    | S category     |
| S lrz    | S keyword      |
| S lsr    | N max          |
| S lsub   | T chem         |
| S mark   | N arguments    |
| S mov    | F direct       |
| S move   | T closecomplex |
| S number | S direct       |
| S off    | T eight        |
| S r      | S category     |
| S rb     | T keys         |
| S rbd    | S ad           |
| S rd     | S adj          |
| S rh     | S arrow        |
| S rn     | S au           |
| S rr     | S b            |
| S rrb    | S c            |
| S rrbd   | S cc           |
| S rrd    | S ccd          |
| S rrh    | S cd           |
| S rrn    | S crz          |
| S rrt    | S dash         |
| S rrz    | S db           |
| S rsr    | S diff         |
| S rsub   | S dr           |

|          |                  |
|----------|------------------|
| S eb     | S rz             |
| S ed     | S s              |
| S er     | S sb             |
| S hb     | S sr             |
| S label  | S ss             |
| S line   | S sub            |
| S lr     | S z              |
| S lrb    | S zln            |
| S lrbd   | S zlt            |
| S lrd    | S zn             |
| S lrh    | S zrn            |
| S lrn    | S zrt            |
| S lrt    | S zt             |
| S lrz    | S keyword        |
| S lsr    | N max            |
| S lsub   | T equals         |
| S mark   | S direct         |
| S mid    | T equilibrium    |
| S mids   | N arguments      |
| S midz   | F direct         |
| S mir    | T five           |
| S mov    | S category       |
| S move   | S keyword        |
| S number | N max            |
| S off    | T fivefront      |
| S r      | S category       |
| S rb     | S keyword        |
| S rbd    | N max            |
| S rd     | T four           |
| S rh     | S category       |
| S rm     | S keyword        |
| S rn     | N max            |
| S rot    | T gives          |
| S rr     | N arguments      |
| S rrb    | F direct         |
| S rrbd   | T mesomeric      |
| S rrd    | N arguments      |
| S rrh    | F direct         |
| S rrn    | T minus          |
| S rrt    | S direct         |
| S rrz    | T mp             |
| S rsr    | N arguments      |
| S rsub   | F direct         |
| S rt     | T newmaneclipsed |

|                 |                 |          |
|-----------------|-----------------|----------|
| S category      | S direct        |          |
| S keyword       | T restore       |          |
| N max           | S direct        |          |
| T newmanstagger | T save          |          |
| S category      | S direct        |          |
| S keyword       | T seven         |          |
| N max           | S category      |          |
| T nine          | S keyword       |          |
| S category      | N max           |          |
| S keyword       | T six           |          |
| N max           | S category      |          |
| T one           | S keyword       |          |
| S category      | N max           |          |
| S keyword       | T sixfront      |          |
| N max           | S category      |          |
| T opencomplex   | S keyword       |          |
| S direct        | N max           |          |
| T pb            | T space         |          |
| S direct        | S direct        |          |
| T pe            | T three         |          |
| S direct        | S category      |          |
| T plus          | S keyword       |          |
| S direct        | N max           |          |
| T reset         | <b>undefine</b> | function |

## containers (context)

**cleannname** function  
**content** function  
**define** function  
**is\_usable** function

**is\_valid** function  
**read** function  
**usecache** boolean  
**write** function



## converters (context)

|                            |          |                            |            |
|----------------------------|----------|----------------------------|------------|
| <b>A</b>                   | function | <b>cn-a</b>                | function   |
| <b>A:es</b>                | function | <b>cn-c</b>                | function   |
| <b>A:ru</b>                | function | <b>cn-d</b>                | function   |
| <b>A:sl</b>                | function | <b>convert</b>             | function   |
| <b>AK</b>                  | function | <b>decimals</b>            | function   |
| <b>Alphabetic</b>          | function | <b>define</b>              | function   |
| <b>Alphabetic:es</b>       | function | <b>gregorian_to_hebrew</b> | function   |
| <b>Alphabetic:ru</b>       | function | <b>gregorian_to_jalali</b> | function   |
| <b>Alphabetic:sl</b>       | function | <b>hebrewnumerals</b>      | function   |
| <b>Character</b>           | function | <b>helpers</b>             | table      |
| <b>Characters</b>          | function | <b>hexstringtonumber</b>   | function   |
| <b>I</b>                   | function | <b>i</b>                   | function   |
| <b>KA</b>                  | function | <b>isleapyear</b>          | function   |
| <b>KR</b>                  | function | <b>jalali_to_gregorian</b> | function   |
| <b>R</b>                   | function | <b>lchexnumber</b>         | function   |
| <b>RK</b>                  | function | <b>lchexnumbers</b>        | function   |
| <b>Romannumerals</b>       | function | <b>leapyear</b>            | function   |
| <b>Russiannumerals</b>     | function | <b>lower</b>               | function   |
| <b>Sloveniannumerals</b>   | function | <b>max</b>                 | function   |
| <b>Spanishnumerals</b>     | function | <b>maxchrs</b>             | function   |
| <b>Word</b>                | function | <b>nicenumber</b>          | function   |
| <b>Words</b>               | function | <b>nofdays</b>             | function   |
| <b>a</b>                   | function | <b>number</b>              | function   |
| <b>a:es</b>                | function | <b>numbers</b>             | function   |
| <b>a:ru</b>                | function | <b>octnumber</b>           | function   |
| <b>a:sl</b>                | function | <b>octstringtonumber</b>   | function   |
| <b>abjadnodotnumerals</b>  | function | <b>ordinal</b>             | function   |
| <b>abjadnumerals</b>       | function | <b>r</b>                   | function   |
| <b>alphabetic</b>          | function | <b>romannumerals</b>       | function   |
| <b>alphabetic:es</b>       | function | <b>russiannumerals</b>     | function   |
| <b>alphabetic:ru</b>       | function | <b>sequences</b>           | data table |
| <b>alphabetic:sl</b>       | function | <b>settime</b>             | function   |
| <b>character</b>           | function | <b>sloveniannumerals</b>   | function   |
| <b>characters</b>          | function | <b>spaceddigits</b>        | function   |
| <b>chineseallnumerals</b>  | function | <b>spanishnumerals</b>     | function   |
| <b>chinesecapnumerals</b>  | function | <b>textime</b>             | function   |
| <b>chinesedatenumerals</b> | function | <b>toabjad</b>             | function   |
| <b>chinesenumerals</b>     | function | <b>tochinese</b>           | function   |
| <b>chr</b>                 | function | <b>tohebrew</b>            | function   |
| <b>chrs</b>                | function | <b>toroman</b>             | function   |
| <b>cn</b>                  | function | <b>totime</b>              | function   |

**uchexnumber**  
**uchexnumbers**  
**upper**  
**verbose**

D data

function  
function  
function  
table

F translate

**weekday**  
**word**  
**words**

function  
function  
function

## coroutine (basic lua)

|                    |          |
|--------------------|----------|
| <b>close</b>       | function |
| <b>create</b>      | function |
| <b>isyieldable</b> | function |
| <b>resume</b>      | function |

|                |          |
|----------------|----------|
| <b>running</b> | function |
| <b>status</b>  | function |
| <b>wrap</b>    | function |
| <b>yield</b>   | function |

# ctxrunner (context)

|                     |          |                  |            |
|---------------------|----------|------------------|------------|
| <b>load</b>         | function | <b>prepfiles</b> | data table |
| <b>preparedfile</b> | function | <b>resolve</b>   | function   |

## debug (basic lua)

**getinfo**    function

**sethook**    function

**traceback**   function

## dir (context)

|                       |          |
|-----------------------|----------|
| <b>collectpattern</b> | function |
| <b>current</b>        | function |
| <b>expandlink</b>     | function |
| <b>expandname</b>     | function |
| <b>found</b>          | function |
| <b>glob</b>           | function |
| <b>globdirs</b>       | function |

|                    |          |
|--------------------|----------|
| <b>globfiles</b>   | function |
| <b>globpattern</b> | function |
| <b>ls</b>          | function |
| <b>makedirs</b>    | function |
| <b>mkdirs</b>      | function |
| <b>pop</b>         | function |
| <b>push</b>        | function |

## directives (context)

|                |            |
|----------------|------------|
| <b>data</b>    | data table |
| <b>default</b> | function   |
| <b>disable</b> | function   |
| <b>done</b>    | table      |
| <b>enable</b>  | function   |
| <b>list</b>    | function   |
| <b>name</b>    | string     |

|                 |          |
|-----------------|----------|
| <b>register</b> | function |
| <b>report</b>   | function |
| <b>reset</b>    | function |
| <b>show</b>     | function |
| <b>value</b>    | function |
| <b>values</b>   | function |

# document (context)

|                            |            |                              |            |
|----------------------------|------------|------------------------------|------------|
| <b>arguments</b>           | data table | <b>pdf_cleanup_procsets</b>  | function   |
| <b>files</b>               | table      | <b>pdf_identify_content</b>  | function   |
| <b>functions</b>           | table      | <b>pdf_serialize_content</b> | function   |
| <b>getargument</b>         | function   | <b>pdf_strip_content</b>     | function   |
| <b>getfilename</b>         | function   | <b>pdf_strip_page</b>        | function   |
| <b>options</b>             | table      | <b>setargument</b>           | function   |
| T cmdline                  |            | <b>setcommandline</b>        | function   |
| D environments             |            | <b>setdefaultargument</b>    | function   |
| D modes                    |            | <b>setenvironments</b>       | function   |
| D modules                  |            | <b>setfilename</b>           | function   |
| T ctxfile                  |            | <b>setfilenames</b>          | function   |
| D environments             |            | <b>setmodes</b>              | function   |
| D modes                    |            | <b>setmodules</b>            | function   |
| D modules                  |            | <b>variables</b>             | data table |
| <b>pdf_cleanup_cidsets</b> | function   |                              |            |



# documentdata (context)

**variable** function

# drivers (context)

|                         |          |                   |
|-------------------------|----------|-------------------|
| <b>applyanchor</b>      | function | F rule            |
| <b>applyorientation</b> | function | F simplerule      |
| <b>convert</b>          | function | F specialrule     |
| <b>converters</b>       | table    | F updatefontstate |
| F analyze               |          | T Whatsit         |
| F engine                |          | S name            |
| F lmtx                  |          | T json            |
| F resyncbox             |          | T actions         |
| <b>enable</b>           | function | F cleanup         |
| <b>finalize</b>         | function | F convert         |
| <b>gethpos</b>          | function | F finalize        |
| <b>gethvrpos</b>        | function | F initialize      |
| <b>getpos</b>           | function | F outputfilename  |
| <b>getrpos</b>          | function | F prepare         |
| <b>getvpos</b>          | function | F wrapup          |
| <b>helpers</b>          | table    | T flushers        |
| <b>initialize</b>       | function | F character       |
| <b>install</b>          | function | F popleaderlevel  |
| <b>instances</b>        | table    | F pushleaderlevel |
| T empty                 |          | F rule            |
| T actions               |          | F setstate        |
| F convert               |          | F simplerule      |
| F prepare               |          | F specialrule     |
| T flushers              |          | F updatefontstate |
| F popleaderlevel        |          | T Whatsit         |
| F pushleaderlevel       |          | S name            |
| T Whatsit               |          | T lua             |
| S name                  |          | T actions         |
| T js                    |          | F cleanup         |
| T actions               |          | F convert         |
| F cleanup               |          | F finalize        |
| F convert               |          | F initialize      |
| F finalize              |          | F outputfilename  |
| F initialize            |          | F prepare         |
| F outputfilename        |          | F wrapup          |
| F prepare               |          | T flushers        |
| F wrapup                |          | F character       |
| T flushers              |          | F popleaderlevel  |
| F character             |          | F pushleaderlevel |
| F popleaderlevel        |          | F rule            |
| F pushleaderlevel       |          | F simplerule      |

|                   |                       |          |
|-------------------|-----------------------|----------|
| F specialrule     | T actions             |          |
| F updatefontstate | F cleanup             |          |
| T whatsit         | F convert             |          |
| S name            | F finalize            |          |
| T mpo             | F initialize          |          |
| T actions         | F outputfilename      |          |
| F finalize        | F prepare             |          |
| F initialize      | F wrapup              |          |
| T flushers        | T flushers            |          |
| F character       | F character           |          |
| F popleaderlevel  | F flushimage          |          |
| F pushleaderlevel | F fontchar            |          |
| F rule            | F group               |          |
| F simplerule      | F literal             |          |
| F specialrule     | F popgroup            |          |
| T whatsit         | F popleaderlevel      |          |
| S name            | F poporientation      |          |
| T mps             | F pushgroup           |          |
| T actions         | F pushleaderlevel     |          |
| F cleanup         | F pushorientation     |          |
| F convert         | F restore             |          |
| F finalize        | F rule                |          |
| F initialize      | F save                |          |
| F outputfilename  | F setmatrix           |          |
| F prepare         | F setstate            |          |
| F wrapup          | F simplerule          |          |
| T flushers        | F space               |          |
| F character       | F specialrule         |          |
| F popleaderlevel  | F startclipping       |          |
| F pushleaderlevel | F startmatrix         |          |
| F rule            | F startmirroring      |          |
| F simplerule      | F startrotation       |          |
| F specialrule     | F startscaling        |          |
| F updatefontstate | F stopclipping        |          |
| T whatsit         | F stopmatrix          |          |
| S name            | F stopmirroring       |          |
| T none            | F stoprotation        |          |
| T actions         | F stopscaling         |          |
| T flushers        | F updatefontstate     |          |
| F popleaderlevel  | T whatsit             |          |
| F pushleaderlevel | S name                |          |
| T whatsit         | <b>lmtxversion</b>    | number   |
| S name            | <b>outputfilename</b> | function |
| T pdf             | <b>report</b>         | function |

## effects (context)

|                          |          |
|--------------------------|----------|
| <b>brownianmotion</b>    | function |
| <b>browniannoise</b>     | function |
| <b>getoctaveloops</b>    | function |
| <b>getoctavemethods</b>  | function |
| <b>getoctavestatus</b>   | function |
| <b>getoctavevariants</b> | function |
| <b>newoctave</b>         | function |
| <b>octave</b>            | function |
| <b>octavebytemap</b>     | function |

|                      |          |
|----------------------|----------|
| <b>perlinmotion</b>  | function |
| <b>perlinnoise</b>   | function |
| <b>simplexangle</b>  | function |
| <b>simplexdetail</b> | function |
| <b>simplexnoise</b>  | function |
| <b>stepoctave</b>    | function |
| <b>worleybounds</b>  | function |
| <b>worleymotion</b>  | function |
| <b>worleynoise</b>   | function |

## environment (context)

|                            |            |                               |            |
|----------------------------|------------|-------------------------------|------------|
| <b>argument</b>            | function   | <b>luafilechunk</b>           | function   |
| <b>arguments</b>           | data table | <b>luatexengine</b>           | string     |
| <b>basicengines</b>        | data table | <b>luatexfunctionality</b>    | number     |
| <b>currentrun</b>          | number     | <b>luatexverbose</b>          | string     |
| <b>definefilefallback</b>  | function   | <b>luatexversion</b>          | number     |
| <b>definefilesynonym</b>   | function   | <b>maxnofruns</b>             | number     |
| <b>engineflags</b>         | data table | <b>oldhome</b>                | string     |
| <b>filename</b>            | string     | <b>originalarguments</b>      | data table |
| <b>filenames</b>           | table      | <b>outputfilename</b>         | string     |
| <b>files</b>               | table      | <b>ownbin</b>                 | string     |
| <b>filesynonyms</b>        | data table | <b>ownmain</b>                | string     |
| <b>forcedruns</b>          | number     | <b>ownname</b>                | string     |
| <b>getargument</b>         | function   | <b>ownpath</b>                | string     |
| <b>globfiles</b>           | function   | <b>prerollrun</b>             | boolean    |
| <b>homedir</b>             | string     | <b>rawarguments</b>           | data table |
| <b>initex</b>              | boolean    | <b>reconstructcommandline</b> | function   |
| <b>initexmode</b>          | boolean    | <b>relativepath</b>           | function   |
| <b>initializearguments</b> | function   | <b>setargument</b>            | function   |
| <b>initializefilenames</b> | function   | <b>splitarguments</b>         | function   |
| <b>inputfilebarename</b>   | string     | <b>suffix</b>                 | string     |
| <b>inputfilename</b>       | string     | <b>texfile</b>                | function   |
| <b>inputfilesuffix</b>     | string     | <b>texmfos</b>                | string     |
| <b>jobfilefullname</b>     | string     | <b>texos</b>                  | string     |
| <b>jobfilename</b>         | string     | <b>texroot</b>                | string     |
| <b>jobfilesuffix</b>       | string     | <b>truefilename</b>           | function   |
| <b>kindofrun</b>           | number     | <b>uselua module</b>          | function   |
| <b>lmtxmode</b>            | number     | <b>usemodules</b>             | function   |
| <b>loadexamodes</b>        | function   | <b>validengines</b>           | data table |
| <b>loadluafile</b>         | function   | <b>version</b>                | string     |
| <b>luafile</b>             | function   |                               |            |

# experiments (context)

|                |            |                 |          |
|----------------|------------|-----------------|----------|
| <b>data</b>    | data table | <b>register</b> | function |
| <b>default</b> | function   | <b>report</b>   | function |
| <b>disable</b> | function   | <b>reset</b>    | function |
| <b>enable</b>  | function   | <b>show</b>     | function |
| <b>list</b>    | function   | <b>value</b>    | function |
| <b>name</b>    | string     | <b>values</b>   | function |

# figures (context)

|                         |            |                        |            |
|-------------------------|------------|------------------------|------------|
| <b>applyratio</b>       | function   | <b>F resetdata</b>     |            |
| <b>badname</b>          | function   | <b>F resetstore</b>    |            |
| <b>bases</b>            | data table | <b>F scan</b>          |            |
| <b>bitmapimage</b>      | function   | <b>F serialize</b>     |            |
| <b>boxnumber</b>        | number     | <b>T sizes</b>         |            |
| <b>bytepalette</b>      | function   | <b>B art</b>           |            |
| <b>cachepaths</b>       | data table | <b>B bleed</b>         |            |
| <b>check</b>            | function   | <b>B crop</b>          |            |
| <b>checkers</b>         | data table | <b>B media</b>         |            |
| <b>conversions</b>      | data table | <b>B none</b>          |            |
| <b>converters</b>       | data table | <b>B trim</b>          |            |
| <b>current</b>          | function   | <b>F storedata</b>     |            |
| <b>defaultdepth</b>     | number     | <b>F totable</b>       |            |
| <b>defaultheight</b>    | number     | <b>T types</b>         |            |
| <b>defaultsearch</b>    | boolean    | <b>N jbig2</b>         |            |
| <b>defaultwidth</b>     | number     | <b>N jp2</b>           |            |
| <b>done</b>             | function   | <b>N jpg</b>           |            |
| <b>dummy</b>            | function   | <b>N none</b>          |            |
| <b>existsers</b>        | data table | <b>N pdf</b>           |            |
| <b>exists</b>           | function   | <b>N png</b>           |            |
| <b>formats</b>          | data table | <b>N stream</b>        |            |
| <b>found</b>            | data table | <b>F wrap</b>          |            |
| <b>get</b>              | function   | <b>include</b>         | function   |
| <b>getinfo</b>          | function   | <b>includers</b>       | data table |
| <b>getpdfinfo</b>       | function   | <b>initialize</b>      | function   |
| <b>getrealpage</b>      | function   | <b>loaded</b>          | data table |
| <b>guess</b>            | function   | <b>localpaths</b>      | data table |
| <b>guessfromstring</b>  | function   | <b>magics</b>          | data table |
| <b>hash</b>             | function   | <b>mergegoodies</b>    | function   |
| <b>identifiers</b>      | data table | <b>nofmissing</b>      | number     |
| <b>identify</b>         | function   | <b>nofprocessed</b>    | number     |
| <b>images</b>           | table      | <b>order</b>           | data table |
| <b>F check</b>          |            | <b>palettes</b>        | table      |
| <b>F checksize</b>      |            | <b>paths</b>           | data table |
| <b>F copy</b>           |            | <b>patterns</b>        | data table |
| <b>F create</b>         |            | <b>pop</b>             | function   |
| <b>F embed</b>          |            | <b>preferquality</b>   | boolean    |
| <b>F embedfromstore</b> |            | <b>programs</b>        | data table |
| <b>F fetchdata</b>      |            | <b>push</b>            | function   |
| <b>T keys</b>           |            | <b>qrcode</b>          | function   |
| <b>F print</b>          |            | <b>registerpattern</b> | function   |

|                         |            |                      |            |
|-------------------------|------------|----------------------|------------|
| <b>registerresource</b> | function   | <b>setconversion</b> | function   |
| <b>registersuffix</b>   | function   | <b>setlookups</b>    | function   |
| <b>remappers</b>        | data table | <b>setorder</b>      | function   |
| <b>resources</b>        | data table | <b>setpaths</b>      | function   |
| <b>scale</b>            | function   | <b>suffixes</b>      | data table |
| <b>set</b>              | function   | <b>used</b>          | data table |



## file (basic lua)

|                          |          |
|--------------------------|----------|
| <b>addsuffix</b>         | function |
| <b>basename</b>          | function |
| <b>checksum</b>          | function |
| <b>collapsepath</b>      | function |
| <b>copy</b>              | function |
| <b>dirname</b>           | function |
| <b>expandname</b>        | function |
| <b>extname</b>           | function |
| <b>is_qualified_path</b> | function |
| <b>is_readable</b>       | function |
| <b>is_rootbased_path</b> | function |
| <b>is_writable</b>       | function |
| <b>isreadable</b>        | function |
| <b>iswritable</b>        | function |
| <b>join</b>              | function |
| <b>joinpath</b>          | function |
| <b>loadchecksum</b>      | function |
| <b>nameonly</b>          | function |
| <b>nametotable</b>       | function |
| <b>needs Updating</b>    | function |

|                      |          |
|----------------------|----------|
| <b>needsupdating</b> | function |
| <b>pathpart</b>      | function |
| <b>readdata</b>      | function |
| <b>removesuffix</b>  | function |
| <b>replacesuffix</b> | function |
| <b>reslash</b>       | function |
| <b>robustname</b>    | function |
| <b>savechecksum</b>  | function |
| <b>savedata</b>      | function |
| <b>size</b>          | function |
| <b>splitbase</b>     | function |
| <b>splitname</b>     | function |
| <b>splitpath</b>     | function |
| <b>suffix</b>        | function |
| <b>suffixes</b>      | function |
| <b>suffixesonly</b>  | function |
| <b>suffixonly</b>    | function |
| <b>syncmtimes</b>    | function |
| <b>withinbase</b>    | function |

## fio (extra lua)

|                          |          |
|--------------------------|----------|
| <b>getposition</b>       | function |
| <b>read2dot14</b>        | function |
| <b>readbytes</b>         | function |
| <b>readbytetable</b>     | function |
| <b>readcardinal1</b>     | function |
| <b>readcardinal1le</b>   | function |
| <b>readcardinal2</b>     | function |
| <b>readcardinal2le</b>   | function |
| <b>readcardinal3</b>     | function |
| <b>readcardinal3le</b>   | function |
| <b>readcardinal4</b>     | function |
| <b>readcardinal4le</b>   | function |
| <b>readcardinaltable</b> | function |
| <b>readcline</b>         | function |
| <b>readcstring</b>       | function |
| <b>readdouble</b>        | function |
| <b>readdoublele</b>      | function |
| <b>readfixed2</b>        | function |
| <b>readfixed4</b>        | function |
| <b>readfloat</b>         | function |
| <b>readfloatle</b>       | function |
| <b>readinteger1</b>      | function |

|                         |          |
|-------------------------|----------|
| <b>readinteger1le</b>   | function |
| <b>readinteger2</b>     | function |
| <b>readinteger2le</b>   | function |
| <b>readinteger3</b>     | function |
| <b>readinteger3le</b>   | function |
| <b>readinteger4</b>     | function |
| <b>readinteger4le</b>   | function |
| <b>readintegertable</b> | function |
| <b>setposition</b>      | function |
| <b>skipposition</b>     | function |
| <b>writecardinal1</b>   | function |
| <b>writecardinal1le</b> | function |
| <b>writecardinal2</b>   | function |
| <b>writecardinal2le</b> | function |
| <b>writecardinal3</b>   | function |
| <b>writecardinal3le</b> | function |
| <b>writecardinal4</b>   | function |
| <b>writecardinal4le</b> | function |
| <b>writedouble</b>      | function |
| <b>writedoublele</b>    | function |
| <b>writefloat</b>       | function |
| <b>writefloatle</b>     | function |

# floats (context)

|                         |          |                   |          |
|-------------------------|----------|-------------------|----------|
| <b>analysemethod</b>    | function | <b>nofstacked</b> | function |
| <b>checkedpagefloat</b> | function | <b>pop</b>        | function |
| <b>collect</b>          | function | <b>push</b>       | function |
| <b>consult</b>          | function | <b>resave</b>     | function |
| <b>flush</b>            | function | <b>save</b>       | function |
| <b>getvariable</b>      | function | <b>stacked</b>    | function |
| <b>hasstacked</b>       | function |                   |          |

## font (basic tex)

|                               |          |                               |          |
|-------------------------------|----------|-------------------------------|----------|
| <b>addcharacters</b>          | function | <b>getmathspec</b>            | function |
| <b>addquality</b>             | function | <b>getparameterfields</b>     | function |
| <b>current</b>                | function | <b>gettextcharacterfields</b> | function |
| <b>define</b>                 | function | <b>id</b>                     | function |
| <b>getfont</b>                | function | <b>max</b>                    | function |
| <b>getfontdimen</b>           | function | <b>nextid</b>                 | function |
| <b>getfontfields</b>          | function | <b>setfont</b>                | function |
| <b>getfontspec</b>            | function | <b>setfontdimen</b>           | function |
| <b>getmathcharacterfields</b> | function | <b>xscaled</b>                | function |
| <b>getmathindex</b>           | function | <b>yscaled</b>                | function |

# fonts (context)

|                           |       |                          |
|---------------------------|-------|--------------------------|
| <b>analyzers</b>          | table | F beforepassingfonttotex |
| D features                |       | D cache                  |
| D initializers            |       | F calculatescale         |
| D methods                 |       | F checkedfeatures        |
| F setstate                |       | F checkedfilename        |
| D states                  |       | F collectprocessors      |
| B useunicodemarks         |       | F currentfonthasfeature  |
| <b>callbacks</b>          | table | D designsizes            |
| F compile                 |       | F enhanceparameters      |
| F devirtualize            |       | T enhancers              |
| <b>checkers</b>           | table | T afm                    |
| F getmissing              |       | F apply                  |
| D mapping                 |       | F patch                  |
| F missing                 |       | T patches                |
| F placeholder             |       | F register               |
| <b>cid</b>                | table | F report                 |
| F getmap                  |       | F register               |
| F loadfile                |       | F report                 |
| <b>collections</b>        | table | T otf                    |
| F checkenabled            |       | F addfeature             |
| F clonevector             |       | F apply                  |
| F define                  |       | F enhance                |
| T definitions             |       | F patch                  |
| F direct                  |       | T patches                |
| F found                   |       | F register               |
| F prepare                 |       | F report                 |
| F process                 |       | F register               |
| F register                |       | F report                 |
| F registermain            |       | T tfm                    |
| F report                  |       | F apply                  |
| F reset                   |       | F patch                  |
| T rscales                 |       | T patches                |
| T vectors                 |       | F register               |
| <b>constructors</b>       | table | F report                 |
| F addcharacters           |       | F register               |
| F addcoreunicodes         |       | F report                 |
| F aftercopyingcharacters  |       | N factor                 |
| F applyfinalizers         |       | T features               |
| F applymanipulators       |       | D afm                    |
| F assignmathparameters    |       | D otf                    |
| F beforecopyingcharacters |       | D tfm                    |

|                            |                                                     |
|----------------------------|-----------------------------------------------------|
| F finalize                 | S defaultbasealternate                              |
| F getfeatureaction         | S defaultnodealternate                              |
| F getmathparameter         | D features                                          |
| F getprivate               | F featuresinitializer                               |
| T handlers                 | F featuresprocessor                                 |
| T afm                      | T fixers                                            |
| B autoprefixed             | F getactualtext                                     |
| D cache                    | F getalternate                                      |
| D features                 | F getgsub                                           |
| F getkern                  | F getkern                                           |
| T helpdata                 | F getmultiple                                       |
| D bothkerned               | F getstreamhash                                     |
| D leftkerned               | F getsubstitution                                   |
| D ligatures                | T handlers                                          |
| D rightkerned              | F ctx_camelcasing                                   |
| D texligatures             | F devanagari_remove_joiners                         |
| F load                     | F devanagari_reorder_matras                         |
| T readers                  | F devanagari_reorder_pre_base_reordering_consonants |
| F getinfo                  | F devanagari_reorder_reph                           |
| F loadfont                 | F gpos_context                                      |
| F loadshapes               | F gpos_contextchain                                 |
| F setfeatures              | F gpos_cursive                                      |
| B syncspace                | F gpos_mark2base                                    |
| N version                  | F gpos_mark2ligature                                |
| T otf                      | F gpos_mark2mark                                    |
| F addfeature               | F gpos_pair                                         |
| F basemodeinitializer      | F gpos_single                                       |
| D cache                    | F gsub_alternate                                    |
| F collectlookups           | F gsub_context                                      |
| T coverup                  | F gsub_contextchain                                 |
| T actions                  | F gsub_ligature                                     |
| F alternate                | F gsub_multiple                                     |
| F chainposition            | F gsub_reversecontextchain                          |
| F chainsubstitution        | F gsub_single                                       |
| F kern                     | F trigger_space_kerns                               |
| F ligature                 | T helpers                                           |
| F multiple                 | F blockligatures                                    |
| F pair                     | F pardirstate                                       |
| F single                   | F txtfirststate                                     |
| F substitution             | F load                                              |
| F register                 | F loadoutlinedata                                   |
| S stepkey                  | F loadstreamdata                                    |
| F dataset                  | F modeinitializer                                   |
| F datasetpositionprocessor | D mpscache                                          |

|                     |                               |
|---------------------|-------------------------------|
| D pdfcache          | F gsub                        |
| F plugininitializer | T gsubhandlers                |
| F pluginprocessor   | F alternate                   |
| T plugins           | F chainedcontext              |
| T text              | F context                     |
| D pngcache          | F extension                   |
| B pngenabled        | F ligature                    |
| T readers           | F multiple                    |
| F addunicodetable   | F reversechainedcontextsingle |
| F avar              | F single                      |
| F cbdt              | F gvar                        |
| F cblc              | F head                        |
| F cff               | T helpers                     |
| F cff2              | F axistofactors               |
| T cffencoder        | F cleanname                   |
| F checkhash         | F getaxisscale                |
| F cmap              | F getfactors                  |
| F colr              | F getscales                   |
| F compact           | F gotodatatable               |
| F condense          | F normalizedaxis              |
| F cpal              | T normalizedaxisishash        |
| F dsig              | F readdevice                  |
| F ebdt              | F readvariationdata           |
| F eblc              | F reportskippedtable          |
| F ebsc              | F setvariabledata             |
| F expand            | F hhea                        |
| F extend            | F hmtx                        |
| F fvar              | F hvar                        |
| F gasp              | F kern                        |
| F gdef              | F loadfont                    |
| F getcomponents     | F loadshapes                  |
| F getinfo           | F loca                        |
| F glyf              | F math                        |
| F gpos              | F maxp                        |
| T gposhandlers      | F mvar                        |
| F chainedcontext    | F name                        |
| F context           | F os/2                        |
| F cursive           | F pack                        |
| F extension         | F parsecharstrings            |
| F marktobase        | F post                        |
| F marktoligature    | F potracetocff                |
| F marktomark        | F registerextender            |
| F pair              | F rehash                      |
| F single            | F sbix                        |

|                     |                     |
|---------------------|---------------------|
| F stat              | F skipshort         |
| T streamreader      | F writebyte         |
| F close             | F writecardinal1    |
| F getposition       | F writecardinal1le  |
| F getsize           | F writecardinal2    |
| F look              | F writecardinal2le  |
| F open              | F writecardinal3    |
| F read2dot14        | F writecardinal3le  |
| F readbyte          | F writecardinal4    |
| F readbytes         | F writecardinal4le  |
| F readbytetable     | F writeinteger1     |
| F readcardinal      | F writeinteger1le   |
| F readcardinal1     | F writeinteger2     |
| F readcardinal1le   | F writeinteger2le   |
| F readcardinal2     | F writeinteger3     |
| F readcardinal2le   | F writeinteger3le   |
| F readcardinal3     | F writeinteger4     |
| F readcardinal3le   | F writeinteger4le   |
| F readcardinal4     | F writestring       |
| F readcardinal4le   | F stripredundant    |
| F readcardinaltable | F svg               |
| F readchar          | N tableversion      |
| F readcline         | F unifymissing      |
| F readcstring       | F unpack            |
| F readfixed2        | F vhea              |
| F readfixed4        | F vmtx              |
| F readinteger       | F vorg              |
| F readinteger1      | F vvar              |
| F readinteger1le    | F woff2otf          |
| F readinteger2      | F registerpalette   |
| F readinteger2le    | F registerplugin    |
| F readinteger3      | F reorderlookups    |
| F readinteger3le    | F scriptandlanguage |
| F readinteger4      | F setdynamics       |
| F readinteger4le    | F setfeatures       |
| F readintegertable  | T statistics        |
| F readsignedbyte    | D usedfeatures      |
| F readstring        | D svgcache          |
| F readtag           | B svgenabled        |
| F setposition       | T tables            |
| F size              | D baselines         |
| F skip              | D features          |
| F skiplong          | D featuretypes      |
| F skipposition      | D languages         |



- D scripts
- F texthandler
- T vectors
- T arab
  - T gpos
    - N curs
    - N kern
    - N mark
    - N mkmk
  - T gsub
    - N calt
    - N ccmp
    - N cswh
    - N dlig
    - N fina
    - N init
    - N isol
    - N liga
    - N medi
    - N mset
    - N rclt
    - N rlig
- N version
- T pfb
  - F filterencoding
  - F filtermetadata
  - F loadvector
  - N version
- T tfm
  - D cache
  - D features
  - N maxnestingdepth
  - N maxnestingsize
  - T readers
    - F loadpk
    - F loadtfm
    - F loadtfmvf
    - F loadvf
    - F pktopdf
    - F potracetopdf
    - F showpk
  - F reencode
  - F setfeatures
  - N version

- T vf
  - T combinations
  - F combine
  - T combiner
    - D commands
    - D helpers
    - D math
    - D predefined
    - N version
    - D whatever
  - F hashfeatures
  - F hashinstance
  - T hashmethods
    - F normal
    - F virtual
  - F initializefeatures
  - D loadedfonts
  - N noffontsloaded
  - N nofsharedfonts
  - N nofsharedhashes
  - N nofsharedvectors
  - N privateoffset
  - D privateslots
  - F readanddefine
  - F registerfeature
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  - F resetfeature
  - B resolvevirtualtoo
  - F scale
  - F scaled
  - F setfactor
  - F setfeature
  - F setmathparameter
  - F setname
  - B sharefonts
  - F trytosharefont
  - N version
- current**                      function
- currentid**                    function
- definers**                    table
  - F addlookup
  - F analyze
  - F applypostprocessors
  - F checkedmode

|                        |          |                        |            |
|------------------------|----------|------------------------|------------|
| F checkedscript        |          | D math                 |            |
| S defaultlookup        |          | N version              |            |
| F define               |          | <b>enhanceid</b>       | function   |
| F getspecification     |          | <b>expansions</b>      | data table |
| F internal             |          | <b>formats</b>         | data table |
| F loadfont             |          | <b>getalternate</b>    | function   |
| F makespecification    |          | <b>getindices</b>      | function   |
| T methods              |          | <b>getkern</b>         | function   |
| F install              |          | <b>getmultiple</b>     | function   |
| D variants             |          | <b>getsubstitution</b> | function   |
| F read                 |          | <b>getters</b>         | table      |
| F register             |          | T alternate            |            |
| F registered           |          | F opentype             |            |
| F registerfallbackfont |          | T kern                 |            |
| F registersplit        |          | F opentype             |            |
| F resetnullfont        |          | F type1                |            |
| F resolve              |          | F type3                |            |
| T resolvers            |          | T multiple             |            |
| F file                 |          | F opentype             |            |
| F name                 |          | T substitution         |            |
| F spec                 |          | F opentype             |            |
| <b>definetypeface</b>  | function | <b>goodies</b>         | table      |
| <b>dropins</b>         | table    | D colorschemes         |            |
| F clone                |          | T data                 |            |
| F getshape             |          | T common-math.lfg      |            |
| F getshapes            |          | S author               |            |
| F nextid               |          | S comment              |            |
| F provide              |          | S copyright            |            |
| F registerglyph        |          | B initialized          |            |
| F registerglyphs       |          | T mathematics          |            |
| F swap                 |          | T tweaks               |            |
| F swapone              |          | T presets              |            |
| <b>effects</b>         | table    | F addmathkerns         |            |
| <b>encodings</b>       | table    | F braille              |            |
| T agl                  |          | F eulertocalligraphic  |            |
| D ctxcodes             |          | F fallbacks            |            |
| D extras               |          | F moderntocalligraphic |            |
| D synonyms             |          | F moveintegrals        |            |
| D unicodes             |          | F moveitalics          |            |
| D cache                |          | F pagellaradical       |            |
| F is_known             |          | F rsfstoscript         |            |
| D known                |          | F rsfsuprighttoscript  |            |
| F load                 |          | F scripttocalligraphic |            |
| F make_unicode_vector  |          | F smallldigits         |            |

- F wipeanchors
- F wipeitalics
- F xitsarabic
- S name
- S version
- T dejavu-math
- S author
- S comment
- S copyright
- B initialized
- T mathematics
- T alternates
  - T dotless
    - S comment
    - S feature
    - N value
- T bigslots
- T parameters
  - N AccentBaseDepth
  - N AccentTopShiftUp
  - N DelimiterDisplayPercent
  - N DelimiterDisplayShortfall
  - N DelimiterPercent
  - N DelimiterShortfall
  - N DisplayOperatorMinHeight
  - N FlattenedAccentTopShiftUp
  - N FractionRuleThickness
  - S PrimeBaselineDropMax
  - S PrimeShiftUp
  - S PrimeShiftUpCramped
  - N SubscriptShiftDown
  - S SubscriptShiftDownWithSuperscript
  - N SuperscriptShiftUp
- T tweaks
  - T aftercopying
- S name
- S version
- T dejavu-text
- S author
- S comment
- S copyright
- T extensions
- B initialized
- S name

- S version
- D designsizes
- D filenames
- F getcompositions
- T list
- F load
- F prepare\_features
- F register
- F registerpostprocessor
- F report
- T suffixes
- handlers**
  - table
- hashes**
  - table
- D characters
- D classes
- D csnames
- D descriptions
- D dynamics
- D emwidths
- D exheights
- D features
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- D properties
- D quads
- D resources
- T setdynamics
- T shapes
- D spaces
- D spacings
- T streams
- D unicodes
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- D variants
- D xheights
- helpers**
  - table
- F addextraprivate

F addprivate  
 F appendcommands  
 F appendcommandtable  
 F checkflags  
 F checkmerge  
 F checksteps  
 F collectanchors  
 F combineglyphs  
 D commands  
 F composecharacters  
 F descriptiontoslot  
 F dimenfactor  
 F expandglyph  
 D extraprivates  
 F fontpatternhassize  
 F getcoloredglyphs  
 F getdigitwidth  
 F getinstancespec  
 F getparameters  
 F getprivatecharornode  
 F getprivatenode  
 F getprivates  
 F getprivateslot  
 F glyphdimenfactor  
 F hasprivate  
 F indextoslot  
 F isprivate  
 F mathscriptslots  
 F name  
 F nametoslot  
 F newprivateslot  
 F prependcommands  
 F prependcommandtable  
 F privateslot  
 F registerglyphbox  
 F scalecommands  
 F setboxdirectly  
 F setboxglyph  
 F setboxglyphs  
 F setdigitwidth  
 F setmetaglyph  
 F setmetaglyphs  
 D sharedprivates  
 F splitfontpattern

F tochar  
 T vfinjectors  
 F startcolor  
 F stopcolor  
 T vfspecials  
 T backgrounds  
 T outlines

**infofont** function

**iterators** table

F characters  
 F descriptions  
 F glyphs

**loggers** table

F format  
 F onetimemessage  
 F register  
 F reportdefinedfonts  
 F reportusedfeatures

**mappings** table

F addtounicode  
 F fromunicode16  
 F getentry  
 F loadfile  
 F loadline  
 F makenameparser  
 D overloads  
 F reset  
 F tounicode  
 F tounicode16  
 F tounicode16sequence

**metapost** table

F baseline  
 F boundingbox  
 F boxtomp  
 F boxtompdimensions  
 F boxtompposition  
 F draw  
 F fill  
 F glyph  
 F maxbounds  
 F paths  
 F widthline  
 F zeroline

**mode** function

|                          |            |                           |       |
|--------------------------|------------|---------------------------|-------|
| <b>names</b>             | data table | F mathfallback            |       |
| <b>nulldata</b>          | data table | D methods                 |       |
| <b>privateoffsets</b>    | table      | D presets                 |       |
| B keepnames              |            | F registerfallback        |       |
| N mathbase               |            | F registerfiles           |       |
| N mathextrabase          |            | F registerfontalternative |       |
| N textbase               |            | F registerfontfamily      |       |
| N textextrabase          |            | F textfallback            |       |
| <b>protrusions</b>       | data table | F typeface                |       |
| <b>readers</b>           | table      | F typescript              |       |
| F afm                    |            | F userdata                |       |
| F check_lua              |            | <b>specifiers</b>         | table |
| F check_tfm              |            | D contextmerged           |       |
| F lua                    |            | F contextnumber           |       |
| F ofm                    |            | D contextnumbers          |       |
| F opentype               |            | D contextsetups           |       |
| F otf                    |            | F contexttostring         |       |
| F pfb                    |            | F definecontext           |       |
| D sequence               |            | F mergecontext            |       |
| F tfm                    |            | F presetcontext           |       |
| F ttc                    |            | F registercontext         |       |
| F ttf                    |            | T scriptsets              |       |
| F woff                   |            | T af                      |       |
| F woff2                  |            | S language                |       |
| <b>registersanitizer</b> | function   | N number                  |       |
| <b>reserveid</b>         | function   | S script                  |       |
| <b>select</b>            | table      | T always                  |       |
| D alternatives           |            | S autolanguage            |       |
| F bodyfont               |            | S autoscript              |       |
| D data                   |            | B checkmissing            |       |
| F default                |            | B compoundhyphen          |       |
| D defaults               |            | B curs                    |       |
| F definefallbackfamily   |            | B kern                    |       |
| F definefontfamily       |            | B mark                    |       |
| F definefontfamilypreset |            | B mkmk                    |       |
| D extras                 |            | S mode                    |       |
| F fallback               |            | N number                  |       |
| D fallbacks              |            | S script                  |       |
| F features               |            | S textcontrol             |       |
| F filefallback           |            | B wipemath                |       |
| F fontdata               |            | T arabic                  |       |
| F fontfallback           |            | B analyze                 |       |
| F fontsynonym            |            | S autolanguage            |       |
| F goodies                |            | S autoscript              |       |

|                 |               |
|-----------------|---------------|
| B calt          | B nukt        |
| B ccmp          | N number      |
| B clig          | B pres        |
| B curs          | B pstf        |
| B dlig          | B psts        |
| B fina          | B rphf        |
| B init          | S script      |
| B isol          | B vatu        |
| B kern          | T bengali-two |
| S language      | B abvm        |
| B liga          | B abvs        |
| B mark          | B akhn        |
| B medi          | B blwf        |
| B mkmk          | B blwm        |
| S mode          | B blws        |
| N number        | B calt        |
| B rlig          | B cjct        |
| S script        | B dist        |
| T ba            | B half        |
| S language      | B haln        |
| N number        | S indic       |
| S script        | B init        |
| T bandw:overlay | B kern        |
| B ccmp          | S language    |
| B dist          | B locl        |
| N number        | S mode        |
| T bengali-one   | B nukt        |
| B abvm          | N number      |
| B abvs          | B pres        |
| B akhn          | B pstf        |
| B blwf          | B psts        |
| B blwm          | B rphf        |
| B blws          | S script      |
| B calt          | B vatu        |
| B cjct          | T boldened    |
| B dist          | S effect      |
| B half          | N number      |
| B haln          | T boldened-10 |
| S indic         | S effect      |
| B init          | N number      |
| B kern          | T boldened-15 |
| S language      | S effect      |
| B locl          | N number      |
| S mode          | T boldened-20 |

|                 |                  |
|-----------------|------------------|
| S effect        | B compoundhyphen |
| N number        | B curs           |
| T boldened-25   | B kern           |
| S effect        | B liga           |
| N number        | B mark           |
| T boldened-30   | B mkmk           |
| S effect        | S mode           |
| N number        | N number         |
| T ca            | S script         |
| S language      | S textcontrol    |
| N number        | B wipemath       |
| S script        | T cy             |
| T camelcasing   | S language       |
| B camelcasing   | N number         |
| N number        | S script         |
| T cn            | T cz             |
| S language      | S language       |
| N number        | N number         |
| S script        | S script         |
| T color         | T de             |
| B ccmp          | S language       |
| S color         | N number         |
| B dist          | S script         |
| N number        | T default        |
| T color:bitmap  | S autolanguage   |
| B ccmp          | S autoscript     |
| B dist          | B checkmissing   |
| N number        | B compoundhyphen |
| B sbix          | B curs           |
| T color:overlay | B itlc           |
| B ccmp          | B kern           |
| B colr          | B liga           |
| B dist          | B mark           |
| N number        | B mkmk           |
| T color:svg     | S mode           |
| B ccmp          | N number         |
| B dist          | S script         |
| N number        | S textcontrol    |
| B svg           | B wipemath       |
| T complete      | T deo            |
| S autolanguage  | S language       |
| S autoscript    | N number         |
| B checkmissing  | S script         |
| B compose       | T devanagari-one |

|                  |               |
|------------------|---------------|
| B abvm           | B psts        |
| B abvs           | B rkrf        |
| B akhn           | B rphf        |
| B blwf           | S script      |
| B blwm           | B vatu        |
| B blws           | T dingbats    |
| B calt           | S goodies     |
| B cjct           | S mode        |
| B dist           | N number      |
| B half           | B unicoding   |
| B haln           | T es          |
| S indic          | S language    |
| B kern           | N number      |
| S language       | S script      |
| B locl           | T et          |
| S mode           | S language    |
| B nukt           | N number      |
| N number         | S script      |
| B pres           | T f:fractions |
| B psts           | B frac        |
| B rkrf           | N number      |
| B rphf           | T f:inferiors |
| S script         | N number      |
| B vatu           | B subs        |
| T devanagari-two | T f:kern      |
| B abvm           | B kern        |
| B abvs           | N number      |
| B akhn           | T f:kerns     |
| B blwf           | B kern        |
| B blwm           | N number      |
| B blws           | T f:oldstyle  |
| B calt           | N number      |
| B cjct           | B onum        |
| B dist           | T f:smallcaps |
| B half           | N number      |
| B haln           | B smcp        |
| S indic          | T f:superiors |
| B kern           | N number      |
| S language       | B sups        |
| B locl           | T f:tabular   |
| S mode           | N number      |
| B nukt           | B tnum        |
| N number         | T fi          |
| B pres           | S language    |



|                  |                |
|------------------|----------------|
| N number         | T gujarati-two |
| S script         | B abvm         |
| T followtext     | B abvs         |
| B liga           | B akhn         |
| N number         | B blwf         |
| T fr             | B blwm         |
| S language       | B blws         |
| N number         | B calt         |
| S script         | B cjct         |
| T fullprotrusion | B dist         |
| N number         | B half         |
| S protrusion     | B haln         |
| T gr             | S indic        |
| S language       | B kern         |
| N number         | S language     |
| S script         | B locl         |
| T grek           | S mode         |
| N number         | B nukt         |
| S script         | N number       |
| T gujarati-one   | B pres         |
| B abvm           | B psts         |
| B abvs           | B rkrf         |
| B akhn           | B rphf         |
| B blwf           | S script       |
| B blwm           | B watu         |
| B blws           | T gurmukhi-one |
| B calt           | B abvm         |
| B cjct           | B abvs         |
| B dist           | B akhn         |
| B half           | B blwf         |
| B haln           | B blwm         |
| S indic          | B blws         |
| B kern           | B calt         |
| S language       | B cjct         |
| B locl           | B dist         |
| S mode           | B half         |
| B nukt           | B haln         |
| N number         | S indic        |
| B pres           | B kern         |
| B psts           | S language     |
| B rkrf           | B locl         |
| B rphf           | S mode         |
| S script         | B nukt         |
| B watu           | N number       |

|                |                    |
|----------------|--------------------|
| B pres         | S language         |
| B pstf         | B liga             |
| B psts         | B mark             |
| B rphf         | B medi             |
| S script       | B mkmk             |
| B watu         | S mode             |
| T gurmukhi-two | N number           |
| B abvm         | B reordercombining |
| B abvs         | B rlig             |
| B akhn         | S script           |
| B blwf         | T hr               |
| B blwm         | S language         |
| B blws         | N number           |
| B calt         | S script           |
| B cjct         | T hu               |
| B dist         | S language         |
| B half         | N number           |
| B haln         | S script           |
| S indic        | T indic-common     |
| B kern         | B abvm             |
| S language     | B abvs             |
| B locl         | B akhn             |
| S mode         | B blwm             |
| B nukt         | B blws             |
| N number       | B calt             |
| B pres         | B dist             |
| B pstf         | B haln             |
| B psts         | S indic            |
| B rphf         | B kern             |
| S script       | S language         |
| B watu         | B locl             |
| T hebrew       | S mode             |
| B analyze      | B nukt             |
| S autolanguage | N number           |
| S autoscript   | B pres             |
| B calt         | B psts             |
| B ccmp         | B rphf             |
| B clig         | T inlinenumbers    |
| B curs         | N number           |
| B dlig         | B pnum             |
| B fina         | B tnum             |
| B init         | T is               |
| B isol         | S language         |
| B kern         | N number           |

|               |                  |
|---------------|------------------|
| S script      | B abvm           |
| T it          | B abvs           |
| S language    | B akhn           |
| N number      | B blwf           |
| S script      | B blwm           |
| T ja          | B blws           |
| S language    | B calt           |
| N number      | B cjct           |
| S script      | B dist           |
| T jamoforms   | B half           |
| B ljmo        | B haln           |
| N number      | S indic          |
| B tjmo        | B kern           |
| B vjmo        | S language       |
| T just-os     | B locl           |
| S mode        | S mode           |
| N number      | B nuk            |
| B onum        | N number         |
| T kannada-one | B pref           |
| B abvm        | B pres           |
| B abvs        | B pstf           |
| B akhn        | B psts           |
| B blwf        | B rphf           |
| B blwm        | S script         |
| B blws        | T la             |
| B calt        | S language       |
| B cjct        | N number         |
| B dist        | S script         |
| B half        | T latn           |
| B haln        | N number         |
| S indic       | S script         |
| B kern        | T letterspacing  |
| S language    | B ccmp           |
| B locl        | B clig           |
| S mode        | B dlig           |
| B nuk         | S keepligatures  |
| N number      | B liga           |
| B pref        | N number         |
| B pres        | B rlig           |
| B pstf        | T ligatures      |
| B psts        | S autolanguage   |
| B rphf        | S autoscript     |
| S script      | B checkmissing   |
| T kannada-two | B compoundhyphen |

B curs  
 B kern  
 B liga  
 B mark  
 B mkmk  
 S mode  
 N number  
 S script  
 S textcontrol  
 B wipemath  
 T lm-math-bold  
 S effect  
 N number  
 T lowercasing  
 B lowercasing  
 N number  
 T malayalam-one  
 B abvm  
 B abvs  
 B akhn  
 B blwf  
 B blwm  
 B blws  
 B calt  
 B cjct  
 B dist  
 B half  
 B haln  
 S indic  
 B kern  
 S language  
 B locl  
 S mode  
 B nukt  
 N number  
 B pref  
 B pres  
 B pstf  
 B psts  
 B rphf  
 S script  
 T malayalam-two  
 B abvm  
 B abvs

B akhn  
 B blwf  
 B blwm  
 B blws  
 B calt  
 B cjct  
 B dist  
 B half  
 B haln  
 S indic  
 B kern  
 S language  
 B locl  
 S mode  
 B nukt  
 N number  
 B pref  
 B pres  
 B pstf  
 B psts  
 B rphf  
 S script  
 T math  
 B compactmath  
 B flattenaccents  
 B kern  
 S language  
 B mathalternates  
 S mathexpansion  
 S mode  
 N number  
 S script  
 B ssty  
 B wipemath  
 T math-script  
 B compactmath  
 B flattenaccents  
 B kern  
 S language  
 B mathalternates  
 S mathexpansion  
 S mode  
 N number  
 S script

|                     |                   |
|---------------------|-------------------|
| N ssty              | B curs            |
| B wipemath          | B fina            |
| T math-scriptscript | B init            |
| B compactmath       | B isol            |
| B flattenaccents    | B itlc            |
| B kern              | B kern            |
| S language          | B liga            |
| B mathalternates    | B mark            |
| S mathexpansion     | B medi            |
| S mode              | B mkmk            |
| N number            | S mode            |
| S script            | B mset            |
| N ssty              | N number          |
| B wipemath          | B rclt            |
| T math-text         | B rlig            |
| B compactmath       | S script          |
| B flattenaccents    | S textcontrol     |
| B kern              | B wipemath        |
| S language          | T nl              |
| B mathalternates    | S language        |
| S mathexpansion     | N number          |
| S mode              | S script          |
| N number            | T none            |
| S script            | B features        |
| B ssty              | S mode            |
| B wipemath          | N number          |
| T mathematics       | T oldstyle        |
| B compactmath       | S autolanguage    |
| B flattenaccents    | S autoscript      |
| B kern              | B checkmissing    |
| S language          | B compoundhyphen  |
| B mathalternates    | B curs            |
| S mathexpansion     | B kern            |
| S mode              | B liga            |
| N number            | B mark            |
| S script            | B mkmk            |
| B wipemath          | S mode            |
| T mixed             | N number          |
| T mongolian         | B onum            |
| S autolanguage      | S script          |
| S autoscript        | S textcontrol     |
| B calt              | B wipemath        |
| B checkmissing      | T oldstylenumbers |
| B compoundhyphen    | N number          |

|                  |                  |
|------------------|------------------|
| B onum           | B blwf           |
| T original       | B blwm           |
| S autolanguage   | B blws           |
| S autoscript     | B calt           |
| B checkmissing   | B cjct           |
| B compoundhyphen | B dist           |
| B curs           | B haln           |
| B itlc           | S indic          |
| B kern           | B kern           |
| B liga           | S language       |
| B mark           | B locl           |
| B mkmk           | S mode           |
| S mode           | B nuk            |
| N number         | N number         |
| S script         | B pres           |
| S textcontrol    | B pstf           |
| B wipemath       | B psts           |
| T oriya-one      | B rphf           |
| B abvm           | S script         |
| B abvs           | T pl             |
| B akhn           | S language       |
| B blwf           | N number         |
| B blwm           | S script         |
| B blws           | T primes         |
| B calt           | N number         |
| B cjct           | B primes         |
| B dist           | T pt             |
| B haln           | S language       |
| S indic          | N number         |
| B kern           | S script         |
| S language       | T quality        |
| B locl           | S expansion      |
| S mode           | N number         |
| B nuk            | S protrusion     |
| N number         | T recommended    |
| B pres           | S autolanguage   |
| B pstf           | S autoscript     |
| B psts           | B calt           |
| B rphf           | B checkmissing   |
| S script         | B clig           |
| T oriya-two      | B compoundhyphen |
| B abvm           | B curs           |
| B abvs           | B itlc           |
| B akhn           | B kern           |

|                    |                |
|--------------------|----------------|
| B liga             | S autoscript   |
| B mark             | B calt         |
| B mkmk             | B ccmp         |
| S mode             | B curs         |
| N number           | B fina         |
| B rlig             | B init         |
| S script           | B isol         |
| S textcontrol      | B kern         |
| B wipemath         | S language     |
| T reordercombining | B mark         |
| N number           | B medi         |
| B reordercombining | B mkmk         |
| T ro               | S mode         |
| S language         | N number       |
| N number           | B rlig         |
| S script           | T simplearabic |
| T ru               | B analyze      |
| S language         | S autolanguage |
| N number           | S autoscript   |
| S script           | B calt         |
| T semitic-complete | B ccmp         |
| B analyze          | B curs         |
| S autolanguage     | B fina         |
| S autoscript       | B init         |
| B calt             | B isol         |
| B ccmp             | B kern         |
| B clig             | S language     |
| B curs             | B mark         |
| B dlig             | B medi         |
| B fina             | B mkmk         |
| B init             | S mode         |
| B isol             | N number       |
| B kern             | B rlig         |
| S language         | S script       |
| B liga             | T simplehebrew |
| B mark             | B analyze      |
| B medi             | S autolanguage |
| B mkmk             | S autoscript   |
| S mode             | B calt         |
| N number           | B ccmp         |
| B rlig             | B curs         |
| T semitic-simple   | B fina         |
| B analyze          | B init         |
| S autolanguage     | B isol         |

|                  |                  |
|------------------|------------------|
| B kern           | B ccmp           |
| S language       | B clig           |
| B mark           | B curs           |
| B medi           | B dlig           |
| B mkmk           | B fin2           |
| S mode           | B fin3           |
| N number         | B fina           |
| B rlig           | B init           |
| S script         | B isol           |
| T sk             | B kern           |
| S language       | S language       |
| N number         | B liga           |
| S script         | B mark           |
| T sl             | B med2           |
| S language       | B medi           |
| N number         | B mkmk           |
| S script         | S mode           |
| T slanted        | N number         |
| N number         | B rlig           |
| N slant          | S script         |
| T smallcaps      | T system:tabnum  |
| S autolanguage   | B lnum           |
| S autoscript     | N number         |
| B checkmissing   | B tnum           |
| B compoundhyphen | T tableddigits   |
| B curs           | N number         |
| B kern           | B tableddigits   |
| B mark           | B tnum           |
| B mkmk           | T tabularnumbers |
| S mode           | N number         |
| N number         | B pnum           |
| S script         | B tnum           |
| B smcp           | T tamil-one      |
| S textcontrol    | B abvm           |
| B wipemath       | B abvs           |
| T sv             | B akhn           |
| S language       | B blwm           |
| N number         | B blws           |
| S script         | B calt           |
| T syriac         | B dist           |
| B analyze        | B half           |
| S autolanguage   | B haln           |
| S autoscript     | S indic          |
| B calt           | B kern           |



|              |              |
|--------------|--------------|
| S language   | S indic      |
| B locl       | B kern       |
| S mode       | S language   |
| B nukt       | B locl       |
| N number     | S mode       |
| B pref       | B nukt       |
| B pres       | N number     |
| B psts       | B pref       |
| B rphf       | B pres       |
| S script     | B pstf       |
| T tamil-two  | B psts       |
| B abvm       | B rphf       |
| B abvs       | S script     |
| B akhn       | T telugu-two |
| B blwm       | B abvm       |
| B blws       | B abvs       |
| B calt       | B akhn       |
| B dist       | B blwf       |
| B half       | B blwm       |
| B haln       | B blws       |
| S indic      | B calt       |
| B kern       | B cjct       |
| S language   | B dist       |
| B locl       | B half       |
| S mode       | B haln       |
| B nukt       | S indic      |
| N number     | B kern       |
| B pref       | S language   |
| B pres       | B locl       |
| B psts       | S mode       |
| B rphf       | B nukt       |
| S script     | N number     |
| T telugu-one | B pref       |
| B abvm       | B pres       |
| B abvs       | B pstf       |
| B akhn       | B psts       |
| B blwf       | B rphf       |
| B blwm       | S script     |
| B blws       | T th         |
| B calt       | S language   |
| B cjct       | N number     |
| B dist       | S script     |
| B half       | T tibetan    |
| B haln       | B abvm       |

- B abvs
- S autolanguage
- S autoscript
- B blwm
- B blws
- B calt
- B ccmp
- B checkmissing
- B compoundhyphen
- B curs
- B kern
- S language
- B liga
- B locl
- B mark
- B mkmk
- S mode
- N number
- S script
- S textcontrol
- B wipemath
- T tr
  - S language
  - N number
  - S script
- T uk
  - S language
  - N number
  - S script
- T uppercasing
  - N number
  - B uppercasing
- T us
  - S language
  - N number
  - S script
- T virtualmath

- B compactmath
- B flattenaccents
- B kern
- S language
- B mathalternates
- S mathexpansion
- S mode
- N number
- S script
- B wipemath
- T vn
  - S language
  - N number
  - S script
- T zero
  - N number
  - B zero
- F showcontext
- F splitcontext
- D synonyms
- D variants

## **symbols** table

- T emoji
- F checked
- F resolved
- F uselibrary

## **tables** table

## **toeffect** function

## **tracers** table

## **treatments** table

- F applyfixes
- D data
- S filename
- F ignoredfile
- F report
- F trace

## **typefaces** table

# functions (context)

**dummy** function

# globaldata (context)

|                            |          |                            |
|----------------------------|----------|----------------------------|
| <b>CONTEXT</b>             | number   | F colorcomponents          |
| <b>LUAFORMAT</b>           | number   | D counts                   |
| <b>LUAMAJORVERSION</b>     | number   | B couple                   |
| <b>LUAMINORVERSION</b>     | number   | F currentmodel             |
| <b>LUATEXENGINE</b>        | string   | F currentnamedmodel        |
| <b>LUATEXFORMATID</b>      | number   | D data                     |
| <b>LUATEXFUNCTIONALITY</b> | number   | N default                  |
| <b>LUATEXVERBOSE</b>       | string   | F defineintermediatecolor  |
| <b>LUATEXVERSION</b>       | number   | F definemultitonecolor     |
| <b>LUAVERSION</b>          | number   | F defineprocesscolor       |
| <b>MP</b>                  | function | F defineprocesscolordirect |
| <b>SAVEDREQUIRE</b>        | function | F definesimplegray         |
| <b>aes</b>                 | table    | F definespotcolor          |
| F decode                   |          | F definetransparency       |
| F encode                   |          | F enable                   |
| F random                   |          | F filter                   |
| <b>arg</b>                 | table    | N forced                   |
| <b>assert</b>              | function | F forcedmodel              |
| <b>attributes</b>          | table    | F forcesupport             |
| T alternates               |          | F formatcolor              |
| N attribute                |          | F formatgray               |
| D data                     |          | F formatluminance          |
| F enable                   |          | F formatluminanceratio     |
| F handler                  |          | F getpagecolormodel        |
| F register                 |          | F gray                     |
| D registered               |          | F graytohsv                |
| D values                   |          | F handler                  |
| F applyoverloads           |          | F hsvtorgb                 |
| F checkvalues              |          | F hsvtorgb                 |
| T colorintents             |          | F hwborgb                  |
| N attribute                |          | F iccprofile               |
| D data                     |          | F isblack                  |
| F enable                   |          | F labtolch                 |
| F handler                  |          | F labtorgb                 |
| F register                 |          | F labtoxyz                 |
| D registered               |          | F lchtolab                 |
| T colors                   |          | F lchtorgb                 |
| N attribute                |          | N main                     |
| F cmyk                     |          | S model                    |
| F cmyktorgb                |          | D models                   |
| F cmyktorgb                |          | F mpcolor                  |

|                          |                     |
|--------------------------|---------------------|
| F mnamedcolor            | F handler           |
| F mpoptions              | F register          |
| F mprgbcolor             | D registered        |
| F namedcolorattributes   | D values            |
| N none                   | F getvalue          |
| F popset                 | F getvalues         |
| F processcolorcomponents | T handlers          |
| F pushset                | F hasvalues         |
| F register               | D list              |
| D registered             | F markasoverload    |
| F registerspotcolor      | D names             |
| F rgb                    | T negatives         |
| F rgbbytes               | N attribute         |
| F rgbtocmyk              | D data              |
| F rgbtorgay              | F enable            |
| F rgbtohsv               | F handler           |
| F rgbtoxyz               | F register          |
| N selector               | D registered        |
| F setlist                | F set               |
| F setmodel               | D numbers           |
| F setpagecolormodel      | F ofnode            |
| D sets                   | F private           |
| F spec                   | F public            |
| F spot                   | F registervalue     |
| F spotcolorname          | F restore           |
| F spotcolorparent        | F save              |
| F spotcolorvalue         | F setcleaner        |
| B supported              | F showcurren        |
| F toattributes           | T states            |
| F transparencycomponents | F disabletriggering |
| B triggering             | F enabletriggering  |
| F usecolors              | F finalize          |
| D valid                  | F initialize        |
| F value                  | F process           |
| D values                 | F selective         |
| B weightgray             | F simple            |
| F xyztolab               | F stacked           |
| F xyztorgb               | F stacker           |
| D counts                 | F system            |
| F define                 | F tooverloads       |
| T effects                | T transparencies    |
| N attribute              | N attribute         |
| D data                   | D data              |
| F enable                 | F enable            |

|                          |                    |       |
|--------------------------|--------------------|-------|
| F forcesupport           | D tables           |       |
| F handler                | F getcallbackstate |       |
| N none                   | T included         |       |
| F register               | B context          |       |
| D registered             | B date             |       |
| B supported              | B id               |       |
| F toattribute            | B metadata         |       |
| B triggering             | B pdf              |       |
| F value                  | F initialize       |       |
| D values                 | F latelua          |       |
| N unsetvalue             | T nodeinjections   |       |
| D values                 | F nothing          |       |
| T viewerlayers           | F openout          |       |
| N attribute              | T registered       |       |
| F cleanup                | T registrations    |       |
| D data                   | F setencryption    |       |
| F define                 | T tables           |       |
| F definedlayoutcomponent | F writeout         |       |
| F enable                 | <b>basexx</b>      | table |
| F forcesupport           | F decode16         |       |
| F handler                | F decode64         |       |
| B hasorder               | F decode85         |       |
| D listwise               | F decodeLZW        |       |
| F register               | F decodeRL         |       |
| D registered             | F encode16         |       |
| F resolve_begin          | F encode64         |       |
| F resolve_end            | F encode85         |       |
| F resolve_reset          | F encodeLZW        |       |
| F resolve_step           | F encodeRL         |       |
| D scopes                 | <b>bibtex</b>      | table |
| F setfeatures            | <b>bit32</b>       | table |
| F setup                  | F arshift          |       |
| F start                  | F band             |       |
| F stop                   | F bnot             |       |
| B supported              | F bor              |       |
| D values                 | F btest            |       |
| <b>backends</b>          | F bxor             |       |
| F closeout               | F extract          |       |
| T codeinjections         | F lrotate          |       |
| F registerboxresource    | F lshift           |       |
| F registerfont           | F replace          |       |
| F restoreboxresource     | F rrotate          |       |
| S current                | F rshift           |       |
| T defaults               | <b>bitset</b>      | table |

|                  |              |                           |              |
|------------------|--------------|---------------------------|--------------|
| F asstring       |              | F reset                   |              |
| F get            |              | F save                    |              |
| F new            |              | F prepend                 |              |
| F reset          |              | F raw                     |              |
| F set            |              | F run                     |              |
| F totable        |              | F samplefile              |              |
| F traverse       |              | F splitsamplefile         |              |
| F wipe           |              | F undent                  |              |
| <b>blobs</b>     | <b>table</b> | <b>builders</b>           | <b>table</b> |
| F append         |              | F checkparcontext         |              |
| F dimensions     |              | F checksidefloat          |              |
| F dispose        |              | T hpack                   |              |
| F new            |              | F qualityactions          |              |
| F pack           |              | F report                  |              |
| F strdp          |              | F show                    |              |
| F strhd          |              | T kernel                  |              |
| F strht          |              | F collapsing              |              |
| F strwd          |              | F hyphenating             |              |
| F tohpack        |              | F kerning                 |              |
| F tonodes        |              | F ligaturing              |              |
| F tovpack        |              | F mlisttohlist            |              |
| F write          |              | T paragraphs              |              |
| <b>boolean</b>   | <b>table</b> | F keptogether             |              |
| F tocontext      |              | F registertogether        |              |
| F tonumber       |              | F tag                     |              |
| <b>buffers</b>   | <b>table</b> | F parpassidentifier       |              |
| F append         |              | D passnames               |              |
| F assign         |              | T profiling               |              |
| F collectcontent |              | F add                     |              |
| F empty          |              | F adddrivers              |              |
| F erase          |              | F boxlinehandler          |              |
| F exists         |              | F distance                |              |
| F get            |              | F fixedprofile            |              |
| F getcontent     |              | F get                     |              |
| F getctxlua      |              | F getrivers               |              |
| F getlines       |              | F list                    |              |
| F getmkvi        |              | F pagehandler             |              |
| F gettex         |              | F pagelinehandler         |              |
| F loadcontent    |              | F profilebox              |              |
| T parallel       |              | F profilelines            |              |
| F define         |              | F set                     |              |
| F hassomecontent |              | F setlines                |              |
| F next           |              | F registerpaspasscallback |              |
| F place          |              | T vpack                   |              |

|                            |              |                        |              |
|----------------------------|--------------|------------------------|--------------|
| F qualityactions           |              | F hasbytegray          |              |
| F report                   |              | F hasbyterange         |              |
| T vspacing                 |              | F hasbytergb           |              |
| D categories               |              | F new                  |              |
| F checkstrutdepth          |              | F newdomain            |              |
| F collapsevbox             |              | F overlay              |              |
| F definesnapmethod         |              | F process              |              |
| B fixed                    |              | F reduce               |              |
| F flushcontributions       |              | F reset                |              |
| F forcepageflush           |              | F setgray              |              |
| F forcestrutdepth          |              | F setgrayadd           |              |
| F getnofpreviouslines      |              | F setgraymin           |              |
| F inject                   |              | F setrgb               |              |
| F injectdisable            |              | F slicegray            |              |
| F injectnowhite            |              | F slicerange           |              |
| F injectpenalty            |              | F slicergb             |              |
| F injectskip               |              | F totable              |              |
| F interceptsamepagecontent |              | <b>caches</b>          | <b>table</b> |
| F pagehandler              |              | S base                 |              |
| F pageoverflow             |              | F collapsecontent      |              |
| F popatsame                |              | F configfiles          |              |
| F pushatsame               |              | F contentstate         |              |
| F setmap                   |              | T defaults             |              |
| F setskip                  |              | F getfirstreadablefile |              |
| F snapbox                  |              | F getreadablepaths     |              |
| F synchronizepage          |              | F getwritablepath      |              |
| F tocategories             |              | F hashed               |              |
| F tocategory               |              | F is_writable          |              |
| F vboxhandler              |              | F loadcontent          |              |
| <b>bytemap</b>             | <b>table</b> | F loaddata             |              |
| F bounds                   |              | S more                 |              |
| F clip                     |              | F savecontent          |              |
| F copy                     |              | F savedata             |              |
| F dimensions               |              | F setfirstwritablefile |              |
| F domain                   |              | F setluenames          |              |
| F downgrade                |              | F treehash             |              |
| F downsample               |              | F usedpaths            |              |
| F fillgray                 |              | <b>callback</b>        | <b>table</b> |
| F fillrgb                  |              | F find                 |              |
| F filter                   |              | F getcontracts         |              |
| F getbyte                  |              | F getcontractvalues    |              |
| F getbytes                 |              | F getindex             |              |
| F getluminance             |              | F getoptions           |              |
| F getvalue                 |              | F getoptionvalues      |              |



|                  |                       |   |                      |
|------------------|-----------------------|---|----------------------|
| F                | getstate              | N | insert_distance      |
| F                | getstatevalues        | N | insert_par           |
| F                | known                 | N | intercept_lua_error  |
| F                | list                  | N | intercept_tex_error  |
| F                | names                 | N | italic_correction    |
| F                | register              | N | kerning              |
| F                | setoptions            | N | ligaturing           |
| F                | setstate              | N | linebreak            |
| F                | testonly              | N | linebreak_check      |
| F                | usage                 | N | linebreak_quality    |
| <b>callbacks</b> | <b>table</b>          | N | local_box            |
| F                | freeze                | N | make_extensible      |
| T                | functions             | N | math_rule            |
|                  | F start_page_number   | N | missing_character    |
|                  | F stop_page_number    | N | mlist_to_hlist       |
| F                | getcomment            | N | open_data_file       |
| F                | getstates             | N | packed_vbox          |
| F                | known                 | N | page_boundary        |
| T                | list                  | N | paragraph_context    |
|                  | N alignment           | N | paragraph_pass       |
|                  | N append_adjust       | N | post_linebreak       |
|                  | N append_line         | N | pre_dump             |
|                  | N append_migrate      | N | pre_linebreak        |
|                  | N append_to_vlist     | N | pre_output           |
|                  | N balance_boundary    | N | process_character    |
|                  | N balance_check       | N | process_jobname      |
|                  | N balance_insert      | N | quality_font         |
|                  | N begin_paragraph     | N | register_extensible  |
|                  | N buildpage           | N | show_build           |
|                  | N define_font         | N | show_error_message   |
|                  | N find_format_file    | N | show_loners          |
|                  | N find_log_file       | N | show_lua_call        |
|                  | N get_attribute       | N | show_vsplit          |
|                  | N get_math_dictionary | N | show_warning_message |
|                  | N get_noad_class      | N | show_whatsit         |
|                  | N glyph_run           | N | start_file           |
|                  | N handle_overload     | N | start_run            |
|                  | N handle_uinsert      | N | stop_file            |
|                  | N handle_uleader      | N | stop_run             |
|                  | N hpack               | N | tail_append          |
|                  | N hpack_quality       | N | test_only            |
|                  | N hyphenate           | N | trace_memory         |
|                  | N insert_boundary     | N | vpack                |
|                  | N insert_check_split  | N | vpack_quality        |

|                      |                  |
|----------------------|------------------|
| N wrapup_run         | F description    |
| T names              | D descriptions   |
| F optimizer          | D detailtags     |
| T optionbits         | T directions     |
| T optioncodes        | T emoji          |
| N direct             | F compact        |
| N trace              | F known          |
| F pop                | F resolve        |
| F push               | F emojiized      |
| F register           | D encodes        |
| F report             | D entities       |
| T statebits          | D fallbacks      |
| T statecodes         | D filters        |
| N disabled           | T fractions      |
| N frozen             | F fromnumber     |
| N fundamental        | D fschars        |
| N private            | D fscodes        |
| N selective          | F getrange       |
| N set                | D graphemes      |
| N touched            | D hangul         |
| N tracing            | T indicgroups    |
| <b>catcodes</b>      | F initialize     |
| D names              | D is_character   |
| D numbers            | D is_command     |
| F register           | D is_hyphenator  |
| <b>characterkern</b> | D is_letter      |
| <b>characters</b>    | D is_mark        |
| F addcharacternames  | D is_nothing     |
| F adobename          | D is_punctuation |
| D bidi               | D is_spacing     |
| F blockrange         | D is_symbol      |
| D blocks             | D lcchars        |
| D can_have_space     | F lccode         |
| T casemap            | D lccodes        |
| D categories         | F lettered       |
| F category           | D lhash          |
| D categorytags       | D linebreaks     |
| T classifiers        | F lower          |
| D collapsed          | D mathlists      |
| D combined           | T mirrors        |
| F contextname        | D otfscripts     |
| D csletters          | T packdata       |
| D data               | N luasize        |
| D decomposed         | N newsize        |

|                        |                           |          |
|------------------------|---------------------------|----------|
| N oldsize              | F inlinechemical          |          |
| N version              | S instance                |          |
| D ranges               | S method                  |          |
| F registerentities     | F molecule                |          |
| F remap_hangul_syllabe | U moleculeparser          |          |
| T scripthash           | F start                   |          |
| F setactivecatcodes    | F stop                    |          |
| F setlettercatcodes    | T syntax                  |          |
| F setothercatcodes     | F undefine                |          |
| F shape                | <b>collectgarbage</b>     | function |
| F shaped               | <b>commands</b>           | table    |
| D shash                | F Alphabetic              |          |
| D shchars              | F MPLIBconvertfollowtext  |          |
| D shcodes              | F MPLIBconvertoutlinetext |          |
| F showstring           | F MPc                     |          |
| D specialchars         | F MPd                     |          |
| D specials             | F MPfromhash              |          |
| T splits               | F MPH                     |          |
| T subscripts           | F MPLl                    |          |
| T superscripts         | F MPLr                    |          |
| D synonyms             | F MPLs                    |          |
| T tex                  | F MPn                     |          |
| D accentmapping        | F MPp                     |          |
| D commandmapping       | F MPpadata                |          |
| F defineaccents        | F MPplus                  |          |
| F safechar             | F MPpos                   |          |
| F strttextoutf         | F MPr                     |          |
| F strtoutfpattern      | F MPrest                  |          |
| F toutf                | F MPrs                    |          |
| F toutfpattern         | F MPul                    |          |
| T textclasses          | F MPur                    |          |
| F tonumber             | F MPw                     |          |
| D ucchars              | F MPwhd                   |          |
| F uccode               | F MPx                     |          |
| D uccodes              | F MPxy                    |          |
| D uhash                | F MPxywhd                 |          |
| F unicodechar          | F MPy                     |          |
| F upper                | F Romannumerals           |          |
| D verbosemarks         | F actualday               |          |
| <b>chemistry</b>       | F actualmonth             |          |
| F component            | F actualyear              |          |
| F define               | F adaptfontfeature        |          |
| D definitions          | F adaptivecheckbox        |          |
| S format               | F addbtxentry             |          |

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|-----------------------------|-------------------------|
| F addcounter                | F autorule              |
| F adddefaultsuffix          | F autosetups            |
| F addfeature                | F autostarttext         |
| F addfontpath               | F autostoptext          |
| F addgrapheme               | F badnessmodecodestring |
| F addlinenumbers            | F balanceshapeextra     |
| F addlinklistelement        | F benchmarktimer        |
| F addreplacements           | F bf                    |
| F addreplacementslist       | F bi                    |
| F addrivers                 | F bitmapimage           |
| F addtocolumnset            | F bitwise               |
| F addtocolumnsetmvl         | F bitwiseand            |
| F addtojavascriptpreamble   | F bitwiseflip           |
| F addtolist                 | F bitwisenil            |
| F align_point_delta         | F bitwisenot            |
| F allinputpaths             | F bitwiseor             |
| F alphabetic                | F bitwiset              |
| F analysefloatmethod        | F bitwiseshift          |
| F analyzeitemgroup          | F bitwisexor            |
| F analyzelist               | F blockcolumnset        |
| F analyzelocalanchors       | F blockligatures        |
| F analyzeregister           | F boolcase              |
| F analyzerotate             | F boxdir                |
| F analyzeupcomingitemgroup  | F boxline               |
| F applypattern              | F boxlinedp             |
| F applytobox                | F boxlineht             |
| F applytofirstcharacter     | F boxlineil             |
| F arraycompare              | F boxlinein             |
| F arrayequals               | F boxlineir             |
| F arraymaxvalue             | F boxlinelh             |
| F arrayminvalue             | F boxlinelp             |
| F arraysize                 | F boxlinels             |
| F arrayvalue                | F boxlinemaxdp          |
| F ascender                  | F boxlinemaxht          |
| F askedreference            | F boxlinemaxwd          |
| F assignbuffer              | F boxlinend             |
| F async                     | F boxlinenh             |
| F attachinsertion           | F boxlinew              |
| F attachinsertions          | F boxlinerh             |
| F atwordboundary            | F boxlinerp             |
| F autofinishstructurelevels | F boxliners             |
| F autoinsertnextspace       | F boxlines              |
| F automigrationcodestring   | F boxlinewd             |
| F autonextstructurelevel    | F boxoptioncodestring   |

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|-----------------------------|------------------------------|
| F boxrangedp                | F btxlastofrange             |
| F boxrangeht                | F btxlistvariant             |
| F boxrangewd                | F btxloaddefinitionfile      |
| F boxtostring               | F btxloadjournallist         |
| F bposcolumn                | F btxloadreplacementfile     |
| F bposcolumnregistered      | F btxmissing                 |
| F bs                        | F btxoneorrange              |
| F btxabbreviatedjournal     | F btxpreparelistentries      |
| F btxaddentry               | F btxremapauthor             |
| F btxaddjournal             | F btxsavedataset             |
| F btxauthor                 | F btxsavejournallist         |
| F btxauthorfield            | F btxsetdataset              |
| F btxauthortoregister       | F btxsetentry                |
| F btxcmdstring              | F btxsetregister             |
| F btxcollectlistentries     | F btxsetspecification        |
| F btxcurrentfirstnames      | F btxshortcut                |
| F btxcurrentinitials        | F btxshowauthorremapping     |
| F btxcurrentjuniors         | F btxsingularorplural        |
| F btxcurrentsurnames        | F btxtoregister              |
| F btxcurrentvons            | F btxusedataset              |
| F btxdetail                 | F btxuservariable            |
| F btxdirect                 | F build_page_after_shipout   |
| F btxdoif                   | F build_page_before_shipout  |
| F btxdoifelse               | F build_page_setup           |
| F btxdoifelsecitedone       | F capheight                  |
| F btxdoifelsecombiinlist    | F charactercontrolcodestring |
| F btxdoifelsematches        | F chardescription            |
| F btxdoifelsesameasprevious | F checkcolumnset             |
| F btxdoifelseuservariable   | F checkcolumnsetmvl          |
| F btxdoifnot                | F checkcountersetup          |
| F btxentrytobuffer          | F checkedconversion          |
| F btxentrytostring          | F checkedemoji               |
| F btxexpandedjournal        | F checkedpagefloat           |
| F btxfetchlistentries       | F checkinjector              |
| F btxfield                  | F checkinterfacechain        |
| F btxfirstofrange           | F checkmarkedpages           |
| F btxflush                  | F checknesting               |
| F btxflushlistcombi         | F checkspacingavailable      |
| F btxflushlistentry         | F checkstrutdepth            |
| F btxflushlisttag           | F chemicalcomponent          |
| F btxflushmarked            | F chinesenumerals            |
| F btxflushpages             | F chr                        |
| F btxhandlecite             | F classoptioncodestring      |
| F btxhandlenocite           | F cleancolumnset             |

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|---------------------------------|----------------------------|
| F cleanfontname                 | F columnsetstate           |
| F cleanupattributes             | F columnsetstatus          |
| F cleanupbox                    | F combineglyphs            |
| F cleanupplayers                | F component                |
| F clonefield                    | F concatcommalist          |
| F closein                       | F consultfloat             |
| F closeout                      | F continueifinputfile      |
| F codedef                       | F convertbtxdatasettoxml   |
| F collectedmath                 | F converteddimen           |
| F collectfloat                  | F converttmpstopdf         |
| F collecthelp                   | F copyboxfromcache         |
| F columnsetbalancefirst         | F copyboxline              |
| F columnsetbalancelast          | F copyposition             |
| F columnsetbalancepage          | F cos                      |
| F columnsetcolumnwidth          | F cosd                     |
| F columnsetdisablewide          | F ctxfunction              |
| F columnsetenablewide           | F ctxluamatch              |
| F columnsetgoto                 | F ctxluamatchfile          |
| F columnsethascontent           | F ctxresetter              |
| F columnsethspan                | F ctxscanner               |
| F columnsetlastfuture           | F currentcomponent         |
| F columnsetlastmvl              | F currentdate              |
| F columnsetlimit                | F currentdesignsize        |
| F columnsetnoto                 | F currentenvironment       |
| F columnsetresetdirect          | F currentfontinstancespec  |
| F columnsetresetshapeextra      | F currentposthyphenchar    |
| F columnsetreshape              | F currentprehyphenchar     |
| F columnsetsblockrows           | F currentproduct           |
| F columnsetsetshapeextra        | F currentproject           |
| F columnsetshape                | F currentsectiontolist     |
| F columnsetshapebalancecheck    | F datasetsize              |
| F columnsetshapebalanceextend   | F datasetvariable          |
| F columnsetshapecolumn          | F day                      |
| F columnsetshapecount           | F daynmem                  |
| F columnsetshapeindex           | F dayname                  |
| F columnsetshapepage            | F decimals                 |
| F columnsetshapesetbalancerange | F decrementcounter         |
| F columnsetshapeslots           | F decrementsubcounter      |
| F columnsetshapespread          | F deferredenhancelist      |
| F columnsetshapevsize           | F deferredenhancereference |
| F columnsetshapeworst           | F deferredenhanceregister  |
| F columnsetshowgrid             | F defineaccents            |
| F columnsetshowshape            | F defineattribute          |
| F columnsetshowtracedgrid       | F defineboundary           |

F definebreakpoint  
F definebreakpoints  
F definecharacterspacing  
F definechemical  
F definecolumnset  
F defineconversion  
F definecounter  
F definedlayoutcomponent  
F definefallbackfamily  
F definefield  
F definefieldset  
F definefilefallback  
F definefilesynonym  
F definefont\_one  
F definefont\_two  
F definefontcolorpalette  
F definefontfamily  
F definefontfamilypreset  
F definefontfeature  
F definefontscriptset  
F definefontsolution  
F definehyphenationfeatures  
F defineinsertion  
F defineintermediatecolor  
F definelabcolorglobal  
F definelabcolorlocal  
F definelabels  
F definelanguage  
F definelinkedlist  
F definemathpreset  
F definemultitonecolorglobal  
F definenamespace  
F definenote  
F defineparallel  
F defineparbuilder  
F defineprocesscolordummy  
F defineprocesscolorglobal  
F defineprocesscolorlocal  
F defineprogram  
F definereference  
F defineregister  
F definesectionlevels  
F definespotcolorglobal  
F definestructureblock

F definestructureset  
F definesubcolumnset  
F definetransparency  
F definetransparencyglobal  
F definetypeface  
F defineviewerlayer  
F delayedcolumnsetsheet  
F depthnumber  
F depthofstring  
F descender  
F detachinsertions  
F diagnostic  
F digits\_normal  
F digits\_reverse  
F directboxfromcache  
F disableautofontscript  
F disablecollectingmath  
F disabledirectives  
F disableexperiments  
F disablejobsave  
F disableparbuilder  
F disableregime  
F disablespellchecking  
F disablestatetriggering  
F disablestream  
F disabletrackers  
F discardfromlist  
F discptioncodestring  
F disposeboxesincache  
F disposeuatable  
F doanalyzeframedbox  
F doif  
F doifcounter  
F doifelse  
F doifelseblack  
F doifelseboxincache  
F doifelsebuffer  
F doifelsebufferempty  
F doifelsecharinfont  
F doifelseconverter  
F doifelsecounter  
F doifelsecounteractive  
F doifelsecurrentfontasfeature  
F doifelseminenstring

F doifelsedrawingblack  
F doifelsefieldcategory  
F doifelsefieldset  
F doifelsefiledefined  
F doifelsefileexist  
F doifelsefirstchar  
F doifelsefirstsectionpage  
F doifelsefontfeature  
F doifelsefontpresent  
F doifelseindented  
F doifelseinelement  
F doifelseinputfile  
F doifelseleapyear  
F doifelselisthasentry  
F doifelselisthasnumber  
F doifelselisthaspage  
F doifelselisthasitle  
F doifelsemarkedcontent  
F doifelsemarkedpage  
F doifelsemessage  
F doifelsenoteonsamepage  
F doifelseobjectfound  
F doifelseobjectreferencefound  
F doifelseolderversion  
F doifelseoverlapping  
F doifelseparallel  
F doifelseparentfile  
F doifelsepathexist  
F doifelsependingpagecontent  
F doifelseposition  
F doifelsepositiononpage  
F doifelsepositiononsamepage  
F doifelsepositionsonthispage  
F doifelsepositionsused  
F doifelserangeonpage  
F doifelserreference  
F doifelserreferencechecked  
F doifelserreferenced  
F doifelserighttoleftinbox  
F doifelsesame  
F doifelsesomething  
F doifelsespaces  
F doifelsestackedfloats  
F doifelsestructureblockempty

F doifelsesynonymshown  
F doifelsesynonymused  
F doifelsetaggednoteonsamepage  
F doifelseurldefined  
F doifelseutfmathaccent  
F doifelseutfmathfiller  
F doifelseutfmathlimop  
F doifelsevalidhashedfiles  
F doifelsevisualizer  
F doifmarkedpage  
F doifnot  
F doifnotcounter  
F doifnotecontent  
F doifnotsame  
F doifnothing  
F doifposition  
F doifsame  
F doifsomething  
F doifunknownfontfeature  
F domainattribute  
F doreshapeframedbox  
F doresyncframedbox  
F dosaveposition  
F dosavepositionplus  
F dosavepositionwhd  
F dosetposition  
F dosetpositionbox  
F dosetpositionplus  
F dosetpositionstrut  
F dosetpositionstrutkind  
F dosetpositionwhd  
F dostarttextfile  
F dostoptextfile  
F doublescriptmodecodestring  
F droptablegroup  
F elapsedsteptime  
F elapsedtime  
F embedimagefromstore  
F enablealignmentcharacter  
F enablealignskips  
F enableasciimode  
F enableautofences  
F enableautofontscript  
F enablebackgroundalign



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|---------------------------|---------------------------------|
| F enablebackgroundboxes   | F feedback                      |
| F enablecollectingmath    | F fetchlanguagetext             |
| F enablecolor             | F fetchmultipar                 |
| F enablecolorintents      | F fetchposboxes                 |
| F enablecrlf              | F fetchterminaldata             |
| F enabledirectives        | F figure_flush_registered_pages |
| F enabledriver            | F figure_mergegoodies           |
| F enableexperiments       | F figure_nof_registered_pages   |
| F enablefontcolorsschemes | F figure_register_page          |
| F enablehidecontent       | F figuresetdimensions           |
| F enablemathalignrelocate | F filebasename                  |
| F enablematrixornaments   | F filedirname                   |
| F enablematrixrules       | F filenameonly                  |
| F enableparbuilder        | F filesuffix                    |
| F enableregime            | F filterdefaultreference        |
| F enablescriptunstacking  | F filterreference               |
| F enablesidefloatchecker  | F finalizeobjectbox             |
| F enablespellchecking     | F finalizeshipoutbox            |
| F enablestacking          | F findregisterinternal          |
| F enablestatetriggering   | F finishcolumnsetflush          |
| F enablestream            | F finishexport                  |
| F enableswapping          | F firstinset                    |
| F enabletighten           | F firstsectionpage              |
| F enabletrackers          | F firstsubcountervalue          |
| F enabletransparency      | F flagcodestring                |
| F enableutf               | F flushcolumnsetareas           |
| F enhancelinkedlist       | F flushcolumnsetcolumn          |
| F enhancelist             | F flushfloat                    |
| F enhanceregister         | F flushlabeledfloat             |
| F enhancesynonym          | F flushnoterange                |
| F environment             | F flushnotes                    |
| F eposcolumn              | F flushpageblocks               |
| F eposcolumnregistered    | F flushpageinjectionsafter      |
| F erasebuffer             | F flushpageinjectionsbefore     |
| F execute                 | F flushpostponedblocks          |
| F exportbox               | F flushpostponednotes           |
| F exportformdata          | F flushstream                   |
| F exportreferences        | F flushstreamcopy               |
| F expression              | F fontchar                      |
| F extendregister          | F fontcharbyindex               |
| F extendregisterlate      | F fontcollectionclone           |
| F extensiblecode          | F fontcollectiondefine          |
| F fakeword                | F fontcollectionprepare         |
| F featureattribute        | F fontcollectionregister        |

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|-------------------------------|----------------------------------|
| F fontcollectionreport        | F getfile                        |
| F fontcollectionreset         | F getfirstcharacter              |
| F fontlookupgetkey            | F getfloatvariable               |
| F fontlookupgetkeyofindex     | F getfromluatable                |
| F fontlookupinitialize        | F getfullstructurenumber         |
| F fontlookupnofound           | F getinterlinepenalties          |
| F forceendjob                 | F getinternalcachedlistreference |
| F forcepageflush              | F getinternallistreference       |
| F forcequitjob                | F getinternalorderreference      |
| F forcestrutdepth             | F getmathblobindex               |
| F format                      | F getmathblobmapping             |
| F formatcolor                 | F getmathcharone                 |
| F formatgray                  | F getmathchartwo                 |
| F formatluminance             | F getmathmathblob                |
| F formatluminanceratio        | F getmathtextblob                |
| F formatone                   | F getmathvariant                 |
| F framedoutline               | F getmessage                     |
| F from                        | F getmpextensions                |
| F frozenparcodestring         | F getnaturaldimensions           |
| F frule                       | F getnaturalwd                   |
| F getMPdata                   | F getnofpreviouslines            |
| F getbidimode                 | F getnotestate                   |
| F getbodyfontsizes            | F getnotestorecount              |
| F getboxcountfromcache        | F getnotestoreflushed            |
| F getboxdpfromcache           | F getobjectdimensions            |
| F getboxfromcache             | F getobjectreference             |
| F getboxhtdpfromcache         | F getobjectreferencepage         |
| F getboxhtfromcache           | F getparshape                    |
| F getboxwdfromcache           | F getparwrapper                  |
| F getbuffer                   | F getprivatechar                 |
| F getbuffercontent            | F getprivateglyphslot            |
| F getbufferctxlua             | F getprivatemathchar             |
| F getbuffermkvi               | F getprivateslot                 |
| F getbuffertex                | F getprogram                     |
| F getcharactercasingcategory  | F getrandomdimen                 |
| F getclubpenalties            | F getrandomfloat                 |
| F getcurrentprefixspec        | F getrandomnumber                |
| F getcurrentreferenceLabel    | F getrandomseed                  |
| F getcurrentreferencemetadata | F getreadfilename                |
| F getdefaultfieldvalue        | F getreferencesectionnumber      |
| F getdisplaywidowpenalties    | F getrivers                      |
| F getdocumentargument         | F getsetupstring                 |
| F getdocumentfilename         | F getshapeparameter              |
| F getfieldcategory            | F getsomefullstructurenumber     |

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|-----------------------------|------------------------------|
| F getsomestructurenumber    | F ifinset                    |
| F getspecificstructuretitle | F iflua                      |
| F getstructurelevel         | F ifmode                     |
| F getstructurenumber        | F ifmodes                    |
| F getsubstring              | F ifmvl                      |
| F getsynchronizemarking     | F ifpositunum                |
| F geturl                    | F ifsetups                   |
| F getvisual                 | F ifsystemmode               |
| F getwidowpenalties         | F iftext                     |
| F globalpopmacro            | F ifxml                      |
| F globalpushmacro           | F ifxmlatt                   |
| F gluetolinesinbox          | F ifxmlattempty              |
| F glyphoptioncodestring     | F ifxmlempty                 |
| F gobblenested              | F ifxmlselfempty             |
| F gotocolumnsetslot         | F ifxmltext                  |
| F grabbuffer                | F ignorefile                 |
| F groupcodestring           | F ignoretagsinexport         |
| F groupsetgroup             | F immediatemessage           |
| F grph_nodes_initialize     | F includesvgbuffer           |
| F grph_nodes_register       | F includesvgfile             |
| F grph_nodes_reset          | F incrementcounter           |
| F grph_nodes_resolve        | F incrementsubcounter        |
| F handlemathhang            | F inhibitprimitive           |
| F hboxtovbox                | F initializeexport           |
| F heightofstring            | F initializejob              |
| F helpaction                | F initializemathcollapsing   |
| F helpreference             | F initializemathfencing      |
| F hour                      | F initializemathitalics      |
| F hpackbox                  | F initializemathkernpairs    |
| F hpackedbox                | F initializemathnumbers      |
| F hsplit                    | F initializemathspacing      |
| F hyphenateddigits          | F initializestream           |
| F hyphenatedlist            | F initializetexdirectives    |
| F hyphenationcodestring     | F initializetexexperiments   |
| F ic                        | F initializetextrackers      |
| F idxfromluatable           | F injectcurrentreference     |
| F if_common                 | F injectcurrentreferencehtdp |
| F ifallmodes                | F injectdisable              |
| F ifbitwiseand              | F injectnowhite              |
| F ifblobindexdone           | F injectreference            |
| F ifboxhaslist              | F injectvpenalty             |
| F ifcommon                  | F injectvskip                |
| F ifeof                     | F injectvspacing             |
| F ifgluelineinbox           | F injectzerobaselineskip     |

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|-------------------------------|-------------------------|
| F inlinechemical              | F lastypos              |
| F inputfilebarename           | F lateindent            |
| F inputfilename               | F latelua               |
| F inputfilerealsuffix         | F lateundent            |
| F inputfilesuffix             | F lchexnumber           |
| F insertattachment            | F lchexnumbers          |
| F insertcategorycodestring    | F leapyear              |
| F insertcomment               | F limited               |
| F insertfield                 | F linebuffering         |
| F insertionnumber             | F linedir               |
| F insertoptioncodestring      | F lineprofilebox        |
| F insertrenderingwindow       | F linesegment           |
| F inspectluatable             | F listanchorcodestring  |
| F installedlanguages          | F listbookmark          |
| F installtexdirective         | F listexternal          |
| F installtexexperiment        | F listgroupindex        |
| F installtextracker           | F listlabel             |
| F interceptsamepagecontent    | F listlocation          |
| F it                          | F listnamespaces        |
| F jobfilefullname             | F listprefixednumber    |
| F jobfilename                 | F listprefixedpage      |
| F jobfilesuffix               | F listrawnumber         |
| F jobposd                     | F listrawrealpage       |
| F jobposh                     | F listrawuserpage       |
| F jobposp                     | F listsigncodestring    |
| F jobposw                     | F listsize              |
| F jobposx                     | F listtitle             |
| F jobposy                     | F listuserdata          |
| F languagecharwidth           | F lmt_parameters_define |
| F languagenumber              | F lmt_parameters_preset |
| F lastblobindex               | F lmt_parameters_reset  |
| F lastnodeequals              | F loadcldfile           |
| F lastnodeid                  | F loadcldfileonce       |
| F lastnodeidstring            | F loadexamodes          |
| F lastnodesubtypestring       | F loadfontgoodies       |
| F lastparwrapper              | F loadlibrary           |
| F lastrpos                    | F loadluafile           |
| F lastsavedboxresourceindex   | F loadluafileonce       |
| F lastsavedimageresourceindex | F loadluamodule         |
| F lastsavedimageresourcepages | F loadmapfile           |
| F lastsectionpage             | F loadmapline           |
| F lastsubcountervalue         | F loadspellchecklist    |
| F lastunitstring              | F loadtexfile           |
| F lastxpos                    | F loadtexfileonce       |

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|----------------------------|---------------------------|
| F loadtypescriptfile       | F markedpageparameter     |
| F loadvisualizer           | F markedpages             |
| F loadxmlfile              | F markingnumber           |
| F loadxmlfileonce          | F markingtitle            |
| F localanchorx             | F markinjector            |
| F localanchory             | F markpage                |
| F localboxindent           | F markregionbox           |
| F localboxindex            | F markregionboxcorrected  |
| F localboxlefthang         | F markregionboxtagged     |
| F localboxleftoffset       | F markregionboxtaggedkind |
| F localboxleftskip         | F markregionboxtaggedn    |
| F localboxlinenumber       | F markvisualfonts         |
| F localboxlinewidth        | F mathclassvalue          |
| F localboxlocalwidth       | F mathcontrolcodestring   |
| F localboxmarkonce         | F mathhorizontalcode      |
| F localboxovershoot        | F mathoptioncodestring    |
| F localboxparfillleftskip  | F mathrenderset           |
| F localboxparfillrightskip | F mathstrutcode           |
| F localboxparinitleftskip  | F mathvariantcode         |
| F localboxparinitrightskip | F mathvariantcount        |
| F localboxprogress         | F mathvariantslot         |
| F localboxrighthang        | F maxbufferwidth          |
| F localboxrightoffset      | F message                 |
| F localboxrightskip        | F metapostescaped         |
| F localpopmacro            | F metapostformatted       |
| F localpushmacro           | F metapostgraphformat     |
| F locatefilepath           | F minute                  |
| F loopcs                   | F mixcleanup              |
| F loopcsn                  | F mixfinalize             |
| F lousiness                | F mixflushlist            |
| F luacardinal              | F mixflushrest            |
| F luacount                 | F mixgetsplit             |
| F luadimen                 | F mixsetsplit             |
| F luaexpression            | F mixstate                |
| F luafloat                 | F modecodestring          |
| F luainteger               | F molecule                |
| F luanumber                | F month                   |
| F luarandom                | F monthmnem               |
| F luascript                | F monthname               |
| F luathread                | F mpblobtext              |
| F lxmlid                   | F mpcolor                 |
| F manipulated              | F mpgetblob               |
| F markblobindexdone        | F mpgettext               |
| F markcontent              | F mpgraphic               |

|                              |                               |
|------------------------------|-------------------------------|
| F mpgraphicstxtoks           | F notedeltapage               |
| F mpoptions                  | F noteinternal                |
| F mpoutlinetoks              | F notelistindex               |
| F mppopvariables             | F noteprefixednumber          |
| F mppushvariables            | F notetitle                   |
| F mprunset                   | F ntb_split_section_check     |
| F mpruntab                   | F ntb_split_section_fetch     |
| F mprunvar                   | F ntb_split_section_locate    |
| F mpsetclippath              | F ntb_split_section_reset     |
| F mpsetoutercolor            | F ntimes                      |
| F mpsetsxxy                  | F number                      |
| F mpsettext                  | F numbers                     |
| F mptexget                   | F objectreference             |
| F mptexreset                 | F objectreferencepage         |
| F mptexset                   | F octnumber                   |
| F mptexsetfrombuffer         | F openin                      |
| F mptexttoks                 | F openout                     |
| F mvloptioncodestring        | F operatingsystem             |
| F namedstructureuservariable | F ordinal                     |
| F namedstructurevariable     | F outputfilename              |
| F naturalwd                  | F overloadbookmark            |
| F negatedcolorcomponent      | F overloadlowercase           |
| F newarray                   | F overloadsattribute          |
| F newhashedtable             | F overloaduppercase           |
| F newindexedtable            | F pageblockrealpage           |
| F newluatable                | F pageofinternal              |
| F newparwrapper              | F pageoverflow                |
| F newrepeatablerandom        | F pagestaterealpage           |
| F newstacking                | F pagestaterealpageorder      |
| F nextparallel               | F paragraphfillmodecodestring |
| F nextsubcountervalue        | F paragraphoptioncodestring   |
| F nodimen                    | F parcontextcodestring        |
| F nofbufferlines             | F pardir                      |
| F nofconversions             | F parpasscallback             |
| F nofdays                    | F parpassidentifier           |
| F nofdetachedinsertions      | F parpos                      |
| F noflinesinbox              | F partriggercodestring        |
| F nofmathvariants            | F pdfbackendcurrentresources  |
| F nofstackedfloats           | F pdfbackendsetcatalog        |
| F nologbuffering             | F pdfbackendsetcolorspace     |
| F normalizecodestring        | F pdfbackendsetextgstate      |
| F normalizedbodyfontsize     | F pdfbackendsetinfo           |
| F normalizeparcodestring     | F pdfbackendsetname           |
| F nospacesmodecodestring     | F pdfbackendsetpageattribute  |

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|-------------------------------|-------------------------------|
| F pdfbackendsetpageresource   | F popfloat                    |
| F pdfbackendsetpagesattribute | F pophyphenation              |
| F pdfbackendsetpattern        | F poplist                     |
| F pdfbackendsetshade          | F poplocalanchors             |
| F pdfcolor                    | F poplogfile                  |
| F pdfextension                | F popparagraphtweak           |
| F pdffeedback                 | F poppath                     |
| F pdflastobj                  | F poprandomseed               |
| F pdfliteral                  | F popreferenceprefix          |
| F pdfobj                      | F popregime                   |
| F pdfrefobj                   | F poprunstate                 |
| F pdfsetuserdata              | F popsectionblock             |
| F pdfvariable                 | F positunum                   |
| F percentageof                | F positunumdef                |
| F pfabs                       | F postlistbox                 |
| F pfacos                      | F postlistcopy                |
| F pfacosh                     | F postponenotes               |
| F pfasin                      | F prefixedconverted           |
| F pfasinh                     | F prelistbox                  |
| F pfatan                      | F prelistcopy                 |
| F pfatanh                     | F preparecolumnsetflush       |
| F pfatantwo                   | F prerolledpages              |
| F pfceil                      | F presetcolumnsetsheets       |
| F pfcos                       | F presetcolumnsetspreadsheets |
| F pfcosh                      | F presetmathematics           |
| F pfdeg                       | F presetmathfontalternate     |
| F pfexp                       | F presetsymbollist            |
| F pffloor                     | F previoussubcountervalue     |
| F pflog                       | F privatecharacter            |
| F pfmod                       | F privatecharactercode        |
| F pfpow                       | F processbodyfontsizes        |
| F pfrad                       | F processbuffer               |
| F pfrem                       | F processcollectedmath        |
| F pfround                     | F processedfile               |
| F pfsin                       | F processedfiles              |
| F pfsinh                      | F processfile                 |
| F pfsqrt                      | F processfilemany             |
| F pftan                       | F processfilenone             |
| F pftanh                      | F processfileonce             |
| F pgfxcolorspec               | F processjob                  |
| F pickupbuffer                | F processlist                 |
| F pickupmarkedcontent         | F processregister             |
| F pickuppunctuation           | F processsplit                |
| F placeparallel               | F processsynonyms             |

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|-------------------------|---------------------------------|
| F product               | F registerbodyfontsize          |
| F profilebox            | F registerbookmark              |
| F profilemacro          | F registerboxglyph              |
| F project               | F registerbuttons               |
| F pruneboxesincache     | F registercolorintent           |
| F purefontname          | F registercolumnsetarea         |
| F pushfloat             | F registercolumnsetsheet        |
| F pushhyphenation       | F registercolumnsetspreadsheets |
| F pushlist              | F registerengineclass           |
| F pushlocalanchors      | F registerfallbackfont          |
| F pushlogfile           | F registerfiguremask            |
| F pushparagraphtweak    | F registerfigurepattern         |
| F pushpath              | F registerfigurerange           |
| F pushrandomseed        | F registerfiguresuffix          |
| F pushreferenceprefix   | F registerfilescheme            |
| F pushregime            | F registerfontalternative       |
| F pushrunstate          | F registerfontfamily            |
| F pushsectionblock      | F registerforcedbreak           |
| F pushstream            | F registergroupset              |
| F putboxincache         | F registerhashedfiles           |
| F putincolumnset        | F registerhelp                  |
| F putincolumnsetdirect  | F registerhyphenationexception  |
| F putnextboxincache     | F registerhyphenationpattern    |
| F qrcodebuffer          | F registeritemgroup             |
| F qrcodestring          | F registerlabel                 |
| F rawMPdata             | F registerlanguagefeatures      |
| F rawcharacter          | F registerlinenumbering         |
| F rawsetupstring        | F registermaintextcolor         |
| F rawsubcountervalue    | F registermpextensible          |
| F read                  | F registernamedlua              |
| F readfromterminal      | F registernamespace             |
| F readline              | F registerobject                |
| F readlinedirect        | F registerpattern               |
| F recolorbox            | F registerpostponedblock        |
| F referencecolumn       | F registerreferencedobject      |
| F referencepagedetail   | F registerrendering             |
| F referencepagestate    | F registersection               |
| F referenceposx         | F registershownsynonym          |
| F referenceposy         | F registersnaprange             |
| F referencerealpage     | F registerstructureprocessor    |
| F registeranchorbox     | F registersynchronize           |
| F registerattachment    | F registersynonym               |
| F registerbackendsymbol | F registertransliteration       |
| F registerbackground    | F registerunit                  |



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|----------------------------|----------------------------|
| F registerunitshortcut     | F restartsubcounter        |
| F registerunknownbodysize  | F restorecounter           |
| F registerusedsynonym      | F restorecurrentattributes |
| F registeruserdata         | F restorenote              |
| F registerverbosemathlabel | F restoreobject            |
| F removelastrline          | F reuserandomseed          |
| F renderingvar             | F reversevboxcontent       |
| F reorderfloats            | F revivefeature            |
| F replacefeature           | F rightmathkern            |
| F replacepospxywhd         | F romannumerals            |
| F replacewildcard          | F rotateparshape           |
| F reportstructure          | F rounded                  |
| F resavefloat              | F roundedfill              |
| F reservedautoregiontag    | F roundedoutline           |
| F resetMPinstance          | F ruby                     |
| F resetalternate           | F ruleoptioncodestring     |
| F resetbackgroundrowdata   | F runcontextjob            |
| F resetboxesincache        | F safechar                 |
| F resetcharacteralign      | F samplefile               |
| F resetcolumnset           | F saveboxresource          |
| F resetcomparingtext       | F savebuffer               |
| F resetcounter             | F savecounter              |
| F reseteffect              | F savecurrentattributes    |
| F resetfeature             | F savecurrentvalue         |
| F resetfontfeature         | F savedlistnumber          |
| F resetfontsolution        | F savedlistprefixednumber  |
| F resetforcedbreak         | F savedlisttitle           |
| F resetinjector            | F savefloat                |
| F resetlogfile             | F saveimageresource        |
| F resetmapfiles            | F savemargindata           |
| F resetmathattributes      | F savenote                 |
| F resetnullfont            | F savepagedata             |
| F resetparallel            | F savepageinjections       |
| F resetpath                | F saveparallel             |
| F resetreference           | F savepos                  |
| F resetscript              | F savespecificfloat        |
| F resetstructureprocessor  | F savestructureblock       |
| F resetsubcounter          | F scangivendimensions      |
| F resetsynchronizemarking  | F scriptsmodecodestring    |
| F resetsynonyms            | F second                   |
| F resettimer               | F selectstructureblock     |
| F resettrackers            | F set_extensible_data      |
| F resolvedemoji            | F setadaptive              |
| F restartcounter           | F setalternate             |

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|------------------------------|------------------------------|
| F setautoprefix              | F setfigurepaths             |
| F setbackgroundrowdata       | F setfirstline               |
| F setboxglyph                | F setfontfeature             |
| F setboxline                 | F setfontofid                |
| F setbreakpoints             | F setfontsolution            |
| F setcharacteralign          | F setformat                  |
| F setcharacteraligndetail    | F setformsmethod             |
| F setcharactercasing         | F sethsizecolumnset          |
| F setcharactercleaning       | F sethsizecolumnspan         |
| F setcharacterkerning        | F sethyphenatedurlcharacters |
| F setcharacterspacing        | F sethyphenationfeatures     |
| F setcharacterstripping      | F sethyphenationmethod       |
| F setclosedocumentaction     | F setidentityvariable        |
| F setclosepageaction         | F setinitial                 |
| F setcolormodel              | F setinitialsectionblock     |
| F setcolumnsetarea           | F setinjector                |
| F setcolumnsetlines          | F setinsertionlocation       |
| F setcolumnsetmvl            | F setinsertmigration         |
| F setcolumnsetpresent        | F setinterfaceconstant       |
| F setcolumnsetproperties     | F setinterfaceelement        |
| F setcolumnsetstart          | F setinterfacemessage        |
| F setcomparetext             | F setinterfacemessages       |
| F setcomparingtext           | F setinterfacevariable       |
| F setcounter                 | F setinternalreference       |
| F setdataset                 | F setitaliccorrection        |
| F setdestinationattribute    | F setjavascriptpreamble      |
| F setdigitsmanipulation      | F setjobcomment              |
| F setdirection               | F setlanguageexceptions      |
| F setdocumentargument        | F setlanguageoptions         |
| F setdocumentcommandline     | F setlanguagepatterns        |
| F setdocumentctxfile         | F setlanguagesynonym         |
| F setdocumentdefaultargument | F setlinefiller              |
| F setdocumentenvironments    | F setlineprofile             |
| F setdocumentfilename        | F setlocalalignanchor        |
| F setdocumentfilenames       | F setlocalmarkanchor         |
| F setdocumentmodes           | F setlocalsyncanchor         |
| F setdocumentmodules         | F setluatable                |
| F seteffect                  | F setmathalphabet            |
| F setelementbackendtag       | F setmathattribute           |
| F setelementnature           | F setmathfontalternate       |
| F setextrafontkerns          | F setmathfontstyle           |
| F setfieldcalculationset     | F setmathsnapping            |
| F setfigureconversion        | F setmathvariant             |
| F setfigurelookuporder       | F setmetaglyph               |

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|-----------------------------|---------------------------------|
| F setmpextensions           | F setsubcounter                 |
| F setnaturalwd              | F settagcodepoint               |
| F setnegative               | F settagcombination             |
| F setnextinternalreferences | F settagcombinationpair         |
| F setnextreferenceorder     | F settagconstruct               |
| F setnotestate              | F settagdelimitedsymbol         |
| F setopendocumentaction     | F settagdescriptionindex        |
| F setopenpageaction         | F settagfigure                  |
| F setowncounter             | F settagfloat                   |
| F setownsubcounter          | F settagformulacontent          |
| F setpagecolormodel         | F settaggedmetadata             |
| F setpagestate              | F settaggingcomment             |
| F setpagestaterealpageno    | F settaggingoption              |
| F setpagetransition         | F settaggingpresets             |
| F setparagraphmetashape     | F settaghighlight               |
| F setparagraphshape         | F settagitem                    |
| F setparbuilder             | F settagitemgroup               |
| F setparwrapper             | F settaglist                    |
| F setpdfcompression         | F settagmarginanchor            |
| F setperiodkerning          | F settagmargintext              |
| F setpostlistbox            | F settagmpgraphic               |
| F setprelistbox             | F settagnotation                |
| F setprofile                | F settagnotationsymbol          |
| F setrandomseed             | F settagparagraph               |
| F setrealign                | F settagproperty                |
| F setrealspaces             | F settagpublication             |
| F setreferencelinkmethod    | F settagregister                |
| F setregionbox              | F settagsetreferencedescription |
| F setregionboxtagged        | F settagsorting                 |
| F setregistermethod         | F settagsubsentencesymbol       |
| F setreplacements           | F settagsynonym                 |
| F setruby                   | F settagtablecell               |
| F setrule                   | F settagtabulatecell            |
| F setscript                 | F settransliteration            |
| F setsectionblock           | F setupbookmarks                |
| F setsectionentry           | F setupcanvas                   |
| F setshadow                 | F setupcharacterspacing         |
| F setshift                  | F setupcomparingtext            |
| F setsofthyphens            | F setupexport                   |
| F setsplitlistthdp          | F setupfontexpansion            |
| F setstacking               | F setupfontprotrusion           |
| F setstructureblockstate    | F setupinsertion                |
| F setstructurelevel         | F setuplinenumbering            |
| F setstructurenumber        | F setupsynctex                  |

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|------------------------------|---------------------------------|
| F setupusermetadata          | F silliness                     |
| F setupviewerlayer           | F simplealign                   |
| F setvisual                  | F simplecases                   |
| F setvisualfont              | F simplematrix                  |
| F setvisuallayer             | F sin                           |
| F setvisuals                 | F sind                          |
| F setvsizecolumnset          | F sl                            |
| F setxmpfile                 | F spaceddigits                  |
| F shiftmatrixornaments       | F spacefactorcodestring         |
| F shiftparshape              | F spaceskipcodestring           |
| F shipoutpage                | F special                       |
| F showarray                  | F specificationoptioncodestring |
| F showassignerror            | F specifiedfont                 |
| F showattributes             | F specifiedfontsize             |
| F showbits                   | F specifiedfontspec             |
| F showboxinbuffer            | F splitfilename                 |
| F showbtxdatasetauthors      | F splitsamplefile               |
| F showbtxdatasetcompleteness | F spotcolorname                 |
| F showbtxdatasetfields       | F spotcolorparent               |
| F showbtxentry               | F spotcolorvalue                |
| F showbtxfields              | F squid_check                   |
| F showbtxhashedauthors       | F startactualtext               |
| F showbtxtables              | F startalternativetext          |
| F showbtxvariants            | F startchemical                 |
| F showchardata               | F startclipping                 |
| F showcounter                | F startcollectingmath           |
| F showdirectives             | F startcolorset                 |
| F showdiscretionaries        | F startcomponent                |
| F showexperiments            | F startenvironment              |
| F showfontparameters         | F startfontsolution             |
| F showhyphenatedinlist       | F startgraphicgroup             |
| F showhyphenationtrace       | F startlocalcomponent           |
| F showinjector               | F startmarkpages                |
| F showjustification          | F startmatrix                   |
| F showlogcategories          | F startmirroring                |
| F showluatables              | F startpageblock                |
| F showluatokens              | F startparbuilder               |
| F showmacrostack             | F startprerollpageblock         |
| F showmessage                | F startproduct                  |
| F showpagelist               | F startproject                  |
| F showparagraphshape         | F startregime                   |
| F showprofiledmacros         | F startrotation                 |
| F showstructure              | F startruby                     |
| F showtrackers               | F startscaling                  |

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|-----------------------------------------|------------------------------|
| F startsectionlevel                     | F strc_tags_start_detail     |
| F startstream                           | F strc_tags_start_ignore     |
| F starttext                             | F strc_tags_start_userdata   |
| F startviewerlayer                      | F strc_tags_stop             |
| F step                                  | F stringtoorientation        |
| F stopactualtext                        | F stripinserts               |
| F stopalternativetext                   | F structureautocatcodedget   |
| F stopchemical                          | F structureblockuservariable |
| F stopclipping                          | F structurecatcodedget       |
| F stopcollectingmath                    | F structuregivencatcodedget  |
| F stopcolorset                          | F structurenumber            |
| F stopcomponent                         | F structuretitle             |
| F stopenvironment                       | F structureuservariable      |
| F stopfontsolution                      | F structurevariable          |
| F stopgraphicgroup                      | F subcountervalue            |
| F stoplocalcomponent                    | F subsubcountervalues        |
| F stopmarkpages                         | F subtocolumnsetmvl          |
| F stopmatrix                            | F subtocolumnsetmvlbalance   |
| F stopmirroring                         | F subtractfeature            |
| F stoppageblock                         | F svghashed                  |
| F stopparbuilder                        | F sync                       |
| F stopprerollpageblock                  | F synccolor                  |
| F stopproduct                           | F synccolorclone             |
| F stopproject                           | F synccolorcount             |
| F stopregime                            | F synchronizelocalanchors    |
| F stoprotation                          | F synchronizemarking         |
| F stopscaling                           | F synchronizesteps           |
| F stopsectionlevel                      | F synchronizestream          |
| F stopstream                            | F syntexblockfilename        |
| F stoptext                              | F syntexdisable              |
| F stopviewerlayer                       | F syntexpause                |
| F storejavascriptcode                   | F syntexpopline              |
| F storejavascriptpreamble               | F syntexpushline             |
| F storenote                             | F syntexresetfilename        |
| F storeregister                         | F syntexresume               |
| F strc_tags_document_pop                | F syntexsetfilename          |
| F strc_tags_document_push               | F synonymfirstuserpage       |
| F strc_tags_get_element_tag             | F synonymmeaning             |
| F strc_tags_parcounter_pop              | F synonymname                |
| F strc_tags_parcounter_push             | F synonympages               |
| F strc_tags_set_aspect                  | F synonymuserpages           |
| F strc_tags_set_element_user_properties | F systemlog                  |
| F strc_tags_start                       | F tabl_ntb_get_sec           |
| F strc_tags_start_chained               | F tabl_ntb_set_sec           |

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|----------------------------|-------------------------|
| F tagmfunctionlab          | F unsupportedcs         |
| F tagmfunctionlabattribute | F useMPLibrary          |
| F tan                      | F useanyfile            |
| F tand                     | F useanyfileonce        |
| F textdir                  | F useboxresource        |
| F textime                  | F usecolors             |
| F tf                       | F usecomponent          |
| F thebox                   | F useenvironment        |
| F thefirstcharacter        | F usefigurebase         |
| F themaxboxwidth           | F usefile               |
| F theremainingcharacters   | F useimageresource      |
| F tightfitcompensation     | F usejavascriptpreamble |
| F timestring               | F usejavascriptscripts  |
| F tobigpoints              | F uselibrary            |
| F tobit                    | F uselist               |
| F tobits                   | F useluamodule          |
| F tochar                   | F useluatable           |
| F tohexa                   | F usemathpluginlibrary  |
| F toorientation            | F usemodules            |
| F totalofstring            | F usepath               |
| F tounicharnum             | F useproduct            |
| F towholebigpoints         | F useproject            |
| F transliterate            | F usereferences         |
| F transliteratebuffer      | F useregister           |
| F transliterated           | F usesubpath            |
| F transliteratedbuffer     | F usesymbols            |
| F transliterationexception | F setransliteration     |
| F traverseblock            | F useurl                |
| F traversemath             | F usezipfile            |
| F triggerpagebuilder       | F utfchar               |
| F truefilename             | F utflower              |
| F truefontname             | F utfmathclass          |
| F type                     | F utfmathcommand        |
| F typebuffer               | F utfmathcommandfiller  |
| F typefile                 | F utfmathfiller         |
| F uchar                    | F utfmathstretch        |
| F uchexnumber              | F utfshape              |
| F uchexnumbers             | F utfstrip              |
| F undefinechemical         | F utfupper              |
| F unicodechar              | F uuid                  |
| F unihex                   | F vboxlisttohbox        |
| F unit                     | F verbose               |
| F unloadlanguage           | F vpackbox              |
| F unstripinserts           | F vpackedbox            |

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|-------------------------------------|--------------------------|
| F way                               | F xmlconcatrange         |
| F weekday                           | F xmlcontext             |
| F weekdayname                       | F xmlcount               |
| F widthofstring                     | F xmlcssmappedstylevalue |
| F widthuptohere                     | F xmlcssstylevalue       |
| F write                             | F xmldelete              |
| F writeline                         | F xmldepth               |
| F writestatus                       | F xmldirect              |
| F writestring                       | F xmldirectives          |
| F x_table_c                         | F xmldirectivesafter     |
| F x_table_cleanup                   | F xmldirectivesbefore    |
| F x_table_construct                 | F xmldisplayverbatim     |
| F x_table_create                    | F xmldoif                |
| F x_table_finish_row                | F xmldoifatt             |
| F x_table_flush                     | F xmldoifelse            |
| F x_table_init_construct            | F xmldoifelseatt         |
| F x_table_init_reflow_height        | F xmldoifelseempty       |
| F x_table_init_reflow_height_option | F xmldoifelseselfempty   |
| F x_table_init_reflow_width         | F xmldoifelsetext        |
| F x_table_init_reflow_width_option  | F xmldoifempty           |
| F x_table_next_row                  | F xmldoifnot             |
| F x_table_next_row_option           | F xmldoifnotatt          |
| F x_table_r                         | F xmldoifnotempty        |
| F x_table_reflow_height             | F xmldoifnotselfempty    |
| F x_table_reflow_width              | F xmldoifnottext         |
| F x_table_set_construct             | F xmldoifselfempty       |
| F x_table_set_reflow_height         | F xmldoiftext            |
| F x_table_set_reflow_width          | F xmlelement             |
| F xmladdindex                       | F xmlfileline            |
| F xmlafterssetup                    | F xmlfilename            |
| F xmlall                            | F xmlfilter              |
| F xmlappendsetup                    | F xmlfilterlist          |
| F xmlapplyselectors                 | F xmlfirst               |
| F xmlatt                            | F xmlflush               |
| F xmlattdef                         | F xmlflushcontext        |
| F xmlattribute                      | F xmlflushlinewise       |
| F xmlattributedef                   | F xmlflushpure           |
| F xmlbadinclusions                  | F xmlflushsetups         |
| F xmlbeforesetup                    | F xmlflushspacewise      |
| F xmlchainatt                       | F xmlflushstext          |
| F xmlchainattdef                    | F xmlformulatobuffer     |
| F xmlchecknamespace                 | F xmlfunction            |
| F xmlcommand                        | F xmlgetindex            |
| F xmlconcat                         | F xmlinclude             |

- F xmlincludeoptions
- F xmlinclusion
- F xmlinclusionbase
- F xmlinclusions
- F xmlindex
- F xmlinlineverbatim
- F xmlinstalldirective
- F xmllast
- F xmllastatt
- F xmllastmatch
- F xmllastpar
- F xmlloadbuffer
- F xmlloaddata
- F xmlloaddirectives
- F xmlloadentities
- F xmlloadfile
- F xmlmain
- F xmlmatch
- F xmlname
- F xmlnamespace
- F xmlnonspace
- F xmlpar
- F xmlparam
- F xmlpath
- F xmlpopmatch
- F xmlpos
- F xmlprependsetup
- F xmlpure
- F xmlpushmatch
- F xmlraw
- F xmlrawtex
- F xmlrefatt
- F xmlregisters
- F xmlremapname
- F xmlremapnamespace
- F xmlremovesetup
- F xmlresetinjectors
- F xmlresetsetups
- F xmlsave
- F xmlsetatt
- F xmlsetattribute
- F xmlsetcommandtonone
- F xmlsetcommandtotext
- F xmlsetentity

- F xmlsetinjectors
- F xmlsetpar
- F xmlsetparam
- F xmlsetsetup
- F xmlsnippet
- F xmlstartrow
- F xmlstarttiming
- F xmlstoprow
- F xmlstoptiming
- F xmlstrip
- F xmlstripanywhere
- F xmlstripeverything
- F xmlstripnolines
- F xmlstripped
- F xmlstrippednolines
- F xmltag
- F xmltexatt
- F xmltexentity
- F xmltext
- F xmltobuffer
- F xmltobuffertextonly
- F xmltobufferverbose
- F xmltofile
- F xmltoparameters
- F xmlverbatim
- F xmlwithindex
- F year

## containers

- F cleanname
- F content
- F define
- F is\_usable
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- F bgroup
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## table

## function



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| F LuaSnippetCommentText      | F XmlSnippetComment            |
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| F LuaSnippetString           | F beginhbox                    |
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| F MetapostSnippetNameMetafun | F btxhandlelistentry           |
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| F btxsetauthorvariant    | F chemicalinline       |
| F btxsetbacklink         | F chemicalspecial      |
| F btxsetbefore           | F color                |
| F btxsetcategory         | F colordefagc          |
| F btxsetcombis           | F colordefagt          |
| F btxsetconcat           | F colordefalc          |
| F btxsetcount            | F colordefalt          |
| F btxsetcurrentlistentry | F colordeffgc          |
| F btxsetcurrentlistindex | F colordeffgt          |
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| F btxsetfirst            | F colordefflt          |
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| F btxsetinitials         | F colordefrlt          |
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| F setruby                  | F stopreadingfile         |
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| F robustname        |              | F setposition       |              |
| F savechecksum      |              | F skipposition      |              |
| F savedata          |              | F writecardinal1    |              |
| F size              |              | F writecardinal1lle |              |
| F splitbase         |              | F writecardinal2    |              |
| F splitname         |              | F writecardinal2le  |              |
| F splitpath         |              | F writecardinal3    |              |
| F suffix            |              | F writecardinal3le  |              |
| F suffixes          |              | F writecardinal4    |              |
| F suffixesonly      |              | F writecardinal4le  |              |

F writedouble  
 F writedoublele  
 F writefloat  
 F writefloatle  
**floats**                      table

F analysemethod  
 F checkedpagefloat  
 F collect  
 F consult  
 F flush  
 F getvariable  
 F hasstacked  
 F nofstacked  
 F pop  
 F push  
 F resave  
 F save  
 F stacked

**font**                              table

F addcharacters  
 F addquality  
 F current  
 F define  
 F getfont  
 F getfontdimen  
 F getfontfields  
 F getfontspec  
 F getmathcharacterfields  
 F getmathindex  
 F getmathspec  
 F getparameterfields  
 F gettextcharacterfields  
 F id  
 F max  
 F nextid  
 F setfont  
 F setfontdimen  
 F xscaled  
 F yscaled

**fonts**                              table

T analyzers  
 D features  
 D initializers  
 D methods

F setstate  
 D states  
 B useunicodemarks  
 T callbacks  
 F compile  
 F devirtualize  
 T checkers  
 F getmissing  
 D mapping  
 F missing  
 F placeholder  
 T cid  
 F getmap  
 F loadfile  
 T collections  
 F checkenabled  
 F clonevector  
 F define  
 F direct  
 F found  
 F prepare  
 F process  
 F register  
 F registermain  
 F report  
 F reset  
 T constructors  
 F addcharacters  
 F addcoreunicodes  
 F aftercopyingcharacters  
 F applyfinalizers  
 F applymanipulators  
 F assignmathparameters  
 F beforecopyingcharacters  
 F beforepassingfonttotex  
 D cache  
 F calculatescale  
 F checkedfeatures  
 F checkedfilename  
 F collectprocessors  
 F currentfonthasfeature  
 D designsizes  
 F enhanceparameters  
 N factor

|                       |                         |
|-----------------------|-------------------------|
| F finalize            | F registered            |
| F getfeatureaction    | F registerfallbackfont  |
| F getmathparameter    | F registersplit         |
| F getprivate          | F resetnullfont         |
| F hashfeatures        | F resolve               |
| F hashinstance        | F definetypesface       |
| F initializefeatures  | T dropins               |
| D loadedfonts         | F clone                 |
| N noffontsloaded      | F getshape              |
| N nofsharedfonts      | F getshapes             |
| N nofsharedhashes     | F nextid                |
| N nofsharedvectors    | F provide               |
| N privateoffset       | F registerglyph         |
| D privateslots        | F registerglyphs        |
| F readanddefine       | F swap                  |
| F registerfeature     | F swapone               |
| F registerscriptset   | T effects               |
| F resetfeature        | T encodings             |
| B resolvevirtualtoo   | D cache                 |
| F scale               | F is_known              |
| F scaled              | D known                 |
| F setfactor           | F load                  |
| F setfeature          | F make_unicode_vector   |
| F setmathparameter    | D math                  |
| F setname             | N version               |
| B sharefonts          | F enhanceid             |
| F trytosharefont      | D expansions            |
| N version             | D formats               |
| F current             | F getalternate          |
| F currentid           | F getindices            |
| T definers            | F getkern               |
| F addlookup           | F getmultiple           |
| F analyze             | F getsubstitution       |
| F applypostprocessors | T getters               |
| F checkedmode         | T goodies               |
| F checkedscript       | D colorschemes          |
| S defaultlookup       | D designsizes           |
| F define              | D filenames             |
| F getspecification    | F getcompositions       |
| F internal            | F load                  |
| F loadfont            | F prepare_features      |
| F makespecification   | F register              |
| F read                | F registerpostprocessor |
| F register            | F report                |

|                      |                        |
|----------------------|------------------------|
| T handlers           | F getcoloredglyphs     |
| T hashes             | F getdigitwidth        |
| D characters         | F getinstancespec      |
| D classes            | F getparameters        |
| D csnames            | F getprivatecharornode |
| D descriptions       | F getprivatenode       |
| D dynamics           | F getprivates          |
| D emwidths           | F getprivateslot       |
| D exheights          | F glyphdimenfactor     |
| D features           | F hasprivate           |
| D identifiers        | F indextoslot          |
| D italics            | F isprivate            |
| D lastmathids        | F mathscriptslots      |
| D marks              | F name                 |
| D mathparameters     | F nametoslot           |
| D modes              | F newprivateslot       |
| D originals          | F prependcommands      |
| D parameters         | F prependcommandtable  |
| D properties         | F privateslot          |
| D quads              | F registerglyphbox     |
| D resources          | F scalecommands        |
| D spaces             | F setboxdirectly       |
| D spacings           | F setboxglyph          |
| D unicodes           | F setboxglyphs         |
| D unislots           | F setdigitwidth        |
| D variants           | F setmetaglyph         |
| D xheights           | F setmetaglyphs        |
| T helpers            | D sharedprivates       |
| F addextraprivate    | F splitfontpattern     |
| F addprivate         | F tochar               |
| F appendcommands     | F infofont             |
| F appendcommandtable | T iterators            |
| F checkflags         | F characters           |
| F checkmerge         | F descriptions         |
| F checksteps         | F glyphs               |
| F collectanchors     | T loggers              |
| F combineglyphs      | F format               |
| D commands           | F onetimemessage       |
| F composecharacters  | F register             |
| F descriptiontoslot  | F reportdefinedfonts   |
| F dimenfactor        | F reportusedfeatures   |
| F expandglyph        | T mappings             |
| D extraprivates      | F addtounicode         |
| F fontpatternhassize | F fromunicode16        |

|                       |                           |
|-----------------------|---------------------------|
| F getentry            | F ttf                     |
| F loadfile            | F woff                    |
| F loadline            | F woff2                   |
| F makenameparser      | F registersanitizer       |
| D overloads           | F reserveid               |
| F reset               | T select                  |
| F tounicode           | D alternatives            |
| F tounicode16         | F bodyfont                |
| F tounicode16sequence | D data                    |
| T metapost            | F default                 |
| F baseline            | D defaults                |
| F boundingbox         | F definefallbackfamily    |
| F boxtomp             | F definefontfamily        |
| F boxtompdimensions   | F definefontfamilypreset  |
| F boxtompposition     | D extras                  |
| F draw                | F fallback                |
| F fill                | D fallbacks               |
| F glyph               | F features                |
| F maxbounds           | F filefallback            |
| F paths               | F fontdata                |
| F widthline           | F fontfallback            |
| F zeroline            | F fontsynonym             |
| F mode                | F goodies                 |
| D names               | F mathfallback            |
| D nulldata            | D methods                 |
| T privateoffsets      | D presets                 |
| B keepnames           | F registerfallback        |
| N mathbase            | F registerfiles           |
| N mathextrabase       | F registerfontalternative |
| N textbase            | F registerfontfamily      |
| N textextrabase       | F textfallback            |
| D protrusions         | F typeface                |
| T readers             | F typescript              |
| F afm                 | F userdata                |
| F check_lua           | T specifiers              |
| F check_tfm           | D contextmerged           |
| F lua                 | F contextnumber           |
| F ofm                 | D contextnumbers          |
| F opentype            | D contextsetups           |
| F otf                 | F contexttostring         |
| F pfb                 | F definecontext           |
| D sequence            | F mergecontext            |
| F tfm                 | F presetcontext           |
| F ttc                 | F registercontext         |

|                      |                    |
|----------------------|--------------------|
| F showcontext        | F getvalue         |
| F splitcontext       | F getvalues        |
| D synonyms           | F hasvalues        |
| D variants           | D list             |
| T symbols            | F markasoverload   |
| F uselibrary         | D names            |
| T tables             | D numbers          |
| F toeffect           | F ofnode           |
| T tracers            | F private          |
| T treatments         | F public           |
| F applyfixes         | F registervalue    |
| D data               | F restore          |
| S filename           | F save             |
| F ignoredfile        | F setcleaner       |
| F report             | F showcurrent      |
| F trace              | F system           |
| T typefaces          | F tooverloads      |
| <b>functions</b>     | N unsetvalue       |
| F dummy              | D values           |
| <b>getmetatable</b>  | T backends         |
| <b>globaldata</b>    | F closeout         |
| N CONTEXTLMTXMODE    | S current          |
| N LUAFORMAT          | F getcallbackstate |
| N LUAMAJORVERSION    | F initialize       |
| N LUAMINORVERSION    | F latelua          |
| S LUAEXENGINE        | F nothing          |
| N LUAEXFORMATID      | F openout          |
| N LUAEXFUNCTIONALITY | F setencryption    |
| S LUAEXVERBOSE       | F writeout         |
| N LUAEXVERSION       | T basexx           |
| N LUAVERSION         | F decode16         |
| T MP                 | F decode64         |
| F SAVEDREQUIRE       | F decode85         |
| T aes                | F decodeLZW        |
| F decode             | F decodeRL         |
| F encode             | F encode16         |
| F random             | F encode64         |
| T arg                | F encode85         |
| F assert             | F encodeLZW        |
| T attributes         | F encodeRL         |
| F applyoverloads     | T bibtex           |
| F checkvalues        | T bit32            |
| D counts             | F arshift          |
| F define             | F band             |



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|------------------|----------------------------|
| F bnot           | F getcontent               |
| F bor            | F getctxlua                |
| F btest          | F getlines                 |
| F bxor           | F getmkvi                  |
| F extract        | F gettex                   |
| F lrotate        | F loadcontent              |
| F lshift         | F prepend                  |
| F replace        | F raw                      |
| F rrotate        | F run                      |
| F rshift         | F samplefile               |
| T bitset         | F splitsamplefile          |
| F asstring       | F undent                   |
| F get            | T builders                 |
| F new            | F checkparcontext          |
| F reset          | F checksidefloat           |
| F set            | F parpassidentifier        |
| F totable        | D passnames                |
| F traverse       | F registerpasspasscallback |
| F wipe           | T bytemap                  |
| T blobs          | F bounds                   |
| F append         | F clip                     |
| F dimensions     | F copy                     |
| F dispose        | F dimensions               |
| F new            | F domain                   |
| F pack           | F downgrade                |
| F strdp          | F downsample               |
| F strhd          | F fillgray                 |
| F strht          | F fillrgb                  |
| F strwd          | F filter                   |
| F tohpack        | F getbyte                  |
| F tonodes        | F getbytes                 |
| F tovpack        | F getluminance             |
| F write          | F getvalue                 |
| T boolean        | F hasbytegray              |
| F tocontext      | F hasbyterange             |
| F tonumber       | F hasbytergb               |
| T buffers        | F new                      |
| F append         | F newdomain                |
| F assign         | F overlay                  |
| F collectcontent | F process                  |
| F empty          | F reduce                   |
| F erase          | F reset                    |
| F exists         | F setgray                  |
| F get            | F setgrayadd               |

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|------------------------|---------------------|
| F setgraymin           | F getcomment        |
| F setrgb               | F getstates         |
| F slicegray            | F known             |
| F slicerange           | F optimizer         |
| F slicergb             | F pop               |
| F totable              | F push              |
| T caches               | F register          |
| S base                 | F report            |
| F collapsecontent      | T catcodes          |
| F configfiles          | D names             |
| F contentstate         | D numbers           |
| F getfirstreadablefile | F register          |
| F getreadablepaths     | F characterkern     |
| F getwritablepath      | T characters        |
| F hashed               | F addcharacternames |
| F is_writable          | F adobename         |
| F loadcontent          | D bidi              |
| F loaddata             | F blockrange        |
| S more                 | D blocks            |
| F savecontent          | D can_have_space    |
| F savedata             | D categories        |
| F setfirstwritablefile | F category          |
| F setluaname           | D categorytags      |
| F treehash             | D collapsed         |
| F usedpaths            | D combined          |
| T callback             | F contextname       |
| F find                 | D csletters         |
| F getcontracts         | D data              |
| F getcontractvalues    | D decomposed        |
| F getindex             | F description       |
| F getoptions           | D descriptions      |
| F getoptionvalues      | D detailltags       |
| F getstate             | F emojiized         |
| F getstatevalues       | D encodes           |
| F known                | D entities          |
| F list                 | D fallbacks         |
| F names                | D filters           |
| F register             | F fromnumber        |
| F setoptions           | D fschars           |
| F setstate             | D fscodes           |
| F testonly             | F getrange          |
| F usage                | D graphemes         |
| T callbacks            | D hangul            |
| F freeze               | F initialize        |

|                         |                           |
|-------------------------|---------------------------|
| D is_character          | D definitions             |
| D is_command            | S format                  |
| D is_hyphenator         | F inlinechemical          |
| D is_letter             | S instance                |
| D is_mark               | S method                  |
| D is_nothing            | F molecule                |
| D is_punctuation        | U moleculeparser          |
| D is_spacing            | F start                   |
| D is_symbol             | F stop                    |
| D lcchars               | F undefine                |
| F lccode                | F collectgarbage          |
| D lccodes               | T commands                |
| F lettered              | F Alphabetic              |
| D lhash                 | F MPLIBconvertfollowtext  |
| D linebreaks            | F MPLIBconvertoutlinetext |
| F lower                 | F MPc                     |
| D mathlists             | F MPd                     |
| D otfscripts            | F MPfromhash              |
| D ranges                | F MPH                     |
| F registerentities      | F MPLl                    |
| F remap_hangul_syllable | F MPLr                    |
| F setactivecatcodes     | F MPLs                    |
| F setlettercatcodes     | F MPn                     |
| F setothercatcodes      | F MPp                     |
| F shape                 | F MPpadata                |
| F shaped                | F MPplus                  |
| D shash                 | F MPpos                   |
| D shchars               | F MPr                     |
| D shcodes               | F MPrest                  |
| F showstring            | F MPrs                    |
| D specialchars          | F MPul                    |
| D specials              | F MPur                    |
| D synonyms              | F MPw                     |
| F tonumber              | F MPwhd                   |
| D ucchars               | F MPx                     |
| F uccode                | F MPxy                    |
| D uccodes               | F MPxywhd                 |
| D uhash                 | F MPy                     |
| F unicodechar           | F Romannumerals           |
| F upper                 | F actualday               |
| D verbosemarks          | F actualmonth             |
| T chemistry             | F actualyear              |
| F component             | F adaptfontfeature        |
| F define                | F adaptivecheckbox        |

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|-----------------------------|--------------------------|
| F addbtxentry               | F autonextstructurelevel |
| F addcounter                | F autorule               |
| F adddefaultsuffix          | F autosetups             |
| F addfeature                | F autostarttext          |
| F addfontpath               | F autostoptext           |
| F addgrapheme               | F badnessmodecodestring  |
| F addlinenumbers            | F balanceshapeextra      |
| F addlinklistelement        | F benchmarktimer         |
| F addreplacements           | F bf                     |
| F addreplacementslist       | F bi                     |
| F addrivers                 | F bitmapimage            |
| F addtocolumnset            | F bitwise                |
| F addtocolumnsetmvl         | F bitwiseand             |
| F addtojavascriptpreamble   | F bitwiseflip            |
| F addtolist                 | F bitwisenil             |
| F align_point_delta         | F bitwisenot             |
| F allinputpaths             | F bitwiseor              |
| F alphabetic                | F bitwiset               |
| F analysefloatmethod        | F bitwiseshift           |
| F analyzeitemgroup          | F bitwisexor             |
| F analyzelist               | F blockcolumnset         |
| F analyzelocalanchors       | F blockligatures         |
| F analyzeregister           | F boolcase               |
| F analyzerotate             | F boxdir                 |
| F analyzeupcomingitemgroup  | F boxline                |
| F applypattern              | F boxlinedp              |
| F applytobox                | F boxlineht              |
| F applytofirstcharacter     | F boxlineil              |
| F arraycompare              | F boxlinein              |
| F arrayequals               | F boxlineir              |
| F arraymaxvalue             | F boxlinelh              |
| F arrayminvalue             | F boxlinelp              |
| F arraysizes                | F boxlinels              |
| F arrayvalue                | F boxlinemaxdp           |
| F ascender                  | F boxlinemaxht           |
| F askedreference            | F boxlinemaxwd           |
| F assignbuffer              | F boxlinend              |
| F async                     | F boxlinenh              |
| F attachinsertion           | F boxlinenw              |
| F attachinsertions          | F boxlinerh              |
| F atwordboundary            | F boxlinerp              |
| F autofinishstructurelevels | F boxliners              |
| F autoinsertnextspace       | F boxlines               |
| F automigrationcodestring   | F boxlinewd              |

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|-----------------------------|------------------------------|
| F boxoptioncodestring       | F btchandlenocite            |
| F boxrangedp                | F btlastofrange              |
| F boxrangeht                | F btlistvariant              |
| F boxrangewd                | F btloaddefinitionfile       |
| F boxtostring               | F btloadjournalist           |
| F bposcolumn                | F btloadreplacementfile      |
| F bposcolumnregistered      | F btmissing                  |
| F bs                        | F btoneorrange               |
| F btabbreviatedjournal      | F btpreparelistentries       |
| F btaddentry                | F btremapauthor              |
| F btaddjournal              | F btsavedataset              |
| F btauthor                  | F btsavejournalist           |
| F btauthorfield             | F btsetdataset               |
| F btauthorstoregister       | F btsetentry                 |
| F btcmdstring               | F btsetregister              |
| F btcollectlistentries      | F btsetspecification         |
| F btcurrentfirstnames       | F btshortcut                 |
| F btcurrentinitials         | F btshowauthorremapping      |
| F btcurrentjuniors          | F btsingularorplural         |
| F btcurrentsurnames         | F btxtoregister              |
| F btcurrentvons             | F btusedataset               |
| F btcdetail                 | F btuservariable             |
| F btcdirect                 | F build_page_after_shipout   |
| F btcdoif                   | F build_page_before_shipout  |
| F btcdoifelse               | F build_page_setup           |
| F btcdoifelsecitedone       | F capheight                  |
| F btcdoifelsecombiinlist    | F charactercontrolcodestring |
| F btcdoifelsematches        | F chardescription            |
| F btcdoifelsesameasprevious | F checkcolumnset             |
| F btcdoifelseuservariable   | F checkcolumnsetmvl          |
| F btcdoifnot                | F checkcountersetup          |
| F btcdentrytobuffer         | F checkedconversion          |
| F btcdentrytostring         | F checkedemoji               |
| F btcdexpandedjournal       | F checkedpagefloat           |
| F btcdfetchlistentries      | F checkinjector              |
| F btcdfield                 | F checkinterfacechain        |
| F btcdfirstofrange          | F checkmarkedpages           |
| F btcdflush                 | F checknesting               |
| F btcdflushlistcombi        | F checkspacingavailable      |
| F btcdflushlistentry        | F checkstrutdepth            |
| F btcdflushlisttag          | F chemicalcomponent          |
| F btcdflushmarked           | F chinesenumerals            |
| F btcdflushpages            | F chr                        |
| F btcdhandlecite            | F classoptioncodestring      |

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|-------------------------------|----------------------------|
| F cleancolumnset              | F columnsetshowtracedgrid  |
| F cleanfontname               | F columnsetstate           |
| F cleanupattributes           | F columnsetstatus          |
| F cleanupbox                  | F combineglyphs            |
| F cleanuplayers               | F component                |
| F clonefield                  | F concatcommalist          |
| F closein                     | F consultfloat             |
| F closeout                    | F continueifinputfile      |
| F codedef                     | F convertbtxdatasettoxml   |
| F collectedmath               | F converteddimen           |
| F collectfloat                | F converttmpstopdf         |
| F collecthelp                 | F copyboxfromcache         |
| F columnsetbalancefirst       | F copyboxline              |
| F columnsetbalancelast        | F copyposition             |
| F columnsetbalancepage        | F cos                      |
| F columnsetcolumnwidth        | F cosd                     |
| F columnsetdisablewide        | F ctxfunction              |
| F columnsetenablewide         | F ctxluamatch              |
| F columnsetgoto               | F ctxluamatchfile          |
| F columnsethascontent         | F ctxresetter              |
| F columnsethspan              | F ctxscanner               |
| F columnsetlastfuture         | F currentcomponent         |
| F columnsetlastmvl            | F currentdate              |
| F columnsetlimit              | F currentdesignsize        |
| F columnsetnoto               | F currentenvironment       |
| F columnsetresetdirect        | F currentfontinstancespec  |
| F columnsetresetshapeextra    | F currentposthyphenchar    |
| F columnsetreshape            | F currentprehyphenchar     |
| F columnsetsblockrows         | F currentproduct           |
| F columnsetsetshapeextra      | F currentproject           |
| F columnsetshape              | F currentsectiontolist     |
| F columnsetshapebalancecheck  | F datasetsize              |
| F columnsetshapebalanceextend | F datasetvariable          |
| F columnsetshapecolumn        | F day                      |
| F columnsetshapecount         | F daynm                    |
| F columnsetshapeindex         | F dayname                  |
| F columnsetshapepage          | F decimals                 |
| F columnsetshapetbalancerange | F decrementcounter         |
| F columnsetshapelots          | F decrementsubcounter      |
| F columnsetshapspread         | F deferredenhancelist      |
| F columnsetshapevsize         | F deferredenhancereference |
| F columnsetshapeworst         | F deferredenhanceregister  |
| F columnsetshowgrid           | F defineaccents            |
| F columnsetshowshape          | F defineattribute          |

|                              |                                 |
|------------------------------|---------------------------------|
| F defineboundary             | F definestructureblock          |
| F definebreakpoint           | F definestructureset            |
| F definebreakpoints          | F definesubcolumnset            |
| F definecharacterspacing     | F definetransparency            |
| F definechemical             | F definetransparencyglobal      |
| F definecolumnset            | F definetypface                 |
| F defineconversion           | F defineviewerlayer             |
| F definecounter              | F delayedcolumnsetsheet         |
| F definedlayoutcomponent     | F depthnumber                   |
| F definefallbackfamily       | F depthofstring                 |
| F definefield                | F descender                     |
| F definefieldset             | F detachinsertions              |
| F definefilefallback         | F diagnostic                    |
| F definefilesynonym          | F digits_normal                 |
| F definefont_one             | F digits_reverse                |
| F definefont_two             | F directboxfromcache            |
| F definefontcolorpalette     | F disableautofontscript         |
| F definefontfamily           | F disablecollectingmath         |
| F definefontfamilypreset     | F disabledirectives             |
| F definefontfeature          | F disableexperiments            |
| F definefontscriptset        | F disablejobsave                |
| F definefontsolution         | F disableparbuilder             |
| F definehyphenationfeatures  | F disableregime                 |
| F defineinsertion            | F disablespellchecking          |
| F defineintermediatecolor    | F disablestatetriggering        |
| F definelabcolorglobal       | F disablestream                 |
| F definelabcolorlocal        | F disabletrackers               |
| F definelabels               | F discardfromlist               |
| F definelanguage             | F discoptioncodestring          |
| F definelinkedlist           | F disposeboxesincache           |
| F definemathpreset           | F disposelutable                |
| F definemultitonecolorglobal | F doanalyzeframedbox            |
| F definenamespace            | F doif                          |
| F definenote                 | F doifcounter                   |
| F defineparallel             | F doifelse                      |
| F defineparbuilder           | F doifelseblack                 |
| F defineprocesscolordummy    | F doifelseboxincache            |
| F defineprocesscolorglobal   | F doifelsebuffer                |
| F defineprocesscolorlocal    | F doifelsebufferempty           |
| F defineprogram              | F doifelsecharinfont            |
| F definereference            | F doifelseconverter             |
| F defineregister             | F doifelsecounter               |
| F definesectionlevels        | F doifelsecounteractive         |
| F definespotcolorglobal      | F doifelsecurrentfonthasfeature |

F doifelsedimensionstring  
F doifelsedrawingblack  
F doifelsefieldcategory  
F doifelsefieldset  
F doifelsefiledefined  
F doifelsefileexist  
F doifelsefirstchar  
F doifelsefirstsectionpage  
F doifelsefontfeature  
F doifelsefontpresent  
F doifelseindented  
F doifelseinelement  
F doifelseinputfile  
F doifelseleapyear  
F doifelselisthasentry  
F doifelselisthasnumber  
F doifelselisthaspage  
F doifelselisthasitle  
F doifelsemarkedcontent  
F doifelsemarkedpage  
F doifelsemessage  
F doifelsenoteonsamepage  
F doifelseobjectfound  
F doifelseobjectreferencefound  
F doifelseolderversion  
F doifelseoverlapping  
F doifelseparallel  
F doifelseparentfile  
F doifelsepathexist  
F doifelsependingpagecontent  
F doifelseposition  
F doifelsepositiononpage  
F doifelsepositiononsamepage  
F doifelsepositionsonthispage  
F doifelsepositionsused  
F doifelserangeonpage  
F doifelserreference  
F doifelserreferencechecked  
F doifelserreferenced  
F doifelserighttopleftinbox  
F doifelsesame  
F doifelsesomething  
F doifelsespaces  
F doifelsestackedfloats

F doifelsestructureblockempty  
F doifelsesynonymshown  
F doifelsesynonymused  
F doifelse>taggednoteonsamepage  
F doifelseurldefined  
F doifelseutfmathaccent  
F doifelseutfmathfiller  
F doifelseutfmathlimop  
F doifelsevalidhashedfiles  
F doifelsevisualizer  
F doifmarkedpage  
F doifnot  
F doifnotcounter  
F doifnotecontent  
F doifnotsame  
F doifnothing  
F doifposition  
F doifsame  
F doifsomething  
F doifunknownfontfeature  
F domainattribute  
F doreshapeframedbox  
F doresyncframedbox  
F dosaveposition  
F dosavepositionplus  
F dosavepositionwhd  
F dosetposition  
F dosetpositionbox  
F dosetpositionplus  
F dosetpositionstrut  
F dosetpositionstrutkind  
F dosetpositionwhd  
F dostarttextfile  
F dostoptextfile  
F doublescriptmodecodestring  
F droptablegroup  
F elapsedsteptime  
F elapsedtime  
F embedimagefromstore  
F enablealignmentcharacter  
F enablealignskips  
F enableasciimode  
F enableautofences  
F enableautofontscript



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|---------------------------|---------------------------------|
| F enablebackgroundalign   | F featureattribute              |
| F enablebackgroundboxes   | F feedback                      |
| F enablecollectingmath    | F fetchlanguage                 |
| F enablecolor             | F fetchmultipar                 |
| F enablecolorintents      | F fetchposboxes                 |
| F enablecrlf              | F fetchterminaldata             |
| F enabledirectives        | F figure_flush_registered_pages |
| F enabledriver            | F figure_mergegoodies           |
| F enableexperiments       | F figure_nof_registered_pages   |
| F enablefontcolor         | F figure_register_page          |
| F enablehidecontent       | F figuresetdimensions           |
| F enablemathalignrelocate | F filebasename                  |
| F enablematrixornaments   | F filedirname                   |
| F enablematrixrules       | F filenameonly                  |
| F enableparbuilder        | F filesuffix                    |
| F enableregime            | F filterdefaultreference        |
| F enablescriptunstacking  | F filterreference               |
| F enablesidefloatchecker  | F finalizeobjectbox             |
| F enablespellchecking     | F finalizeshipoutbox            |
| F enablestacking          | F findregisterinternal          |
| F enablestatetriggering   | F finishcolumnsetflush          |
| F enablestream            | F finishexport                  |
| F enableswapping          | F firstinset                    |
| F enabletighten           | F firstsectionpage              |
| F enabletrackers          | F firstsubcountervalue          |
| F enabletransparency      | F flagcodestring                |
| F enableutf               | F flushcolumnsetareas           |
| F enhancelinkedlist       | F flushcolumnsetcolumn          |
| F enhancelist             | F flushfloat                    |
| F enhanceregister         | F flushlabeledfloat             |
| F enhancesynonym          | F flushnoterange                |
| F environment             | F flushnotes                    |
| F eposcolumn              | F flushpageblocks               |
| F eposcolumnregistered    | F flushpageinjectionsafter      |
| F erasebuffer             | F flushpageinjectionsbefore     |
| F execute                 | F flushpostponedblocks          |
| F exportbox               | F flushpostponednotes           |
| F exportformdata          | F flushstream                   |
| F exportreferences        | F flushstreamcopy               |
| F expression              | F fontchar                      |
| F extendregister          | F fontcharbyindex               |
| F extendregisterlate      | F fontcollectionclone           |
| F extensiblecode          | F fontcollectiondefine          |
| F fakeword                | F fontcollectionprepare         |

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|-------------------------------|----------------------------------|
| F fontcollectionregister      | F getfieldcategory               |
| F fontcollectionreport        | F getfile                        |
| F fontcollectionreset         | F getfirstcharacter              |
| F fontlookupgetkey            | F getfloatvariable               |
| F fontlookupgetkeyofindex     | F getfromluatable                |
| F fontlookupinitialize        | F getfullstructurenumber         |
| F fontlookupnofound           | F getinterlinepenalties          |
| F forceendjob                 | F getinternalcachedlistreference |
| F forcepageflush              | F getinternallistreference       |
| F forcequitjob                | F getinternalorderreference      |
| F forcestrutdepth             | F getmathblobindex               |
| F format                      | F getmathblobmapping             |
| F formatcolor                 | F getmathcharone                 |
| F formatgray                  | F getmathchartwo                 |
| F formatluminance             | F getmathmathblob                |
| F formatluminanceratio        | F getmathtextblob                |
| F formatone                   | F getmathvariant                 |
| F framedoutline               | F getmessage                     |
| F from                        | F getmpextensions                |
| F frozenparcodestring         | F getnaturaldimensions           |
| F frule                       | F getnaturalwd                   |
| F getMPdata                   | F getnofpreviouslines            |
| F getbidimode                 | F getnotestate                   |
| F getbodyfontsizes            | F getnotestorecount              |
| F getboxcountfromcache        | F getnotestoreflushed            |
| F getboxdpfromcache           | F getobjectdimensions            |
| F getboxfromcache             | F getobjectreference             |
| F getboxhtdpfromcache         | F getobjectreferencepage         |
| F getboxhtfromcache           | F getparshape                    |
| F getboxwdfromcache           | F getparwrapper                  |
| F getbuffer                   | F getprivatechar                 |
| F getbuffercontent            | F getprivateglyphslot            |
| F getbufferctxlua             | F getprivatemathchar             |
| F getbuffermkvi               | F getprivateslot                 |
| F getbuffertex                | F getprogram                     |
| F getcharactercasingcategory  | F getrandomdimen                 |
| F getclubpenalties            | F getrandomfloat                 |
| F getcurrentprefixspec        | F getrandomnumber                |
| F getcurrentreferencelabel    | F getrandomseed                  |
| F getcurrentreferencemetadata | F getreadfilename                |
| F getdefaultfieldvalue        | F getreferencesectionnumber      |
| F getdisplaywidowpenalties    | F getrivers                      |
| F getdocumentargument         | F getsetupstring                 |
| F getdocumentfilename         | F getshapeparameter              |

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|------------------------------|------------------------------|
| F getsomefullstructurenumber | F ifgluelineinbox            |
| F getsomestructurenumber     | F ifinset                    |
| F getspecificstructuretitle  | F iflua                      |
| F getstructurelevel          | F ifmode                     |
| F getstructurenumber         | F ifmodes                    |
| F getsubstring               | F ifmvl                      |
| F getsynchronizemarking      | F ifpositunum                |
| F geturl                     | F ifsetups                   |
| F getvisual                  | F ifsystemmode               |
| F getwidowpenalties          | F iftext                     |
| F globalpopmacro             | F ifxml                      |
| F globalpushmacro            | F ifxmlatt                   |
| F gluetolinesinbox           | F ifxmlattempty              |
| F glyphoptioncodestring      | F ifxmlempty                 |
| F gobblenested               | F ifxmlselfempty             |
| F gotocolumnsetslot          | F ifxmltext                  |
| F grabbuffer                 | F ignorefile                 |
| F groupcodestring            | F ignoretagsinexport         |
| F groupsetgroup              | F immediatemessage           |
| F grph_nodes_initialize      | F includesvgbuffer           |
| F grph_nodes_register        | F includesvgfile             |
| F grph_nodes_reset           | F incrementcounter           |
| F grph_nodes_resolve         | F incrementsubcounter        |
| F handlemathhang             | F inhibitprimitive           |
| F hboxtovbox                 | F initializeexport           |
| F heightofstring             | F initializejob              |
| F helpaction                 | F initializemathcollapsing   |
| F helpreference              | F initializemathfencing      |
| F hour                       | F initializemathitalics      |
| F hpackbox                   | F initializemathkernpairs    |
| F hpackedbox                 | F initializemathnumbers      |
| F hsplit                     | F initializemathspacing      |
| F hyphenateddigits           | F initializestream           |
| F hyphenatedlist             | F initializetexdirectives    |
| F hyphenationcodestring      | F initializetexexperiments   |
| F ic                         | F initializetextrackers      |
| F idxfromluatable            | F injectcurrentreference     |
| F if_common                  | F injectcurrentreferencehtdp |
| F ifallmodes                 | F injectdisable              |
| F ifbitwiseand               | F injectnowhite              |
| F ifblobindexdone            | F injectreference            |
| F ifboxhaslist               | F injectvpenalty             |
| F ifcommon                   | F injectvskip                |
| F ifeof                      | F injectvspacing             |

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|-------------------------------|-------------------------|
| F injectzerobaselineskip      | F lastxpos              |
| F inlinechemical              | F lastypos              |
| F inputfilebarename           | F lateindent            |
| F inputfilename               | F latelua               |
| F inputfilerealsuffix         | F lateundent            |
| F inputfilesuffix             | F lchexnumber           |
| F insertattachment            | F lchexnumbers          |
| F insertcategorycodestring    | F leapyear              |
| F insertcomment               | F limited               |
| F insertfield                 | F linebuffering         |
| F insertionnumber             | F linedir               |
| F insertoptioncodestring      | F lineprofilebox        |
| F insertrenderingwindow       | F linesegment           |
| F inspectluatable             | F listanchorcodestring  |
| F installedlanguages          | F listbookmark          |
| F installtexdirective         | F listexternal          |
| F installtexexperiment        | F listgroupindex        |
| F installtextracker           | F listlabel             |
| F interceptsamepagecontent    | F listlocation          |
| F it                          | F listnamespaces        |
| F jobfilefullname             | F listprefixednumber    |
| F jobfilename                 | F listprefixedpage      |
| F jobfilesuffix               | F listrawnumber         |
| F jobposd                     | F listrawrealpage       |
| F jobposh                     | F listrawuserpage       |
| F jobposp                     | F listsigncodestring    |
| F jobposw                     | F listsize              |
| F jobposx                     | F listtitle             |
| F jobposy                     | F listuserdata          |
| F languagecharwidth           | F lmt_parameters_define |
| F languagenumber              | F lmt_parameters_preset |
| F lastblobindex               | F lmt_parameters_reset  |
| F lastnodeequals              | F loadcldfile           |
| F lastnodeid                  | F loadcldfileonce       |
| F lastnodeidstring            | F loadexamodes          |
| F lastnodesubtypestring       | F loadfontgoodies       |
| F lastparwrapper              | F loadlibrary           |
| F lastrpos                    | F loadluafile           |
| F lastsavedboxresourceindex   | F loadluafileonce       |
| F lastsavedimageresourceindex | F loadluamodule         |
| F lastsavedimageresourcepages | F loadmapfile           |
| F lastsectionpage             | F loadmapline           |
| F lastsubcountervalue         | F loadspellchecklist    |
| F lastunitstring              | F loadtexfile           |

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|----------------------------|---------------------------|
| F loadtexfileonce          | F markcontent             |
| F loadtypescriptfile       | F markedpageparameter     |
| F loadvisualizer           | F markedpages             |
| F loadxmlfile              | F markingnumber           |
| F loadxmlfileonce          | F markingtitle            |
| F localanchorx             | F markinjector            |
| F localanchory             | F markpage                |
| F localboxindent           | F markregionbox           |
| F localboxindex            | F markregionboxcorrected  |
| F localboxlefthang         | F markregionboxtagged     |
| F localboxleftoffset       | F markregionboxtaggedkind |
| F localboxleftskip         | F markregionboxtaggedn    |
| F localboxlinenumber       | F markvisualfonts         |
| F localboxlinewidth        | F mathclassvalue          |
| F localboxlocalwidth       | F mathcontrolcodestring   |
| F localboxmarkonce         | F mathhorizontalcode      |
| F localboxovershoot        | F mathoptioncodestring    |
| F localboxparfillleftskip  | F mathrenderset           |
| F localboxparfillrightskip | F mathstrutcode           |
| F localboxparinitleftskip  | F mathvariantcode         |
| F localboxparinitrightskip | F mathvariantcount        |
| F localboxprogress         | F mathvariantslot         |
| F localboxrighthang        | F maxbufferwidth          |
| F localboxrightoffset      | F message                 |
| F localboxrightskip        | F metapostescaped         |
| F localpopmacro            | F metapostformatted       |
| F localpushmacro           | F metapostgraphformat     |
| F locatefilepath           | F minute                  |
| F loopcs                   | F mixcleanup              |
| F loopcsn                  | F mixfinalize             |
| F lousiness                | F mixflushlist            |
| F luacardinal              | F mixflushrest            |
| F luacount                 | F mixgetsplit             |
| F luadimen                 | F mixsetsplit             |
| F luaexpression            | F mixstate                |
| F luafloat                 | F modecodestring          |
| F luainteger               | F molecule                |
| F luanumber                | F month                   |
| F luarandom                | F monthnmn                |
| F luascript                | F monthname               |
| F luathread                | F mpblobtext              |
| F lxmlid                   | F mpcolor                 |
| F manipulated              | F mpgetblob               |
| F markblobindexdone        | F mpgettext               |

F mpgraphic  
F mpgraphictexttoks  
F mpoptions  
F mpoutlinetoks  
F mppopvariables  
F mppushvariables  
F mprunset  
F mpruntab  
F mprunvar  
F mpsetclippath  
F mpsetoutercolor  
F mpsetsxsy  
F mpsettext  
F mptexget  
F mptexreset  
F mptexset  
F mptexsetfrombuffer  
F mptexttoks  
F mvloptioncodestring  
F namedstructureuservariable  
F namedstructurevariable  
F naturalwd  
F negatedcolorcomponent  
F newarray  
F newhashedtable  
F newindexedtable  
F newluatable  
F newparwrapper  
F newrepeatablerandom  
F newstacking  
F nextparallel  
F nextsubcountervalue  
F nodimen  
F nobufferlines  
F nofconversions  
F nofdays  
F nofdetachedinsertions  
F noflinesinbox  
F nofmathvariants  
F nofstackedfloats  
F nologbuffering  
F normalizecodestring  
F normalizedbodyfontsize  
F normalizeparcodestring

F nospacesmodecodestring  
F notedeltapage  
F noteinternal  
F notelistindex  
F noteprefixednumber  
F notetitle  
F ntb\_split\_section\_check  
F ntb\_split\_section\_fetch  
F ntb\_split\_section\_locate  
F ntb\_split\_section\_reset  
F ntimes  
F number  
F numbers  
F objectreference  
F objectreferencepage  
F octnumber  
F openin  
F openout  
F operatingsystem  
F ordinal  
F outputfilename  
F overloadbookmark  
F overloadlowercase  
F overloadsattribute  
F overloaduppercase  
F pageblockrealpage  
F pageofinternal  
F pageoverflow  
F pagestaterealpage  
F pagestaterealpageorder  
F paragraphfillmodecodestring  
F paragraphoptioncodestring  
F parcontextcodestring  
F pardir  
F parpasscallback  
F parpassidentifier  
F parpos  
F partriggercodestring  
F pdfbackendcurrentresources  
F pdfbackendsetcatalog  
F pdfbackendsetcolorspace  
F pdfbackendsetextgstate  
F pdfbackendsetinfo  
F pdfbackendsetname

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|-------------------------------|-------------------------------|
| F pdfbackendsetpageattribute  | F placeparallel               |
| F pdfbackendsetpageresource   | F popfloat                    |
| F pdfbackendsetpagesattribute | F pophyphenation              |
| F pdfbackendsetpattern        | F poplist                     |
| F pdfbackendsetshade          | F poplocalanchors             |
| F pdfcolor                    | F poplogfile                  |
| F pdfextension                | F popparagraphtweak           |
| F pdffeedback                 | F poppath                     |
| F pdflastobj                  | F poprandomseed               |
| F pdfliteral                  | F popreferenceprefix          |
| F pdfobj                      | F popregime                   |
| F pdfrefobj                   | F poprunstate                 |
| F pdfsetuserdata              | F popsectionblock             |
| F pdfvariable                 | F positunum                   |
| F percentageof                | F positunumdef                |
| F pfabs                       | F postlistbox                 |
| F pfacos                      | F postlistcopy                |
| F pfacosh                     | F postponeotes                |
| F pfasin                      | F prefixedconverted           |
| F pfasinh                     | F prelistbox                  |
| F pfatan                      | F prelistcopy                 |
| F pfatanh                     | F preparecolumnsetflush       |
| F pfatantwo                   | F prerolledpages              |
| F pfceil                      | F presetcolumnsetsheets       |
| F pfcos                       | F presetcolumnsetspreadsheets |
| F pfcosh                      | F presetmathematics           |
| F pfdeg                       | F presetmathfontalternate     |
| F pfexp                       | F presetsymbolist             |
| F pffloor                     | F previoussubcountervalue     |
| F pflog                       | F privatecharacter            |
| F pfmod                       | F privatecharactercode        |
| F pfpow                       | F processbodyfontsizes        |
| F pfrad                       | F processbuffer               |
| F pfrem                       | F processcollectedmath        |
| F pfround                     | F processedfile               |
| F pfsin                       | F processedfiles              |
| F pfsinh                      | F processfile                 |
| F pfsqrt                      | F processfilemany             |
| F pftan                       | F processfilenone             |
| F pftanh                      | F processfileonce             |
| F pgfxcolorspec               | F processjob                  |
| F pickupbuffer                | F processlist                 |
| F pickupmarkedcontent         | F processregister             |
| F pickuppunctuation           | F processsplit                |

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|-------------------------|---------------------------------|
| F processsynonyms       | F registerbackground            |
| F product               | F registerbodyfontsize          |
| F profilebox            | F registerbookmark              |
| F profilemacro          | F registerboxglyph              |
| F project               | F registerbuttons               |
| F pruneboxesincache     | F registercolorintent           |
| F purefontname          | F registercolumnsetarea         |
| F pushfloat             | F registercolumnsetsheet        |
| F pushhyphenation       | F registercolumnsetspreadsheets |
| F pushlist              | F registerengineclass           |
| F pushlocalanchors      | F registerfallbackfont          |
| F pushlogfile           | F registerfiguremask            |
| F pushparagraphtweak    | F registerfigurepattern         |
| F pushpath              | F registerfigurerange           |
| F pushrandomseed        | F registerfiguresuffix          |
| F pushreferenceprefix   | F registerfilescheme            |
| F pushregime            | F registerfontalternative       |
| F pushrunstate          | F registerfontfamily            |
| F pushsectionblock      | F registerforcedbreak           |
| F pushstream            | F registergroupset              |
| F putboxincache         | F registerhashedfiles           |
| F putincolumnset        | F registerhelp                  |
| F putincolumnsetdirect  | F registerhyphenationexception  |
| F putnextboxincache     | F registerhyphenationpattern    |
| F qrcodebuffer          | F registeritemgroup             |
| F qrcodestring          | F registerlabel                 |
| F rawMPdata             | F registerlanguagefeatures      |
| F rawcharacter          | F registerlinenumbers           |
| F rawsetupstring        | F registermaintextcolor         |
| F rawsubcountervalue    | F registermpextensible          |
| F read                  | F registernamedlua              |
| F readfromterminal      | F registernamespace             |
| F readline              | F registerobject                |
| F readlinedirect        | F registerpattern               |
| F recolorbox            | F registerpostponedblock        |
| F referencecolumn       | F registerreferencedobject      |
| F referencepagedetail   | F registerrendering             |
| F referencepagestate    | F registersection               |
| F referenceposx         | F registershownsynonym          |
| F referenceposy         | F registersnaprange             |
| F referencerealpage     | F registerstructureprocessor    |
| F registeranchorbox     | F registersynchronize           |
| F registerattachment    | F registersynonym               |
| F registerbackendsymbol | F registertransliteration       |



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|----------------------------|----------------------------|
| F registerunit             | F restartcounter           |
| F registerunitshortcut     | F restartsubcounter        |
| F registerunknownbodysize  | F restorecounter           |
| F registerusedsynonym      | F restorecurrentattributes |
| F registeruserdata         | F restorenote              |
| F registerverbosemathlabel | F restoreobject            |
| F removelastring           | F reuserandomseed          |
| F renderingvar             | F reverseboxcontent        |
| F reorderfloats            | F revivefeature            |
| F replacefeature           | F rightmathkern            |
| F replacepospxywhd         | F romannumerals            |
| F replacewildcard          | F rotateparshape           |
| F reportstructure          | F rounded                  |
| F resavefloat              | F roundedfill              |
| F reservedautoregiontag    | F roundedoutline           |
| F resetMPinstance          | F ruby                     |
| F resetalternate           | F ruleoptioncodestring     |
| F resetbackgroundrowdata   | F runcontextjob            |
| F resetboxesincache        | F safechar                 |
| F resetcharacteralign      | F samplefile               |
| F resetcolumnset           | F saveboxresource          |
| F resetcomparingtext       | F savebuffer               |
| F resetcounter             | F savecounter              |
| F reseteffect              | F savecurrentattributes    |
| F resetfeature             | F savecurrentvalue         |
| F resetfontfeature         | F savedlistnumber          |
| F resetfontsolution        | F savedlistprefixednumber  |
| F resetforcedbreak         | F savedlisttitle           |
| F resetinjector            | F savefloat                |
| F resetlogfile             | F saveimageresource        |
| F resetmapfiles            | F savemargindata           |
| F resetmathattributes      | F savenote                 |
| F resetnullfont            | F savepagedata             |
| F resetparallel            | F savepageinjections       |
| F resetpath                | F saveparallel             |
| F resetreference           | F savepos                  |
| F resetscript              | F savespecificfloat        |
| F resetstructureprocessor  | F savestructureblock       |
| F resetsubcounter          | F scangivendimensions      |
| F resetsynchronizemarking  | F scriptsmodecodestring    |
| F resetsynonyms            | F second                   |
| F resettimer               | F selectstructureblock     |
| F resettrackers            | F set_extensible_data      |
| F resolvedemoji            | F setadaptive              |

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|------------------------------|------------------------------|
| F setalternate               | F setfigurelookuporder       |
| F setautoprefix              | F setfigurepaths             |
| F setbackgroundrowdata       | F setfirstline               |
| F setboxglyph                | F setfontfeature             |
| F setboxline                 | F setfontofid                |
| F setbreakpoints             | F setfontsolution            |
| F setcharacteralign          | F setformat                  |
| F setcharacteralignndetail   | F setformsmethod             |
| F setcharactercasing         | F sethsizecolumnset          |
| F setcharactercleaning       | F sethsizecolumnspan         |
| F setcharacterkerning        | F sethyphenatedurlcharacters |
| F setcharacterspacing        | F sethyphenationfeatures     |
| F setcharacterstripping      | F sethyphenationmethod       |
| F setcloseddocumentaction    | F setidentityvariable        |
| F setclosepageaction         | F setinitial                 |
| F setcolormodel              | F setinitialsectionblock     |
| F setcolumnsetarea           | F setinjector                |
| F setcolumnsetlines          | F setinsertionlocation       |
| F setcolumnsetmvl            | F setinsertmigration         |
| F setcolumnsetpresent        | F setinterfaceconstant       |
| F setcolumnsetproperties     | F setinterfaceelement        |
| F setcolumnsetstart          | F setinterfacemessage        |
| F setcomparetext             | F setinterfacemessages       |
| F setcomparingtext           | F setinterfacevariable       |
| F setcounter                 | F setinternalreference       |
| F setdataset                 | F setitaliccorrection        |
| F setdestinationattribute    | F setjavascriptpreamble      |
| F setdigitsmanipulation      | F setjobcomment              |
| F setdirection               | F setlanguageexceptions      |
| F setdocumentargument        | F setlanguageoptions         |
| F setdocumentcommandline     | F setlanguagepatterns        |
| F setdocumentctxfile         | F setlanguagesynonym         |
| F setdocumentdefaultargument | F setlinefiller              |
| F setdocumentenvironments    | F setlineprofile             |
| F setdocumentfilename        | F setlocalalignanchor        |
| F setdocumentfilenames       | F setlocalmarkanchor         |
| F setdocumentmodes           | F setlocalsyncanchor         |
| F setdocumentmodules         | F setluatable                |
| F seteffect                  | F setmathalphabet            |
| F setelementbackendtag       | F setmathattribute           |
| F setelementnature           | F setmathfontalternate       |
| F setextrafontkerns          | F setmathfontstyle           |
| F setfieldcalculationset     | F setmathsnapping            |
| F setfigureconversion        | F setmathvariant             |

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| F setmetaglyph              | F setstructurenumber            |
| F setmpextensions           | F setsubcounter                 |
| F setnaturalwd              | F settagcodepoint               |
| F setnegative               | F settagcombination             |
| F setnextinternalreferences | F settagcombinationpair         |
| F setnextreferenceorder     | F settagconstruct               |
| F setnotestate              | F settagdelimitedsymbol         |
| F setopendocumentaction     | F settagdescriptionindex        |
| F setopenpageaction         | F settagfigure                  |
| F setowncounter             | F settagfloat                   |
| F setownsubcounter          | F settagformulacontent          |
| F setpagecolormodel         | F settaggedmetadata             |
| F setpagestate              | F settaggingcomment             |
| F setpagestaterealpageno    | F settaggingoption              |
| F setpagetransition         | F settaggingpresets             |
| F setparagraphmetashape     | F settaghighlight               |
| F setparagraphshape         | F settagitem                    |
| F setparbuilder             | F settagitemgroup               |
| F setparwrapper             | F settaglist                    |
| F setpdfcompression         | F settagmarginanchor            |
| F setperiodkerning          | F settagmargintext              |
| F setpostlistbox            | F settagmpgraphic               |
| F setprelistbox             | F settagnotation                |
| F setprofile                | F settagnotationsymbol          |
| F setrandomseed             | F settagparagraph               |
| F setrealign                | F settagproperty                |
| F setrealspaces             | F settagpublication             |
| F setreferencelinkmethod    | F settagregister                |
| F setregionbox              | F settagsetreferencedescription |
| F setregionboxtagged        | F settagsorting                 |
| F setregistermethod         | F settagssentencesymbol         |
| F setreplacements           | F settagssynonym                |
| F setruby                   | F settagtablecell               |
| F setrule                   | F settagtabulatecell            |
| F setscript                 | F settransliteration            |
| F setsectionblock           | F setupbookmarks                |
| F setsectionentry           | F setupcanvas                   |
| F setshadow                 | F setupcharacterspacing         |
| F setshift                  | F setupcomparingtext            |
| F setsofthyphens            | F setupexport                   |
| F setsplitlistthdp          | F setupfontexpansion            |
| F setstacking               | F setupfontprotrusion           |
| F setstructureblockstate    | F setupinsertion                |
| F setstructurelevel         | F setuplinenumbering            |

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|------------------------------|---------------------------------|
| F setupsynctex               | F showtrackers                  |
| F setupusermetadata          | F silliness                     |
| F setupviewerlayer           | F simplealign                   |
| F setvisual                  | F simplecases                   |
| F setvisualfont              | F simplematrix                  |
| F setvisuallayer             | F sin                           |
| F setvisuals                 | F sind                          |
| F setvsizecolumnset          | F sl                            |
| F setxmpfile                 | F spaceddigits                  |
| F shiftmatrixornaments       | F spacefactorcodestring         |
| F shiftparshape              | F spaceskipcodestring           |
| F shipoutpage                | F special                       |
| F showarray                  | F specificationoptioncodestring |
| F showassignerror            | F specifiedfont                 |
| F showattributes             | F specifiedfontsize             |
| F showbits                   | F specifiedfontspec             |
| F showboxinbuffer            | F splitfilename                 |
| F showbtxdatasetauthors      | F splitsamplefile               |
| F showbtxdatasetcompleteness | F spotcolorname                 |
| F showbtxdatasetfields       | F spotcolorparent               |
| F showbtxentry               | F spotcolorvalue                |
| F showbtxfields              | F squid_check                   |
| F showbtxhashedauthors       | F startactualtext               |
| F showbtxtables              | F startalternativetext          |
| F showbtxxvariants           | F startchemical                 |
| F showchardata               | F startclipping                 |
| F showcounter                | F startcollectingmath           |
| F showdirectives             | F startcolorset                 |
| F showdiscretionaries        | F startcomponent                |
| F showexperiments            | F startenvironment              |
| F showfontparameters         | F startfontsolution             |
| F showhyphenatedinlist       | F startgraphicgroup             |
| F showhyphenationtrace       | F startlocalcomponent           |
| F showinjector               | F startmarkpages                |
| F showjustification          | F startmatrix                   |
| F showlogcategories          | F startmirroring                |
| F showluatables              | F startpageblock                |
| F showluatokens              | F startparbuilder               |
| F showmacrostack             | F startprerollpageblock         |
| F showmessage                | F startproduct                  |
| F showpagelist               | F startproject                  |
| F showparagraphshape         | F startregime                   |
| F showprofiledmacros         | F startrotation                 |
| F showstructure              | F startruby                     |

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|-----------------------------------------|------------------------------|
| F startscaling                          | F strc_tags_start_chained    |
| F startsectionlevel                     | F strc_tags_start_detail     |
| F startstream                           | F strc_tags_start_ignore     |
| F starttext                             | F strc_tags_start_userdata   |
| F startviewerlayer                      | F strc_tags_stop             |
| F step                                  | F stringtoorientation        |
| F stopactualtext                        | F stripinserts               |
| F stopalternativetext                   | F structureautocatcodedget   |
| F stopchemical                          | F structureblockuservariable |
| F stopclipping                          | F structurecatcodedget       |
| F stopcollectingmath                    | F structuregivencatcodedget  |
| F stopcolorset                          | F structurenumber            |
| F stopcomponent                         | F structuretitle             |
| F stopenvironment                       | F structureuservariable      |
| F stopfontsolution                      | F structurevariable          |
| F stopgraphicgroup                      | F subcountervalue            |
| F stoplocalcomponent                    | F subsubcountervalue         |
| F stopmarkpages                         | F subtocolumnsetmvl          |
| F stopmatrix                            | F subtocolumnsetmvlbalance   |
| F stopmirroring                         | F subtractfeature            |
| F stoppageblock                         | F svghashed                  |
| F stopparbuilder                        | F sync                       |
| F stopprerollpageblock                  | F synccolor                  |
| F stopproduct                           | F synccolorclone             |
| F stopproject                           | F synccolorcount             |
| F stopregime                            | F synchronizelocalanchors    |
| F stoprotation                          | F synchronizemarking         |
| F stopscaling                           | F synchronizesteps           |
| F stopsectionlevel                      | F synchronizestream          |
| F stopstream                            | F syntexblockfilename        |
| F stoptext                              | F syntexdisable              |
| F stopviewerlayer                       | F syntexpause                |
| F storejavascriptcode                   | F syntexpopline              |
| F storejavascriptpreamble               | F syntexpushline             |
| F storenote                             | F syntexresetfilename        |
| F storeregister                         | F syntexresume               |
| F strc_tags_document_pop                | F syntexsetfilename          |
| F strc_tags_document_push               | F synonymfirstuserpage       |
| F strc_tags_get_element_tag             | F synonymmeaning             |
| F strc_tags_parcounter_pop              | F synonymname                |
| F strc_tags_parcounter_push             | F synonympages               |
| F strc_tags_set_aspect                  | F synonymuserpages           |
| F strc_tags_set_element_user_properties | F systemlog                  |
| F strc_tags_start                       | F tabl_ntb_get_sec           |

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|----------------------------|-------------------------|
| F tabl_ntb_set_sec         | F unstripinserts        |
| F tagmfunctionlab          | F unsupportedcs         |
| F tagmfunctionlabattribute | F useMPLibrary          |
| F tan                      | F useanyfile            |
| F tand                     | F useanyfileonce        |
| F textdir                  | F useboxresource        |
| F textime                  | F usecolors             |
| F tf                       | F usecomponent          |
| F thebox                   | F useenvironment        |
| F thefirstcharacter        | F usefigurebase         |
| F themaxboxwidth           | F usefile               |
| F theremainingcharacters   | F useimageresource      |
| F tightfitcompensation     | F usejavascriptpreamble |
| F timestring               | F usejavascriptscripts  |
| F tobigpoints              | F uselibrary            |
| F tobit                    | F uselist               |
| F tobits                   | F useluamodule          |
| F tochar                   | F useluatable           |
| F tohexa                   | F usemathpluginlibrary  |
| F toorientation            | F usemodules            |
| F totalofstring            | F usepath               |
| F tounicharnum             | F useproduct            |
| F towholebigpoints         | F useproject            |
| F transliterate            | F usereferences         |
| F transliteratebuffer      | F useregister           |
| F transliterated           | F usesubpath            |
| F transliteratedbuffer     | F usesymbols            |
| F transliterationexception | F settransliteration    |
| F traverseblock            | F useurl                |
| F traversemath             | F usezipfile            |
| F triggerpagebuilder       | F utfchar               |
| F truefilename             | F utflower              |
| F truefontname             | F utfmathclass          |
| F type                     | F utfmathcommand        |
| F typebuffer               | F utfmathcommandfiller  |
| F typefile                 | F utfmathfiller         |
| F uchar                    | F utfmathstretch        |
| F uchexnumber              | F utfshape              |
| F uchexnumbers             | F utfstrip              |
| F undefinechemical         | F utfupper              |
| F unicodechar              | F uuid                  |
| F unihex                   | F vboxlisttohbox        |
| F unit                     | F verbose               |
| F unloadlanguage           | F vpackbox              |

|                                     |                          |
|-------------------------------------|--------------------------|
| F vpackedbox                        | F xmlconcat              |
| F way                               | F xmlconcatrange         |
| F weekday                           | F xmlcontext             |
| F weekdayname                       | F xmlcount               |
| F widthofstring                     | F xmlcssmappedstylevalue |
| F widthuptohere                     | F xmlcssstylevalue       |
| F write                             | F xmldelete              |
| F writeline                         | F xmldepth               |
| F writestatus                       | F xmldirect              |
| F writestring                       | F xmldirectives          |
| F x_table_c                         | F xmldirectivesafter     |
| F x_table_cleanup                   | F xmldirectivesbefore    |
| F x_table_construct                 | F xmldisplayverbatim     |
| F x_table_create                    | F xmldoif                |
| F x_table_finish_row                | F xmldoifatt             |
| F x_table_flush                     | F xmldoifelse            |
| F x_table_init_construct            | F xmldoifelseatt         |
| F x_table_init_reflow_height        | F xmldoifelseempty       |
| F x_table_init_reflow_height_option | F xmldoifelseselfempty   |
| F x_table_init_reflow_width         | F xmldoifelsestext       |
| F x_table_init_reflow_width_option  | F xmldoifempty           |
| F x_table_next_row                  | F xmldoifnot             |
| F x_table_next_row_option           | F xmldoifnotatt          |
| F x_table_r                         | F xmldoifnotempty        |
| F x_table_reflow_height             | F xmldoifnotselfempty    |
| F x_table_reflow_width              | F xmldoifnottext         |
| F x_table_set_construct             | F xmldoifselfempty       |
| F x_table_set_reflow_height         | F xmldoiftext            |
| F x_table_set_reflow_width          | F xmlelement             |
| F xmladdindex                       | F xmlfileline            |
| F xmlafterssetup                    | F xmlfilename            |
| F xmlall                            | F xmlfilter              |
| F xmlappendsetup                    | F xmlfilterlist          |
| F xmlapplyselectors                 | F xmlfirst               |
| F xmlatt                            | F xmlflush               |
| F xmlattdef                         | F xmlflushcontext        |
| F xmlattribute                      | F xmlflushlinewise       |
| F xmlattributedef                   | F xmlflushpure           |
| F xmlbadinclusions                  | F xmlflushsetups         |
| F xmlbeforesetup                    | F xmlflushspacewise      |
| F xmlchainatt                       | F xmlflushstext          |
| F xmlchainattdef                    | F xmlformatobuffer       |
| F xmlchecknamespace                 | F xmlfunction            |
| F xmlcommand                        | F xmlgetindex            |

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|-----------------------|-----------------------|
| F xmlinclude          | F xmlsetentity        |
| F xmlincludeoptions   | F xmlsetinjectors     |
| F xmlinclusion        | F xmlsetpar           |
| F xmlinclusionbase    | F xmlsetparam         |
| F xmlinclusions       | F xmlsetsetup         |
| F xmlindex            | F xmlsnippet          |
| F xmlinlineverbatim   | F xmlstartraw         |
| F xmlinstalldirective | F xmlstarttiming      |
| F xmllast             | F xmlstopraw          |
| F xmllastatt          | F xmlstoptiming       |
| F xmllastmatch        | F xmlstrip            |
| F xmllastpar          | F xmlstripanywhere    |
| F xmlloadbuffer       | F xmlstripeverything  |
| F xmlloaddata         | F xmlstripnolines     |
| F xmlloaddirectives   | F xmlstripped         |
| F xmlloadentities     | F xmlstrippednolines  |
| F xmlloadfile         | F xmltag              |
| F xmlmain             | F xmltexatt           |
| F xmlmatch            | F xmltextentity       |
| F xmlname             | F xmltext             |
| F xmlnamespace        | F xmltobuffer         |
| F xmlnonspace         | F xmltobuffertextonly |
| F xmlpar              | F xmltobufferverbose  |
| F xmlparam            | F xmltofile           |
| F xmlpath             | F xmltoparameters     |
| F xmlpopmatch         | F xmlverbatim         |
| F xmlpos              | F xmlwithindex        |
| F xmlprependsetup     | F year                |
| F xmlpure             | T containers          |
| F xmlpushmatch        | F cleanname           |
| F xmlraw              | F content             |
| F xmlrawtex           | F define              |
| F xmlrefatt           | F is_usable           |
| F xmlregisterns       | F is_valid            |
| F xmlremapname        | F read                |
| F xmlremapnamespace   | B usecache            |
| F xmlremovesetup      | F write               |
| F xmlresetinjectors   | T context             |
| F xmlresetsetup       | F bc                  |
| F xmlsave             | F bgroup              |
| F xmlsetatt           | F char                |
| F xmlsetattribute     | F concat              |
| F xmlsetcommandtonone | D conditionals        |
| F xmlsetcommandtotext | F cssdimension        |



F ctxescape  
 F ctxescaped  
 F direct  
 F egroup  
 F escape  
 F escaped  
 F firstofoneargument  
 F firstoftwoarguments  
 F fprint  
 F getlogger  
 F glyph  
 F gobbleoneargument  
 F latelua  
 F loadfile  
 F loadviafile  
 D modes  
 F newindexer  
 F newtexthandler  
 F newverbosehandler  
 F nr  
 F par  
 F popcatcodes  
 F poplogger  
 F printlines  
 F printtable  
 F protect  
 F pushcatcodes  
 F pushlogger  
 F rule  
 F runfile  
 F runstring  
 F safechar  
 F secondoftwoarguments  
 F setconditional  
 F setfontid  
 F setmode  
 F setsystemmode  
 F settracing  
 F settrialtypesettingmethod  
 F space  
 F sprint  
 F startcollecting  
 F step  
 F stepwise

F stopcollecting  
 D systemmodes  
 F tobuffer  
 D tokenizedcs  
 F tolines  
 F trialtypesetting  
 F unprotect  
 F utfchar  
 F viafile  
 T converters  
 F A  
 F A:es  
 F A:ru  
 F A:sl  
 F AK  
 F Alphabetic  
 F Alphabetic:es  
 F Alphabetic:ru  
 F Alphabetic:sl  
 F Character  
 F Characters  
 F I  
 F KA  
 F KR  
 F R  
 F RK  
 F Romannumerals  
 F Russiannumerals  
 F Sloveniannumerals  
 F Spanishnumerals  
 F Word  
 F Words  
 F a  
 F a:es  
 F a:ru  
 F a:sl  
 F abjadnodotnumerals  
 F abjadnumerals  
 F alphabetic  
 F alphabetic:es  
 F alphabetic:ru  
 F alphabetic:sl  
 F character  
 F characters

|                       |                  |
|-----------------------|------------------|
| F chineseallnumerals  | F tohebrew       |
| F chineseapnumerals   | F toroman        |
| F chinesedatenumerals | F totime         |
| F chinesenumerals     | F uchexnumber    |
| F chr                 | F uchexnumbers   |
| F chrs                | F upper          |
| F cn                  | F weekday        |
| F cn-a                | F word           |
| F cn-c                | F words          |
| F cn-d                | T coroutine      |
| F convert             | F close          |
| F decimals            | F create         |
| F define              | F isyieldable    |
| F gregorian_to_hebrew | F resume         |
| F gregorian_to_jalali | F running        |
| F hebrewnumerals      | F status         |
| F hexstringtonumber   | F wrap           |
| F i                   | F yield          |
| F isleapyear          | T ctxrunner      |
| F jalali_to_gregorian | F load           |
| F lchexnumber         | F preparedfile   |
| F lchexnumbers        | D prepfiles      |
| F leapyear            | F resolve        |
| F lower               | T debug          |
| F max                 | F getinfo        |
| F maxchrs             | F sethook        |
| F nicenumber          | F traceback      |
| F nofdays             | F dimen          |
| F number              | T dir            |
| F numbers             | F collectpattern |
| F octnumber           | F current        |
| F octstringtonumber   | F expandlink     |
| F ordinal             | F expandname     |
| F r                   | F found          |
| F romannumerals       | F glob           |
| F russiannumerals     | F globdirs       |
| D sequences           | F globfiles      |
| F settime             | F globpattern    |
| F sloveniannumerals   | F ls             |
| F spaceddigits        | F makedirs       |
| F spanishnumerals     | F mkdirs         |
| F textime             | F pop            |
| F toabjad             | F push           |
| F tochinese           | T directives     |

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|-------------------------|-----------------------|
| D data                  | F getvpos             |
| F default               | F initialize          |
| F disable               | F install             |
| F enable                | N lmtxversion         |
| F list                  | F outputfilename      |
| S name                  | F report              |
| F register              | T effects             |
| F report                | F brownianmotion      |
| F reset                 | F browniannoise       |
| F show                  | F getoctaveloops      |
| F value                 | F getoctavemethods    |
| F values                | F getoctavestatus     |
| T document              | F getoctavevariants   |
| D arguments             | F newoctave           |
| F getargument           | F octave              |
| F getfilename           | F octavebytemap       |
| F pdf_cleanup_cidsets   | F perlinmotion        |
| F pdf_cleanup_procsets  | F perlinnoise         |
| F pdf_identify_content  | F simplexangle        |
| F pdf_serialize_content | F simplexdetail       |
| F pdf_strip_content     | F simplexnoise        |
| F pdf_strip_page        | F stepoctave          |
| F setargument           | F worleybounds        |
| F setcommandline        | F worleymotion        |
| F setdefaultargument    | F worleynoise         |
| F setenvironments       | T environment         |
| F setfilename           | F argument            |
| F setfilenames          | D arguments           |
| F setmodes              | D basicengines        |
| F setmodules            | N currentrun          |
| D variables             | F definefilefallback  |
| T documentdata          | F definefilesynonym   |
| F variable              | D engineflags         |
| F dofile                | S filename            |
| T drivers               | D filesynonyms        |
| F applyanchor           | N forcedruns          |
| F applyorientation      | F getargument         |
| F convert               | F globfiles           |
| F enable                | S homedir             |
| F finalize              | B initex              |
| F gethpos               | B initexmode          |
| F gethvrpos             | F initializearguments |
| F getpos                | F initializefilenames |
| F getrpos               | S inputfilebarename   |

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|--------------------------|-------------------|
| S inputfilename          | F enable          |
| S inputfilesuffix        | F list            |
| S jobfilefullname        | S name            |
| S jobfilename            | F register        |
| S jobfilesuffix          | F report          |
| N kindofrun              | F reset           |
| N lmtxmode               | F show            |
| F loadexamodes           | F value           |
| F loadluafile            | F values          |
| F luafile                | T figures         |
| F luafilechunk           | F applyratio      |
| S luatexengine           | F badname         |
| N luatexfunctionality    | D bases           |
| S luatexverbose          | F bitmapimage     |
| N luatexversion          | N boxnumber       |
| N maxnofruns             | F bytepalette     |
| S oldhome                | D cachepaths      |
| D originalarguments      | F check           |
| S outputfilename         | D checkers        |
| S ownbin                 | D conversions     |
| S ownmain                | D converters      |
| S ownname                | F current         |
| S ownpath                | N defaultdepth    |
| B prerollrun             | N defaultheight   |
| D rawarguments           | B defaultsearch   |
| F reconstructcommandline | N defaultwidth    |
| F relativepath           | F done            |
| F setargument            | F dummy           |
| F splitarguments         | D exists          |
| S suffix                 | F exists          |
| F texfile                | D formats         |
| S texmfos                | D found           |
| S texos                  | F get             |
| S texroot                | F getinfo         |
| F truefilename           | F getpdfinfo      |
| F useluamodule           | F getrealpage     |
| F usemodules             | F guess           |
| D validengines           | F guessfromstring |
| S version                | F hash            |
| F error                  | D identifiers     |
| T experiments            | F identify        |
| D data                   | F include         |
| F default                | D includers       |
| F disable                | F initialize      |

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|---------------------|---------------------|
| D loaded            | F loadchecksum      |
| D localpaths        | F nameonly          |
| D magics            | F nametotable       |
| F mergegoodies      | F needs_updating    |
| N nofmissing        | F needsupdating     |
| N nofprocessed      | F pathpart          |
| D order             | F readdata          |
| D paths             | F removesuffix      |
| D patterns          | F replacesuffix     |
| F pop               | F reslash           |
| B preferquality     | F robustname        |
| D programs          | F savechecksum      |
| F push              | F savedata          |
| F qrcode            | F size              |
| F registerpattern   | F splitbase         |
| F registerresource  | F splitname         |
| F registersuffix    | F splitpath         |
| D remappers         | F suffix            |
| D resources         | F suffixes          |
| F scale             | F suffixesonly      |
| F set               | F suffixonly        |
| F setconversion     | F syncmtimes        |
| F setlookups        | F withinbase        |
| F setorder          | T fio               |
| F setpaths          | F getposition       |
| D suffixes          | F read2dot14        |
| D used              | F readbytes         |
| T file              | F readbytetable     |
| F addsuffix         | F readcardinal1     |
| F basename          | F readcardinal11e   |
| F checksum          | F readcardinal2     |
| F collapsepath      | F readcardinal21e   |
| F copy              | F readcardinal3     |
| F dirname           | F readcardinal31e   |
| F expandname        | F readcardinal4     |
| F extname           | F readcardinal41e   |
| F is_qualified_path | F readcardinaltable |
| F is_readable       | F readcline         |
| F is_rootbased_path | F readcstring       |
| F is_writable       | F readdouble        |
| F isreadable        | F readdoublele      |
| F iswritable        | F readfixed2        |
| F join              | F readfixed4        |
| F joinpath          | F readfloat         |

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|--------------------|--------------------------|
| F readfloatle      | F getfontdimen           |
| F readinteger1     | F getfontfields          |
| F readinteger1le   | F getfontspec            |
| F readinteger2     | F getmathcharacterfields |
| F readinteger2le   | F getmathindex           |
| F readinteger3     | F getmathspec            |
| F readinteger3le   | F getparameterfields     |
| F readinteger4     | F gettextcharacterfields |
| F readinteger4le   | F id                     |
| F readintegertable | F max                    |
| F setposition      | F nextid                 |
| F skipposition     | F setfont                |
| F writecardinal1   | F setfontdimen           |
| F writecardinal1le | F xscaled                |
| F writecardinal2   | F yscaled                |
| F writecardinal2le | T fonts                  |
| F writecardinal3   | F current                |
| F writecardinal3le | F currentid              |
| F writecardinal4   | F definetypeface         |
| F writecardinal4le | F enhanceid              |
| F writedouble      | D expansions             |
| F writedoublele    | D formats                |
| F writefloat       | F getalternate           |
| F writefloatle     | F getindices             |
| T floats           | F getkern                |
| F analysemethod    | F getmultiple            |
| F checkedpagefloat | F getsubstitution        |
| F collect          | F infofont               |
| F consult          | F mode                   |
| F flush            | D names                  |
| F getvariable      | D nulldata               |
| F hasstacked       | D protrusions            |
| F nofstacked       | F registersanitizer      |
| F pop              | F reserveid              |
| F push             | F toeffect               |
| F resave           | T functions              |
| F save             | F dummy                  |
| F stacked          | F getmetatable           |
| T font             | T globaldata             |
| F addcharacters    | N CONTEXTLMTXMODE        |
| F addquality       | N LUAFORMAT              |
| F current          | N LUAMAJORVERSION        |
| F define           | N LUAMINORVERSION        |
| F getfont          | S LUAEXENGINE            |

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|------------------------|-----------------------|
| N LUAFORMATID          | F getbottom           |
| N LUAEXFUNCTIONALITY   | F getd                |
| S LUAEXVERBOSE         | F geth                |
| N LUAEXVERSION         | F getkind             |
| N LUAVERSION           | F getsizes            |
| F SAVEDREQUIRE         | F gettask             |
| F assert               | F gettop              |
| F characterkern        | F getw                |
| F collectgarbage       | F getx                |
| F dimen                | F gety                |
| F dofile               | F prune               |
| F error                | T gzip                |
| F getmetatable         | F compress            |
| T graphics             | F compressed          |
| T backgrounds          | F decompress          |
| F fetchmultipar        | F load                |
| F pair                 | F save                |
| F path                 | F suffix              |
| F point                | T helper              |
| T bitmaps              | F getprofile          |
| F flush                | F mathnested          |
| F load                 | F mathprocess         |
| T loaders              | F mathstep            |
| F jpeg                 | F snapindeed          |
| F jpg                  | F snaplist            |
| F png                  | F traversesearmethod  |
| F new                  | F traverseglyphmethod |
| F placeholder          | F traversehorizontal  |
| F tocontext            | F traversevertical    |
| F bpsize               | F inspect             |
| T colorspace           | T interactions        |
| T identifiers          | D attachments         |
| F bitmap               | T fields              |
| F jp2                  | F clone               |
| F jpeg                 | F define              |
| F jpg                  | F defineset           |
| F png                  | F insert              |
| F identify             | T general             |
| F registerpdfcompactor | F getidentity         |
| F registerpdffont      | F getidentityvariable |
| T synchronizers        | T help                |
| F collapse             | F collect             |
| F collect              | F reference           |
| F extend               | F register            |

|                           |                     |
|---------------------------|---------------------|
| T javascripts             | T integers          |
| F addtopreamble           | F interfacedcommand |
| F code                    | T interfaces        |
| D codes                   | F knownscanner      |
| F flushpreambles          | F message           |
| D functions               | F nameofscanner     |
| D preambles               | T namesofscanners   |
| F setpreamble             | T namespaces        |
| F storecode               | F define            |
| F storepreamble           | F list              |
| F usepreamblenow          | T numbers           |
| F usescripts              | F registernamespace |
| D linkedlists             | F registerscanner   |
| T pages                   | D scanners          |
| F setpagetransition       | F setconstant       |
| F setupcanvas             | F setcount          |
| T references              | F setdimen          |
| F setcloseddocumentaction | F setelement        |
| F setclosepageaction      | F setformat         |
| F setopendocumentaction   | F setmacro          |
| F setopenpageaction       | F setmessage        |
| D renderings              | F setmessages       |
| T interfaces              | F settranslation    |
| T arrays                  | D setupstrings      |
| D cachedsetups            | F setvariable       |
| F cachesetup              | F showmessage       |
| T cardinals               | D storedscanners    |
| F checkedspecification    | F tolist            |
| D complete                | D translations      |
| D constants               | D usednamespaces    |
| D corenamespaces          | D variables         |
| S currentinterface        | F writestatus       |
| S currentresponse         | T io                |
| F definecommand           | F ask               |
| F defined                 | F bytes             |
| F doifelsemessage         | F characters        |
| D elements                | F close             |
| T floats                  | F copydata          |
| D formats                 | F exists            |
| F getmessage              | S fileseparator     |
| F getnamespace            | F flush             |
| F getsetupstring          | F gobble            |
| D hashedsetups            | F input             |
| F implement               | F lines             |



|                     |                 |
|---------------------|-----------------|
| F loadchunk         | F initialize    |
| F loaddata          | T linkedlists   |
| F loadlines         | D collected     |
| F newreader         | D tobesaved     |
| F noflines          | F load          |
| F open              | F loadother     |
| F output            | T meshedup      |
| S pathseparator     | D collected     |
| F popen             | D tobesaved     |
| F read              | T objects       |
| F readall           | D collected     |
| F readnumber        | F get           |
| F readstring        | F number        |
| F savedata          | F page          |
| F size              | F save          |
| U stderr            | F set           |
| U stdin             | D tobesaved     |
| U stdout            | B pack          |
| F tmpfile           | N packversion   |
| F type              | T pagestates    |
| F write             | D collected     |
| F ipairs            | F countervalue  |
| T job               | F extend        |
| F comment           | F realpage      |
| T datasets          | F realpageorder |
| D collected         | F setstate      |
| F extend            | D tobesaved     |
| F getdata           | T parwrappers   |
| F setdata           | D collected     |
| D tobesaved         | D tobesaved     |
| F disablesave       | T positions     |
| T embedded          | F b_column      |
| D collected         | F b_region      |
| D tobesaved         | F c             |
| T fileobjreferences | D collected     |
| D collected         | F column        |
| D tobesaved         | F columnofpos   |
| T files             | F copy          |
| D collected         | F depth         |
| F context           | F e_column      |
| B forcerun          | F e_region      |
| F run               | F enhance       |
| D tobesaved         | F extra         |
| F getrandomseed     | F get           |

|                    |                    |
|--------------------|--------------------|
| F getcolumndata    | F width            |
| F getfree          | F x                |
| F gethpos          | F xy               |
| F gethvrpos        | F y                |
| F getpos           | F register         |
| F getreserved      | F save             |
| F getrpos          | T structure        |
| F getsync          | D collected        |
| F gettobesaved     | T components       |
| F getvpos          | T namestack        |
| F hangafter        | T tobesaved        |
| F hangindent       | T branches         |
| F height           | S name             |
| F hsize            | S type             |
| N lastr            | T variables        |
| N lastx            | D checksums        |
| N lasty            | D collected        |
| F leftskip         | F getchecksum      |
| F lowerleft        | F makechecksum     |
| F lowerright       | F restore          |
| F markregionbox    | F save             |
| F n                | F setchecksum      |
| F onsamepage       | D tobesaved        |
| F overlapping      | N version          |
| F p                | T kdtree           |
| F page             | F delete           |
| F paragraph        | F found            |
| F parindent        | F insert           |
| F position         | F nearest          |
| F r                | F nearestindex     |
| F region           | F new              |
| F registerhandlers | F newweighted      |
| F replace          | F reset            |
| F rightskip        | F totable          |
| F set              | F traverse         |
| F setall           | T kpse             |
| F setregionbox     | T language         |
| F setspec          | F clean            |
| F settobesaved     | F clearhyphenation |
| D tobesaved        | F clearpatterns    |
| F upperleft        | F current          |
| F upperright       | F gethjcode        |
| F used             | F haslanguage      |
| F whd              | F hyphenate        |

|                    |                          |
|--------------------|--------------------------|
| F hyphenation      | F traditional            |
| F hyphenationmin   | F popmethod              |
| F id               | F pushmethod             |
| F new              | F setmethod              |
| F patterns         | F setup                  |
| F postexhyphenchar | F showhyphenationtrace   |
| F posthyphenchar   | T traditional            |
| F preexhyphenchar  | F definefeatures         |
| F prehyphenchar    | F gettrace               |
| F sethcode         | F hyphenate              |
| F setwordhandler   | F injecthyphens          |
| T languages        | F installmethod          |
| F addgoodiesdata   | F lasttrace              |
| F addgoodiesfile   | F loadpatterns           |
| T appliedoptions   | F registerexception      |
| F associate        | F registerpattern        |
| D associated       | F setfeatures            |
| F association      | F installed              |
| D counters         | F installhandler         |
| F current          | T labels                 |
| D data             | F define                 |
| D decimals         | F split                  |
| F define           | F loadable               |
| F explicithyphen   | F loadwords              |
| F exportlanguage   | T logger                 |
| F flatten          | F report                 |
| T frequencies      | F noflattened            |
| F averagecharwidth | D numbers                |
| F getdata          | F pdflanguage            |
| F getdata          | F postexhyphenchar       |
| F getmetadata      | F posthyphenchar         |
| F getnumber        | F preexhyphenchar        |
| F goodiefiles      | F prehyphenchar          |
| T hjcounts         | D registered             |
| F hyphenate        | T registeredactions      |
| T hyphenators      | T replacements           |
| T dictionaries     | F add                    |
| D featuresets      | F addlist                |
| F handler          | F handler                |
| T methods          | F set                    |
| F expanded         | F serializediscretionary |
| B none             | F setexceptions          |
| F original         | F setgoodieshandler      |
| F tex              | F setoptions             |

|                            |                   |
|----------------------------|-------------------|
| F setpatterns              | F currentdir      |
| F setsynonym               | F dir             |
| F setwordhandler           | F expandpath      |
| F showdiscretionaries      | F isdir           |
| F strippedgoodiewords      | F isfile          |
| F tolang                   | F isfound         |
| T transliteration          | F isreadabledir   |
| F exception                | F isreadablefile  |
| F handler                  | F iswriteabledir  |
| F register                 | F iswriteablefile |
| F set                      | F link            |
| F use                      | F mkdir           |
| F unload                   | F makedirs        |
| T urls                     | F modification    |
| D characters               | F readlink        |
| N lefthyphenmin            | F rmdir           |
| D mapping                  | F setexecutable   |
| B packslashes              | F symlink         |
| N righthyphenmin           | F symlinktarget   |
| F setcharacters            | F touch           |
| N version                  | F walkdir         |
| F visualizediscretionaries | T libraries       |
| T words                    | D basiclua        |
| F check                    | D basictex        |
| T data                     | D builtin         |
| F disable                  | D extralua        |
| F enable                   | D extratex        |
| T enablers                 | D functions       |
| B enables                  | D globals         |
| F found                    | D obsolete        |
| F load                     | D optional        |
| T methods                  | T lmx             |
| N threshold                | F color           |
| T layouts                  | F convert         |
| F getcanvas                | F convertfile     |
| F getpagedimensions        | F convertstring   |
| F setupcanvas              | F einject         |
| T status                   | F eprint          |
| F isleftpage               | F escape          |
| F leftorrightpageaction    | F finalized       |
| T lfs                      | F finject         |
| F attributes               | F fprint          |
| F canonicalize             | F get             |
| F chdir                    | F getvariables    |

|                           |                        |
|---------------------------|------------------------|
| F htmfile                 | F application          |
| T html                    | F categories           |
| F a                       | F direct               |
| F td                      | F disable              |
| F th                      | F enable               |
| F include                 | F finalactions         |
| T inherited               | F flush                |
| S color-background-four   | F loggingerrors        |
| S color-background-one    | F messenger            |
| S color-background-three  | F new                  |
| S color-background-two    | F newline              |
| S title                   | F poplogfile           |
| F initialize              | F poptarget            |
| F inject                  | T private              |
| F lmxfile                 | F enablepagetiming     |
| F loadedfile              | F getpagetiming        |
| F make                    | F pushlogfile          |
| F new                     | F pushtarget           |
| F popupfile               | F registerfinalactions |
| F print                   | F report               |
| F pv                      | F reporter             |
| F report                  | F resetlogfile         |
| F reset                   | F setformats           |
| F result                  | F setformatters        |
| F set                     | F setlogfile           |
| F show                    | F setmessenger         |
| F stripcss                | F setprocessor         |
| F tv                      | F settarget            |
| F type                    | F settimedlog          |
| F urlescape               | F settranslations      |
| F variable                | F show                 |
| T variables               | F start_page_number    |
| S color-background-blue   | F start_run            |
| S color-background-body   | F startererrorlogging  |
| S color-background-green  | F startfilelogging     |
| S color-background-main   | F status               |
| S color-background-purple | F stop_page_number     |
| S color-background-yellow | F stoperrorlogging     |
| S title-default           | F stopfilelogging      |
| F load                    | F subdirect            |
| D loaders                 | F subreport            |
| F loadfile                | F system               |
| F loadstring              | F texerrormessage      |
| T logs                    | F writer               |

|                           |                            |
|---------------------------|----------------------------|
| T lpdf                    | F deferredobject           |
| F action                  | F delayedobject            |
| F adddocumentcolorspace   | F destinationspecification |
| F adddocumenttextgstate   | F dictionary               |
| F adddocumentpattern      | F docstring                |
| F adddocumentshade        | T encodingvalues           |
| F addtocatalog            | N base16                   |
| F addtoinfo               | N base85                   |
| F addtonames              | N decoded                  |
| F addtopageattributes     | N encoded                  |
| F addtopageresources      | N plain                    |
| F addtopagesattributes    | N utf16be                  |
| F addxmpinfo              | N utf16le                  |
| F annotationspecification | T endpoints                |
| F array                   | S Div                      |
| F attachmentsymbols       | S Part                     |
| F boolean                 | S Sect                     |
| F border                  | T epdf                     |
| F checkedkey              | F allcontent               |
| F checkedresources        | F analyzefonts             |
| F checkedvalue            | F contentplugin            |
| F circularshade           | F contenttostring          |
| F close                   | F contenttotext            |
| F cmykcode                | D encryptioncodes          |
| F collectedresources      | F expanded                 |
| F color                   | F expandwidths             |
| F colorspace              | F fontplugin               |
| D colorspaceconstants     | F getpagecontent           |
| F colorspec               | F getstructure             |
| F colorvalue              | T helpers                  |
| F colorvalues             | F tounicodetable           |
| F commentsymbols          | T image                    |
| F compressdata            | F close                    |
| F compresslevel           | F copy                     |
| F constant                | F new                      |
| F copyarray               | F open                     |
| F copyboolean             | F query                    |
| F copyconstant            | F load                     |
| F copydictionary          | F mergewidths              |
| F copyfloat               | D objectcodes              |
| F copyinteger             | F objectrange              |
| F copynumber              | F objecttype               |
| F copystring              | F pageplugin               |
| F copyunicode             | F parsecontent             |

|                          |                         |
|--------------------------|-------------------------|
| F producer               | N PopUp                 |
| T resolvers              | N PushButton            |
| F destinations           | N Radio                 |
| F embeddedfiles          | N RadiosInUnison        |
| F javascripts            | N ReadOnly              |
| F layers                 | N Required              |
| F pages                  | N RichText              |
| F structure              | N Sort                  |
| F widgets                | N file                  |
| F unload                 | N fixed                 |
| F verboseobject          | N nocheck               |
| F escaped                | N protected             |
| F finalizeddocument      | N readonly              |
| F finalizepage           | N required              |
| F finalizesignature      | N sorted                |
| F finishtransparencycode | N unavailable           |
| T flags                  | F flushfonts            |
| T annotations            | F flushgroup            |
| N AutoView               | F flushobject           |
| N Hidden                 | F flushstreamfileobject |
| N Invisible              | F flushstreamobject     |
| N Locked                 | T fonts                 |
| N LockedContents         | F color_direct          |
| N NoRotate               | F color_indirect        |
| N NoView                 | D formats               |
| N NoZoom                 | F frombytes             |
| N Print                  | F fromeight             |
| N Printable              | F frompdfdoc            |
| N ReadOnly               | F fromsixteen           |
| N ToggleNoView           | F getcatalog            |
| N auto                   | F getcompression        |
| N hidden                 | F getfontname           |
| N printable              | F getfontobjectnumber   |
| T widgets                | F getformatoption       |
| N Comb                   | F getinfo               |
| N CommitOnSelChange      | F getmatrix             |
| N DoNotScroll            | F getnofpages           |
| N DoNotSpellCheck        | F getpageproperties     |
| N Edit                   | F getpagereference      |
| N FileSelect             | F getpagetags           |
| N MultiLine              | F getpos                |
| N NoExport               | F gettaggingstatus      |
| N NoToggleToOff          | F gettrailerid          |
| N Password               | F getversion            |

|                       |                             |
|-----------------------|-----------------------------|
| F getxformname        | N assemble                  |
| F graycode            | N extract                   |
| T h_hex_2             | N fillin                    |
| S                     | N modify                    |
| S                     | N print                     |
| S                     | N quality                   |
| S                     | F preparesignature          |
| S                     | F print                     |
| S (                   | F procset                   |
| S )                   | B protectresources          |
| S \                   | F quads                     |
| T h_hex_4             | F rectangle                 |
| F hasmatrix           | F reference                 |
| F immediateobject     | F registerannotation        |
| F includeimage        | F registerdestination       |
| F initialize          | F registerdocumentfinalizer |
| T injectors           | F registerfontmethod        |
| F bitmap              | F registerinitializer       |
| F jp2                 | F registerpagefinalizer     |
| F jpg                 | F registerpattern           |
| F png                 | F registersignature         |
| F injectxmpinfo       | F replacexmpinfo            |
| F insertxmpinfo       | F reserveobject             |
| F lastreferredpage    | F rgbcode                   |
| F layerreference      | F rotationcm                |
| F limited             | F setcompression            |
| F linearshade         | F setencryption             |
| F literal             | F setpageorder              |
| F majorversion        | F setpagesstate             |
| F makenametree        | F setpagetag                |
| F minorversion        | F setutfstringmode          |
| F noffontregistries   | F setversion                |
| F nopages             | F sharedobject              |
| F null                | F shareobject               |
| F number              | F shareobjectreference      |
| F object              | F splitsixteen              |
| F objectcompresslevel | F spotcode                  |
| F pagereference       | F stream                    |
| F patternstream       | F string                    |
| F pdfcolor            | F supportedformats          |
| F pdftimestamp        | F tocidset                  |
| T permissions         | F toeight                   |
| N access              | F topdfdoc                  |
| N add                 | F topermissions             |



|                               |                         |
|-------------------------------|-------------------------|
| F tosixteen                   | F endstripper           |
| F transparency                | F finder                |
| F transparencycode            | F firstofsplit          |
| T transparencygroups          | F frontstripper         |
| B none                        | F instringchecker       |
| F transparencyvalue           | F is_lpeg               |
| F unicode                     | F keeper                |
| T usedcharacters              | F locale                |
| T usedfontnames               | F match                 |
| N dejavusansmonoboldttf-1-0-0 | F oneof                 |
| T usedfontobjects             | D patterns              |
| N dejavusansmonoboldttf-1-0-0 | F pcode                 |
| D usedfontstreams             | F print                 |
| T usedindices                 | F ptree                 |
| T dejavusansmonoboldttf-1-0-0 | F replacer              |
| T dejavusansmonottf-1-0-0     | F secondofsplit         |
| F verbose                     | F setmaxstack           |
| T lpeg                        | F setutfcasers          |
| F B                           | F split                 |
| F C                           | F splitat               |
| F Carg                        | F splitter              |
| F Cb                          | F stripper              |
| F Cc                          | F times                 |
| F Cf                          | F tsplitat              |
| F Cg                          | F tsplitter             |
| F Cmt                         | F type                  |
| F Cp                          | F utfR                  |
| F Cs                          | F utfchartabletopattern |
| F Ct                          | F utfreplacer           |
| F P                           | S version               |
| F R                           | T lua                   |
| F S                           | F addnibble             |
| F UP                          | T bytecode              |
| F UR                          | D bytedata              |
| F US                          | T bytedone              |
| F V                           | F callbytecode          |
| F afterprefix                 | F comparekeys           |
| F anywhere                    | F finalizeinitex        |
| F append                      | N firstbytecode         |
| F balancer                    | F get_functions_table   |
| F beforesuffix                | F getallkeys            |
| F checkedsplit                | F getbytecode           |
| F containsws                  | F getcheckedkeys        |
| F counter                     | F getcurrenttime        |

|                           |                        |
|---------------------------|------------------------|
| F getdebuginfo            | F registerdumpactions  |
| F getexitcode             | F registerpageactions  |
| F getfunctionstable       | F registerstartactions |
| F getnibble               | F registerstopactions  |
| F getnumerickeys          | F registertempfile     |
| F getpreciseseconds       | N starttime            |
| F getpreciseticks         | T syntex               |
| F getruntime              | F blockfilename        |
| F getstacktop             | F collect              |
| F getstartupfile          | F disable              |
| F getstringkeys           | F enable               |
| F getversion              | F finish               |
| N lastbytecode            | F isenabled            |
| F luastringcall           | F pause                |
| D messages                | F popline              |
| F mpsstringcall           | F pushline             |
| F newindex                | F registerdisabler     |
| F newtable                | F registerenabler      |
| D numbers                 | F resetfilename        |
| F protectglobal           | F resume               |
| F registercode            | F setfilename          |
| F registeredcodes         | F setup                |
| F registerglobal          | F start                |
| F registerinitexfinalizer | F stop                 |
| F registername            | F wrapup               |
| F setbytecode             | F wrapup               |
| F setexitcode             | T lxml                 |
| F setnibble               | F addindex             |
| S startupfile             | F all                  |
| F subnibble               | F applyfunction        |
| F usestringcall           | F applyselectors       |
| S version                 | F att                  |
| T luac                    | F attribute            |
| F inspect                 | F badinclusions        |
| F print                   | F chainattribute       |
| T luatex                  | F checkindex           |
| F abort                   | F collected            |
| F cleanup                 | F command              |
| F cleanuptempfiles        | F concat               |
| F currentfile             | F concatrange          |
| F currentlevel            | F content              |
| F currenttotal            | F context              |
| F poprunstate             | F convert              |
| F pushrunstate            | F count                |

|                   |                    |
|-------------------|--------------------|
| F delete          | F last             |
| F depth           | F lastatt          |
| F direct          | F lastmatch        |
| T directives      | F lastpar          |
| F after           | F load             |
| F before          | F loadbuffer       |
| F handle          | F loaddata         |
| F load            | T loaded           |
| F setup           | F main             |
| F displayverbatim | F match            |
| F doif            | F name             |
| F doifelse        | F namespace        |
| F doifelseempty   | F nonspace         |
| F doifelsetext    | T obsolete         |
| F doifempty       | F get_id           |
| F doifnot         | F par              |
| F doifnotempty    | F param            |
| F doifnottext     | F parameter        |
| F doiftext        | F path             |
| F element         | F popmatch         |
| F elements        | F pos              |
| D entities        | F position         |
| F fileline        | F pure             |
| F filename        | F pushmatch        |
| F filter          | F raw              |
| F filterlist      | F rawroot          |
| F first           | F rawtex           |
| F flush           | F refatt           |
| F flushlinewise   | F register         |
| F flushsetups     | F registerentity   |
| F flushspacewise  | F removesetup      |
| F get_id          | F replace          |
| F getid           | F resetcdata       |
| F getindex        | F resetsetups      |
| F id              | F resolvedentity   |
| U idsplitter      | F root             |
| F ifatt           | F save             |
| F ifattempty      | F serialize        |
| F include         | F setaction        |
| F inclusion       | F setatt           |
| F inclusions      | F setattribute     |
| F index           | F setcdata         |
| F inlineverbatim  | F setcommandtonone |
| F installsetup    | F setcommandtotext |

|                  |                          |
|------------------|--------------------------|
| F setpar         | F exp                    |
| F setparam       | F floor                  |
| F setparameter   | F fmod                   |
| F setsetup       | F frexp                  |
| F snippet        | N huge                   |
| T sorters        | N initialseed            |
| F add            | F ldexp                  |
| F compare        | F log                    |
| F flush          | F log10                  |
| F reset          | F max                    |
| F show           | N maxinteger             |
| F sort           | F min                    |
| F splitid        | N mininteger             |
| F startraw       | F mod                    |
| F startunescaped | F modf                   |
| F stopraw        | F odd                    |
| F stopunescaped  | N pi                     |
| F store          | F pow                    |
| F strip          | F rad                    |
| F stripped       | F random                 |
| F tag            | F randomseed             |
| F texatt         | F round                  |
| F text           | F setrandomseedi         |
| F tobuffer       | F sin                    |
| F tofile         | F sind                   |
| F toparameters   | F sinh                   |
| F toverbatim     | F sqrt                   |
| F verbatim       | F tan                    |
| F withinindex    | F tand                   |
| D xmltexhandler  | F tanh                   |
| T math           | F tointeger              |
| F abs            | F type                   |
| F acos           | F ult                    |
| F asin           | T mathematics            |
| F atan           | F addfallbacks           |
| F atan2          | F addvariant             |
| F ceil           | D alphabets              |
| F ceiling        | D alternatives           |
| F cos            | F beforepassingfonttotex |
| F cosd           | F big                    |
| F cosh           | D boldmap                |
| F deg            | T categories             |
| F div            | D functions              |
| F even           | D charactersets          |

|                         |                          |
|-------------------------|--------------------------|
| F checkaftercopyingfont | F initializeparameters   |
| D classes               | F makefont               |
| D classnames            | D mapremap               |
| D codes                 | F markblobindexdone      |
| F define                | F overloadparameters     |
| T dictionaries          | F preset                 |
| D data                  | T presets                |
| D defaults              | T less                   |
| D groups                | T more                   |
| F groupset              | T zero                   |
| F groupsetgroup         | N privatebase            |
| F name                  | F registerextensible     |
| D names                 | F registersimplematrix   |
| F registercharacter     | F registerverboselabel   |
| F registergroup         | F remapalphabets         |
| F registergroupset      | F renderextensible       |
| D sets                  | T renderings             |
| D variants              | D mappings               |
| F dimensions            | D numbers                |
| F disablecollecting     | D sets                   |
| T domainlabels          | F renderset              |
| T domainnames           | F resetattributes        |
| N default               | F scaleparameters        |
| F enablecollecting      | F setextensibledata      |
| F extensiblecode        | F setmathsymbol          |
| D extensibles           | D sets                   |
| N extrabase             | F setsnapping            |
| T fallbacks             | F setvariant             |
| F apply                 | F strutcode              |
| F install               | D styles                 |
| F fallbackstyleattr     | F syncboth               |
| F familyname            | F syncname               |
| F functiontype          | F syncstyle              |
| D gaps                  | T tagging                |
| F getblobindex          | B mfenced                |
| F getboth               | F toengineclass          |
| F getdomainname         | F toname                 |
| F getextensibledata     | F tostyle                |
| F getmathblob           | F tweakaftercopyingfont  |
| F getoptionallabel      | F tweakbeforecopyingfont |
| F getstyle              | T tweaks                 |
| F gettextblob           | F accentdimensions       |
| F getverboselabel       | F action                 |
| F horizontalcode        | F addactuarian           |

|                    |                        |
|--------------------|------------------------|
| F addarrows        | T .,                   |
| F addbars          | F margins              |
| F addcomposites    | F message              |
| F addequals        | F moveitalics          |
| F addfourier       | F movelimits           |
| F addmirrors       | F parameters           |
| F addparts         | F radicalbodymargins   |
| F addprivates      | F radicaldegreeanchors |
| F addrules         | F replace              |
| F addscripts       | F replacealphabets     |
| F bigslots         | F replacerules         |
| F checkaccents     | F setoptions           |
| F checkspacing     | F setovershoots        |
| F copyaccents      | F showinfo             |
| T datasets         | F simplifykerns        |
| T accentdimensions | F sortvariants         |
| T accent           | F staircase            |
| T over             | F strut                |
| T under            | T subsets              |
| T addarrows        | T acenorsuvxz          |
| T addscripts       | T bhklt                |
| T checkspacing     | T d                    |
| T copyaccents      | T dotless              |
| T fixprimes        | T f                    |
| T fixslashes       | T gjqy                 |
| T wipecues         | T horizontalfences     |
| F dimensions       | T i                    |
| F extendaccents    | T integrals            |
| F fallbacks        | T letterlike           |
| F fixaccents       | T mw                   |
| F fixellipses      | T p                    |
| F fixintegrals     | F substitute           |
| F fixoldschool     | F topanchors           |
| F fixprimes        | F tweaks               |
| F fixradicals      | F version              |
| F fixslashes       | F wipeanchors          |
| F fixvariants      | F wipecues             |
| F flattenaccents   | F wipeitalics          |
| F fontname         | F wipevariants         |
| F inspect          | F utfmathaccent        |
| F keepbases        | F utfmathclass         |
| F kernpairs        | F utfmathcommand       |
| F kerns            | F utfmathfiller        |
| T kernsets         | F utfmathstretch       |

|                      |                   |
|----------------------|-------------------|
| F variantcode        | N leftbracket     |
| F variantcount       | N leftdelimiter   |
| F virtualize         | N let             |
| D virtualized        | N macrodef        |
| T md5                | N macrospecial    |
| F HEX                | N maketext        |
| F dec                | N message         |
| F hex                | N mode            |
| F sum                | N newbytemap      |
| F sumHEXA            | N newinternal     |
| F sumhexa            | N nullary         |
| T metapost           | N numeric         |
| F checkformat        | N of              |
| F checktracingonline | N ofbinary        |
| F cm                 | N onlyset         |
| T codes              | N parametertype   |
| N addto              | N pathconnect     |
| N ampersand          | N pathjoin        |
| N and                | N plusorminus     |
| N assignment         | N primarybinary   |
| N atleast            | N primarydef      |
| N begingroup         | N property        |
| N btex               | N protection      |
| N capsule            | N relax           |
| N colon              | N repeatloop      |
| N comma              | N rightbrace      |
| N controls           | N rightbracket    |
| N curl               | N rightdelimiter  |
| N cycle              | N runscript       |
| N definedmacro       | N save            |
| N delimiters         | N scantokens      |
| N endgroup           | N secondarybinary |
| N equals             | N secondarydef    |
| N etex               | N semicolon       |
| N everyjob           | N setbounds       |
| N exittest           | N shipout         |
| N expandafter        | N show            |
| N fiorelse           | N slash           |
| N if                 | N step            |
| N input              | N stop            |
| N interim            | N str             |
| N internal           | N string          |
| N iteration          | N tag             |
| N leftbrace          | N tension         |

|                    |                      |
|--------------------|----------------------|
| N tertiarybinary   | F getextradata       |
| N tertiarydef      | F gethashentries     |
| N thingstoadd      | F getinstancebyname  |
| N to               | F getinstanceinfo    |
| N typename         | F getinstancenames   |
| N unary            | F getjobdata         |
| N undefinedcs      | F getmovetolerance   |
| N until            | F getparameter       |
| N void             | F getparameterpreset |
| N with             | F getparameterset    |
| N within           | F getrecord          |
| N write            | F getstatistics      |
| F colorinitializer | F gettext            |
| F convert          | F gettolerance       |
| F currentmpx       | F graphic            |
| F currentmpxstatus | F graphic_base_pass  |
| S defaultformat    | F hascurvature       |
| S defaultinstance  | F includesvgbuffer   |
| S defaultmethod    | F includesvgfile     |
| F directrun        | F installplugin      |
| F escaped          | F isobject           |
| T exectime         | T knownparameters    |
| D extensiondata    | S lastlog            |
| F filter           | F load               |
| F filterclippath   | F maketext           |
| F flush            | F mesh               |
| T flushers         | F mesher             |
| T pdf              | F meshupper          |
| F flushfigure      | T metafonts          |
| F startfigure      | T metafun            |
| F stopfigure       | F interpolate        |
| F textfigure       | F topath             |
| F tocomment        | T mptopdf            |
| F flushnormalpath  | F convertmpstopdf    |
| T fonts            | N nofconverted       |
| F define           | N n                  |
| F format_number    | F newlogger          |
| F format_string    | N nofruns            |
| F formatnumber     | F nofsciptruns       |
| F getbackendoption | F nvformat           |
| F getbendtolerance | F pdf literals       |
| F getbuffer        | F pluginactions      |
| F getclippath      | F points             |
| F getextensions    | F pointset           |



|                        |                    |
|------------------------|--------------------|
| F popformat            | F setttext         |
| F popscriptrunner      | F settolerance     |
| F popvariable          | F setvariable      |
| F popvariables         | B showlog          |
| F preparetexttextsdata | F showsvgpage      |
| F process              | F simple           |
| F processing           | F splitprescript   |
| F processplugins       | F startgraphic     |
| T properties           | F startsvghashing  |
| F pushformat           | F stopgraphic      |
| F pushscriptrunner     | F stopsvghashing   |
| F pushvariable         | F svformat         |
| F pushvariables        | B svgcachefile     |
| B quitonerror          | F svgcolorremapper |
| B reducetogray         | T svgfactors       |
| F registerdirect       | N %                |
| F registerfilter       | N bp               |
| F registermesher       | N cm               |
| F registermeshupper    | N em               |
| F registerpointset     | N ex               |
| F registerscript       | N in               |
| F registertokens       | N mm               |
| F remappedtext         | N pc               |
| F remaptext            | N pt               |
| F reporterror          | N px               |
| F reportsvgfonts       | F svgglyptomp      |
| F reset                | F svgshapestomp    |
| F resetbackendoptions  | F svgshapestopdf   |
| F resetlastlog         | F svgsplitcolor    |
| F resetplugins         | F svgtomp          |
| F run                  | F svgtopdf         |
| F runinternal          | F sxsy             |
| F runscrip             | T tex              |
| F scanparameters       | F get              |
| F scriptindex          | F reset            |
| F setbendtolerance     | F set              |
| F setbuffer            | F setfrombuffer    |
| F setextensions        | B texerrors        |
| F setlogging           | F texexp           |
| F setmovetolerance     | F theclippath      |
| F setoutercolor        | F tographic        |
| F setparameter         | F totable          |
| F setparameterset      | T types            |
| F setrecord            | N boolean          |

|                    |                       |
|--------------------|-----------------------|
| N cmykcolor        | F report              |
| N color            | F stuff               |
| N dependent        | F unb64               |
| N independent      | F unqp                |
| N known            | F wrap                |
| N nep              | T wrapt               |
| N numeric          | F base64              |
| N pair             | F default             |
| N path             | F quoted-printable    |
| N pen              | F text                |
| N picture          | F wrp                 |
| N protodependent   | T moduledata          |
| N string           | T modules             |
| N structured       | D anch-loc            |
| N suffixedmacro    | D anch-pgr            |
| N tokenlist        | D anch-snc            |
| N transform        | D back-exp            |
| N unknownboolean   | D back-exp-imp-fnt    |
| N unknownnep       | D back-exp-imp-mth    |
| N unknownpath      | D back-exp-imp-ref    |
| N unknownpen       | D back-exp-imp-tag    |
| N unknownpicture   | D back-imp-lua        |
| N unknownstring    | D back-imp-mps        |
| N unsuffixedmacro  | D back-imp-pdf        |
| N vacuous          | D back-imp-pdp        |
| F typesvgpage      | D blob-ini            |
| F unload           | D buff-imp-default    |
| F untagvariable    | D buff-imp-escaped    |
| T variables        | D buff-imp-lua        |
| T mime             | D buff-imp-mp         |
| F b64              | D buff-imp-nested     |
| F decode           | D buff-imp-parsed-xml |
| T decodet          | D buff-imp-tex        |
| F base64           | D buff-imp-xml        |
| F quoted-printable | D buff-ini            |
| F dot              | D buff-par            |
| F encode           | D buff-ver            |
| T encodet          | D char-enc            |
| F base64           | D chem-ini            |
| F quoted-printable | D chem-str            |
| F eol              | D cldf-com            |
| F normalize        | D cldf-ver            |
| F qp               | D colo-icc            |
| F qpwrp            | D colo-ini            |

|                       |            |
|-----------------------|------------|
| D cont-run            | D font-ini |
| D core-con            | D font-lig |
| D core-ctx            | D font-log |
| D core-dat            | D font-lua |
| D core-sys            | D font-map |
| D driv-shp            | D font-mpf |
| D file-mod            | D font-mps |
| D file-syn            | D font-nod |
| D font-afk            | D font-ogr |
| D font-agl            | D font-one |
| D font-aux            | D font-onr |
| D font-cff            | D font-osd |
| D font-chk            | D font-ota |
| D font-cid            | D font-otc |
| D font-col            | D font-otd |
| D font-con            | D font-oth |
| D font-ctx            | D font-oti |
| D font-def            | D font-otj |
| D font-dsp            | D font-otl |
| D font-enc            | D font-oto |
| D font-enh            | D font-otr |
| D font-fbk            | D font-ots |
| D font-fmp            | D font-ott |
| D font-hsh            | D font-oup |
| D font-imp-checks     | D font-pat |
| D font-imp-combining  | D font-prv |
| D font-imp-compound   | D font-san |
| D font-imp-digits     | D font-sel |
| D font-imp-dimensions | D font-shp |
| D font-imp-effects    | D font-sol |
| D font-imp-italics    | D font-syn |
| D font-imp-ligatures  | D font-tfm |
| D font-imp-math       | D font-tpk |
| D font-imp-notused    | D font-trt |
| D font-imp-properties | D font-ttf |
| D font-imp-quality    | D font-txt |
| D font-imp-reorder    | D font-vfc |
| D font-imp-scripts    | D font-vir |
| D font-imp-spacekerns | D good-ctx |
| D font-imp-tex        | D good-gen |
| D font-imp-text       | D good-ini |
| D font-imp-tracing    | D good-mth |
| D font-imp-tweaks     | D grph-bmp |
| D font-imp-unicode    | D grph-con |

|                           |                |
|---------------------------|----------------|
| D grph-epd                | D lpdf-tag     |
| D grph-fil                | D lpdf-u3d     |
| D grph-img                | D lpdf-wid     |
| D grph-inc                | D lpdf-xmp     |
| D grph-pat                | D lxml-css     |
| D grph-raw                | D lxml-dir     |
| D grph-rul                | D lxml-ent     |
| D grph-trf                | D lxml-ini     |
| D grph-u3d                | D lxml-mms     |
| D java-ini                | D lxml-sor     |
| D lang-cnt                | D lxml-tex     |
| D lang-def                | D math-act     |
| D lang-dis                | D math-ali     |
| D lang-exc                | D math-dim     |
| D lang-hup                | D math-ext     |
| D lang-hyp                | D math-fbk     |
| D lang-ini                | D math-fnt     |
| D lang-lab                | D math-frc     |
| D lang-rep                | D math-inc     |
| D lang-tra                | D math-ini     |
| D lang-txt                | D math-map     |
| D lang-url                | D math-mps     |
| D lang-wrd                | D math-noa     |
| D libs-imp-curl           | D math-pre     |
| D libs-imp-ghostscript    | D math-rad     |
| D libs-imp-graphicsmagick | D math-ren     |
| D libs-ini                | D math-spa     |
| D lpdf-ano                | D math-tag     |
| D lpdf-col                | D math-txt     |
| D lpdf-epa                | D math-vfu     |
| D lpdf-epd                | D meta-blb     |
| D lpdf-fix                | D meta-fnt     |
| D lpdf-fix-imp-contents   | D meta-fun     |
| D lpdf-fix-imp-fonts      | D meta-imp-txt |
| D lpdf-fld                | D meta-ini     |
| D lpdf-fmt                | D meta-lua     |
| D lpdf-grp                | D meta-nod     |
| D lpdf-img                | D meta-pdf     |
| D lpdf-ini                | D meta-tex     |
| D lpdf-lmt                | D mlib-ctx     |
| D lpdf-mis                | D mlib-fio     |
| D lpdf-ren                | D mlib-int     |
| D lpdf-rul                | D mlib-lmp     |
| D lpdf-sig                | D mlib-lmt     |

|            |            |
|------------|------------|
| D mlib-lua | D publ-oth |
| D mlib-mat | D publ-reg |
| D mlib-mpf | D publ-sor |
| D mlib-msh | D publ-tra |
| D mlib-pdf | D publ-usr |
| D mlib-pos | D regi-ini |
| D mlib-pps | D scrn-but |
| D mlib-ptr | D scrn-flt |
| D mlib-pts | D scrn-hlp |
| D mlib-ran | D scrn-ini |
| D mlib-run | D scrn-pag |
| D mlib-scn | D scrn-ref |
| D mlib-snc | D scrn-wid |
| D mlib-svg | D scrp-cjk |
| D node-ali | D scrp-eth |
| D node-bck | D scrp-ini |
| D node-fnt | D scrp-tha |
| D node-par | D scrp-tib |
| D node-ref | D sort-ini |
| D node-rul | D sort-lan |
| D node-tra | D spac-ali |
| D node-typ | D spac-brk |
| D pack-obj | D spac-chr |
| D pack-ori | D spac-hor |
| D pack-rul | D spac-ovl |
| D page-blk | D spac-par |
| D page-cst | D spac-pas |
| D page-flt | D spac-prf |
| D page-ini | D spac-twi |
| D page-inj | D spac-ver |
| D page-ins | D strc-bkm |
| D page-lin | D strc-blk |
| D page-mix | D strc-con |
| D page-otr | D strc-doc |
| D page-pst | D strc-flt |
| D page-sel | D strc-ini |
| D page-str | D strc-itm |
| D phys-dim | D strc-lev |
| D publ-aut | D strc-lst |
| D publ-dat | D strc-mar |
| D publ-fnd | D strc-mat |
| D publ-inc | D strc-not |
| D publ-ini | D strc-num |
| D publ-jrn | D strc-pag |

|            |                          |
|------------|--------------------------|
| D strc-ref | D typo-pag               |
| D strc-reg | D typo-par               |
| D strc-rsc | D typo-pnc               |
| D strc-syn | D typo-prc               |
| D strc-tag | D typo-rep               |
| D strc-usr | D typo-rub               |
| D symb-emj | D typo-spa               |
| D symb-ini | D typo-stc               |
| D syst-cmp | D typo-sus               |
| D tabl-ali | D typo-syn               |
| D tabl-ntb | D typo-tal               |
| D tabl-tbl | D typo-wrp               |
| D tabl-xtb | D unic-ini               |
| D task-ini | T mp                     |
| D trac-bld | F autoinject             |
| D trac-jus | T aux                    |
| D trac-par | F boolean                |
| D trac-riv | F color                  |
| D trac-sig | F direct                 |
| D trac-vis | F draw                   |
| D type-ini | F fill                   |
| D typo-adj | F flush                  |
| D typo-bld | F fprint                 |
| D typo-brk | F integer                |
| D typo-cap | F number                 |
| D typo-chr | F numeric                |
| D typo-cln | F pair                   |
| D typo-dha | F pairpoints             |
| D typo-dif | F path                   |
| D typo-dig | F pathpoints             |
| D typo-dir | F points                 |
| D typo-drp | F print                  |
| D typo-duc | F quadruple              |
| D typo-fkr | F quadruplepoints        |
| D typo-fln | F quoted                 |
| D typo-hid | F size                   |
| D typo-inj | F string                 |
| D typo-itc | F transform              |
| D typo-krn | F triplet                |
| D typo-lan | F tripletpoints          |
| D typo-lbx | F vprint                 |
| D typo-man | F boolean                |
| D typo-mar | F bytemap_load_from_file |
| D typo-ovl | F cleaned                |

|                     |                              |
|---------------------|------------------------------|
| F color             | F integer                    |
| F dataset           | F lmt_function_x             |
| T datasets          | F lmt_glyphshape Accent      |
| F load              | F lmt_glyphshape baseline    |
| F direct            | F lmt_glyphshape boundingbox |
| F downgradebytemap  | F lmt_glyphshape depth       |
| F downsamplebytemap | F lmt_glyphshape height      |
| F draw              | F lmt_glyphshape italic      |
| F expandtex         | F lmt_glyphshape llx         |
| F fill              | F lmt_glyphshape n           |
| F fillbytemap       | F lmt_glyphshape path        |
| F flatten           | F lmt_glyphshape start       |
| F flush             | F lmt_glyphshape stop        |
| F format            | F lmt_glyphshape usedbox     |
| F formatted         | F lmt_glyphshape usedline    |
| F fprint            | F lmt_glyphshape width       |
| T get               | F lmt_mesh_set               |
| F hashentry         | F lmt_mesh_update            |
| F getbytemap        | F lmt_poisson_generate       |
| F getbytemapdata    | F lmt_start_potrace          |
| F getbytemapdirect  | F lmt_step_potrace           |
| F getlastaddtype    | F lmt_stop_potrace           |
| F graphformat       | F lmt_svg_include            |
| T inject            | F loadbytemap                |
| F boolean           | F loadbytemapfile            |
| F cmykcolor         | F mesh_register              |
| F color             | F mf_blob_add                |
| F integer           | F mf_blob_dimensions         |
| F number            | F mf_blob_index              |
| F numeric           | F mf_blob_new                |
| F pair              | F mf_blob_raw_content        |
| F path              | F mf_blob_raw_dimensions     |
| F points            | F mf_blob_raw_reset          |
| F pt                | F mf_blob_raw_toutf          |
| F quadruplet        | F mf_blob_raw_wipe           |
| F string            | F mf_blob_size               |
| F tokens            | F mf_blob_width              |
| F transform         | F mf_external_figure         |
| F triplet           | F mf_finish_saving_data      |
| F vector            | F mf_formatted_text          |
| F vectors           | F mf_get_outline_text        |
| F whatever          | F mf_graphic_text            |
| F whd               | F mf_inject_blob             |
| F xy                | F mf_map_move                |

|                        |                        |
|------------------------|------------------------|
| F mf_map_text          | T set                  |
| F mf_named_color       | F setbytemap           |
| F mf_outline_text      | F setbytemapoctave     |
| F mf_save_data         | F size                 |
| F mf_script_index      | T skip                 |
| F mf_start_saving_data | F token                |
| F mf_stop_saving_data  | F solve                |
| F mode                 | F string               |
| F n                    | F systemmode           |
| F newbytemap           | F transform            |
| F number               | F triplet              |
| F numeric              | F tripletpoints        |
| F pair                 | F vprint               |
| F pairpoints           | T mplib                |
| F path                 | F downgradebytemap     |
| F pathpoints           | F downsamplebytemap    |
| F points               | F execute              |
| F points_register      | F expandtex            |
| F processbytemap       | F fillbytemap          |
| F processingmode       | F finder               |
| F quadruple            | T finders              |
| F quadruplepoints      | F file                 |
| F quoted               | F mpstemplate          |
| F report               | F finish               |
| T scan                 | F getbytemap           |
| F boolean              | F getbytemapdata       |
| F cmykcolor            | F getbytemapdirect     |
| F color                | F getcallbackstate     |
| F expression           | F getcodes             |
| F integer              | F getcolormodels       |
| F next                 | F getfields            |
| F number               | F gethashentries       |
| F numeric              | F gethashentry         |
| F pair                 | F getinternalactions   |
| F path                 | F getknotstates        |
| F pen                  | F getlastaddtype       |
| F property             | F getlogtargets        |
| F string               | F getobjectdata        |
| F symbol               | F getobjectpeninfo     |
| F token                | F getobjectstacking    |
| T tokenscanners        | F getobjecttypes       |
| F transform            | F getprimitivelegacies |
| T typescanners         | F getprimitiveorigins  |
| F whatever             | F getprimitives        |



|                   |                    |
|-------------------|--------------------|
| F getpruneoptions | F scansymbol       |
| F getresultstates | F scantoken        |
| F getscaledbits   | F scantransform    |
| F getscantypes    | F setbytemap       |
| F getstates       | F setbytemapoctave |
| F getstatistics   | F settolerance     |
| F getstatus       | F showcontext      |
| F gettolerance    | F skiptoken        |
| F gettype         | F solvepath        |
| F gettypes        | F version          |
| F injectboolean   | T nanojpeg         |
| F injectcmkcolor  | F decode           |
| F injectcolor     | F next             |
| F injectinteger   | T noads            |
| F injectnumber    | F getmathmode      |
| F injectnumeric   | T handlers         |
| F injectpair      | F alternates       |
| F injectpath      | F autofences       |
| F injectpoints    | F classes          |
| F injectstring    | F collapse         |
| F injecttokens    | F dictionaries     |
| F injecttransform | F export           |
| F injectvector    | F families         |
| F injectvectors   | F fencing          |
| F injectwhatever  | F intervals        |
| F new             | F kernpairs        |
| F newbytemap      | F makeup           |
| F peninfo         | F normalize        |
| F processbytemap  | F numbers          |
| T propertycodes   | F relocate         |
| B realtimelogging | F render           |
| F scanboolean     | F showtree         |
| F scancmkcolor    | F snap             |
| F scancolor       | F spacing          |
| F scanexpression  | F suspicious       |
| F scaninteger     | F tags             |
| F scannext        | F unscript         |
| F scannumber      | F unstack          |
| F scannumeric     | F variants         |
| F scanpair        | F process          |
| F scanpath        | F processnested    |
| F scanpen         | T processors       |
| F scanproperty    | T dictionaries     |
| F scanstring      | T intervals        |

|                          |                           |
|--------------------------|---------------------------|
| T suspicious             | F free                    |
| T unscript               | F freeze                  |
| T unstack                | F getanchors              |
| F processouter           | F getattribute            |
| T node                   | F getattributelist        |
| F checkdiscretionaries   | F getattributes           |
| F checkdiscretionary     | F getboth                 |
| F copy                   | F getbottom               |
| F copyleft               | F getbottomdelimiter      |
| F count                  | F getbox                  |
| F currentattributes      | F getcachestate           |
| F dimensions             | F getchar                 |
| T direct                 | F getchardict             |
| F addmargins             | F getcharspec             |
| F addxoffset             | F getchoice               |
| F addxymargins           | F getclass                |
| F addyoffset             | F getcontinuation         |
| F appendaftertail        | F getcontrol              |
| F appendcurrenttail      | F getcornerkerns          |
| F beginofmath            | F getcurrenttail          |
| F checkdiscretionaries   | F getdata                 |
| F checkdiscretionary     | F getdegree               |
| F collapsing             | F getdelimiter            |
| F copy                   | F getdenominator          |
| F copyleft               | F getdepth                |
| F copyonly               | F getdirection            |
| F count                  | F getdisc                 |
| F currentattributes      | F getdiscpart             |
| F dimensions             | F getexcept               |
| F effectiveglue          | F getexpansion            |
| F endofmath              | F getfam                  |
| F exchange               | F getfield                |
| F fields                 | F getfielderror           |
| F findattribute          | F getfirstdirectioninlist |
| F findattributerange     | F getfont                 |
| F findnode               | F getfontcheck            |
| F firstchar              | F getgeometry             |
| F firstglyph             | F getglue                 |
| F firstglyphnode         | F getglyphdata            |
| F firstitalicglyph       | F getglyphdimensions      |
| F flattendiscretionaries | F getheight               |
| F flattenleaders         | F getid                   |
| F flushlist              | F getidsubtype            |
| F flushnode              | F getidsubtypenext        |

|                       |                     |
|-----------------------|---------------------|
| F getindex            | F getsup            |
| F getinputfields      | F getsuppre         |
| F getkern             | F gettop            |
| F getkerndimension    | F gettopdelimiter   |
| F getlanguage         | F gettotal          |
| F getleader           | F getusage          |
| F getleftdelimiter    | F getusedattributes |
| F getlist             | F getweight         |
| F getlistdimensions   | F getwhd            |
| F getlocalparboxstate | F getwidth          |
| F getmvl              | F getwordrange      |
| F getnaturalwhd       | F getxscale         |
| F getnext             | F getxyscales       |
| F getnodeerrorvalues  | F getyscale         |
| F getnodes            | F gluetostring      |
| F getnormalizedline   | F hasattribute      |
| F getnucleus          | F hasdimensions     |
| F getnumerator        | F hasdiscoption     |
| F getoffsets          | F hasfield          |
| F getoptions          | F hasgeometry       |
| F getorientation      | F hasglyph          |
| F getparstate         | F hasglyphoption    |
| F getpenalty          | F hasidsubtype      |
| F getpost             | F hasoptions        |
| F getpre              | F hasusage          |
| F getprev             | F hpack             |
| F getprime            | F hyphenating       |
| F getpropertystable   | F id                |
| F getproperty         | F ignoremathskip    |
| F getreplace          | F insertafter       |
| F getrightdelimiter   | F insertbefore      |
| F getruledimensions   | F insertcurrenthead |
| F getscale            | F isboth            |
| F getscales           | F ischar            |
| F getscrip            | F ischarcheck       |
| F getscripts          | F ischardisc        |
| F getshift            | F isdirect          |
| F getshort            | F isfontcheck       |
| F getslant            | F isglyph           |
| F getspecialist       | F isitalicglyph     |
| F getstate            | F isloop            |
| F getsub              | F isnext            |
| F getsubpre           | F isnextchar        |
| F getsubtype          | F isnextcharcheck   |

|                        |                       |
|------------------------|-----------------------|
| F isnextglyph          | F setattributelist    |
| F isnode               | F setattributes       |
| F isprev               | F setattributesinlist |
| F isprevchar           | F setboth             |
| F isprevcharcheck      | F setbottom           |
| F isprevglyph          | F setbottomdelimiter  |
| F issimilarglyph       | F setbox              |
| F issnapped            | F setchar             |
| F isspeciallist        | F setchardict         |
| F isvalid              | F setchoice           |
| F iszeroglue           | F setclass            |
| F kerning              | F setcontinuation     |
| F lastnode             | F setcontrol          |
| F length               | F setcurrenttail      |
| F ligaturing           | F setdata             |
| F makeextensible       | F setdegree           |
| F markglyphprocessing  | F setdelimiter        |
| F migrate              | F setdenominator      |
| F mlisttohlist         | F setdepth            |
| F naturalhsize         | F setdirection        |
| F naturalwidth         | F setdisc             |
| F new                  | F setdiscpart         |
| F newcontinuationatom  | F setexcept           |
| F newfontcheck         | F setexpansion        |
| F newmathglyph         | F setfam              |
| F newtextglyph         | F setfield            |
| F patchattributes      | F setfielderror       |
| F patchparshape        | F setfont             |
| F prependbeforehead    | F setfontcheck        |
| F prependcurrenttail   | F setgeometry         |
| F protectglyph         | F setglue             |
| F protectglyphs        | F setglyphdata        |
| F protectglyphsbase    | F setheight           |
| F protectglyphsnone    | F setindex            |
| F protrusionskipppable | F setinputfields      |
| F rangedimensions      | F setkern             |
| F remove               | F setlanguage         |
| F removefromlist       | F setleader           |
| F repack               | F setleftdelimiter    |
| F reverse              | F setlink             |
| F samefontcheck        | F setlist             |
| F serialized           | F setnext             |
| F setanchors           | F setnucleus          |
| F setattribute         | F setnumerator        |

|                     |                          |
|---------------------|--------------------------|
| F setoffsets        | F traverse               |
| F setoptions        | F traversechar           |
| F setorientation    | F traversecontent        |
| F setpenalty        | F traverseglyph          |
| F setpost           | F traverseid             |
| F setpre            | F traverseitalic         |
| F setprev           | F traverseleader         |
| F setprime          | F traverselist           |
| F setproperty       | F traversepossible       |
| F setreplace        | F type                   |
| F setrightdelimiter | F types                  |
| F setruledimensions | F unprotectglyph         |
| F setscale          | F unprotectglyphs        |
| F setscales         | F unsetattribute         |
| F setscript         | F unsetattributes        |
| F setscripts        | F updatefirstandbotmark  |
| F setshift          | F updatefirstmarks       |
| F setslant          | F updatemarks            |
| F setsnapped        | F updatetopmarks         |
| F setspeciallist    | F usedlist               |
| F setsplit          | F usesfont               |
| F setstate          | F vbalance               |
| F setsub            | F verticalbreak          |
| F setsubpre         | F vpack                  |
| F setsubtype        | F write                  |
| F setup             | F xscaled                |
| F setsuppre         | F yscaled                |
| F settop            | F effectiveglue          |
| F settopdelimiter   | F endofmath              |
| F settotal          | F family_font            |
| F setweight         | F fields                 |
| F setwhd            | F findattribute          |
| F setwidth          | F flattendiscretionaries |
| F show              | F flushlist              |
| F size              | F flushnode              |
| F slide             | F free                   |
| F softenhyphens     | F getattribute           |
| F startofpar        | F getcachestate          |
| F subtypes          | F getfield               |
| F tail              | F getfielderror          |
| F todirect          | F getglue                |
| F tonode            | F getnodeerrorvalues     |
| F toString          | F getpropertystable      |
| F tovaliddirect     | F getproperty            |

F gluetostr  
 F hasattribute  
 F hasfield  
 F hasglyph  
 F hpack  
 F hyphenating  
 F id  
 F insertafter  
 F insertbefore  
 F instock  
 F intracking  
 F inuse  
 F isnode  
 F iszeroglue  
 F kerning  
 F lastnode  
 F length  
 F ligaturing  
 F makeextensible  
 F mlisttohtml  
 F new  
 F protectglyph  
 F protectglyphs  
 F protrusionskipable  
 F rangedimensions  
 F remove  
 F serialized  
 F setattribute  
 F setfield  
 F setfielderror  
 F setglue  
 F setproperty  
 F show  
 F size  
 F slide  
 F subtypes  
 F tail  
 F todirect  
 F tonode  
 F toString  
 F traverse  
 F traverseid  
 F type  
 F types

F unprotectglyph  
 F unprotectglyphs  
 F unsetattribute  
 F usedlist  
 F usesfont  
 F vpack  
 F write  
 T nodes  
 D accentcodes  
 T adaptive  
 F handlehorizontal  
 F handlevertical  
 T methods  
 F set  
 F aligned  
 F append  
 F apply  
 F applyvisuals  
 F astable  
 F basepoints  
 D boundarycodes  
 T boxes  
 F count  
 F dimensions  
 F direct  
 F dispose  
 F found  
 F prune  
 F reset  
 F restore  
 F save  
 F savenode  
 T builders  
 D choicecodes  
 F concat  
 F copy  
 F copy\_node  
 F copylist  
 F countall  
 F currentattributes  
 F delete  
 T destinations  
 N attribute  
 T done

|                      |                        |
|----------------------|------------------------|
| F handler            | F cleanupbox           |
| T stack              | F cleanuppage          |
| D dircodes           | F export               |
| D directioncodes     | F finalizebox          |
| D directionvalues    | F finalizelist         |
| D dirvalues          | F flatten              |
| D disccodes          | F flattenline          |
| D discretionarycodes | T linebreakchecks      |
| D fencecodes         | F mathalignrelocate    |
| F fields             | F mathmatrixornaments  |
| T fieldtypes         | F mathmatrixrules      |
| D fillcodes          | F migrate              |
| D fillvalues         | F protectglyphs        |
| F firstdirinbox      | F recolor              |
| F flush              | F show                 |
| F flushlist          | F showhyphenation      |
| F flushnode          | F showpreamble         |
| D fractioncodes      | F stripping            |
| F fullhpack          | F tabulateskip         |
| F getattr            | F textbackgrounds      |
| F getattribute       | F unprotectglyphs      |
| F getboth            | F visualizehyphenation |
| F getbox             | F wipe                 |
| F getchar            | F hasattribute         |
| F getfield           | F hasfield             |
| F getfont            | F hpack                |
| F getid              | F idsandsubtypes       |
| F getleader          | F idstostring          |
| F getlist            | T injections           |
| F getnext            | F copy                 |
| F getprev            | F getligaindex         |
| F getproperty        | F getthreshold         |
| F getsubtype         | F handler              |
| D gluecodes          | F installgetspaceboth  |
| D glyphcodes         | F installnewkern       |
| T handlers           | F isspace              |
| F adjusters          | F keepcounts           |
| F aligncharacter     | F reset                |
| F backgrounds        | F resetcounts          |
| F backgroundspage    | F setcursive           |
| F backgroundsvbox    | F setkern              |
| F characters         | F setligaindex         |
| F checkglyphs        | F setmark              |
| F checkparcounter    | F setmove              |

|                           |                        |
|---------------------------|------------------------|
| F setposition             | F nopts                |
| F setspacekerns           | T nuts                 |
| F insertafter             | F addmargins           |
| F insertbefore            | F addxoffset           |
| F insertlistafter         | F addxymargins         |
| F installattributehandler | F addyoffset           |
| F is_display_math         | F append               |
| F isdirect                | F appendaftertail      |
| F isnode                  | F apply                |
| F isnut                   | F applyvisuals         |
| D kerncodes               | F beginofmath          |
| F leftmarginwidth         | F checkdiscretionaries |
| T linefillers             | F checkdiscretionary   |
| F filler                  | F checkslide           |
| F handler                 | F collapsing           |
| F set                     | T colors               |
| T lines                   | F set                  |
| T boxed                   | T components           |
| F addlinenumbers          | F copynocomponents     |
| F register                | F copyonlyglyphs       |
| F setup                   | F count                |
| D data                    | F flush                |
| F finalize                | F get                  |
| F number                  | F set                  |
| F link                    | F concat               |
| F linked                  | F copy                 |
| F list                    | F copy_properties      |
| D listcodes               | F copypoint            |
| F listtutf                | F copynode             |
| D literalvalues           | F copyonly             |
| F locate                  | F copyvisual           |
| D markcodes               | F count                |
| T markers                 | F countall             |
| F get                     | F currentattributes    |
| F set                     | F delete               |
| D mathcodes               | F dimensions           |
| D mathvalues              | F dirdimensions        |
| F maxboxwidth             | F effectiveglue        |
| T migrations              | F endofmath            |
| F setinserts              | F exchange             |
| F setmarks                | F findattribute        |
| F new                     | F findattributerange   |
| D noadcodes               | F findnode             |
| D nodecodes               | F firstchar            |



|                          |                           |
|--------------------------|---------------------------|
| F firstglyph             | F getfam                  |
| F firstglyphnode         | F getfield                |
| F firstitalicglyph       | F getfirstdirectioninlist |
| F flattendiscretionaries | F getfont                 |
| F flattenleaders         | F getfontcheck            |
| F flush                  | F getgeometry             |
| F flushlist              | F getglue                 |
| F flushnode              | F getglyphdata            |
| F free                   | F getglyphdimensions      |
| F freeze                 | F getheight               |
| F fullhpack              | F getid                   |
| F fullhpackrl            | F getidsubtype            |
| F getanchors             | F getidsubtypenext        |
| F getattr                | F getindex                |
| F getattribute           | F getinputfields          |
| F getattributelist       | F getkern                 |
| F getattributes          | F getkerndimension        |
| F getattrlist            | F getlang                 |
| F getattrs               | F getlanguage             |
| F getboth                | F getleader               |
| F getbottom              | F getleftdelimiter        |
| F getbottomdelimiter     | F getlist                 |
| F getbox                 | F getlistdimensions       |
| F getboxglue             | F getlocalparboxstate     |
| F getchar                | F getmiddle               |
| F getchardict            | F getmiddledelimiter      |
| F getcharspec            | F getmvllist              |
| F getchoice              | F getnaturalwhd           |
| F getclass               | F getnext                 |
| F getcomponents          | F getnormalizedline       |
| F getcontinuation        | F getnucleus              |
| F getcontrol             | F getnumerator            |
| F getcornerkerns         | F getoffsets              |
| F getdata                | F getoptions              |
| F getdegree              | F getorientation          |
| F getdelimiter           | F getparstate             |
| F getdenominator         | F getpenalty              |
| F getdepth               | F getpost                 |
| F getdirection           | F getpre                  |
| F getdisc                | F getprev                 |
| F getdiscpart            | F getprime                |
| F getdiscretionary       | F getprop                 |
| F getexcept              | F getreplace              |
| F getexpansion           | F getrightdelimiter       |

|                     |                       |
|---------------------|-----------------------|
| F getruledata       | F insertlistafter     |
| F getruledimensions | F is_display_math     |
| F getscale          | F isboth              |
| F getscales         | F ischar              |
| F getscrip          | F ischarcheck         |
| F getscripts        | F isdirect            |
| F getshift          | F isdone              |
| F getshort          | F isglyph             |
| F getslant          | F isitalicglyph       |
| F getspeciallist    | F isnext              |
| F getstate          | F isnextchar          |
| F getsub            | F isnextcharcheck     |
| F getsubpre         | F isnextglyph         |
| F getsubtype        | F isnode              |
| F getsup            | F isnut               |
| F getsuppre         | F isprev              |
| F getsurround       | F isprevchar          |
| F gettop            | F isprevcharcheck     |
| F gettopdelimiter   | F isprevglyph         |
| F gettotal          | F issimilarglyph      |
| F getusedattributes | F issnapped           |
| F getvalue          | F iszeroglue          |
| F getweight         | F kerning             |
| F getwhd            | F lastnode            |
| F getwidth          | F length              |
| F getwordrange      | F ligaturing          |
| F getxscale         | F link                |
| F getxyscales       | F linked              |
| F getyscale         | F locate              |
| F gluetostring      | F migrate             |
| F hasattribute      | F migratebox          |
| F hasdimensions     | F mlisttohtml         |
| F hasdiscoption     | F naturalhsize        |
| F hasfield          | F naturalwidth        |
| F hasgeometry       | F nestedtracedslide   |
| F hasglyph          | F new                 |
| F hasglyphoption    | F newcontinuationatom |
| F hasidsubtype      | F newfontcheck        |
| F hasoptions        | F newmathglyph        |
| F hpack             | F openup              |
| F hyphenating       | F patchattributes     |
| F ignoremathskip    | F patchparshape       |
| F insertafter       | T pool                |
| F insertbefore      | F accent              |

|                      |                      |
|----------------------|----------------------|
| F baselineskip       | F radical            |
| F boundary           | F rawliteral         |
| F boxrule            | F register           |
| F characterkern      | F restore            |
| F choice             | F righthangskip      |
| F cleanup            | F rightmarginkern    |
| F delim              | F rightskip          |
| F delimiter          | F rule               |
| F direction          | F save               |
| F directliteral      | F savepos            |
| F disc               | F setmatrix          |
| F discretionaryglue  | F setstate           |
| F emptyrule          | F shrink             |
| F fence              | F spacefontkern      |
| F fontkern           | F spacingrule        |
| F fraction           | F startrotation      |
| F glue               | F startscaling       |
| F gluespec           | F stock              |
| F glyph              | F stoprotation       |
| F hlist              | F stopscaling        |
| F imagerule          | F stretch            |
| F indentskip         | F strutrue           |
| F intercharacterglue | F subbox             |
| F italiccorrection   | F submlist           |
| F italickern         | F temp               |
| F kern               | F usage              |
| F latelua            | F userdefined        |
| F leader             | T userids            |
| F lefthangskip       | F usernode           |
| F leftmarginkern     | F userrule           |
| F leftskip           | F virtualrule        |
| F lineskip           | F vlist              |
| F literal            | F wordboundary       |
| F luaboundary        | F prepend            |
| F mathchar           | F prependbeforehead  |
| F mathtextchar       | F processranges      |
| F negatedglue        | F processwords       |
| F noad               | F protectglyph       |
| F originliteral      | F protectglyphs      |
| F outlinerule        | F protectglyphsbase  |
| F pageliteral        | F protectglyphsnone  |
| F penalizedglue      | F protrusionskipable |
| F penalizedspace     | F rangedimensions    |
| F penalty            | F rawprop            |

|                       |                       |
|-----------------------|-----------------------|
| F reference           | F setdiscpart         |
| F rehpack             | F setdiscretionary    |
| F remove              | F setexcept           |
| F removefromlist      | F setexpansion        |
| F repack              | F setfam              |
| F repackhlist         | F setfield            |
| F replace             | F setfont             |
| F reverse             | F setfontcheck        |
| T rules               | F setgeometry         |
| T ruleactions         | F setglue             |
| F box                 | F setglyphdata        |
| F draw                | F setheight           |
| F fill                | F setinputfields      |
| F mp                  | F setkern             |
| F stroke              | F setlang             |
| F userrule            | F setlanguage         |
| F serialized          | F setleader           |
| F setanchors          | F setleftdelimitter   |
| F setattr             | F setlink             |
| F setattribute        | F setlist             |
| F setattributelist    | F setmiddledelimitter |
| F setattributes       | F setnext             |
| F setattributesinlist | F setnucleus          |
| F setattrlist         | F setnumerator        |
| F setattrrs           | F setoffsets          |
| F setattrrsinlist     | F setoptions          |
| F setboth             | F setorientation      |
| F setbottom           | F setparproperty      |
| F setbox              | F setpenalty          |
| F setboxglue          | F setpost             |
| F setchar             | F setpre              |
| F setchardict         | F setprev             |
| F setchoice           | F setprime            |
| F setclass            | F setprop             |
| F setcomponents       | F setreplace          |
| F setcontinuation     | F setrightdelimitter  |
| F setcontrol          | F setruledata         |
| F setdata             | F setruledimensions   |
| F setdegree           | F setscale            |
| F setdelimiter        | F setscales           |
| F setdenominator      | F setscript           |
| F setdepth            | F setscripts          |
| F setdirection        | F setshift            |
| F setdisc             | F setslant            |

|                   |                         |
|-------------------|-------------------------|
| F setsnapped      | F traversepossible      |
| F setspeciallist  | T traversers            |
| F setsplit        | F alignrecord           |
| F setstate        | F boundary              |
| F setsub          | F char                  |
| F setsubpre       | F content               |
| F setsubtype      | F dir                   |
| F setup           | F disc                  |
| F setsuppre       | F glue                  |
| F setsurround     | F glyph                 |
| F settop          | F hlist                 |
| F setvalue        | F leader                |
| F setvisual       | F list                  |
| F setvisuals      | F mark                  |
| F setweight       | F next                  |
| F setwhd          | F node                  |
| F setwidth        | F penalty               |
| F show            | F possible              |
| F slide           | F rule                  |
| F softenhyphens   | F unset                 |
| F somepenalty     | F vlist                 |
| F somespace       | F whatsit               |
| F splitbox        | T treversers            |
| F startofpar      | F char                  |
| F striprange      | F content               |
| F tail            | F glyph                 |
| F takeattr        | F leader                |
| F takebox         | F list                  |
| F takelist        | F node                  |
| F theprop         | F possible              |
| F tonode          | T typesetters           |
| F tonodes         | F tohbox                |
| F tonut           | F tohpack               |
| F tosequence      | F tonodes               |
| F tostring        | F tovbox                |
| F tracedslide     | F tovpack               |
| F traverse        | F unprotectglyph        |
| F traversechar    | F unprotectglyphs       |
| F traversecontent | F unsetattribute        |
| F traverseglyph   | F unsetattributes       |
| F traverseid      | F untracedslide         |
| F traverseitalic  | F updatefirstandbotmark |
| F traverseleader  | F updatefirstmarks      |
| F traverselist    | F updatemarks           |

|                     |                     |
|---------------------|---------------------|
| F updatetopmarks    | T properties        |
| F usedlist          | T data              |
| F usesfont          | F pts               |
| F verticalbreak     | D radicalcodes      |
| F vianodes          | T references        |
| F vianuts           | N attribute         |
| F vpack             | T done              |
| F write             | F handler           |
| F xscaled           | T stack             |
| F yscaled           | F rehpac            |
| F packlist          | F remove            |
| D parcodes          | F repackhlist       |
| D penaltycodes      | F replace           |
| T plugindata        | F report            |
| F finalizer         | F rightmarginwidth  |
| F initializer       | D rulecodes         |
| S name              | T rules             |
| F processor         | F handler           |
| F points            | F process           |
| T pool              | F set               |
| F baselineskip      | F userrule          |
| F cleanup           | F serialize         |
| F directliteral     | F serializebox      |
| F glyph             | F serialized        |
| F hlist             | F setattr           |
| F latelua           | F setattrtribute    |
| F leader            | F setattrrlist      |
| F lineskip          | F setboth           |
| F literal           | F setbox            |
| F originliteral     | F setboxtonaturalwd |
| F register          | F setchar           |
| F restore           | F setfield          |
| F save              | F setfont           |
| F savepos           | F setleader         |
| F setmatrix         | F setlink           |
| F stock             | F setlist           |
| F usage             | F setnext           |
| F usernode          | F setprev           |
| F vlist             | F setproperty       |
| F prepend           | T shadows           |
| F print             | F handler           |
| T processors        | F set               |
| F trace_glyph_run_a | T shifts            |
| F trace_glyph_run_b | F handler           |

|                        |                   |
|------------------------|-------------------|
| F set                  | F toinches        |
| F showboxes            | F tomillimeters   |
| F showlist             | F tonode          |
| F showsimplelist       | F tonodes         |
| D skipcodes            | F tonut           |
| T snapshots            | F topics          |
| F getsamples           | F topoints        |
| F resetsamples         | F toscaledpoints  |
| F takesample           | F tosequence      |
| F somepenalty          | F tostring        |
| F somespace            | F totable         |
| T specialskipcodes     | F totree          |
| F splitbox             | F toutf           |
| F startofpar           | T tracers         |
| F stripdiscretionaries | T colors          |
| T stripping            | F reset           |
| T glyphs               | F set             |
| F set                  | F setlist         |
| D subtypes             | F directions      |
| F tail                 | F fontchar        |
| F takeattr             | T frequencies     |
| F takebox              | F register        |
| T tasks                | T pool            |
| F actions              | T nodes           |
| F appendaction         | F rule            |
| F disableaction        | T nuts            |
| F disablegroup         | F rule            |
| F enableaction         | F resetproperties |
| F enablegroup          | F rule            |
| F freeze               | F setlist         |
| F freezegroup          | F setproperties   |
| F new                  | F start           |
| F prependaction        | T steppers        |
| F removeaction         | F check           |
| F replaceaction        | F codes           |
| F restart              | F features        |
| F setaction            | F font            |
| F showactions          | F glyphs          |
| F table                | F message         |
| F tobasepoints         | F messages        |
| F tocentimeters        | F nofsteps        |
| F tociceros            | F register        |
| F todidots             | F reset           |
| F todimen              | F start           |

|                      |                 |
|----------------------|-----------------|
| F stop               | F handler       |
| F stop               | F markfonts     |
| T transparencies     | T modes         |
| F reset              | N boundary      |
| F set                | N cglue         |
| F setlist            | N ckern         |
| F traverse           | N depth         |
| F traverseid         | N dir           |
| T typesetters        | N discretionary |
| F hbox               | N expansion     |
| F hpack              | N fontkern      |
| F tohbox             | N glue          |
| F tohpack            | N glyph         |
| F tonodes            | N hbox          |
| F toinbox            | N hglue         |
| F toinbox            | N hkern         |
| F toinbox            | N hpenalty      |
| F vpack              | N insert        |
| F unsetattribute     | N italic        |
| F upcomingproperties | N kern          |
| F usedlist           | N line          |
| F vianodes           | N marginkern    |
| F vianuts            | N mark          |
| F visualizebox       | N math          |
| T visualizers        | N mathglue      |
| F attribute          | N mathkern      |
| T attributes         | N origin        |
| F box                | N par           |
| F definelayer        | N penalty       |
| F enable             | N pglue         |
| T filters            | N sglue         |
| N cglue              | N short         |
| N ckern              | N simple        |
| N continuation       | N simplehbox    |
| N hglue              | N simplevbox    |
| N hkern              | N simplevtop    |
| N hpenalty           | N space         |
| N pglue              | N strut         |
| N sglue              | N toddler       |
| N short              | N user          |
| N toddler            | N vbox          |
| N vglue              | N vglue         |
| N vkern              | N vkern         |
| N vpenalty           | N vpenalty      |
| F getusedfont        |                 |



|                    |                  |
|--------------------|------------------|
| N vtop             | F toinches       |
| N whatsit          | F tomillimeters  |
| F register         | F tonumberf      |
| F registercache    | F tonumberg      |
| F setfont          | F topicas        |
| F setlayer         | F topoints       |
| F setruledglyph    | F toscaled       |
| F setvisual        | F toscaledpoints |
| F vpack            | F totoves        |
| D whatsitcodes     | F valid          |
| F write            | T objects        |
| T number           | T data           |
| F %                | F dimensions     |
| F basepoints       | F found          |
| F bytetodecimal    | N n              |
| F clamp            | F page           |
| F compactfloat     | F reference      |
| F decimaltobyte    | F register       |
| D dimenfactors     | F restore        |
| F formatted        | T optional       |
| F formattedfloat   | T curl           |
| F formattedinteger | F escape         |
| F fractiontobyte   | F fetch          |
| F idiv             | F getversion     |
| N maxdimen         | F initialize     |
| F nopts            | F unescape       |
| F nubasepoints     | T foreign        |
| F nupoints         | F abivalues      |
| F percent          | F getbuffer      |
| F points           | F initialize     |
| F pts              | F load           |
| F signed           | F newbuffer      |
| F sparseexponent   | F totable        |
| F splitdimen       | F types          |
| F to16dot16        | T ghostscript    |
| F tobasepoints     | F execute        |
| F tobitstring      | F initialize     |
| F tocentimeters    | T graphicsmagick |
| F tociceros        | F execute        |
| F tocontext        | F getnoisetypes  |
| F todidots         | F initialize     |
| F todimen          | T hb             |
| F toediths         | F getshapers     |
| F toevenhex        | F getversion     |

|                    |                   |
|--------------------|-------------------|
| F initialize       | F setseekfile     |
| F loadfont         | T mysql           |
| F shapestring      | F close           |
| T imagemagick      | F execute         |
| F execute          | F getmessage      |
| F initialize       | F initialize      |
| T kpse             | F open            |
| F expand_braces    | T openssl         |
| F expand_path      | F getversion      |
| F expand_var       | F initialize      |
| F find_file        | F sign            |
| F find_files       | F verify          |
| F get_file_types   | T postgres        |
| F initialize       | F close           |
| F readable_file    | F execute         |
| F set_program_name | F getmessage      |
| F var_value        | F initialize      |
| T library          | F open            |
| F load             | T sqlite          |
| T loaded           | F close           |
| T lz4              | F execute         |
| F compress         | F getmessage      |
| F decompresssize   | F initialize      |
| F framecompress    | F open            |
| F framedecompress  | T zint            |
| F initialize       | F execute         |
| T lzma             | F initialize      |
| F compress         | T zstd            |
| F decompress       | F compress        |
| F initialize       | F decompress      |
| T lzo              | F initialize      |
| F compress         | F optionalrequire |
| F decompresssize   | T os              |
| F initialize       | F __getenv__      |
| T mujs             | F __setenv__      |
| F dofile           | S binsuffix       |
| F execute          | T binsuffixes     |
| F initialize       | N bits            |
| F reset            | F clock           |
| F setclosefile     | F converttime     |
| F setconsole       | F date            |
| F setfindfile      | F difftime        |
| F setopenfile      | F enableansi      |
| F setreadfile      | D env             |

|                  |                       |
|------------------|-----------------------|
| F execute        | F weekday             |
| F exit           | F where               |
| F fulltime       | F which               |
| F getcodepage    | T package             |
| F getenv         | S config              |
| F getnamevalues  | S cpath               |
| F gettimeofday   | F extralibpath        |
| F gettypevalues  | F extralibpaths       |
| F getunamefields | F extraluaopath       |
| F isleapyear     | F extraluaopaths      |
| F launch         | T helpers             |
| S libsuffix      | T builtin             |
| F localtime      | F path specification  |
| S name           | F preload table       |
| S newline        | F cleanpath           |
| F nofdays        | F loaded              |
| F now            | F loadedaslib         |
| F pipeto         | F loadedbyformat      |
| S platform       | F loadedbyname        |
| F remove         | F loadedbypath        |
| F rename         | F lualibfile          |
| F resultof       | T methods             |
| F runtime        | F all in one fallback |
| T selfarg        | F already loaded      |
| S selfbin        | F cpath specification |
| S selfcore       | F lib extra list      |
| S selfdir        | F lib variable format |
| S selflink       | F lua extra list      |
| S selfname       | F lua variable format |
| S selfpath       | F not loaded          |
| F serialwrite    | F path specification  |
| F setenv         | F preload table       |
| F setexitcode    | F qualified path      |
| F setlocale      | F reset loaded        |
| F sleep          | F registerpath        |
| F startuptime    | F report              |
| F time           | T sequence            |
| F timezone       | F showused            |
| F tmpname        | B trace               |
| F today          | B traceused           |
| S type           | F unload              |
| F uname          | F libpaths            |
| F uuid           | T loaded              |
| F validate       | B anch-loc            |

|                       |                       |
|-----------------------|-----------------------|
| B anch-pgr            | B file-syn            |
| B anch-snc            | B font-afk            |
| B back-exp            | B font-agl            |
| B back-exp-imp-fnt    | B font-aux            |
| B back-exp-imp-mth    | B font-cff            |
| B back-exp-imp-ref    | B font-chk            |
| B back-exp-imp-tag    | B font-cid            |
| B back-imp-lua        | B font-col            |
| B back-imp-mps        | B font-con            |
| B back-imp-pdf        | B font-ctx            |
| B back-imp-pdp        | B font-def            |
| B blob-ini            | B font-dsp            |
| B buff-imp-default    | B font-enc            |
| B buff-imp-escaped    | B font-enh            |
| B buff-imp-lua        | B font-fbk            |
| B buff-imp-mp         | B font-fmp            |
| B buff-imp-nested     | B font-hsh            |
| B buff-imp-parsed-xml | B font-imp-checks     |
| B buff-imp-tex        | B font-imp-combining  |
| B buff-imp-xml        | B font-imp-compound   |
| B buff-ini            | B font-imp-digits     |
| B buff-par            | B font-imp-dimensions |
| B buff-ver            | B font-imp-effects    |
| B char-enc            | B font-imp-italics    |
| B chem-ini            | B font-imp-ligatures  |
| B chem-str            | B font-imp-math       |
| B cldf-com            | B font-imp-notused    |
| B cldf-ver            | B font-imp-properties |
| B colo-icc            | B font-imp-quality    |
| B colo-ini            | B font-imp-reorder    |
| B cont-run            | B font-imp-scripts    |
| B core-con            | B font-imp-spacekerns |
| B core-ctx            | B font-imp-tex        |
| B core-dat            | B font-imp-text       |
| B core-sys            | B font-imp-tracing    |
| T curl                | B font-imp-tweaks     |
| F escape              | B font-imp-unicode    |
| F fetch               | B font-ini            |
| F getversion          | B font-lig            |
| S libfile             | B font-log            |
| S libpath             | B font-lua            |
| F unescape            | B font-map            |
| B driv-shp            | B font-mpf            |
| B file-mod            | B font-mps            |

|            |                           |
|------------|---------------------------|
| B font-nod | B grph-trf                |
| B font-ogr | B grph-u3d                |
| B font-one | B java-ini                |
| B font-onr | T json                    |
| B font-osd | F load                    |
| B font-ota | F tojson                  |
| B font-otc | F tolua                   |
| B font-otd | F tostring                |
| B font-oth | B lang-cnt                |
| B font-oti | B lang-def                |
| B font-otj | B lang-dis                |
| B font-otl | B lang-hup                |
| B font-oto | B lang-hyp                |
| B font-otr | B lang-ini                |
| B font-ots | B lang-lab                |
| B font-ott | B lang-rep                |
| B font-oup | B lang-tra                |
| B font-pat | B lang-txt                |
| B font-prv | B lang-url                |
| B font-san | B lang-wrd                |
| B font-sel | B libs-imp-curl           |
| B font-shp | B libs-imp-ghostscript    |
| B font-sol | B libs-imp-graphicsmagick |
| B font-syn | B libs-ini                |
| B font-tfm | B lpdf-ano                |
| B font-tpk | B lpdf-col                |
| B font-trt | B lpdf-epa                |
| B font-ttf | B lpdf-epd                |
| B font-txt | B lpdf-fix                |
| B font-vfc | B lpdf-fix-imp-contents   |
| B font-vir | B lpdf-fix-imp-fonts      |
| B good-ctx | B lpdf-fld                |
| B good-gen | B lpdf-fmt                |
| B good-ini | B lpdf-grp                |
| B good-mth | B lpdf-img                |
| B grph-bmp | B lpdf-ini                |
| B grph-con | B lpdf-lmt                |
| B grph-epd | B lpdf-mis                |
| B grph-fil | B lpdf-ren                |
| B grph-img | B lpdf-rul                |
| B grph-inc | B lpdf-sig                |
| B grph-pat | B lpdf-tag                |
| B grph-raw | B lpdf-u3d                |
| B grph-rul | B lpdf-wid                |

|             |                |
|-------------|----------------|
| B lpdf-xmp  | B math-noa     |
| T ltn12     | B math-pre     |
| N BLOCKSIZE | B math-rad     |
| T filter    | B math-ren     |
| F chain     | B math-spa     |
| F cycle     | B math-tag     |
| T pump      | B math-txt     |
| F all       | B math-vfu     |
| F step      | B meta-blb     |
| F report    | B meta-fnt     |
| T sink      | B meta-fun     |
| F chain     | B meta-imp-txt |
| F error     | B meta-ini     |
| F file      | B meta-lua     |
| F null      | B meta-nod     |
| F simplify  | B meta-pdf     |
| F table     | B meta-tex     |
| T source    | B mlib-ctx     |
| F cat       | B mlib-fio     |
| F chain     | B mlib-int     |
| F empty     | B mlib-lmp     |
| F error     | B mlib-lmt     |
| F file      | B mlib-lua     |
| F rewind    | B mlib-mat     |
| F simplify  | B mlib-mpf     |
| F string    | B mlib-msh     |
| B lxml-css  | B mlib-pdf     |
| B lxml-dir  | B mlib-pos     |
| B lxml-ent  | B mlib-pps     |
| B lxml-ini  | B mlib-ptr     |
| B lxml-mms  | B mlib-pts     |
| B lxml-sor  | B mlib-ran     |
| B lxml-tex  | B mlib-run     |
| B math-act  | B mlib-scn     |
| B math-ali  | B mlib-snc     |
| B math-dim  | B mlib-svg     |
| B math-ext  | B node-ali     |
| B math-fbk  | B node-bck     |
| B math-fnt  | B node-fnt     |
| B math-frc  | B node-par     |
| B math-inc  | B node-ref     |
| B math-ini  | B node-rul     |
| B math-map  | B node-tra     |
| B math-mps  | B node-typ     |

|                     |                   |
|---------------------|-------------------|
| B pack-obj          | F getsize         |
| B pack-ori          | F getstatus       |
| B pack-rul          | F getstatusvalues |
| B page-blk          | F getstream       |
| B page-cst          | F getstring       |
| B page-flt          | F gettrailer      |
| B page-ini          | F getversion      |
| B page-inj          | F new             |
| B page-ins          | F open            |
| B page-lin          | F openfile        |
| B page-mix          | F openstream      |
| B page-otr          | F pagestotable    |
| B page-pst          | F readfromstream  |
| B page-sel          | F readwholestream |
| B page-str          | F type            |
| T pdfdecode         | F unencrypt       |
| T pdfe              | B phys-dim        |
| F arraytotable      | T pngdecode       |
| F close             | F applyfilter     |
| F closestream       | F expand          |
| F dictionarytotable | F frompalette     |
| F getarray          | F interlace       |
| F getboolean        | F makemask        |
| F getbox            | F palettemask     |
| F getcatalog        | F reduce          |
| F getdictionary     | F setexperiment   |
| F getencodingvalues | F splitmask       |
| F getfieldtypes     | F tocmyk          |
| F getfromarray      | F togray          |
| F getfromdictionary | F tomask          |
| F getfromobject     | F transparentmask |
| F getfromreference  | T posit           |
| F getfromstream     | F NaN             |
| F getinfo           | F NaR             |
| F getinteger        | F __add           |
| F getmemoryusage    | F __band          |
| F getname           | F __bor           |
| F getnofobjects     | F __bxor          |
| F getnofpages       | F __div           |
| F getnumber         | F __eq            |
| F getobjectrange    | F __le            |
| F getpage           | F __lt            |
| F getpages          | F __mul           |
| F getpermissions    | S __name          |

|              |                    |
|--------------|--------------------|
| F __pow      | F tonumber         |
| F __shl      | F toposit          |
| F __shr      | F tostring         |
| F __sub      | T potrace          |
| F __tostring | F checked          |
| F __unm      | F concat           |
| F abs        | F contourplot      |
| F acos       | F convert          |
| F asin       | F count            |
| F atan       | F flush            |
| F band       | F free             |
| F bor        | F fromfile         |
| F bxor       | F getbitmap        |
| F ceil       | F getnewfields     |
| F conj       | F getpolicyvalues  |
| F copy       | F getprocessfields |
| F cos        | F new              |
| F exp        | F process          |
| F exp2       | F setbitmap        |
| F expand     | F stripped         |
| F floor      | F totable          |
| F frombyte   | F trace            |
| F fromchar   | B publ-aut         |
| F fromposit  | B publ-dat         |
| F integer    | B publ-fnd         |
| F log        | B publ-inc         |
| F log10      | B publ-ini         |
| F log1p      | B publ-jrn         |
| F log2       | B publ-oth         |
| F logb       | B publ-reg         |
| F max        | B publ-sor         |
| F min        | B publ-tra         |
| F modf       | B publ-usr         |
| F new        | T qrcodegen        |
| F pow        | F bitmap           |
| F rotate     | F generate         |
| F round      | B regi-ini         |
| F rounded    | B scrn-but         |
| F shift      | B scrn-fld         |
| F sin        | B scrn-hlp         |
| F sqrt       | B scrn-ini         |
| F tan        | B scrn-pag         |
| F tobyte     | B scrn-ref         |
| F tochar     | B scrn-wid         |



|                   |                     |
|-------------------|---------------------|
| B scrp-cjk        | F readcardinal4le   |
| B scrp-eth        | F readcardinaltable |
| B scrp-ini        | F readcline         |
| B scrp-tha        | F readcstring       |
| B scrp-tib        | F readdouble        |
| T serial          | F readdoublele      |
| F __gc            | F readfixed2        |
| S __name          | F readfixed4        |
| F __tostring      | F readfloat         |
| F close           | F readfloatle       |
| F open            | F readfloatle12     |
| F send            | F readfloatle6      |
| F tostring        | F readinteger1      |
| F write           | F readinteger1le    |
| T sha2            | F readinteger2      |
| F HASH256         | F readinteger2le    |
| F HASH384         | F readinteger3      |
| F HASH512         | F readinteger3le    |
| F HEX256          | F readinteger4      |
| F HEX384          | F readinteger4le    |
| F HEX512          | F readintegertable  |
| F digest256       | F tocardinal1       |
| F digest384       | F tocardinal1le     |
| F digest512       | F tocardinal2       |
| F hash256         | F tocardinal2le     |
| F hash384         | F tocardinal3       |
| F hash512         | F tocardinal3le     |
| F hex256          | F tocardinal4       |
| F hex384          | F tocardinal4le     |
| F hex512          | T socket            |
| F sum256          | N BLOCKSIZE         |
| F sum384          | N _SETSIZE          |
| F sum512          | N _SOCKETINVALID    |
| T sio             | F bind              |
| F read2dot14      | F choose            |
| F readbytes       | F connect           |
| F readbytetable   | F connect4          |
| F readcardinal1   | F connect6          |
| F readcardinal1le | T dns               |
| F readcardinal2   | F getaddrinfo       |
| F readcardinal2le | F gethostname       |
| F readcardinal3   | F getnameinfo       |
| F readcardinal3le | F tohostname        |
| F readcardinal4   | F toip              |

|                             |                         |
|-----------------------------|-------------------------|
| F gettime                   | S if-range              |
| T headers                   | S if-unmodified-since   |
| T canonic                   | S in-reply-to           |
| S accept                    | S keywords              |
| S accept-charset            | S last-attempt-date     |
| S accept-encoding           | S last-modified         |
| S accept-language           | S location              |
| S accept-ranges             | S max-forwards          |
| S action                    | S message-id            |
| S age                       | S mime-version          |
| S allow                     | S original-envelope-id  |
| S alternate-recipient       | S original-recipient    |
| S arrival-date              | S pragma                |
| S authorization             | S proxy-authenticate    |
| S bcc                       | S proxy-authorization   |
| S cache-control             | S range                 |
| S cc                        | S received              |
| S comments                  | S received-from-mta     |
| S connection                | S references            |
| S content-description       | S referer               |
| S content-disposition       | S remote-mta            |
| S content-encoding          | S reply-to              |
| S content-id                | S reporting-mta         |
| S content-language          | S resent-bcc            |
| S content-length            | S resent-cc             |
| S content-location          | S resent-date           |
| S content-md5               | S resent-from           |
| S content-range             | S resent-message-id     |
| S content-transfer-encoding | S resent-reply-to       |
| S content-type              | S resent-sender         |
| S cookie                    | S resent-to             |
| S date                      | S retry-after           |
| S diagnostic-code           | S return-path           |
| S dsn-gateway               | S sender                |
| S etag                      | S server                |
| S expect                    | S smtp-remote-recipient |
| S expires                   | S status                |
| S final-log-id              | S subject               |
| S final-recipient           | S te                    |
| S from                      | S to                    |
| S host                      | S trailer               |
| S if-match                  | S transfer-encoding     |
| S if-modified-since         | S upgrade               |
| S if-none-match             | S user-agent            |

|                    |                         |
|--------------------|-------------------------|
| S vary             | F parse                 |
| S via              | F parse_path            |
| S warning          | F unescape              |
| S will-retry-until | B sort-ini              |
| S www-authenticate | B sort-lan              |
| S x-mailer         | B spac-ali              |
| F lower            | B spac-brk              |
| F normalize        | B spac-chr              |
| T http             | B spac-hor              |
| N TIMEOUT          | B spac-ovl              |
| S USERAGENT        | B spac-par              |
| F genericform      | B spac-pas              |
| F open             | B spac-prf              |
| F request          | B spac-twi              |
| F newtry           | B spac-ver              |
| F protect          | T sparse                |
| F report           | F concat                |
| F select           | F get                   |
| F sink             | F max                   |
| T sinkt            | F min                   |
| F close-when-done  | F new                   |
| F default          | F range                 |
| F http-chunked     | F restore               |
| F keep-open        | F set                   |
| F skip             | F traverse              |
| F sleep            | F wipe                  |
| F source           | T status                |
| T sourcet          | F getbalancestate       |
| F by-length        | F getbufferstate        |
| F default          | F getcallbackstate      |
| F http-chunked     | F getconstants          |
| F until-closed     | F getdumpstate          |
| F tcp              | F geterrorlinestate     |
| F tcp4             | F geterrorstate         |
| F tcp6             | F getexpandstate        |
| T tp               | F getfilestate          |
| N TIMEOUT          | F getfontstate          |
| F connect          | F gethalferrorlinestate |
| F try              | F gethashstate          |
| T url              | F gethyphenationstate   |
| F absolute         | F getinputstate         |
| F build            | F getinsertstate        |
| F build_path       | F getlanguagestate      |
| F escape           | F getlinebreakstate     |

|                     |                   |
|---------------------|-------------------|
| F getlookupstate    | F bytes           |
| F getluastate       | F bytetable       |
| F getmarkstate      | F char            |
| F getmvlstate       | F characterpairs  |
| F getneststate      | F characters      |
| F getnodestate      | F checkedsplit    |
| F getparameterstate | F chrtointeger    |
| F getpoolstate      | F collapsespaces  |
| F getreadstate      | F containsws      |
| F getsavestate      | F count           |
| F getsparsestate    | F dectointeger    |
| F getstringstate    | F dump            |
| F gettexstate       | F escapedpattern  |
| F gettokenstate     | F escapedquotes   |
| F getwarningstate   | F f6              |
| T iocodes           | F find            |
| F list              | F format          |
| F resetmessages     | F formatter       |
| B strc-bkm          | D formatters      |
| B strc-blk          | F fullstrip       |
| B strc-con          | F gmatch          |
| B strc-doc          | F gsub            |
| B strc-flt          | T hashes          |
| B strc-ini          | T X00             |
| B strc-itm          | T spaces          |
| B strc-lev          | F hextocharacters |
| B strc-lst          | F hextointeger    |
| B strc-mar          | F is_boolean      |
| B strc-mat          | F is_empty        |
| B strc-not          | F itself          |
| B strc-num          | F len             |
| B strc-pag          | F limit           |
| B strc-ref          | F linetable       |
| B strc-reg          | F longtostring    |
| B strc-rsc          | F lower           |
| B strc-syn          | F match           |
| B strc-tag          | F nospaces        |
| B strc-usr          | F octtointeger    |
| T string            | F optionalquoted  |
| F autodouble        | F pack            |
| F autosingle        | F packrowscolumns |
| F booleanstring     | F packsize        |
| F byte              | F quote           |
| F bytepairs         | F quoted          |

|                      |                   |
|----------------------|-------------------|
| F rep                | F valid           |
| F replacenewlines    | F wordsplitter    |
| F reverse            | B symb-emj        |
| F split              | B symb-ini        |
| F splitintolines     | B syst-cmp        |
| F splitlines         | B tabl-ali        |
| F splitup            | B tabl-ntb        |
| F strip              | B tabl-tbl        |
| F sub                | B tabl-xtb        |
| F texhashed          | T table           |
| F texnewlines        | F append          |
| F tformat            | F are_equal       |
| F toHEX              | F combine         |
| F toboolean          | F compact         |
| F tobytes            | F concat          |
| F tocontext          | F contains        |
| F todec              | F copy            |
| F todimen            | F count           |
| F tohex              | F create          |
| F tonodes            | F derive          |
| F topattern          | F deserialize     |
| F totable            | F fastcopy        |
| F tounicode16        | F fastserialize   |
| F toutf              | F filtered        |
| F toutf16            | F flattened       |
| F toutf32            | F fromhash        |
| F toutf8             | F getn            |
| F tracedchar         | F has_one_entry   |
| F unescapedquotes    | F hashed          |
| F unpack             | F identical       |
| F unquote            | F imerge          |
| F unquoted           | F imerged         |
| F upper              | F insert          |
| F utf16toutf8        | F is_empty        |
| F utfcharacter       | F is_simple_table |
| F utfcharacters      | F keys            |
| F utfcharacterstable | F load            |
| F utflength          | F loweredkeys     |
| F utfpadd            | F makewweak       |
| F utfpadding         | F merge           |
| F utftabletostring   | F merged          |
| F utfvalue           | F mirrored        |
| F utfvalues          | F move            |
| F utfvaluetable      | F pack            |

|                        |                           |
|------------------------|---------------------------|
| F prepend              | D names                   |
| F print                | D numbers                 |
| F remove               | F system                  |
| F reverse              | T box                     |
| F reversed             | D boxcontinuationcodes    |
| F save                 | D boxoptioncodes          |
| F sequenced            | T boxresources            |
| F serialize            | F getbox                  |
| F setmetatablecall     | F getdimensions           |
| F setmetatableindex    | F save                    |
| F setmetatableindices  | F use                     |
| F setmetatablenewindex | D breakcodes              |
| F sort                 | D buildcodes              |
| F sorted               | T catcode                 |
| F sortedhash           | T cccode                  |
| F sortedhashkeys       | D charactercontrolcodes   |
| F sortedhashonly       | D charactertagcodes       |
| F sortedindexonly      | F chardef                 |
| F sortedkeys           | D classoptioncodes        |
| F sortedpairs          | T codes                   |
| F sparse               | D conditionals            |
| F strip                | D constants               |
| F sub                  | T count                   |
| F swapped              | F cprint                  |
| F tocontext            | N ctdcatcodes             |
| F tocsv                | N ctxcatcodes             |
| F tofile               | F definefont              |
| F tohandle             | T delcode                 |
| F tohash               | T dimen                   |
| F toxml                | F dimensiondef            |
| F unique               | D directioncodes          |
| F unpack               | D directives              |
| B task-ini             | D discoptioncodes         |
| T tex                  | D discpartcodes           |
| T amcode               | F dontcomplain            |
| T attribute            | D doublescriptoptioncodes |
| F attributetracking    | F enableprimitives        |
| D automigrationcodes   | F equal                   |
| D autoparagraphcodes   | F error                   |
| F badness              | F expandasvalue           |
| D badnessmodecodes     | D experiments             |
| F balance              | F extraprimitives         |
| D balancecallbackcodes | F fatalerror              |
| T boundaries           | D fillcodes               |

|                                 |                               |
|---------------------------------|-------------------------------|
| D flagcodes                     | F getdoublescriptoptionvalues |
| F fontidentifier                | F getemptyparagraphmodevalues |
| F fontname                      | F geterrorrecoverymodevalues  |
| F forcehmode                    | F geterrorvalues              |
| F forceinputstatefile           | F getfillvalues               |
| F forceinputstateline           | F getflagvalues               |
| F fprint                        | F getfloat                    |
| D frozenparcodes                | F getfontdimension            |
| F get                           | F getfontoffamily             |
| F getadjustoptionvalues         | F getfrozenparvalues          |
| F getalignmentcontextvalues     | F getglue                     |
| F getamcode                     | F getglueoptionvalues         |
| F getappendadjustcontextvalues  | F getglyphdata                |
| F getappendmigratecontextvalues | F getglyphdiscvalues          |
| F getattribute                  | F getglyphoptionvalues        |
| F getautomigrationvalues        | F getglyphprotectionvalues    |
| F getautoparagraphvalues        | F getglyphscales              |
| F getbadnessmodevalues          | F getglyphscript              |
| F getbalancecallbackvalues      | F getglyphstate               |
| F getbalanceparameterfields     | F getgroupvalues              |
| F getbalanceresultfields        | F gethccode                   |
| F getbalancestepoptionvalues    | F gethelptext                 |
| F getbox                        | F gethmcodes                  |
| F getboxcontinuationvalues      | F gethyphenationvalues        |
| F getboxdir                     | F getiftypes                  |
| F getboxoptionvalues            | F getinputstatefile           |
| F getboxresourcebox             | F getinputstateline           |
| F getboxresourcedimensions      | F getinputstatemode           |
| F getbreakcontextvalues         | F getinsertcategoryvalues     |
| F getbuildcontextvalues         | F getinsertcontent            |
| F getcardinalvalue              | F getinsertdepth              |
| F getcatcode                    | F getinsertdistance           |
| F getcccode                     | F getinsertheight             |
| F getcharactercontrolvalues     | F getinsertlimit              |
| F getcharactertagvalues         | F getinsertmaxplaced          |
| F getcount                      | F getinsertmultiplier         |
| F getcurrentmvl                 | F getinsertoptionvalues       |
| F getdelcode                    | F getinsertsplitvalues        |
| F getdelcodes                   | F getinsertwidth              |
| F getdimen                      | F getintegervalue             |
| F getdimensionvalue             | F getinteraction              |
| F getdirectionvalues            | F getinteractionmodes         |
| F getdiscoptionvalues           | F getiovalues                 |
| F getdiscpartvalues             | F getkerneloptionvalues       |

|                               |                                |
|-------------------------------|--------------------------------|
| F getkernoptionvalues         | F getnoadoptoptionvalues       |
| F getlargestusedmark          | F getnormalizelinevalues       |
| F getlccode                   | F getnormalizeparvalues        |
| F getlinebreakparameterfields | F getnospacesmodevalues        |
| F getlinebreakresultfields    | F getoutputactive              |
| F getlinebreakstatevalues     | F getoverloadstate             |
| F getlinedir                  | F getpacktypevalues            |
| F getlist                     | F getpagecontextvalues         |
| F getlistanchorvalues         | F getpagestate                 |
| F getlistfields               | F getpagestatevalues           |
| F getlistgeometryvalues       | F getparagraphfillmodevalues   |
| F getlistsignvalues           | F getparagraphoptionvalues     |
| F getlocalbox                 | F getparametermodevalues       |
| F getlocalboxlocations        | F getparcontextvalues          |
| F getlocallevel               | F getpardataspecifications     |
| F getmark                     | F getpardir                    |
| F getmarknames                | F getparmodevalues             |
| F getmath                     | F getparstate                  |
| F getmathclassoptionvalues    | F getparstatefields            |
| F getmathcode                 | F getpartriggervalue           |
| F getmathcodes                | F getpenaltyoptionvalues       |
| F getmathcontrolvalues        | F getpositvalue                |
| F getmathdir                  | F getprepoststatevalues        |
| F getmathgluevalues           | F getprimitivelegacies         |
| F getmathoptionvalues         | F getprimitiveorigins          |
| F getmathparametervalues      | F getprotrusionboundaryvalues  |
| F getmathscriptordervalue     | F getregisterindex             |
| F getmathscriptsmodevalues    | F getruleoptionvalues          |
| F getmathstylenamevalues      | F getrunstate                  |
| F getmathstylevalues          | F getrunstatevalues            |
| F getmathstylevariant         | F getsfcode                    |
| F getmathsurroundvalues       | F getshapingpenaltiesvalues    |
| F getmathvariantpresets       | F getskip                      |
| F getmathvariantvalues        | F getsnappingvalues            |
| F getmeasure                  | F getspacefactorvalues         |
| F getmissingcharactervalues   | F getspaceskipvalues           |
| F getmode                     | F getspcode                    |
| F getmodevalues               | F getspecialmathclassvalues    |
| F getmuglue                   | F getspecification             |
| F getmuskip                   | F getspecificationoptionvalues |
| F getmvloptionvalues          | F gettextcontrolvalues         |
| F getnest                     | F gettextdir                   |
| F getnestfields               | F gettoks                      |
| F getnestlevel                | F getuccode                    |



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|----------------------------|-----------------------------|
| F getuleaderlocationvalues | T magicconstants            |
| F getunitclassvalues       | S activecharacternamespace  |
| F getusedmarks             | N allfitnessvalues          |
| F getvsplitcontextvalues   | N assumedmathcontrol        |
| T glue                     | N awfulbad                  |
| D glueoptioncodes          | N decentcriterion           |
| D glyphdisccodes           | N defaultcatcodetable       |
| D glyphoptioncodes         | N defaultcharactercontrol   |
| D groupcodes               | N defaultdeadcycles         |
| F hashtokens               | N defaulteqnogapstep        |
| T hccode                   | N defaulthangafter          |
| T hmcode                   | N defaultoutputbox          |
| N hshcatcodes              | N defaultpredisplaygap      |
| D hyphenationcodes         | N defaultrule               |
| D ifs                      | N defaultspacefactor        |
| N inicatcodes              | N defaulttolerance          |
| D insertcategorycodes      | N deplorable                |
| D insertoptioncodes        | N ejectpenalty              |
| D insertsplitcodes         | N ignoredepth               |
| F integerdef               | N infinitebad               |
| D interactioncodes         | N infinitepenalty           |
| F isattribute              | N infinity                  |
| F isbox                    | N largewidthexcess          |
| F iscount                  | N loosecriterion            |
| F isdefined                | N mathallclass              |
| F isdimen                  | N mathbeginclass            |
| F isfloat                  | N mathdefaultpenalty        |
| F isglue                   | N mathendclass              |
| F ismuglue                 | N mathfirstuserclass        |
| F ismuskip                 | N mathlastuserclass         |
| F isprintable              | N maxattributeregisterindex |
| F isskip                   | N maxboxregisterindex       |
| F istoks                   | N maxbytecodeindex          |
| F lastnodetype             | N maxcalculatedbadness      |
| T lccode                   | N maxcardinal               |
| F less                     | N maxcharactercode          |
| F lessequal                | N maxdatavalue              |
| F linebreak                | N maxdimen                  |
| D linebreakstatecodes      | N maxdimenregisterindex     |
| D listanchorcodes          | N maxdimension              |
| D listgeometrycodes        | N maxdimensionregisterindex |
| T lists                    | N maxendlinecharacter       |
| D listsigncodes            | N maxfloatregisterindex     |
| N luacatcodes              | N maxfontadjustshrinkfactor |

|                              |                          |
|------------------------------|--------------------------|
| N maxfontadjuststep          | N null                   |
| N maxfontadjuststretchfactor | N nullflag               |
| N maxfunctionreference       | N nullfont               |
| N maxglueregisterindex       | N onebp                  |
| N maxhalfvalue               | N presetrulethickness    |
| N maxhalfword                | N runningrule            |
| N maxinteger                 | N saparthigh             |
| N maxintegerregisterindex    | N sapartlow              |
| N maxintregisterindex        | N sapartmiddle           |
| N maxlimitedscale            | N smallstretchability    |
| N maxmarkindex               | N specialspacefactor     |
| N maxmathclasscode           | N texeqtbsize            |
| N maxmathfamilyindex         | N texhashprime           |
| N maxmathscalingfactor       | N texhashsize            |
| N maxmathstylescale          | N texmemorymode          |
| N maxmuglueregisterindex     | N two                    |
| N maxmvindex                 | N undefinedmathparameter |
| N maxnewlinecharacter        | N unity                  |
| N maxnofbytecodes            | N unusedattributevalue   |
| N maxnofcatcodetables        | N unusedmathfamily       |
| N maxnoffitnessvalues        | N unusedmathstyle        |
| N maxnoffonts                | N unusedscriptvalue      |
| N maxnoflanguages            | N unusedstatevalue       |
| N maxnofmarks                | N zeroglu                |
| N maxnofmathfamilies         | F mathchardef            |
| N maxquarterword             | T mathcode               |
| N maxscalefactor             | D mathcontrolcodes       |
| N maxsizeofword              | D mathoptioncodes        |
| N maxspacefactor             | D mathparametercodes     |
| N maxtoksregisterindex       | D mathscriptordercodes   |
| N maxtwinlength              | D mathscriptsmodecodes   |
| N mincardinal                | T modelevels             |
| N mindatavalue               | N horizontal             |
| N mindimen                   | N math                   |
| N mindimension               | N unset                  |
| N minhalfword                | N vertical               |
| N mininfinity                | D modes                  |
| N mininteger                 | F mprint                 |
| N minmvindex                 | T muglu                  |
| N minnoffitnessvalues        | T muskip                 |
| N minquarterword             | D mvloptioncodes         |
| N minscalefactor             | T nest                   |
| N minspacefactor             | N nilcatcodes            |
| N nocatcodetable             | D noadoptioncodes        |

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|---------------------------|-------------------------|
| D normalizelinecodes      | F setconditional        |
| D normalizeparcodes       | F setcount              |
| D nospacesmodecodes       | F setdelcode            |
| N notcatcodes             | F setdimen              |
| F number                  | F setdimensionvalue     |
| D paragraphfillmodecodes  | F setfloat              |
| D paragraphoptioncodes    | F setglue               |
| D parcontextcodes         | F setglyphdata          |
| D partriggercodes         | F setglyphscript        |
| D penaltyoptioncodes      | F setglyphstate         |
| F poplocal                | F sethccode             |
| F popsavelevel            | F sethmcodes            |
| T posit                   | F setinputstatefile     |
| F positdef                | F setinputstateline     |
| F preparelinebreak        | F setinputstatemode     |
| F primitives              | F setinsertcontent      |
| F print                   | F setinsertdistance     |
| D protrusionboundarycodes | F setinsertlimit        |
| N prtcacodes              | F setinsertmaxplaced    |
| F pushlocal               | F setinsertmultiplier   |
| F pushesavelevel          | F setintegervalue       |
| F quitlocal               | F setinteraction        |
| F quittoks                | F setlccode             |
| F resetparagraph          | F setlinedir            |
| N rlnacodes               | F setlist               |
| F romannumeral            | F setlocalbox           |
| F round                   | F setmath               |
| D ruleoptioncodes         | F setmathcode           |
| F runlocal                | F setmathdir            |
| D runstatecodes           | F setmuglue             |
| F runstring               | F setmuskip             |
| F runtoks                 | F setnest               |
| F saveboxresource         | F setoverloadstate      |
| F scale                   | F setpaddir             |
| F scantoks                | F setpositvalue         |
| F set                     | F setrunstate           |
| F setamcode               | F setsfcode             |
| F setattribute            | F setskip               |
| F setbalanceshape         | F setspcode             |
| F setbox                  | F settextdir            |
| F setboxdir               | F settoks               |
| F setcardinalvalue        | F setuccode             |
| F setcatcode              | T sfcode                |
| F setccode                | D shapingpenaltiescodes |

|                            |                    |
|----------------------------|--------------------|
| F shiftparshape            | F write            |
| F shipout                  | F write_nl         |
| F showcontext              | F writenl          |
| T skip                     | F writeselector    |
| D snappingcodes            | F writeselectorlf  |
| F snapshotpar              | F writeselectornl  |
| F sp                       | T token            |
| D spacefactorcodes         | F create           |
| D spaceskipcodes           | F expand           |
| D specialmathclasscodes    | F expand_macro     |
| D specificationoptioncodes | F expandmacro      |
| F splitbox                 | F future_expand    |
| D splitcodes               | F futureexpand     |
| F sprint                   | F get_active       |
| F stringtocodesbitmap      | F get_cmdchrsc     |
| D systemmodes              | F get_cmdname      |
| F takebox                  | F get_command      |
| N texcatcodes              | F get_csname       |
| D textcontrolcodes         | F get_cstoken      |
| T toks                     | F get_dimension    |
| F tonumber                 | F get_expandable   |
| F toscaled                 | F get_fields       |
| N tpatcatcodes             | F get_flags        |
| N tpbcatcodes              | F get_frozen       |
| F tprint                   | F get_id           |
| F trackedattribute         | F get_immutable    |
| D trackers                 | F get_index        |
| F triggerbuildpage         | F get_instance     |
| N txtcatcodes              | F get_integer      |
| T uccode                   | F get_macro        |
| F uprint                   | F get_meaning      |
| F useboxresource           | F get_noaligned    |
| D userdiscoptioncodes      | F get_parameters   |
| D userglyphoptioncodes     | F get_permanent    |
| N vrbcatcodes              | F get_primitive    |
| F write                    | F get_protected    |
| T texio                    | F get_range        |
| F closeinput               | F get_tok          |
| F forceendoffile           | F get_tolerant     |
| F getinputindex            | F getactive        |
| F getselectorvalues        | F getcmdchrsc      |
| F getsourcefilename        | F getcmdname       |
| F setlogfile               | F getcommand       |
| F terminal                 | F getcommandvalues |

|                     |                        |
|---------------------|------------------------|
| F getconstant       | F peek_next            |
| F getcsname         | F peek_next_char       |
| F getcstoken        | F peek_next_expanded   |
| F getdimension      | F peeknext             |
| F getexpandable     | F peeknextchar         |
| F getexpansion      | F peeknextexpanded     |
| F getfields         | F pop_macro            |
| F getflags          | F popmacro             |
| F getfrozen         | F push_macro           |
| F getfunctionvalues | F pushmacro            |
| F getid             | F put_back             |
| F getimmutable      | F put_next             |
| F getindex          | F putback              |
| F getinstance       | F putnext              |
| F getinteger        | F save_lua             |
| F getlength         | F savelua              |
| F getmacro          | F scan_argument        |
| F getmeaning        | F scan_box             |
| F getnoaligned      | F scan_cardinal        |
| F getparameters     | F scan_char            |
| F getpermanent      | F scan_cmdchr          |
| F getprimitive      | F scan_cmdchr_expanded |
| F getprimitives     | F scan_code            |
| F getprotected      | F scan_csname          |
| F getrange          | F scan_delimited       |
| F gettok            | F scan_dimen           |
| F gettolerant       | F scan_float           |
| F gobble            | F scan_glue            |
| F gobble_dimension  | F scan_integer         |
| F gobble_integer    | F scan_key             |
| F gobbledimension   | F scan_keyword         |
| F gobbleinteger     | F scan_keyword_cs      |
| F grab              | F scan_letters         |
| F grabstring        | F scan_luacardinal     |
| F inhibit           | F scan_luainteger      |
| F is_defined        | F scan_luanumber       |
| F is_next_char      | F scan_next            |
| F is_token          | F scan_next_char       |
| F isactivestring    | F scan_next_expanded   |
| F isdefined         | F scan_real            |
| F isnextchar        | F scan_scale           |
| F istoken           | F scan_skip            |
| F locatemacro       | F scan_string          |
| F new               | F scan_token           |

|                         |                      |
|-------------------------|----------------------|
| F scan_token_code       | F scantoks           |
| F scan_tokenlist        | F scanvalue          |
| F scan_toks             | F scanword           |
| F scan_value            | F serialize          |
| F scan_word             | F set_dimension      |
| F scanargument          | F set_integer        |
| F scanbox               | F set_lua            |
| F scancardinal          | F set_macro          |
| F scanchar              | F setdimension       |
| F scancmdchr            | F setinteger         |
| F scancmdchrexposed     | F setlua             |
| F scancode              | F setmacro           |
| F scancsname            | F setmacrofrommark   |
| F scanclistoken         | F setmacrotonothing  |
| F scandelimited         | F skip_next          |
| F scandetokened         | F skip_next_expanded |
| F scandimen             | F skipnext           |
| F scandimenargument     | F skipnextexpanded   |
| F scandimension         | F type               |
| F scandimensionargument | F undefine_macro     |
| F scanfloat             | F undefinemacro      |
| F scanglue              | B trac-bld           |
| F scanint               | B trac-jus           |
| F scaninteger           | B trac-par           |
| F scanintegerargument   | B trac-riv           |
| F scankey               | B trac-sig           |
| F scankeyword           | B trac-vis           |
| F scankeywordcs         | B type-ini           |
| F scanletters           | B typo-adj           |
| F scanluacardinal       | B typo-bld           |
| F scanluainteger        | B typo-brk           |
| F scanluanumber         | B typo-cap           |
| F scannext              | B typo-chr           |
| F scannextchar          | B typo-cln           |
| F scannextexpanded      | B typo-dha           |
| F scanposit             | B typo-dif           |
| F scanreal              | B typo-dig           |
| F scanscale             | B typo-dir           |
| F scanskip              | B typo-drp           |
| F scanstring            | B typo-duc           |
| F scantoken             | B typo-fkr           |
| F scantokencode         | B typo-fln           |
| F scantokenlist         | B typo-hid           |
| F scantokenstring       | B typo-inj           |

|                  |                 |
|------------------|-----------------|
| B typo-itc       | F show          |
| B typo-krn       | F sort          |
| B typo-lan       | F copy          |
| B typo-lbx       | F crossproduct  |
| B typo-man       | F delete        |
| B typo-mar       | F determinant   |
| B typo-ovl       | F div           |
| B typo-pag       | F equal         |
| B typo-par       | F exchange      |
| B typo-pnc       | F floor         |
| B typo-prc       | F get           |
| B typo-rep       | F getdimensions |
| B typo-rub       | F getepsilon    |
| B typo-spa       | F getmemory     |
| B typo-stc       | F getmesh       |
| B typo-sus       | F getmeshtypes  |
| B typo-syn       | F getpointtypes |
| B typo-tal       | F getrc         |
| B typo-wrp       | F getrow        |
| B unic-ini       | F getsize       |
| T utf8           | F getstacking   |
| F char           | F gettype       |
| S charpattern    | T helpers       |
| F codepoint      | F points        |
| F codes          | F transform     |
| F len            | F xrotation     |
| F offset         | F xyzscale      |
| T vector         | F yrotation     |
| F add            | F zrotation     |
| F append         | F homogenize    |
| F average        | F identity      |
| F ceiling        | F inner         |
| F checkoverlap   | F insert        |
| F collectoverlap | F inverse       |
| F concat         | F ismesh        |
| T contour        | F ispoints      |
| F average        | F issingular    |
| F bounds         | F isvector      |
| F checkoverlap   | F iszero        |
| F collectoverlap | T mesh          |
| F getarea        | F copy          |
| F getmesh        | F get           |
| F makemesh       | F getaverage    |
| F overlapping    | F getdimensions |

|                 |                 |
|-----------------|-----------------|
| F gettypevalues | F norm          |
| F ismesh        | F normalize     |
| F new           | F perp          |
| F set           | F perpendicular |
| F setaverage    | F set           |
| F tostring      | F sub           |
| F totable       | F tostring      |
| F type          | F totable       |
| F meshname      | F type          |
| F mul           | T points        |
| F negate        | F copy          |
| F new           | F get           |
| F normalize     | F getbounds     |
| T normals       | F getindex      |
| F flip          | F getnormal     |
| F get           | F getpool       |
| F isnormals     | F gettriangle   |
| F new           | F getxy         |
| F set           | F getxytable    |
| F tostring      | F getxyz        |
| F totable       | F getxyztable   |
| F type          | F grow          |
| F onecolumn     | F index         |
| F onerow        | F ispoints      |
| T point         | F new           |
| F add           | F prune         |
| F copy          | F set           |
| F cross         | F setindex      |
| F crossproduct  | F setnormal     |
| F dist          | F setxy         |
| F distance      | F setxyz        |
| F div           | F size          |
| F dot           | F tostring      |
| F dotproduct    | F totable       |
| F equal         | F type          |
| F get           | F product       |
| F ispoint       | F remove        |
| F iszero        | F replace       |
| F len           | F round         |
| F length        | F rowechelon    |
| F mul           | F set           |
| F neg           | F setepsilon    |
| F negate        | F setnext       |
| F new           | F setrc         |



|               |                |
|---------------|----------------|
| F setstacking | F proj         |
| F slice       | F real         |
| F sort        | F sin          |
| F sub         | F sinh         |
| F swap        | F sqrt         |
| F tostring    | F tan          |
| F totable     | F tanh         |
| F transpose   | F topair       |
| F triangulate | F tostring     |
| F truncate    | F totable      |
| F type        | F voigt        |
| T xcomplex    | F voigt_hwhm   |
| F I           | T xdecimal     |
| F __add       | F __add        |
| F __div       | F __band       |
| F __eq        | F __bor        |
| F __mul       | F __bxor       |
| S __name      | F __div        |
| F __pow       | F __eq         |
| F __sub       | F __idiv       |
| F __tostring  | F __le         |
| F __unm       | F __lt         |
| F abs         | F __mod        |
| F acos        | F __mul        |
| F acosh       | S __name       |
| F arg         | F __pow        |
| F asin        | F __shl        |
| F asinh       | F __shr        |
| F atan        | F __sub        |
| F atanh       | F __tostring   |
| F conj        | F __unm        |
| F cos         | F abs          |
| F cosh        | F band         |
| F dawson      | F bor          |
| F erf         | F bxor         |
| F erfc        | F conj         |
| F erfcx       | F copy         |
| F erfi        | F exp          |
| F exp         | F getprecision |
| F i           | F ln           |
| F imag        | F log          |
| F log         | F max          |
| F new         | F min          |
| F pow         | F minus        |

|                |                |
|----------------|----------------|
| F new          | F hull         |
| F plus         | F intersect    |
| F pow          | F less         |
| F rotate       | F lessequal    |
| F setprecision | F log          |
| F shift        | F log10        |
| F sqrt         | F log1p        |
| F toengstring  | F log2         |
| F tonumber     | F mid          |
| F tostring     | F mod          |
| F trim         | F new          |
| T xinterval    | F neweps       |
| F __add        | F pow          |
| F __div        | F seteq        |
| F __eq         | F setge        |
| F __le         | F setgt        |
| F __lt         | F settle       |
| F __mod        | F settlt       |
| F __mul        | F sin          |
| S __name       | F sinh         |
| F __pow        | F sqr          |
| F __sub        | F sqrt         |
| F __tostring   | F tan          |
| F __unm        | F tanh         |
| F abs          | F topair       |
| F acos         | F tostring     |
| F acosh        | F totable      |
| F asin         | F usedendian   |
| F asinh        | F within       |
| F asstring     | T xmath        |
| F atan         | F acos         |
| F atanh        | F acosh        |
| F cos          | F asin         |
| F cosh         | F asinh        |
| F diameter     | F atan         |
| F equal        | F atan2        |
| F erf          | F atanh        |
| F erfc         | F cbrt         |
| F exp          | F ceil         |
| F exp10        | F copysign     |
| F exp2         | F cos          |
| F expm1        | F cosh         |
| F greater      | F crossproduct |
| F greaterequal | F deg          |

|              |                  |
|--------------|------------------|
| F dotproduct | F sinh           |
| F erf        | F sqrt           |
| F erfc       | F tan            |
| F exp        | F tanh           |
| F exp2       | F tgamma         |
| F expm1      | F trunc          |
| F fabs       | F y0             |
| F fdim       | F y1             |
| F floor      | F yn             |
| F fma        | T xzip           |
| F fmax       | F adler32        |
| F fmin       | F compress       |
| F fmod       | F compresssize   |
| F frexp      | F crc32          |
| F gamma      | F decompress     |
| F hypot      | F decompresssize |
| N inf        | T zbuffer        |
| F isfinite   | F bounds         |
| F isinf      | F composite      |
| F isnan      | F crop           |
| F isnormal   | F edgepoint      |
| F j0         | F flatten        |
| F j1         | F get            |
| F jn         | F getcolor       |
| F ldexp      | F getconstants   |
| F lgamma     | F getdepth       |
| F log        | F getnormal      |
| F log10      | F getnormalmodes |
| F log1p      | F getopacity     |
| F log2       | F getsize        |
| F logb       | F implicit       |
| F modf       | F iszbuffer      |
| N nan        | F meshmesh       |
| F nearbyint  | F new            |
| F nextafter  | F points         |
| F normalize  | F process        |
| N pi         | F project        |
| F pow        | F resolve        |
| F rad        | F set            |
| F remainder  | F setcolor       |
| F remquo     | F setdepth       |
| F round      | F setnormal      |
| F scalbn     | F setopacity     |
| F sin        | F setup          |

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|------------------|------------------------|
| F smoothen       | F noto                 |
| F stipple_0      | F prepareflush         |
| F stipple_1      | F presetsheets         |
| F stipple_2      | F presetspreadsheets   |
| F stipple_3      | F put                  |
| F stipple_4      | F putdirect            |
| F stipple_setup  | F registerarea         |
| F tobytes        | F registersheet        |
| F tostring       | F registerspreadsheets |
| F totable        | F reset                |
| F transform      | F resetdirect          |
| F trianglebounds | F reshape              |
| F triangled      | F setarea              |
| F triangles      | F setextra             |
| F type           | F sethsize             |
| T loaders        | F sethspan             |
| F loadlib        | F setlines             |
| F luapaths       | F setmvl               |
| S path           | F setpresent           |
| F poplibpath     | F setproperties        |
| F popluapath     | F setstart             |
| T preload        | F shape                |
| F pushlibpath    | F showshape            |
| F pushluapath    | F state                |
| F searchpath     | F status               |
| T pagebuilders   | F submvl               |
| T columnsets     | F submvlbalance        |
| F add            | T injections           |
| F addmvl         | F flushafter           |
| F block          | F flushbefore          |
| F blockrows      | F save                 |
| F check          | T mixedcolumns         |
| F checkmvl       | F cleanup              |
| F clean          | F finalize             |
| T data           | F getlist              |
| F define         | F getrest              |
| F definesub      | F getsplit             |
| F delayedsheet   | F setsplit             |
| F finishflush    | F pairs                |
| F flushareas     | T parametersets        |
| F flushcolumn    | F pcall                |
| F goto           | T pdf                  |
| F gotoslot       | F immediateobj         |
| F limit          | T physics              |

|                    |                        |
|--------------------|------------------------|
| T units            | D collected            |
| F registershortcut | T components           |
| F registerunit     | F author               |
| D tables           | F concatstate          |
| F print            | F converttxml          |
| T publications     | T currentspecification |
| F addtexentry      | S author               |
| F analyze          | T categories           |
| F anywhere         | T article              |
| D authorcache      | S category             |
| D authorcollapsers | T extra                |
| T authorhashers    | T fields               |
| F invertedshort    | S author               |
| F normal           | S editor               |
| F normalinverted   | S journal              |
| F normalshort      | S keywords             |
| F sequential       | S note                 |
| F sequentialshort  | S number               |
| F short            | S pages                |
| D authormap        | S title                |
| F authortoregister | S volume               |
| T btx              | S year                 |
| T casters          | T optional             |
| F author           | T required             |
| F keyword          | T sets                 |
| F pagenumber       | T author               |
| F range            | T virtual              |
| F checkeddataset   | T book                 |
| T citevariants     | S category             |
| F author           | T extra                |
| F authornum        | T fields               |
| F authorityyear    | S address              |
| F authorityyears   | S author               |
| F category         | S edition              |
| F default          | S editor               |
| F entry            | S keywords             |
| F index            | S note                 |
| F keywords         | S number               |
| F num              | S publisher            |
| F page             | S subtitle             |
| F short            | S title                |
| F tag              | S volume               |
| F textnum          | S year                 |
| F year             | T optional             |

|                |                   |
|----------------|-------------------|
| T required     | S page            |
| T sets         | S pages           |
| T author       | S volume          |
| T editionset   | S version         |
| T virtual      | D virtual         |
| T demo-a       | T virtualfields   |
| S category     | B authornum       |
| T extra        | B authoryear      |
| T fields       | B authoryears     |
| S author       | B num             |
| S institution  | B suffix          |
| S organization | T datasets        |
| S subtitle     | T detailed        |
| S title        | F endnotes_to_btx |
| S year         | T enhancer        |
| T optional     | S arguments       |
| T required     | T askip           |
| T sets         | T attr            |
| T author       | B dirty           |
| T virtual      | T gskip           |
| T demo-b       | T kind            |
| S category     | T list            |
| T extra        | T system          |
| T fields       | T order           |
| S author       | N steps           |
| S institution  | T enhancers       |
| S organization | F details         |
| S subtitle     | F order           |
| S title        | F suffixes        |
| S year         | F finder          |
| T optional     | F firstofrange    |
| T required     | F found           |
| T sets         | F get             |
| T authors      | F getcasted       |
| T virtual      | F getdetail       |
| S comment      | F getdirect       |
| S copyright    | F getfaster       |
| T fields       | F getfield        |
| S name         | F getfuzzy        |
| T types        | F getindex        |
| S author       | D inspectors      |
| S editor       | T journals        |
| S keywords     | F abbreviated     |
| S number       | F add             |

|                      |                       |
|----------------------|-----------------------|
| F expanded           | F buffer              |
| F load               | F endnote             |
| F save               | F json                |
| F lastofrange        | F lua                 |
| T lists              | F refman              |
| F collectentries     | F registercleaner     |
| F combiinlist        | F xml                 |
| F fetchentries       | F loadreplacementfile |
| F filterall          | T meanings            |
| F flushcombi         | F new                 |
| F flushentry         | F okay                |
| F flushtag           | F oneorrange          |
| T methods            | F parenttag           |
| F dataset            | F permitted           |
| F force              | F prerollcmdstring    |
| F global             | F refman_to_btx       |
| F label              | F resolvedname        |
| F local              | F save                |
| F nofregistered      | T savers              |
| F prepareentries     | F bib                 |
| F register           | F lua                 |
| F sameasprevious     | F xml                 |
| T sorters            | F search              |
| F cite               | F setcategory         |
| F dataset            | F setspecification    |
| F default            | F singularorplural    |
| F index              | F sortedentries       |
| F list               | T sortmethods         |
| F reference          | T authortitle         |
| F short              | T sequence            |
| F used               | T authoryear          |
| F uservariable       | T sequence            |
| T listvariants       | T dataset             |
| F default            | T sequence            |
| F index              | T key                 |
| F num                | T sequence            |
| F short              | T specifications      |
| F tag                | F splitauthor         |
| F load               | F splitoneauthor      |
| F loadbibdata        | T statistics          |
| F loaddefinitionfile | N nofbytes            |
| T loaders            | N nofdatasets         |
| F bbl                | N nofdefinitions      |
| F bib                | N nofshortcuts        |

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|---------------------------|--------------------------------|
| T tables                  | F translate                    |
| D implicits               | F require                      |
| D origins                 | T resolvers                    |
| D privates                | F allprefixes                  |
| D sets                    | F appendhash                   |
| D specials                | F automount                    |
| D types                   | F booleanvariable              |
| D virtuals                | S cacheversion                 |
| F tags                    | F checkedvariable              |
| T tracers                 | F cleanedpathlist              |
| F showdatasetauthors      | D cleaners                     |
| F showdatasetcompleteness | F cleanpath                    |
| F showdatasetfields       | F cleanpathlist                |
| F showentry               | F cleanupbinfile               |
| F showfields              | D concatenators                |
| F showhashedauthors       | S configbanner                 |
| F showtables              | F configurationfiles           |
| F showvariants            | D dangerous                    |
| T typesetters             | F datastate                    |
| F author                  | T decompressors                |
| F string                  | F register                     |
| F usedataset              | F unregister                   |
| F usedentries             | T defaultsuffixes              |
| T writers                 | F dirstatus                    |
| F author                  | F dowithfilesandreport         |
| F keyword                 | F dowithfilesintree            |
| F pagenumber              | F dowithpath                   |
| F range                   | F dowithvariable               |
| F rawequal                | T executers                    |
| F rawget                  | F expandbraces                 |
| F rawlen                  | F expandedpathfromlist         |
| F rawset                  | F expandedpathlist             |
| T regimes                 | F expandedpathlistfromvariable |
| D backmapping             | F expandpath                   |
| F cleanup                 | F expandpathfromvariable       |
| F disable                 | F expansion                    |
| F enable                  | F extendtexmfvariable          |
| F fromregime              | F filtered_from_content        |
| F list                    | F find_file                    |
| D mapping                 | F find_files                   |
| F pop                     | F findbinfile                  |
| F process                 | F findctxfile                  |
| F push                    | D finders                      |
| F toregime                | F findfile                     |



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|---------------------------|---------------------|
| F findfiles               | T libraries         |
| F findgivenfile           | F optionalloaded    |
| F findgivenfiles          | F validoptional     |
| F findlib                 | F load              |
| F findpath                | F loadbinfile       |
| F findpk                  | D loaders           |
| F findtexfile             | F loadlualib        |
| F findwildcardfile        | F loadtexfile       |
| F findwildcardfiles       | F loadtime          |
| F formatofsuffix          | F locateformat      |
| F formatofvariable        | D locators          |
| D formats                 | S luacnffallback    |
| F foundintree             | S luacnfname        |
| F foundintrees            | S luacnfspec        |
| D generators              | S luacnfstate       |
| F get_from_content        | T macros            |
| F getdetails              | F convertfile       |
| F getenv                  | F decodecomment     |
| F getextrapaths           | F enablecomment     |
| F getfilehashes           | F encodecomment     |
| F gethashes               | F preprocessed      |
| F getownpath              | F processmk         |
| F getreadfilename         | F processmklx       |
| D hashers                 | F processmkvi       |
| S homedir                 | F version           |
| F ignorelibrary           | N maxreadlevel      |
| F initialized             | F methodhandler     |
| F installinputfilehandler | F missinglibrary    |
| F installinputlinehandler | F openbinfile       |
| F jobpath                 | D openers           |
| T jobs                    | F openfile          |
| F collectstructure        | F opentexfile       |
| F currentcomponent        | S ownpath           |
| F currentenvironment      | F popextrapath      |
| F currentproduct          | F popinputname      |
| F currentproject          | F poppath           |
| F currentstructure        | D prefixes          |
| F justacomponent          | F prependhash       |
| F namelist                | F pushextrapath     |
| D processors              | F pushinputname     |
| F productcomponent        | F pushpath          |
| F usefile                 | F registerextrapath |
| F joinpath                | F registerfile      |
| F knownvariables          | F registerfilehash  |

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|--------------------------|----------------------------|
| F registerintrees        | F updaterelations          |
| F registermethod         | F updatescript             |
| F registerzipfile        | F uselibrary               |
| D relations              | D usertypes                |
| F renew                  | F usezipfile               |
| F renewcache             | F validatedpath            |
| F repath                 | F variable                 |
| F reset                  | F variableofformat         |
| F resetextrapaths        | F variableofformatorsuffix |
| F resetresolve           | F wildcardpattern          |
| F resolve                | T sandbox                  |
| D savers                 | F blockrequire             |
| F scandata               | F disablelibraries         |
| F scanfiles              | F disablerunners           |
| T schemes                | F enable                   |
| T cleaners               | F filehandlerone           |
| F md5                    | F filehandlertwo           |
| F none                   | F finalizer                |
| F strip                  | F getrunner                |
| F cleanname              | F initializer              |
| F fetchstring            | F iohandler                |
| F fetchtable             | F logcalls                 |
| F install                | F logfiles                 |
| F setdynamic             | F original                 |
| F setenv                 | F overload                 |
| F showdetails            | F possiblepath             |
| F showpath               | F redefine                 |
| F simplescanfiles        | F register                 |
| F splitconfigurationpath | F registerbinary           |
| F splitexpansions        | F registerlibrary          |
| F splitmethod            | F registerroot             |
| F splitpath              | F registerrunner           |
| F stackpath              | F report                   |
| F starttiming            | F setfilenamelogger        |
| F stoptiming             | F setlogger                |
| D suffixes               | F setreporter              |
| D suffixmap              | F settrace                 |
| F suffixofformat         | F validfilename            |
| F texdatablob            | T scripts                  |
| F toppath                | T autofontfeature          |
| D tracers                | F disable                  |
| F unexpandedpath         | F enable                   |
| F unexpandedpathlist     | F handler                  |
| F unresolve              | D colors                   |

|                                     |                     |
|-------------------------------------|---------------------|
| F decomposehangul                   | T comparers         |
| T defaults                          | F basic             |
| N inter_char_half_shrink_factor     | T constants         |
| N inter_char_half_stretch_factor    | S defaultdigits     |
| N inter_char_hangul_penalty         | S defaultlanguage   |
| N inter_char_quarter_shrink_factor  | S defaultmethod     |
| N inter_char_quarter_stretch_factor | N digitmaximum      |
| N inter_char_shrink_factor          | N digitoffset       |
| N inter_char_stretch_factor         | N ignoredoffset     |
| N inter_word_stretch_factor         | N replacementoffset |
| F getdata                           | T validmethods      |
| F getinjector                       | B ch                |
| F getsplitter                       | B mc                |
| F getstatus                         | B mm                |
| D handlers                          | B pc                |
| D helpers                           | B pm                |
| D injectors                         | B uc                |
| T inserters                         | B zc                |
| F nobreakspace_after                | B zm                |
| F nobreakspace_before               | D definitions       |
| F space_after                       | F firstofsplit      |
| F space_before                      | F packch            |
| F zerowidthspace_after              | F packuc            |
| F zerowidthspace_before             | F replacementlist   |
| F installdataset                    | F setlanguage       |
| F installmethod                     | F sort              |
| D numbertodataset                   | T splitters         |
| D numbertohandler                   | F utf               |
| F reset                             | F strip             |
| F set                               | D tracers           |
| F setstatus                         | F update            |
| D splitters                         | T statistics        |
| F select                            | F benchmarktimer    |
| T set                               | F checkfmtstatus    |
| F contains                          | F currenttime       |
| F create                            | F elapsed           |
| F tolist                            | F elapsedindeed     |
| F tonumber                          | F elapsedseconds    |
| F totable                           | F elapsedtime       |
| F setinspector                      | B enable            |
| F setmathparameters                 | T feedback          |
| F setmetatable                      | F info              |
| T sorters                           | F processtates      |
| F basicsorter                       | F register          |

|                   |                  |
|-------------------|------------------|
| F reportstates    | T blocks         |
| F setstate        | D collected      |
| F formatruntime   | F define         |
| F hastiming       | F empty          |
| F memused         | F print          |
| F register        | F save           |
| F resettiming     | F select         |
| F runtime         | F setstate       |
| F savefmtstatus   | D states         |
| F show            | D tobesaved      |
| F showmemory      | F uservariable   |
| F showusage       | T bookmarks      |
| F starttiming     | T extras         |
| F stoptiming      | F get            |
| N threshold       | F mergesections  |
| F timed           | F register       |
| F tracefunction   | F reset          |
| T storage         | F tosections     |
| F allocate        | F finalize       |
| T data            | F flatten        |
| F mark            | F installhandler |
| F marked          | F merge          |
| N max             | S method         |
| N min             | F overload       |
| N nofmodules      | F place          |
| N noftables       | F register       |
| F register        | F setopened      |
| D shared          | F setup          |
| D storedfunctions | T counters       |
| B strip           | F add            |
| N tofmodules      | F analyze        |
| N toftables       | F check          |
| T streams         | D collected      |
| F collect         | F compact        |
| F disable         | F converted      |
| F enable          | F currentvalue   |
| F flush           | D data           |
| F initialize      | F define         |
| F push            | F first          |
| F start           | F get            |
| F stop            | F isset          |
| F synchronize     | F last           |
| T structures      | F next           |
| F atlocation      | F prev           |

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|------------------------|------------------|
| F previous             | F detach         |
| F raw                  | F getdata        |
| F record               | F getlocation    |
| F reset                | F getname        |
| F restart              | F getnumber      |
| F restore              | F nofdetached    |
| F save                 | F setlocation    |
| F set                  | F setup          |
| F setlevel             | D stored         |
| F setoffset            | T itemgroups     |
| F setown               | D collected      |
| F setstate             | D tobesaved      |
| F setvalue             | T lists          |
| T specials             | F addto          |
| F userpage             | F analyze        |
| F subs                 | F bookmark       |
| D tobesaved            | D cached         |
| F value                | D collected      |
| F way                  | F discard        |
| F currentsectionnumber | F enhance        |
| T descriptions         | T enhancers      |
| T documents            | F external       |
| D data                 | F filter         |
| F initialize           | F formulanumber  |
| F preset               | F getresult      |
| F reset                | F groupindex     |
| T floats               | F hasnumberdata  |
| T formulas             | F haspagedata    |
| T helpers              | F hastitledata   |
| F analyze              | F integrate      |
| F formulanumber        | F iscached       |
| F merged               | D kinds          |
| F page                 | F label          |
| F pageofinternal       | F location       |
| F prefix               | D names          |
| F prefixlastpage       | F number         |
| F prefixpage           | D ordered        |
| F simplify             | F page           |
| F title                | F popnesting     |
| F touserdata           | F prefix         |
| T inserts              | F prefixednumber |
| F attach               | F prefixedpage   |
| D data                 | F process        |
| F define               | D pushed         |

|                           |                     |
|---------------------------|---------------------|
| F pushnesting             | D states            |
| F rawnumber               | F store             |
| F rawrealpage             | T pages             |
| F rawuserpage             | F allmarked         |
| F realpage                | F analyze           |
| F reorder                 | D collected         |
| F reordered               | F fraction          |
| T result                  | F getlabels         |
| F savedlistnumber         | F in_body           |
| F savedlistprefixednumber | F is_odd            |
| F savedlisttitle          | F mark              |
| D sectionblocks           | F marked            |
| F size                    | F markedlist        |
| F title                   | N nopages           |
| D tobesaved               | F number            |
| F use                     | F on_right          |
| F userdata                | F save              |
| F userpage                | D tobesaved         |
| F uservalue               | F toranges          |
| T marks                   | T references        |
| F number                  | F analyze           |
| F title                   | F checked           |
| T notes                   | F checkedfile       |
| F checkpagechange         | F checkedfileorurl  |
| F define                  | F checkedprogram    |
| F flushrange              | F checkedrealpage   |
| F get                     | F checkedurl        |
| F getdeltapage            | D collected         |
| F getinternal             | U componentsplitter |
| F getn                    | F currentmetadata   |
| F getstate                | F currentorder      |
| F getstorecount           | D currentset        |
| F getstoreflushed         | F define            |
| F hascontent              | D defined           |
| F internal                | D derived           |
| F internalid              | F enhance           |
| F listindex               | D executers         |
| D numbers                 | F export            |
| F ordered                 | T files             |
| F postpone                | T data              |
| F resetpostponed          | F define            |
| F restore                 | F found             |
| F save                    | F get               |
| F setstate                | F filter            |

|                                  |                                    |
|----------------------------------|------------------------------------|
| F filterdefault                  | D runners                          |
| D filters                        | F set                              |
| D flaginternals                  | F setandgetattribute               |
| F forceinnermode                 | F setinternalreference             |
| F from                           | F setlinkmethod                    |
| D functions                      | F setnextinternal                  |
| F get                            | F setnextorder                     |
| F getcurrentlabel                | F setreferencerealpage             |
| F getcurrentmetadata             | F setstructure                     |
| F getcurrentprefixspec           | D specials                         |
| F getinnermethod                 | F splitcomponent                   |
| F getinternalcachedlistreference | F splitprefix                      |
| F getinternallistreference       | F splitreference                   |
| F getinternalreference           | F synchronizepage                  |
| F getoutermethod                 | T testrunners                      |
| D handlers                       | F inner                            |
| F identify                       | F special                          |
| F import                         | F special operation                |
| F inject                         | F special operation with arguments |
| F injectcurrentset               | T testspecials                     |
| S innermethod                    | F deltapage                        |
| F integrate                      | F i                                |
| D internals                      | F internal                         |
| F load                           | F page                             |
| F loadpresets                    | F realpage                         |
| F mark                           | F section                          |
| S outermethod                    | F userpage                         |
| F pagefromhash                   | D tobereferred                     |
| D pages                          | D tobesaved                        |
| U prefixsplitter                 | D tracedpages                      |
| D productdata                    | T urls                             |
| T programs                       | T data                             |
| T data                           | F define                           |
| F define                         | F found                            |
| F get                            | F get                              |
| F realpageofpage                 | F untex                            |
| F referenced                     | D usedinternals                    |
| U referencesplitter              | D usedviews                        |
| D referred                       | F valid                            |
| F referredpage                   | T registers                        |
| F registerfinalizer              | F analyze                          |
| F registerinitializer            | D collected                        |
| F registerpage                   | F compare                          |
| F reset                          | F define                           |

|                       |                    |
|-----------------------|--------------------|
| F enhance             | F register         |
| F extend              | D registered       |
| F filter              | F reportstructure  |
| F filtercollected     | F save             |
| F finalize            | T sectionblockdata |
| F findinternal        | T bodypart         |
| F flush               | S bookmark         |
| F get                 | F set              |
| F integrate           | F setblock         |
| F prepare             | F setchecker       |
| F process             | F setentry         |
| F setmethod           | F setinitialblock  |
| F sort                | F setlevel         |
| F store               | F setnumber        |
| D tobesaved           | F showstructure    |
| F tweak               | F structuredata    |
| F unique              | F title            |
| F use                 | D tobesaved        |
| F userdata            | F typesetnumber    |
| T sections            | F userdata         |
| F autodepth           | B verbose          |
| F cct                 | T sets             |
| D collected           | F define           |
| F current             | F get              |
| F currentblock        | F getall           |
| F currentid           | D setlist          |
| F currentlevel        | T specials         |
| F currentsectionindex | D collected        |
| F depthnumber         | F retrieve         |
| F finddata            | F store            |
| F findnumber          | D tobesaved        |
| F fullnumber          | T synonyms         |
| F getcurrentlevel     | F analyzed         |
| F getlevel            | D collected        |
| F getnumber           | F compare          |
| F internalreference   | F define           |
| D levelmap            | F enhance          |
| D levels              | F filter           |
| F matchingtilldepth   | F finalize         |
| F numberatdepth       | F firstuserpage    |
| F numbers             | F flush            |
| F popblock            | F isshown          |
| F prefixedconverted   | F isused           |
| F pushblock           | F meaning          |



|                     |                       |
|---------------------|-----------------------|
| F pages             | F mover               |
| F prepare           | F mpgraphic           |
| F process           | F msub                |
| F register          | F msupsub             |
| F registershown     | F msup                |
| F registerused      | F munder              |
| F reset             | F munderover          |
| F resetall          | F paragraph           |
| F resetlist         | F publication         |
| F resetshown        | F reference           |
| F resetused         | F registerlocation    |
| F sort              | F registerpages       |
| F synonym           | F registerseparator   |
| D tobesaved         | F section             |
| F userpages         | F sorting             |
| T tags              | F subsentencesymbol   |
| F analyze           | F synonym             |
| T backend           | F tablecell           |
| T checks            | F tabulate            |
| F marginanchor      | F tabulatecell        |
| F margintext        | T finalizers          |
| F math              | F descriptions        |
| F mrow              | T fixes               |
| F mtable            | F linenumber          |
| T destinationhash   | F fontfaced           |
| T extras            | T listdata            |
| F adddestination    | T referencehash       |
| F addimplicit       | F setattribute        |
| F addinternal       | F tofontface          |
| F addreference      | T usedimages          |
| F combination       | T image               |
| F combinationpair   | T mpgraphic           |
| F delimitedsymbol   | T usedstyles          |
| F description       | T construct           |
| F descriptionsymbol | T highlight           |
| F document          | F checknesting        |
| F float             | F current             |
| F formulacontent    | F elementtag          |
| F image             | F enabled             |
| F item              | F exportbox           |
| F itemgroup         | F finishexport        |
| F link              | F getdescriptionindex |
| F listitem          | F getextradata        |
| F mmultiscripts     | F getid               |

|                              |                           |
|------------------------------|---------------------------|
| F getitemgroup               | F setcodepoint            |
| F getmetadata                | F setcombination          |
| F getoption                  | F setcombinationpair      |
| F getsorting                 | F setconstruct            |
| F getsynonym                 | F setdescriptionindex     |
| F gettablecell               | F setfigure               |
| F gettabulatecell            | F setfloat                |
| F gettagreferencedescription | F setformulacontent       |
| F handler                    | F sethighlight            |
| T ignored                    | F setitem                 |
| F initializeexport           | F setitemgroup            |
| D labels                     | F setmpgraphic            |
| F last                       | F setnotation             |
| F lastinchain                | F setnotationsymbol       |
| F localexport                | F setparagraph            |
| F locatedtag                 | F setproperty             |
| T modes                      | F setpublication          |
| S BlockQuote                 | F setregister             |
| S FENote                     | F setsorting              |
| S H1                         | F setsynonym              |
| S H2                         | F settablecell            |
| S H3                         | F settabulatecell         |
| S H4                         | F settagdelimitedsymbol   |
| S H5                         | F settagsubsentencesymbol |
| S H6                         | F setupexport             |
| S H7                         | F setuserproperties       |
| S H8                         | D specifications          |
| S H9                         | F start                   |
| S Lbl                        | F stop                    |
| S NonStruct                  | F strip                   |
| S P                          | D taglist                 |
| S Part                       | T unsupported             |
| S Sect                       | B Annot                   |
| D overloads                  | B Art                     |
| T patterns                   | B BibEntry                |
| U splitter                   | B Caption                 |
| F pop                        | B Document                |
| D properties                 | B DocumentFragment        |
| F push                       | B Em                      |
| F registerextradata          | B Form                    |
| F registermetadata           | B H                       |
| F restart                    | B Index                   |
| F setaspect                  | B Link                    |
| F setattributehash           | B Note                    |

|                    |                     |
|--------------------|---------------------|
| B Quote            | N size              |
| B Reference        | T filesize          |
| B Strong           | N size              |
| B TOC              | N step              |
| B TOCI             | S firstline         |
| B Title            | T fontsize          |
| T usewithcare      | N size              |
| T where            | N step              |
| S Art              | S formatname        |
| S Artifact         | N functionsize      |
| S Aside            | T halferrorlinesize |
| S BibEntry         | N size              |
| S BlockQuote       | T hashsize          |
| S Code             | N size              |
| S DocumentFragment | N step              |
| S Em               | F init              |
| S FENote           | T inputsize         |
| S H1               | N size              |
| S H2               | N step              |
| S H3               | T insertsize        |
| S H4               | N size              |
| S H5               | N step              |
| S H6               | B kpse_init         |
| S H7               | T languagesize      |
| S H8               | N size              |
| S H9               | N step              |
| S Index            | T marksize          |
| S Note             | N size              |
| S Private          | N step              |
| S Quote            | T nestsize          |
| S Reference        | N size              |
| S Strong           | N step              |
| S Sub              | T nodesize          |
| S TOC              | N size              |
| S TOCI             | N step              |
| S Title            | T parametersize     |
| D texconditionals  | N size              |
| T texconfig        | N step              |
| T buffersize       | T poolsize          |
| N size             | N size              |
| N step             | N step              |
| T errorlinesize    | N propertiessize    |
| N size             | T savesize          |
| T expandsize       | N size              |

|                  |                  |
|------------------|------------------|
| N step           | N mathshift      |
| S shell_escape   | N open           |
| T stringsize     | N other          |
| N size           | N parameter      |
| N step           | N space          |
| T tokensize      | N subscript      |
| N size           | N superscript    |
| N step           | N whitespace     |
| D texconstants   | D cache          |
| D texifs         | D commands       |
| D texisdefined   | F compile        |
| D texmodes       | F constant       |
| D texsystemmodes | T converters     |
| T thirddata      | S toboolean      |
| F toboolean      | S todimen        |
| F todecimal      | S toglue         |
| T tokens         | S tonumber       |
| T accessors      | S tostring       |
| F active         | F create         |
| F cmd            | F defined        |
| F cmdchrcls      | F flags          |
| F cmdname        | F getinteger     |
| F command        | T getters        |
| F csname         | F box            |
| F expandable     | F count          |
| F frozen         | F cstoken        |
| F index          | F dimen          |
| F name           | F glue           |
| F protected      | F macro          |
| F range          | F meaning        |
| F astable        | F skip           |
| T bits           | F token          |
| N active         | F gobble         |
| N alignment      | F isactivestring |
| N begingroup     | F isdefined      |
| N character      | F istoken        |
| N close          | F new            |
| N comment        | F pickup         |
| N endgroup       | F popmacro       |
| N endofline      | T presets        |
| N escape         | T 1 argument     |
| N ignore         | T 1 integer      |
| N invalid        | T 1 optional     |
| N letter         | T 1 string       |

|                     |                   |
|---------------------|-------------------|
| T 2 arguments       | F hbox            |
| T 2 integers        | F integer         |
| T 2 optionals       | F integerargument |
| T 2 strings         | F ischar          |
| T 3 arguments       | F key             |
| T 3 integers        | F keyword         |
| T 3 optionals       | F keywordcs       |
| T 3 strings         | F letters         |
| T 4 arguments       | F list            |
| T 4 integers        | F lua             |
| T 4 optionals       | F luacardinal     |
| T 4 strings         | F luainteger      |
| T 5 strings         | F luanumber       |
| T 6 strings         | F lxmlid          |
| T 7 strings         | F next            |
| T 8 strings         | F nextchar        |
| F pushmacro         | F nextexpanded    |
| T scanners          | F number          |
| F argument          | F optional        |
| F argumentasis      | F peek            |
| F array             | F peekchar        |
| F boolean           | F peekexpanded    |
| F box               | F posit           |
| F bracketed         | F real            |
| F bracketedasis     | F scanclose       |
| F cardinal          | F scanopen        |
| F char              | F skip            |
| F cmdchr            | F skipexpanded    |
| F cmdchrexplanded   | F string          |
| F code              | F table           |
| F conditional       | F token           |
| F count             | F tokencode       |
| F csname            | F tokenlist       |
| F csnameunchecked   | F tokens          |
| F delimited         | F tokenstring     |
| F detokened         | F toks            |
| F dimen             | F value           |
| F dimension         | F vbox            |
| F dimensionargument | F verbatim        |
| F float             | F vtop            |
| F glue              | F whd             |
| F gluespec          | F word            |
| F gluevalues        | F setinteger      |
| F hash              | T setters         |

|                         |                   |
|-------------------------|-------------------|
| F box                   | F scannumber      |
| F char                  | F scanopen        |
| F count                 | F scanoptional    |
| F dimen                 | F scanposit       |
| F glue                  | F scanstring      |
| F lua                   | F scantable       |
| F macro                 | F scantokenlist   |
| F skip                  | F scantokenstring |
| T shortcuts             | F scantoks        |
| N close                 | F scanvbox        |
| F inspect               | F scanverbatim    |
| N open                  | F scanvtop        |
| F report                | F scanword        |
| F scanargument          | F toboolean       |
| F scanargumentasis      | F todimen         |
| F scanarray             | F tonumber        |
| F scanboolean           | F tostring        |
| F scanbox               | D values          |
| F scanbracketed         | F tonumber        |
| F scanbracketedasis     | F tostring        |
| F scancardinal          | T tracers         |
| F scanclose             | F printerror      |
| F scancode              | F printwarning    |
| F scanconditional       | F showlines       |
| F scansname             | T trackers        |
| F scansnameunchecked    | D data            |
| F scandelimited         | F default         |
| F scandetokened         | F disable         |
| F scandimen             | T done            |
| F scandimension         | F dumphashtofile  |
| F scandimensionargument | F enable          |
| F scanfloat             | F list            |
| F scan glue             | S name            |
| F scan glue spec        | F register        |
| F scan glue values      | F report          |
| F scanhash              | F reset           |
| F scanhbox              | F savehash        |
| F scaninteger           | F show            |
| F scanintegerargument   | F value           |
| F scankeyword           | F values          |
| F scankeywordcs         | F type            |
| F scanletters           | T typesetters     |
| F scanlist              | T alignments      |
| F scanlua               | F handler         |

|                       |                      |
|-----------------------|----------------------|
| F set                 | F getglobal          |
| T breakpoints         | F getmethod          |
| F define              | F handler            |
| F handler             | T handlers           |
| D mapping             | F installhandler     |
| T methods             | F set                |
| T numbers             | F setcolor           |
| F set                 | F tomode             |
| F setreplacement      | T firstlines         |
| T cases               | T actions            |
| T actions             | F default            |
| F handler             | F line               |
| F register            | F word               |
| F set                 | F handler            |
| T characteralign      | F set                |
| F handler             | T fontkerns          |
| F resetcharacteralign | F handler            |
| F setcharacteralign   | T initials           |
| T characters          | T actions            |
| F handler             | F default            |
| T methods             | F handler            |
| F replacensbp         | F set                |
| F replacenspaces      | T injectors          |
| T checkers            | F check              |
| F handler             | T list               |
| F reset               | F mark               |
| F set                 | F reset              |
| T cleaners            | F set                |
| F handler             | F show               |
| F set                 | T italics            |
| T compare             | F enablemath         |
| F handler             | F enabletext         |
| F reset               | F handler            |
| F set                 | T kerns              |
| F setpair             | D factors            |
| F setup               | F handler            |
| T corrections         | F keepligature       |
| F handler             | D mapping            |
| T digits              | F set                |
| T actions             | T localboxes         |
| F handler             | F handler            |
| F set                 | T manipulators       |
| T directions          | F apply              |
| F getfences           | F applyspecification |

|                      |                            |
|----------------------|----------------------------|
| T methods            | F define                   |
| F Word               | F handler                  |
| F Words              | D mapping                  |
| F abbreviatedjournal | T numbers                  |
| F expandedjournal    | T frenchpunctuation        |
| F lowercase          | T characters               |
| F stripperperiod     | S name                     |
| F uppercase          | N number                   |
| T patterns           | F reset                    |
| U list               | F set                      |
| U pair               | F setup                    |
| F splitspecification | T stacking                 |
| T margins            | F getindex                 |
| F finalhandler       | F getselection             |
| F globalhandler      | F handler                  |
| F localhandler       | T swapping                 |
| F save               | F handler                  |
| F marksuspects       | T synchronize              |
| T paragraphs         | F handler                  |
| F addanchortoline    | T tighten                  |
| F addtoline          | F handler                  |
| F calculatedelta     | T wrappers                 |
| F checkline          | T xtables                  |
| F moveinline         | F cleanup                  |
| T periodkerns        | F construct                |
| D factors            | F create                   |
| F handler            | F finish_row               |
| D mapping            | F flush                    |
| F set                | F initialize_construct     |
| T processors         | F initialize_reflow_height |
| F apply              | F initialize_reflow_width  |
| F register           | F next_row                 |
| F reset              | F reflow_height            |
| F split              | F reflow_width             |
| F startapply         | F set_construct            |
| F stopapply          | F set_reflow_height        |
| F stripped           | F set_reflow_width         |
| F tostring           | F unpack                   |
| T rubies             | T updaters                 |
| F attach             | F apply                    |
| F check              | F register                 |
| F showinvisibles     | T url                      |
| F showsuspects       | F addscheme                |
| T spacings           | F barepath                 |



|                      |                      |
|----------------------|----------------------|
| F construct          | F utf32_to_utf8_be_t |
| F decode             | F utf32_to_utf8_le   |
| F encode             | F utf32_to_utf8_le_t |
| F escape             | F utf32_to_utf8_t    |
| F filename           | F utf8_to_utf16      |
| F hashed             | F utf8_to_utf16_be   |
| F hasscheme          | F utf8_to_utf16_le   |
| F query              | F utf8_to_utf8_t     |
| F split              | F values             |
| F toquery            | F xstring            |
| F unescape           | T utilities          |
| F unescapeget        | T debugger           |
| T userdata           | F disable            |
| T utf                | F enable             |
| F byte               | F savestats          |
| F bytepairs          | F showstats          |
| F char               | F showtraceback      |
| F characters         | T formatters         |
| F chlren             | F formatcolumns      |
| F count              | F list               |
| F filetype           | F stripzeros         |
| F is_valid           | T ghostscript        |
| F len                | F convert            |
| F length             | F statistics         |
| F lower              | T graphicmagick      |
| F remapper           | F convert            |
| F replacer           | F statistics         |
| F split              | T lua                |
| F splitlines         | B alwaysstripcode    |
| F sub                | F checkmemory        |
| F substituter        | F compile            |
| F tocodes            | F loadedluacode      |
| F toeight            | F loadstring         |
| F toentities         | F loadstripped       |
| F tostring           | N nofstrippedbytes   |
| F totable            | N nofstrippedchunks  |
| F upper              | F registerfinalizer  |
| F ustring            | B stripcode          |
| F utf16_to_utf8_be   | T strippedchunks     |
| F utf16_to_utf8_be_t | F strippedloadstring |
| F utf16_to_utf8_le   | T suffixes           |
| F utf16_to_utf8_le_t | S lmt                |
| F utf16_to_utf8_t    | S lua                |
| F utf32_to_utf8_be   | S luc                |

|                                             |                                 |
|---------------------------------------------|---------------------------------|
| S lui                                       | F mergehashes                   |
| S luj                                       | F options_to_array              |
| S luv                                       | F options_to_hash               |
| S tma                                       | T patterns                      |
| S tmc                                       | U settings_to_array             |
| S tua                                       | U settings_to_hash_a            |
| S tuc                                       | U settings_to_hash_b            |
| T packers                                   | U settings_to_hash_c            |
| F hashed                                    | U settings_to_hash_d            |
| F new                                       | U unittotex                     |
| F pack                                      | F process_stripped_settings     |
| F simplehashed                              | F rfc4180splitter               |
| F strip                                     | F runtime                       |
| F unpack                                    | F settings_to_array             |
| N version                                   | F settings_to_array_obey_fences |
| T parsers                                   | F settings_to_array_with_repeat |
| F add_settings_to_array                     | F settings_to_hash              |
| F arguments_to_table                        | F settings_to_hash_colon_too    |
| F array_to_string                           | F settings_to_hash_strict       |
| F csvsplitter                               | F settings_to_hash_tolerant     |
| F getparameters                             | F settings_to_numbers           |
| F groupedsplitat                            | F settings_to_set               |
| F hash_to_string                            | F simple_hash_to_string         |
| T hashes                                    | F splitmethod                   |
| T settings_to_list                          | F splitthousands                |
| T tablet                                    | F stepper                       |
| T settings_to_set                           | F stripped_settings_to_array    |
| T min,max,fit,broad,auto                    | F totime                        |
| B auto                                      | F unittotex                     |
| B broad                                     | F unittotxml                    |
| B fit                                       | T randomizer                    |
| B max                                       | F get                           |
| B min                                       | F getnumber                     |
| T text,hidden,paper,page,leftpage,rightpage | F getrepeatable                 |
| B hidden                                    | F getrepeatableseed             |
| B leftpage                                  | F getseed                       |
| B page                                      | F newrepeatable                 |
| B paper                                     | F popseed                       |
| B rightpage                                 | F pushseed                      |
| B text                                      | F repeatable                    |
| F iterator                                  | F reuseseed                     |
| F keq_to_hash                               | F setseed                       |
| F listitem                                  | F setseedi                      |
| F make_settings_to_hash_pattern             | T randomizers                   |

|                 |                     |
|-----------------|---------------------|
| F poisson       | T storage           |
| T sequencers    | F allocate          |
| F appendaction  | F checked           |
| F appendgroup   | F mark              |
| F compile       | F marked            |
| F disableaction | F setinitializer    |
| F disablegroup  | F sparse            |
| F enableaction  | T streams           |
| F enablegroup   | F close             |
| D functions     | F getposition       |
| F localize      | F getsize           |
| F new           | F getstring         |
| F nodeprocessor | F look              |
| F prependaction | F open              |
| F prependgroup  | F openstring        |
| F removeaction  | F read2dot14        |
| F replaceaction | F readbyte          |
| F setkind       | F readbytes         |
| F tostring      | F readbytetable     |
| T setters       | F readcardinal      |
| F disable       | F readcardinal1     |
| F enable        | F readcardinal1le   |
| F initialize    | F readcardinal2     |
| F list          | F readcardinal2le   |
| F new           | F readcardinal3     |
| F register      | F readcardinal3le   |
| F report        | F readcardinal4     |
| F reset         | F readcardinal4le   |
| F show          | F readcardinaltable |
| T signals       | F readchar          |
| S configuration | F readcstring       |
| F initialize    | F readdouble        |
| F loadstate     | F readdoublele      |
| F report        | F readfixed2        |
| F savestate     | F readfixed4        |
| F serialclose   | F readfloat         |
| F serialfast    | F readfloatle       |
| F serialopen    | F readfloatle12     |
| S serialprefix  | F readfloatle6      |
| F serialsend    | F readinteger       |
| F serialwrite   | F readinteger1      |
| N version       | F readinteger1le    |
| T stacker       | F readinteger2      |
| F new           | F readinteger2le    |

|                       |                     |
|-----------------------|---------------------|
| F readinteger3        | F definedtable      |
| F readinteger3le      | F definetable       |
| F readinteger4        | F encapsulate       |
| F readinteger4le      | F insertaftervalue  |
| F readintegertable    | F insertbeforevalue |
| F readline            | F migratetable      |
| F readsignedbyte      | F removevalue       |
| F readstring          | F replacevalue      |
| F setposition         | T tar               |
| F size                | T archives          |
| F skip                | F closearchive      |
| F skipbytes           | T files             |
| F skiplong            | F fetch             |
| F skipshort           | F hash              |
| F tocardinal1         | F list              |
| F tocardinal1le       | F pack              |
| F tocardinal2         | F unpack            |
| F tocardinal2le       | F openarchive       |
| F tocardinal3         | T registeredfiles   |
| F tocardinal3le       | T streams           |
| F tocardinal4         | F fetch             |
| F tocardinal4le       | F hash              |
| T strings             | F list              |
| F collapse            | F pack              |
| F collapsecrlf        | F unpack            |
| T formatters          | T strings           |
| F add                 | F fetch             |
| F new                 | F hash              |
| F newcollector        | F list              |
| F newrepeater         | F pack              |
| T striplinepatterns   | F unpack            |
| U collapse            | T templates         |
| U collapse all        | F load              |
| U prune               | F replace           |
| U prune and collapse  | F replacer          |
| U prune and no empty  | F resolve           |
| U prune and to space  | T zipfiles          |
| U retain              | F close             |
| U retain and collapse | F found             |
| U retain and no empty | F hash              |
| F striplines          | F list              |
| F tabtospace          | F open              |
| T tables              | F unzip             |
| F accesstable         | F unzipdir          |

|                         |                      |
|-------------------------|----------------------|
| F zipdir                | B apos               |
| T visualizers           | B gt                 |
| D escapepatterns        | B lt                 |
| F getspecification      | B quot               |
| F getvisualizer         | F cdata              |
| F load                  | F cdatatotext        |
| F makenested            | F checkbom           |
| F makepattern           | F checkedempty       |
| F newgrammar            | F checkerror         |
| F newhandler            | F checknamespace     |
| F parsedxml             | F checkns            |
| F pattern               | F cleansed           |
| F register              | F collect            |
| F registerescapecommand | F collect_elements   |
| F registerescapepline   | F collect_tags       |
| F registerescapepattern | F collect_texts      |
| S signal                | F collected          |
| U signalpattern         | F collecttexts       |
| D specifications        | F compactelement     |
| F visualize             | F content            |
| F visualizebuffer       | F convert            |
| F visualizefile         | F copy               |
| F visualizestring       | F count              |
| F write                 | F cprint             |
| F writeargument         | T css                |
| F writedefault          | F applyselector      |
| F writeemptyline        | F colorspecification |
| F writenewline          | F dimension          |
| F writespace            | F family             |
| F warn                  | F fontspecification  |
| T xml                   | F mappedstylevalue   |
| F addentitiesdoctype    | F padding            |
| F addspecialcode        | F size               |
| F after                 | F style              |
| F all                   | F stylevalue         |
| F applylpath            | F todimension        |
| F applyselector         | F variant            |
| F assign                | F weight             |
| F attribute             | T defaulthandlers    |
| F badinclusions         | D file               |
| F before                | D parsedxml          |
| F body                  | D string             |
| T builtinentities       | D tex                |
| B amp                   | D texline            |

|                   |                  |
|-------------------|------------------|
| D texspace        | F chainpath      |
| D verbose         | F collapsed      |
| S defaultprotocol | F command        |
| F delete          | F concat         |
| F delete_element  | F concatrange    |
| F despecialized   | F context        |
| F each            | F count          |
| F each_element    | F ctxtext        |
| F elements        | F default        |
| F elementtocdata  | F depth          |
| F empty           | F first          |
| D entities        | F function       |
| F erase           | F index          |
| F errorhandler    | F last           |
| F escaped         | F lettered       |
| F expand          | F lower          |
| T expressions     | F lowerall       |
| F attribute       | F match          |
| F boolean         | F name           |
| F child           | F param          |
| F contains        | F parameter      |
| F count           | F position       |
| F error           | F reverse        |
| F find            | F stripped       |
| F hasclass        | F tag            |
| F idstring        | F tags           |
| F lower           | F text           |
| F name            | F upper          |
| F number          | F upperall       |
| F oneof           | F url            |
| F print           | F xml            |
| F quit            | T xml            |
| F tag             | F all            |
| F text            | F att            |
| F undefined       | F attribute      |
| F upper           | F attributes     |
| F fillin          | F cdata          |
| F filter          | F chainattribute |
| T finalizers      | F count          |
| F paragraphs      | F default        |
| T tex             | F elements       |
| F all             | F empty          |
| F attribute       | F first          |
| F chainattribute  | F function       |

|                         |                         |
|-------------------------|-------------------------|
| F index                 | F insertcomment         |
| F last                  | F inspect               |
| F lowerall              | F is_valid              |
| F match                 | F kind                  |
| F name                  | F last                  |
| F position              | F lastmatch             |
| F reverse               | F load                  |
| F table                 | F lpath                 |
| F tag                   | F lpathcached           |
| F tags                  | F lpathcalls            |
| F text                  | F lshow                 |
| F texts                 | F makestandalone        |
| F tomacro               | F match                 |
| F upperall              | T mml                   |
| F url                   | F stripped              |
| F first                 | F verbose               |
| F found                 | F name                  |
| T functions             | F newhandlers           |
| F classes               | F nodesettostring       |
| F hasclass              | T obsolete              |
| F getbuffer             | F collect_elements      |
| F gethandlers           | F collect_texts         |
| F gethandlersfunction   | F delete_element        |
| T helpers               | F each_element          |
| F recursetext           | F inject_element        |
| F include               | F inject_element_after  |
| F inclusion             | F inject_element_before |
| F inclusions            | F insert_element_after  |
| F index                 | F insert_element_before |
| F inheritattributes     | F process_attributes    |
| F inheritedconvert      | F process_elements      |
| F inject                | F remap_name            |
| F inject_element        | F remap_namespace       |
| F inject_element_after  | F remap_tag             |
| F inject_element_before | F replace_element       |
| F injectafter           | F strip_whitespace      |
| F injectbefore          | F originalload          |
| F insert                | F package               |
| F insert_element        | F parent                |
| F insert_element_after  | U parsedentitylpeg      |
| F insert_element_before | T pihandlers            |
| F insertafter           | F injector              |
| F insertbefore          | F tex                   |
| F insertcdata           | T placeholders          |

|                       |                       |
|-----------------------|-----------------------|
| F unknown_any_entity  | T specialhandler      |
| F unknown_dec_entity  | F sprint              |
| F unknown_hex_entity  | F string              |
| F popmatch            | F strip               |
| F position            | F strip_whitespace    |
| D privatecodes        | F stripelement        |
| F privatetoken        | F stripleadingspaces  |
| F process             | F table               |
| F process_attributes  | F text                |
| F process_elements    | F textonly            |
| F processattributes   | F texttocdata         |
| T profiled            | F tocdata             |
| F pure                | F toelement           |
| F pushmatch           | F toString            |
| F raw                 | F totable             |
| F registerentity      | F toxml               |
| F registers           | F tprint              |
| F reindex             | F traverse            |
| F remap_name          | F unescaped           |
| F remap_namespace     | U unescapedentitylpeg |
| F remap_tag           | F unprivatized        |
| F remapname           | F unspecialized       |
| F remapnamespace      | F wipe                |
| F remaptag            | F withelement         |
| F rename              | F withelements        |
| F renamespace         | F wrap                |
| U reparsedentitylpeg  | T xmlns               |
| F replace             | F xpcall              |
| F replace_element     | T zip                 |
| F resetinjectors      | T archives            |
| F resolvns            | F closearchive        |
| F root                | F openarchive         |
| F save                | T registeredfiles     |
| F selection           | F inspect             |
| F separate            | F ipairs              |
| F serialize           | F load                |
| F serializetotext     | D loaders             |
| F setcdata            | F loadfile            |
| F setcomment          | F loadstring          |
| F sethandlersfunction | F next                |
| F setinjectors        | F optionalrequire     |
| F setproperty         | F pairs               |
| F settings            | F pcall               |
| D specialcodes        | F print               |



- F rawequal
- F rawget
- F rawlen
- F rawset
- F require
- F select
- F setinspector
- F setmathparameters
- F setmetatable
- D texconditionals
- D texconstants
- D texifs
- D txisdefined
- D texmodes
- D texsystemmodes
- F toboolean
- F todecimal
- F tonumber
- F tostring
- F type
- F unpack
- F warn
- F xpcall

## globals

## table

- N CONTEXTLMTXMODE
- N LUAFORMAT
- N LUAMAJORVERSION
- N LUAMINORVERSION
- S LUATEXENGINE
- N LUATEXFORMATID
- N LUATEXFUNCTIONALITY
- S LUATEXVERBOSE
- N LUATEXVERSION
- N LUAVERSION
- F SAVEDREQUIRE
- F assert
- F characterkern
- F collectgarbage
- F dimen
- F dofile
- F error
- F getmetatable
- F inspect
- F ipairs

- F load
- D loaders
- F loadfile
- F loadstring
- F next
- F optionalrequire
- F pairs
- F pcall
- F print
- F rawequal
- F rawget
- F rawlen
- F rawset
- F require
- F select
- F setinspector
- F setmathparameters
- F setmetatable
- D texconditionals
- D texconstants
- D texifs
- D txisdefined
- D texmodes
- D texsystemmodes
- F toboolean
- F todecimal
- F tonumber
- F tostring
- F type
- F unpack
- F warn
- F xpcall

## graphics

## table

- F bpsize
- F identify
- F registerpdfcompactor
- F registerpdffont

## gzip

## table

- F compress
- F compressed
- F decompress
- F load
- F save
- F suffix

## helper

F getprofile  
F mathnested  
F mathprocess  
F mathstep  
F snapindeed  
F snaplist  
F traversecharmmethod  
F traverseglyphmethod  
F traversehorizontal  
F traversevertical

## table

## images

F check  
F checksize  
F copy  
F create  
F embed  
F embedfromstore  
F fetchdata  
F print  
F resetdata  
F resetstore  
F scan  
F serialize  
F storedata  
F totable  
F wrap

## table

## inspect

## function

## interactions

## table

D attachments  
D linkedlists  
D renderings

## interfaces

## table

D cachedsetups  
F cachesetup  
F checkedspecification  
D complete  
D constants  
D corenamespaces  
S currentinterface  
S currentresponse  
F definecommand  
F defined  
F doifelsemessage

D elements  
D formats  
F getmessage  
F getnamespace  
F getsetupstring  
D hashedsetups  
F implement  
F interfacedcommand  
F knownscanner  
F message  
F nameofscanner  
F registernamespace  
F registerscanner  
D scanners  
F setconstant  
F setcount  
F setdimen  
F setelement  
F setformat  
F setmacro  
F setmessage  
F setmessages  
F settranslation  
D setupstrings  
F setvariable  
F showmessage  
D storedscanners  
F tolist  
D translations  
D usednamespaces  
D variables  
F writestatus

## io

## table

F ask  
F bytes  
F characters  
F close  
F copydata  
F exists  
S fileseparator  
F flush  
F gobble  
F input  
F lines

|                 |          |                    |       |
|-----------------|----------|--------------------|-------|
| F loadchunk     |          | F traverse         |       |
| F loaddata      |          | <b>kpse</b>        | table |
| F loadlines     |          | <b>language</b>    | table |
| F newreader     |          | F clean            |       |
| F noflines      |          | F clearhyphenation |       |
| F open          |          | F clearpatterns    |       |
| F output        |          | F current          |       |
| S pathseparator |          | F gethjcode        |       |
| F popen         |          | F haslanguage      |       |
| F read          |          | F hyphenate        |       |
| F readall       |          | F hyphenation      |       |
| F readnumber    |          | F hyphenationmin   |       |
| F readstring    |          | F id               |       |
| F savedata      |          | F new              |       |
| F size          |          | F patterns         |       |
| U stderr        |          | F postexhyphenchar |       |
| U stdin         |          | F posthyphenchar   |       |
| U stdout        |          | F preexhyphenchar  |       |
| F tmpfile       |          | F prehyphenchar    |       |
| F type          |          | F sethjcode        |       |
| F write         |          | F setwordhandler   |       |
| <b>ipairs</b>   | function | <b>languages</b>   | table |
| <b>job</b>      | table    | F addgoodiesdata   |       |
| F comment       |          | F addgoodiesfile   |       |
| F disablesave   |          | F associate        |       |
| F getrandomseed |          | D associated       |       |
| F initialize    |          | F association      |       |
| F load          |          | D counters         |       |
| F loadother     |          | F current          |       |
| B pack          |          | D data             |       |
| N packversion   |          | D decimals         |       |
| F register      |          | F define           |       |
| F save          |          | F explicithyphen   |       |
| N version       |          | F exportlanguage   |       |
| <b>kdtree</b>   | table    | F flatten          |       |
| F delete        |          | F getdata          |       |
| F found         |          | F getmetadata      |       |
| F insert        |          | F getnumber        |       |
| F nearest       |          | F goodiefiles      |       |
| F nearestindex  |          | F hyphenate        |       |
| F new           |          | F installed        |       |
| F newweighted   |          | F installhandler   |       |
| F reset         |          | F loadable         |       |
| F totable       |          | F loadwords        |       |

- F noflattened
- D numbers
- F pdflanguage
- F postexhyphenchar
- F posthyphenchar
- F preexhyphenchar
- F prehyphenchar
- D registered
- F serializediscretionary
- F setexceptions
- F setgoodieshandler
- F setoptions
- F setpatterns
- F setsynonym
- F setwordhandler
- F showdiscretionaries
- F strippedgoodiewords
- F tolang
- F unload
- N version
- F visualizediscretionaries

## layouts table

- F getcanvas
- F getpagedimensions
- F setupcanvas

## lfs table

- F attributes
- F canonicalize
- F chdir
- F currentdir
- F dir
- F expandpath
- F isdir
- F isfile
- F isfound
- F isreadabledir
- F isreadablefile
- F iswriteabledir
- F iswriteablefile
- F link
- F mkdir
- F makedirs
- F modification
- F readlink

- F rmdir
- F setexecutable
- F symlink
- F symlinktarget
- F touch
- F walkdir

## libraries table

- D basiclua
- D basictex
- D builtin
- D extralua
- D extratex
- D functions
- D globals
- D obsolete
- D optional

## lmx table

- F color
- F convert
- F convertfile
- F convertstring
- F einject
- F eprint
- F escape
- F finalized
- F finject
- F fprint
- F get
- F getvariables
- F htmfile
- F include
- F initialize
- F inject
- F lmxfile
- F loadedfile
- F make
- F new
- F popupfile
- F print
- F pv
- F report
- F reset
- F result
- F set

|                        |            |                           |       |
|------------------------|------------|---------------------------|-------|
| F show                 |            | F stop_page_number        |       |
| F stripcss             |            | F stoperrorlogging        |       |
| F tv                   |            | F stopfilelogging         |       |
| F type                 |            | F subdirect               |       |
| F urlescape            |            | F subreport               |       |
| F variable             |            | F system                  |       |
| <b>load</b>            | function   | F texerrormessage         |       |
| <b>loaders</b>         | data table | F writer                  |       |
| <b>loadfile</b>        | function   | <b>lpdf</b>               | table |
| <b>loadstring</b>      | function   | F action                  |       |
| <b>logs</b>            | table      | F adddocumentcolorspace   |       |
| F application          |            | F adddocumenttextgstate   |       |
| F categories           |            | F adddocumentpattern      |       |
| F direct               |            | F adddocumentshade        |       |
| F disable              |            | F addtocatalog            |       |
| F enable               |            | F addtoinfo               |       |
| F finalactions         |            | F addtonames              |       |
| F flush                |            | F addtopageattributes     |       |
| F loggingerrors        |            | F addtopageresources      |       |
| F messenger            |            | F addtopagesattributes    |       |
| F new                  |            | F addxmpinfo              |       |
| F newline              |            | F annotationspecification |       |
| F poplogfile           |            | F array                   |       |
| F poptarget            |            | F attachmentsymbols       |       |
| F pushlogfile          |            | F boolean                 |       |
| F pushtarget           |            | F border                  |       |
| F registerfinalactions |            | F checkedkey              |       |
| F report               |            | F checkedresources        |       |
| F reporter             |            | F checkedvalue            |       |
| F resetlogfile         |            | F circularshade           |       |
| F setformats           |            | F close                   |       |
| F setformatters        |            | F cmykcode                |       |
| F setlogfile           |            | F collectedresources      |       |
| F setmessenger         |            | F color                   |       |
| F setprocessor         |            | F colorspace              |       |
| F settarget            |            | D colorspaceconstants     |       |
| F settimedlog          |            | F colorspec               |       |
| F settranslations      |            | F colorvalue              |       |
| F show                 |            | F colorvalues             |       |
| F start_page_number    |            | F commentsymbols          |       |
| F start_run            |            | F compressdata            |       |
| F startererrorlogging  |            | F compresslevel           |       |
| F startfilelogging     |            | F constant                |       |
| F status               |            | F copyarray               |       |

|                            |                             |
|----------------------------|-----------------------------|
| F copyboolean              | F graycode                  |
| F copyconstant             | F hasmatrix                 |
| F copydictionary           | F immediateobject           |
| F copyfloat                | F includeimage              |
| F copyinteger              | F initialize                |
| F copynumber               | F injectxmpinfo             |
| F copystring               | F insertxmpinfo             |
| F copyunicode              | F lastreferredpage          |
| F deferredobject           | F layerreference            |
| F delayedobject            | F limited                   |
| F destinationspecification | F linearshade               |
| F dictionary               | F literal                   |
| F docstring                | F majorversion              |
| F escaped                  | F makenametree              |
| F finalizeddocument        | F minorversion              |
| F finalizepage             | F noffontregistries         |
| F finalizesignature        | F nopages                   |
| F finishtransparencycode   | F null                      |
| F flushfonts               | F number                    |
| F flushgroup               | F object                    |
| F flushobject              | F objectcompresslevel       |
| F flushstreamfileobject    | F pagereference             |
| F flushstreamobject        | F patternstream             |
| D formats                  | F pdfcolor                  |
| F frombytes                | F pdftimestamp              |
| F fromeight                | F preparesignature          |
| F frompdfdoc               | F print                     |
| F fromsixteen              | F procset                   |
| F getcatalog               | B protectresources          |
| F getcompression           | F quads                     |
| F getfontname              | F rectangle                 |
| F getfontobjectnumber      | F reference                 |
| F getformatoption          | F registerannotation        |
| F getinfo                  | F registerdestination       |
| F getmatrix                | F registerdocumentfinalizer |
| F getnofpages              | F registerfontmethod        |
| F getpageproperties        | F registerinitializer       |
| F getpagereference         | F registerpagefinalizer     |
| F getpagetags              | F registerpattern           |
| F getpos                   | F registersignature         |
| F gettaggingstatus         | F replacexmpinfo            |
| F gettrailerid             | F reserveobject             |
| F getversion               | F rgbcode                   |
| F getxformname             | F rotationcm                |

- F setcompression
- F setencryption
- F setpageorder
- F setpagestate
- F setpagetag
- F setutfstringmode
- F setversion
- F sharedobject
- F shareobject
- F shareobjectreference
- F splitsixteen
- F spotcode
- F stream
- F string
- F supportedformats
- F tocidset
- F toeight
- F topdfdoc
- F topermissions
- F tosixteen
- F transparency
- F transparencycode
- F transparencyvalue
- F unicode
- D usedfontstreams
- F verbose

## lpeg table

- F B
- F C
- F Carg
- F Cb
- F Cc
- F Cf
- F Cg
- F Cmt
- F Cp
- F Cs
- F Ct
- F P
- F R
- F S
- F UP
- F UR
- F US

- F V
- F afterprefix
- F anywhere
- F append
- F balancer
- F beforesuffix
- F checkedsplit
- F containsws
- F counter
- F endstripper
- F finder
- F firstofsplit
- F frontstripper
- F instringchecker
- F is\_lpeg
- F keeper
- F locale
- F match
- F oneof
- D patterns
- F pcode
- F print
- F ptree
- F replacer
- F secondofsplit
- F setmaxstack
- F setutfcasers
- F split
- F splitat
- F splitter
- F stripper
- F times
- F tsplitat
- F tsplitter
- F type
- F utfR
- F utfchartabletopattern
- F utfreplacer
- S version

## lua table

- F addnibble
- D bytedata
- F callbytecode
- F comparekeys

|                           |              |                        |              |
|---------------------------|--------------|------------------------|--------------|
| F finalizeinitex          |              | F cleanup              |              |
| N firstbytecode           |              | F cleanuptempfiles     |              |
| F get_functions_table     |              | F currentfile          |              |
| F getallkeys              |              | F currentlevel         |              |
| F getbytecode             |              | F currenttotal         |              |
| F getcheckedkeys          |              | F poprunstate          |              |
| F getcurrenttime          |              | F pushrunstate         |              |
| F getdebuginfo            |              | F registerdumpactions  |              |
| F getexitcode             |              | F registerpageactions  |              |
| F getfunctionstable       |              | F registerstartactions |              |
| F getnibble               |              | F registerstopactions  |              |
| F getnumerickeys          |              | F registertempfile     |              |
| F getpreciseseconds       |              | N starttime            |              |
| F getpreciseticks         |              | F wrapup               |              |
| F getruntime              |              | <b>lxml</b>            | <b>table</b> |
| F getstacktop             |              | F addindex             |              |
| F getstartupfile          |              | F all                  |              |
| F getstringkeys           |              | F applyfunction        |              |
| F getversion              |              | F applyselectors       |              |
| N lastbytecode            |              | F att                  |              |
| F luastringcall           |              | F attribute            |              |
| D messages                |              | F badinclusions        |              |
| F mpsstringcall           |              | F chainattribute       |              |
| F newindex                |              | F checkindex           |              |
| F newtable                |              | F collected            |              |
| D numbers                 |              | F command              |              |
| F protectglobal           |              | F concat               |              |
| F registercode            |              | F concatrange          |              |
| F registeredcodes         |              | F content              |              |
| F registerglobal          |              | F context              |              |
| F registerinitexfinalizer |              | F convert              |              |
| F registername            |              | F count                |              |
| F setbytecode             |              | F delete               |              |
| F setexitcode             |              | F depth                |              |
| F setnibble               |              | F direct               |              |
| S startupfile             |              | F displayverbatim      |              |
| F subnibble               |              | F doif                 |              |
| F usestringcall           |              | F doifelse             |              |
| S version                 |              | F doifelseempty        |              |
| <b>luac</b>               | <b>table</b> | F doifelsetext         |              |
| F inspect                 |              | F doifempty            |              |
| F print                   |              | F doifnot              |              |
| <b>luatex</b>             | <b>table</b> | F doifnotempty         |              |
| F abort                   |              | F doifnottext          |              |



|                  |                    |
|------------------|--------------------|
| F doiftext       | F position         |
| F element        | F pure             |
| F elements       | F pushmatch        |
| D entities       | F raw              |
| F fileline       | F rawroot          |
| F filename       | F rawtex           |
| F filter         | F refatt           |
| F filterlist     | F register         |
| F first          | F registerentity   |
| F flush          | F removesetup      |
| F flushlinewise  | F replace          |
| F flushsetups    | F resetcdata       |
| F flushspacewise | F resetsetups      |
| F get_id         | F resolvedentity   |
| F getid          | F root             |
| F getindex       | F save             |
| F id             | F serialize        |
| U idsplitter     | F setaction        |
| F ifatt          | F setatt           |
| F ifattempty     | F setattribute     |
| F include        | F setcdata         |
| F inclusion      | F setcommandtonone |
| F inclusions     | F setcommandtotext |
| F index          | F setpar           |
| F inlineverbatim | F setparam         |
| F installsetup   | F setparameter     |
| F last           | F setsetup         |
| F lastatt        | F snippet          |
| F lastmatch      | F splitid          |
| F lastpar        | F startraw         |
| F load           | F startunescaped   |
| F loadbuffer     | F stopraw          |
| F loaddata       | F stopunescaped    |
| F main           | F store            |
| F match          | F strip            |
| F name           | F stripped         |
| F namespace      | F tag              |
| F nonspace       | F texatt           |
| F par            | F text             |
| F param          | F tobuffer         |
| F parameter      | F tofile           |
| F path           | F toparameters     |
| F popmatch       | F toverbatim       |
| F pos            | F verbatim         |

F withindex  
D xmltexhandler

## math table

F abs  
F acos  
F asin  
F atan  
F atan2  
F ceil  
F ceiling  
F cos  
F cosd  
F cosh  
F deg  
F div  
F even  
F exp  
F floor  
F fmod  
F frexp  
N huge  
N initialseed  
F ldexp  
F log  
F log10  
F max  
N maxinteger  
F min  
N mininteger  
F mod  
F modf  
F odd  
N pi  
F pow  
F rad  
F random  
F randomseed  
F round  
F setrandomseedi  
F sin  
F sind  
F sinh  
F sqrt  
F tan

F tand  
F tanh  
F tointeger  
F type  
F ult

## mathematics table

F addfallbacks  
F addvariant  
D alphabets  
D alternatives  
F beforepassingfonttotex  
F big  
D boldmap  
D charactersets  
F checkaftercopyingfont  
D classes  
D classnames  
D codes  
F define  
F dimensions  
F disablecollecting  
F enablecollecting  
F extensiblecode  
D extensibles  
N extrabase  
F fallbackstyleattr  
F familyname  
F functiontype  
D gaps  
F getblobindex  
F getboth  
F getdomainname  
F getextensibledata  
F getmathblob  
F getoptionallabel  
F getstyle  
F gettextblob  
F getverboselabel  
F horizontalcode  
F initializeparameters  
F makefont  
D mapremap  
F markblobindexdone  
F overloadparameters

- F preset
- N privatebase
- F registerextensible
- F registersimplematrix
- F registerverboselabel
- F remapalphabets
- F renderextensible
- F renderset
- F resetattributes
- F scaleparameters
- F setextensibledata
- F setmathsymbol
- D sets
- F setsnapping
- F setvariant
- F strutcode
- D styles
- F syncboth
- F syncname
- F syncstyle
- F toengineclass
- F toname
- F tostyle
- F tweakaftercopyingfont
- F tweakbeforecopyingfont
- F utfmathaccent
- F utfmathclass
- F utfmathcommand
- F utfmathfiller
- F utfmathstretch
- F variantcode
- F variantcount
- F virtualize
- D virtualized

## md5 table

- F HEX
- F dec
- F hex
- F sum
- F sumHEXA
- F sumhexa

## metapost table

- F checkformat
- F checktracingonline

- F cm
- F colorinitializer
- F convert
- F currentmpx
- F currentmpxstatus
- S defaultformat
- S defaultinstance
- S defaultmethod
- F directrun
- F escaped
- D extensiondata
- F filter
- F filterclippath
- F flush
- F flushnormalpath
- F format\_number
- F format\_string
- F formatnumber
- F getbackendoption
- F getbendtolerance
- F getbuffer
- F getclippath
- F getextensions
- F getextradata
- F gethashentries
- F getinstancebyname
- F getinstanceinfo
- F getinstancenames
- F getjobdata
- F getmovetolerance
- F getparameter
- F getparameterpreset
- F getparameterset
- F getrecord
- F getstatistics
- F gettext
- F gettolerance
- F graphic
- F graphic\_base\_pass
- F hascurvature
- F includesvgbuffer
- F includesvgfile
- F installplugin
- F isobject

|                        |                    |              |
|------------------------|--------------------|--------------|
| S lastlog              | F run              |              |
| F load                 | F runinternal      |              |
| F maketext             | F runscript        |              |
| F mesh                 | F scanparameters   |              |
| F mesher               | F scriptindex      |              |
| F meshupper            | F setbendtolerance |              |
| N n                    | F setbuffer        |              |
| F newlogger            | F setextensions    |              |
| N nofruns              | F setlogging       |              |
| F nofscriptruns        | F setmovetolerance |              |
| F nvformat             | F setoutercolor    |              |
| F pdfliterals          | F setparameter     |              |
| F pluginactions        | F setparameterset  |              |
| F points               | F setrecord        |              |
| F pointset             | F settext          |              |
| F popformat            | F settolerance     |              |
| F popscriptrunner      | F setvariable      |              |
| F popvariable          | B showlog          |              |
| F popvariables         | F showsvgpage      |              |
| F preparetexttextsdata | F simple           |              |
| F process              | F splitprescript   |              |
| F processing           | F startgraphic     |              |
| F processplugins       | F startsvghashing  |              |
| F pushformat           | F stopgraphic      |              |
| F pushscriptrunner     | F stopsvghashing   |              |
| F pushvariable         | F svformat         |              |
| F pushvariables        | B svgcachefile     |              |
| B quitonerror          | F svgcolorremapper |              |
| B reducetogray         | F svgglyphtomp     |              |
| F registerdirect       | F svgshapestomp    |              |
| F registerfilter       | F svgshapestopdf   |              |
| F registermesher       | F svgsplitcolor    |              |
| F registermeshupper    | F svgtomp          |              |
| F registerpointset     | F svgtopdf         |              |
| F registerscript       | F sxsy             |              |
| F registertokens       | B texerrors        |              |
| F remappedtext         | F texexp           |              |
| F remaptext            | F theclippath      |              |
| F reporterror          | F tographic        |              |
| F reportsvgfonts       | F totable          |              |
| F reset                | F typesvgpage      |              |
| F resetbackendoptions  | F unload           |              |
| F resetlastlog         | F untagvariable    |              |
| F resetplugins         | <b>mime</b>        | <b>table</b> |

|                       |              |                       |
|-----------------------|--------------|-----------------------|
| F b64                 |              | D cldf-ver            |
| F decode              |              | D colo-icc            |
| F dot                 |              | D colo-ini            |
| F encode              |              | D cont-run            |
| F eol                 |              | D core-con            |
| F normalize           |              | D core-ctx            |
| F qp                  |              | D core-dat            |
| F qpwrp               |              | D core-sys            |
| F report              |              | D driv-shp            |
| F stuff               |              | D file-mod            |
| F unb64               |              | D file-syn            |
| F unqp                |              | D font-afk            |
| F wrap                |              | D font-agl            |
| F wrp                 |              | D font-aux            |
| <b>moduledata</b>     | <b>table</b> | D font-cff            |
| <b>modules</b>        | <b>table</b> | D font-chk            |
| D anch-loc            |              | D font-cid            |
| D anch-pgr            |              | D font-col            |
| D anch-snc            |              | D font-con            |
| D back-exp            |              | D font-ctx            |
| D back-exp-imp-fnt    |              | D font-def            |
| D back-exp-imp-mth    |              | D font-dsp            |
| D back-exp-imp-ref    |              | D font-enc            |
| D back-exp-imp-tag    |              | D font-enh            |
| D back-imp-lua        |              | D font-fbk            |
| D back-imp-mps        |              | D font-fmp            |
| D back-imp-pdf        |              | D font-hsh            |
| D back-imp-pdp        |              | D font-imp-checks     |
| D blob-ini            |              | D font-imp-combining  |
| D buff-imp-default    |              | D font-imp-compound   |
| D buff-imp-escaped    |              | D font-imp-digits     |
| D buff-imp-lua        |              | D font-imp-dimensions |
| D buff-imp-mp         |              | D font-imp-effects    |
| D buff-imp-nested     |              | D font-imp-italics    |
| D buff-imp-parsed-xml |              | D font-imp-ligatures  |
| D buff-imp-tex        |              | D font-imp-math       |
| D buff-imp-xml        |              | D font-imp-notused    |
| D buff-ini            |              | D font-imp-properties |
| D buff-par            |              | D font-imp-quality    |
| D buff-ver            |              | D font-imp-reorder    |
| D char-enc            |              | D font-imp-scripts    |
| D chem-ini            |              | D font-imp-spacekerns |
| D chem-str            |              | D font-imp-tex        |
| D cldf-com            |              | D font-imp-text       |

|                    |                           |
|--------------------|---------------------------|
| D font-imp-tracing | D good-mth                |
| D font-imp-tweaks  | D grph-bmp                |
| D font-imp-unicode | D grph-con                |
| D font-ini         | D grph-epd                |
| D font-lig         | D grph-fil                |
| D font-log         | D grph-img                |
| D font-lua         | D grph-inc                |
| D font-map         | D grph-pat                |
| D font-mpf         | D grph-raw                |
| D font-mps         | D grph-rul                |
| D font-nod         | D grph-trf                |
| D font-ogr         | D grph-u3d                |
| D font-one         | D java-ini                |
| D font-onr         | D lang-cnt                |
| D font-osd         | D lang-def                |
| D font-ota         | D lang-dis                |
| D font-otc         | D lang-exc                |
| D font-otd         | D lang-hup                |
| D font-oth         | D lang-hyp                |
| D font-oti         | D lang-ini                |
| D font-otj         | D lang-lab                |
| D font-otl         | D lang-rep                |
| D font-oto         | D lang-tra                |
| D font-otr         | D lang-txt                |
| D font-ots         | D lang-url                |
| D font-ott         | D lang-wrd                |
| D font-oup         | D libs-imp-curl           |
| D font-pat         | D libs-imp-ghostscript    |
| D font-prv         | D libs-imp-graphicsmagick |
| D font-san         | D libs-ini                |
| D font-sel         | D lpdf-ano                |
| D font-shp         | D lpdf-col                |
| D font-sol         | D lpdf-epa                |
| D font-syn         | D lpdf-epd                |
| D font-tfm         | D lpdf-fix                |
| D font-tpk         | D lpdf-fix-imp-contents   |
| D font-trt         | D lpdf-fix-imp-fonts      |
| D font-ttf         | D lpdf-fld                |
| D font-txt         | D lpdf-fmt                |
| D font-vfc         | D lpdf-grp                |
| D font-vir         | D lpdf-img                |
| D good-ctx         | D lpdf-ini                |
| D good-gen         | D lpdf-lmt                |
| D good-ini         | D lpdf-mis                |

|                |             |
|----------------|-------------|
| D lpdf-ren     | D mlib-int  |
| D lpdf-rul     | D mlib-lmp  |
| D lpdf-sig     | D mlib-lmt  |
| D lpdf-tag     | D mlib-lua  |
| D lpdf-u3d     | D mlib-mat  |
| D lpdf-wid     | D mlib-mpf  |
| D lpdf-xmp     | D mlib-msh  |
| D lxml-css     | D mlib-pdf  |
| D lxml-dir     | D mlib-pos  |
| D lxml-ent     | D mlib-pps  |
| D lxml-ini     | D mlib-ptr  |
| D lxml-mms     | D mlib-pts  |
| D lxml-sor     | D mlib-ran  |
| D lxml-tex     | D mlib-run  |
| D math-act     | D mlib-scen |
| D math-ali     | D mlib-snc  |
| D math-dim     | D mlib-svg  |
| D math-ext     | D node-ali  |
| D math-fbk     | D node-bck  |
| D math-fnt     | D node-fnt  |
| D math-frc     | D node-par  |
| D math-inc     | D node-ref  |
| D math-ini     | D node-rul  |
| D math-map     | D node-tra  |
| D math-mps     | D node-typ  |
| D math-noa     | D pack-obj  |
| D math-pre     | D pack-ori  |
| D math-rad     | D pack-rul  |
| D math-ren     | D page-blk  |
| D math-spa     | D page-cst  |
| D math-tag     | D page-flt  |
| D math-txt     | D page-ini  |
| D math-vfu     | D page-inj  |
| D meta-blb     | D page-ins  |
| D meta-fnt     | D page-lin  |
| D meta-fun     | D page-mix  |
| D meta-imp-txt | D page-otr  |
| D meta-ini     | D page-pst  |
| D meta-lua     | D page-sel  |
| D meta-nod     | D page-str  |
| D meta-pdf     | D phys-dim  |
| D meta-tex     | D publ-aut  |
| D mlib-ctx     | D publ-dat  |
| D mlib-fio     | D publ-fnd  |

|            |            |
|------------|------------|
| D publ-inc | D strc-not |
| D publ-ini | D strc-num |
| D publ-jrn | D strc-pag |
| D publ-oth | D strc-ref |
| D publ-reg | D strc-reg |
| D publ-sor | D strc-rsc |
| D publ-tra | D strc-syn |
| D publ-usr | D strc-tag |
| D regi-ini | D strc-usr |
| D scrn-but | D symb-emj |
| D scrn-fld | D symb-ini |
| D scrn-hlp | D syst-cmp |
| D scrn-ini | D tabl-ali |
| D scrn-pag | D tabl-ntb |
| D scrn-ref | D tabl-tbl |
| D scrn-wid | D tabl-xtb |
| D scrp-cjk | D task-ini |
| D scrp-eth | D trac-bld |
| D scrp-ini | D trac-jus |
| D scrp-tha | D trac-par |
| D scrp-tib | D trac-riv |
| D sort-ini | D trac-sig |
| D sort-lan | D trac-vis |
| D spac-ali | D type-ini |
| D spac-brk | D typo-adj |
| D spac-chr | D typo-bld |
| D spac-hor | D typo-brk |
| D spac-ovl | D typo-cap |
| D spac-par | D typo-chr |
| D spac-pas | D typo-cln |
| D spac-prf | D typo-dha |
| D spac-twi | D typo-dif |
| D spac-ver | D typo-dig |
| D strc-bkm | D typo-dir |
| D strc-blk | D typo-drp |
| D strc-con | D typo-duc |
| D strc-doc | D typo-fkr |
| D strc-flt | D typo-fln |
| D strc-ini | D typo-hid |
| D strc-itm | D typo-inj |
| D strc-lev | D typo-itc |
| D strc-lst | D typo-krn |
| D strc-mar | D typo-lan |
| D strc-mat | D typo-lbx |



|                           |                              |
|---------------------------|------------------------------|
| D typo-man                | F lmt_glyphshape_boundingbox |
| D typo-mar                | F lmt_glyphshape_depth       |
| D typo-ovl                | F lmt_glyphshape_height      |
| D typo-pag                | F lmt_glyphshape_italic      |
| D typo-par                | F lmt_glyphshape_llx         |
| D typo-pnc                | F lmt_glyphshape_n           |
| D typo-prc                | F lmt_glyphshape_path        |
| D typo-rep                | F lmt_glyphshape_start       |
| D typo-rub                | F lmt_glyphshape_stop        |
| D typo-spa                | F lmt_glyphshape_usedbox     |
| D typo-stc                | F lmt_glyphshape_usedline    |
| D typo-sus                | F lmt_glyphshape_width       |
| D typo-syn                | F lmt_mesh_set               |
| D typo-tal                | F lmt_mesh_update            |
| D typo-wrp                | F lmt_poisson_generate       |
| D unic-ini                | F lmt_start_potrace          |
| <b>mp</b>                 | F lmt_step_potrace           |
|                           | F lmt_stop_potrace           |
| F autoinject              | F lmt_svg_include            |
| F boolean                 | F loadbytemap                |
| F bytemap_load_from_file  | F loadbytemapfile            |
| F cleaned                 | F mesh_register              |
| F color                   | F mf_blob_add                |
| F dataset                 | F mf_blob_dimensions         |
| F direct                  | F mf_blob_index              |
| F downgradebytemap        | F mf_blob_new                |
| F downsamplebytemap       | F mf_blob_raw_content        |
| F draw                    | F mf_blob_raw_dimensions     |
| F expandtex               | F mf_blob_raw_reset          |
| F fill                    | F mf_blob_raw_toutf          |
| F fillbytemap             | F mf_blob_raw_wipe           |
| F flatten                 | F mf_blob_size               |
| F flush                   | F mf_blob_width              |
| F format                  | F mf_external_figure         |
| F formatted               | F mf_finish_saving_data      |
| F fprint                  | F mf_formatted_text          |
| F getbytemap              | F mf_get_outline_text        |
| F getbytemapdata          | F mf_graphic_text            |
| F getbytemapdirect        | F mf_inject_blob             |
| F getlastaddtype          | F mf_map_move                |
| F graphformat             | F mf_map_text                |
| F integer                 | F mf_named_color             |
| F lmt_function_x          | F mf_outline_text            |
| F lmt_glyphshape_accent   | F mf_save_data               |
| F lmt_glyphshape_baseline |                              |
| function                  |                              |

- F mf\_script\_index
- F mf\_start\_saving\_data
- F mf\_stop\_saving\_data
- F mode
- F n
- F newbytemap
- F number
- F numeric
- F pair
- F pairpoints
- F path
- F pathpoints
- F points
- F points\_register
- F processbytemap
- F processingmode
- F quadruple
- F quadruplepoints
- F quoted
- F report
- F setbytemap
- F setbytemapoctave
- F size
- F solve
- F string
- F systemmode
- F transform
- F triplet
- F tripletpoints
- F vprint

## mplib

## table

- F downgradebytemap
- F downsamplebytemap
- F execute
- F expandtex
- F fillbytemap
- F finder
- F finish
- F getbytemap
- F getbytemapdata
- F getbytemapdirect
- F getcallbackstate
- F getcodes
- F getcolormodels

- F getfields
- F gethashentries
- F gethashentry
- F getinternalactions
- F getknotstates
- F getlastaddtype
- F getlogtargets
- F getobjectdata
- F getobjectpeninfo
- F getobjectstacking
- F getobjecttypes
- F getprimitivelegacies
- F getprimitiveorigins
- F getprimitives
- F getpruneoptions
- F getresultstates
- F getscaledbits
- F getscantypes
- F getstates
- F getstatistics
- F getstatus
- F gettolerance
- F gettype
- F gettypes
- F injectboolean
- F injectcmykcolor
- F injectcolor
- F injectinteger
- F injectnumber
- F injectnumeric
- F injectpair
- F injectpath
- F injectpoints
- F injectstring
- F injecttokens
- F injecttransform
- F injectvector
- F injectvectors
- F injectwhatever
- F new
- F newbytemap
- F peninfo
- F processbytemap
- B realltimelogging

|                        |                          |
|------------------------|--------------------------|
| F scanboolean          | F flattendiscretionaries |
| F scanmykcolor         | F flushlist              |
| F scancolor            | F flushnode              |
| F scanexpression       | F free                   |
| F scaninteger          | F getattribute           |
| F scannext             | F getcachestate          |
| F scannumber           | F getfield               |
| F scannumeric          | F getfielderror          |
| F scanpair             | F getglue                |
| F scanpath             | F getnodeerrorvalues     |
| F scanpen              | F getpropertystable      |
| F scanproperty         | F getproperty            |
| F scanstring           | F gluetostring           |
| F scansymbol           | F hasattribute           |
| F scantoken            | F hasfield               |
| F scantransform        | F hasglyph               |
| F setbyteimap          | F hpack                  |
| F setbyteimapoctave    | F hyphenating            |
| F settolerance         | F id                     |
| F showcontext          | F insertafter            |
| F skiptoken            | F insertbefore           |
| F solvepath            | F instock                |
| F version              | F intracking             |
| <b>nanojpeg</b>        | F inuse                  |
| F decode               | F isnode                 |
| <b>next</b>            | F iszeroglue             |
| <b>noads</b>           | F kerning                |
| F getmathmode          | F lastnode               |
| F process              | F length                 |
| F processnested        | F ligaturing             |
| F processouter         | F makeextensible         |
| <b>node</b>            | F mlisttohlist           |
| F checkdiscretionaries | F new                    |
| F checkdiscretionary   | F protectglyph           |
| F copy                 | F protectglyphs          |
| F copyleft             | F protrusionskipable     |
| F count                | F rangedimensions        |
| F currentattributes    | F remove                 |
| F dimensions           | F serialized             |
| F effectiveglue        | F setattribute           |
| F endofmath            | F setfield               |
| F family_font          | F setfielderror          |
| F fields               | F setglue                |
| F findattribute        | F setproperty            |

F show

F size

F slide

F subtypes

F tail

F todirect

F tonode

F toString

F traverse

F traverseid

F type

F types

F unprotectglyph

F unprotectglyphs

F unsetattribute

F usedlist

F usesfont

F vpack

F write

## nodes

## table

D accentcodes

F aligned

F append

F apply

F applyvisuals

F astable

F basepoints

D boundarycodes

D choicecodes

F concat

F copy

F copy\_node

F copyleft

F countall

F currentattributes

F delete

D dircodes

D directioncodes

D directionvalues

D dirvalues

D disccodes

D discretionarycodes

D fencecodes

F fields

D fillcodes

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F firstdirinbox

F flush

F flushlist

F flushnode

D fractioncodes

F fullhpack

F getattr

F getattribute

F getboth

F getbox

F getchar

F getfield

F getfont

F getid

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F getlist

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F getproperty

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D gluecodes

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F hasattribute

F hasfield

F hpack

F idsandsubtypes

F idstostring

F insertafter

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F insertlistafter

F installattributehandler

F is\_display\_math

F isdirect

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D kerncodes

F leftmarginwidth

F link

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D listcodes

F listtoutf

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| D literalvalues     | F showlist             |              |
| F locate            | F showsimplelist       |              |
| D markcodes         | D skipcodes            |              |
| D mathcodes         | F somepenalty          |              |
| D mathvalues        | F somespace            |              |
| F maxboxwidth       | F splitbox             |              |
| F new               | F startofpar           |              |
| D noadcodes         | F stripdiscretionaries |              |
| D nodecodes         | D subtypes             |              |
| F nopts             | F tail                 |              |
| F packlist          | F takeattr             |              |
| D parcodes          | F takebox              |              |
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| F points            | F tocentimeters        |              |
| F prepend           | F tociceros            |              |
| F print             | F todidots             |              |
| F pts               | F todimen              |              |
| D radicalcodes      | F toinches             |              |
| F rehpick           | F tomillimeters        |              |
| F remove            | F tonode               |              |
| F repackhlist       | F tonodes              |              |
| F replace           | F tonut                |              |
| F report            | F topicas              |              |
| F rightmarginwidth  | F topoints             |              |
| D rulecodes         | F toscaledpoints       |              |
| F serialize         | F tosequence           |              |
| F serializebox      | F toString             |              |
| F serialized        | F totable              |              |
| F setattr           | F totree               |              |
| F setattrtribute    | F toutf                |              |
| F setattrrlist      | F traverse             |              |
| F setboth           | F traverseid           |              |
| F setbox            | F unsetattribute       |              |
| F setboxtonaturalwd | F upcomingproperties   |              |
| F setchar           | F usedlist             |              |
| F setfield          | F vianodes             |              |
| F setfont           | F vianuts              |              |
| F setleader         | F visualizebox         |              |
| F setlink           | F vpack                |              |
| F setlist           | D whatsitcodes         |              |
| F setnext           | F write                |              |
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| F setproperty       | F %                    |              |
| F showboxes         | F basepoints           |              |

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| F bytetodecimal    |       | F page                 |          |
| F clamp            |       | F reference            |          |
| F compactfloat     |       | F register             |          |
| F decimaltobyte    |       | F restore              |          |
| D dimenfactors     |       | <b>optional</b>        | table    |
| F formatted        |       | <b>optionalrequire</b> | function |
| F formattedfloat   |       | <b>os</b>              | table    |
| F formattedinteger |       | F __getenv__           |          |
| F fractiontobyte   |       | F __setenv__           |          |
| F idiv             |       | S binsuffix            |          |
| N maxdimen         |       | N bits                 |          |
| F nopts            |       | F clock                |          |
| F nubasepoints     |       | F converttime          |          |
| F nupoints         |       | F date                 |          |
| F percent          |       | F difftime             |          |
| F points           |       | F enableansi           |          |
| F pts              |       | D env                  |          |
| F signed           |       | F execute              |          |
| F sparseexponent   |       | F exit                 |          |
| F splitdimen       |       | F fulltime             |          |
| F to16dot16        |       | F getcodepage          |          |
| F tobasepoints     |       | F getenv               |          |
| F tobitstring      |       | F getnamevalues        |          |
| F tocentimeters    |       | F gettimeofday         |          |
| F tociceros        |       | F gettypevalues        |          |
| F tocontext        |       | F getunamelfields      |          |
| F todidots         |       | F isleapyear           |          |
| F todimen          |       | F launch               |          |
| F toediths         |       | S libsuffix            |          |
| F toevenhex        |       | F localtime            |          |
| F toinches         |       | S name                 |          |
| F tomillimeters    |       | S newline              |          |
| F tonumberf        |       | F nofdays              |          |
| F tonumberg        |       | F now                  |          |
| F topics           |       | F pipeto               |          |
| F topoints         |       | S platform             |          |
| F toscaled         |       | F remove               |          |
| F toscaledpoints   |       | F rename               |          |
| F totoves          |       | F resultof             |          |
| F valid            |       | F runtime              |          |
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| F dimensions       |       | S selfcore             |          |
| F found            |       | S selfdir              |          |
| N n                |       | S selflink             |          |

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| S selfname           |          | F close             |       |
| S selfpath           |          | F closestream       |       |
| F serialwrite        |          | F dictionarytotable |       |
| F setenv             |          | F getarray          |       |
| F setexitcode        |          | F getboolean        |       |
| F setlocale          |          | F getbox            |       |
| F sleep              |          | F getcatalog        |       |
| F startuptime        |          | F getdictionary     |       |
| F time               |          | F getencodingvalues |       |
| F timezone           |          | F getfieldtypes     |       |
| F tmpname            |          | F getfromarray      |       |
| F today              |          | F getfromdictionary |       |
| S type               |          | F getfromobject     |       |
| F uname              |          | F getfromreference  |       |
| F uuid               |          | F getfromstream     |       |
| F validate           |          | F getinfo           |       |
| F weekday            |          | F getinteger        |       |
| F where              |          | F getmemoryusage    |       |
| F which              |          | F getname           |       |
| <b>package</b>       | table    | F getnofobjects     |       |
| S config             |          | F getnofpages       |       |
| S cpath              |          | F getnumber         |       |
| F extralibpath       |          | F getobjectrange    |       |
| F extralibpaths      |          | F getpage           |       |
| F extralupath        |          | F getpages          |       |
| F extralupaths       |          | F getpermissions    |       |
| F libpaths           |          | F getsizes          |       |
| F loadlib            |          | F getstatus         |       |
| F luapaths           |          | F getstatusvalues   |       |
| S path               |          | F getstream         |       |
| F poplibpath         |          | F getstring         |       |
| F popluapath         |          | F gettrailer        |       |
| F pushlibpath        |          | F getversion        |       |
| F pushluapath        |          | F new               |       |
| F searchpath         |          | F open              |       |
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| <b>pairs</b>         | function | F openstream        |       |
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| <b>pcall</b>         | function | F readfromstream    |       |
| <b>pdf</b>           | table    | F readwholestream   |       |
| F immediateobj       |          | F type              |       |
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| F arraytotable       |          | <b>pngdecode</b>    | table |

- F applyfilter
- F expand
- F frompalette
- F interlace
- F makemask
- F palettemask
- F reduce
- F setexperiment
- F splitmask
- F tocmymk
- F togray
- F tomask
- F transparentmask

## posit

- F NaN
- F NaR
- F \_\_add
- F \_\_band
- F \_\_bor
- F \_\_bxor
- F \_\_div
- F \_\_eq
- F \_\_le
- F \_\_lt
- F \_\_mul
- S \_\_name
- F \_\_pow
- F \_\_shl
- F \_\_shr
- F \_\_sub
- F \_\_tostring
- F \_\_unm
- F abs
- F acos
- F asin
- F atan
- F band
- F bor
- F bxor
- F ceil
- F conj
- F copy
- F cos
- F exp

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- F exp2
- F expand
- F floor
- F frombyte
- F fromchar
- F fromposit
- F integer
- F log
- F log10
- F loglp
- F log2
- F logb
- F max
- F min
- F modf
- F new
- F pow
- F rotate
- F round
- F rounded
- F shift
- F sin
- F sqrt
- F tan
- F tobyte
- F tochar
- F tonumber
- F toposit
- F tostring

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- F checked
- F concat
- F contourplot
- F convert
- F count
- F flush
- F free
- F fromfile
- F getbitmap
- F getnewfields
- F getpolicyvalues
- F getprocessfields
- F new
- F process

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| F setbitmap           |          | F search           |          |
| F stripped            |          | F setcategory      |          |
| F totable             |          | F setspecification |          |
| F trace               |          | F singularorplural |          |
| <b>print</b>          | function | F sortedentries    |          |
| <b>publications</b>   | table    | F splitauthor      |          |
| F addtexentry         |          | F splitoneauthor   |          |
| F analyze             |          | F tags             |          |
| F anywhere            |          | F usedataset       |          |
| D authorcache         |          | F usedentries      |          |
| D authorcollapsers    |          | <b>qrcodegen</b>   | table    |
| D authormap           |          | F bitmap           |          |
| F authortoregister    |          | F generate         |          |
| F checkeddataset      |          | <b>rawequal</b>    | function |
| D collected           |          | <b>rawget</b>      | function |
| F concatstate         |          | <b>rawlen</b>      | function |
| F converttoxml        |          | <b>rawset</b>      | function |
| F endnotes_to_btx     |          | <b>regimes</b>     | table    |
| F finder              |          | D backmapping      |          |
| F firstofrange        |          | F cleanup          |          |
| F found               |          | F disable          |          |
| F get                 |          | F enable           |          |
| F getcasted           |          | F fromregime       |          |
| F getdetail           |          | F list             |          |
| F getdirect           |          | D mapping          |          |
| F getfaster           |          | F pop              |          |
| F getfield            |          | F process          |          |
| F getfuzzy            |          | F push             |          |
| F getindex            |          | F toregime         |          |
| D inspectors          |          | F translate        |          |
| F lastofrange         |          | <b>require</b>     | function |
| F load                |          | <b>resolvers</b>   | table    |
| F loadbibdata         |          | F allprefixes      |          |
| F loaddefinitionfile  |          | F appendhash       |          |
| F loadreplacementfile |          | F automount        |          |
| F new                 |          | F booleanvariable  |          |
| F okay                |          | S cacheversion     |          |
| F oneorrange          |          | F checkedvariable  |          |
| F parenttag           |          | F cleanedpathlist  |          |
| F permitted           |          | D cleaners         |          |
| F prerollcmdstring    |          | F cleanpath        |          |
| F refman_to_btx       |          | F cleanpathlist    |          |
| F resolvedname        |          | F cleanupbinfile   |          |
| F save                |          | D concatinators    |          |

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| S configbanner                 | F gethashes               |
| F configurationfiles           | F getownpath              |
| D dangerous                    | F getreadfilename         |
| F datastate                    | D hashers                 |
| F dirstatus                    | S homedir                 |
| F dowithfilesandreport         | F ignorelibrary           |
| F dowithfilesintree            | F initialized             |
| F dowithpath                   | F installinputfilehandler |
| F dowithvariable               | F installinputlinehandler |
| F expandbraces                 | F jobpath                 |
| F expandedpathfromlist         | F joinpath                |
| F expandedpathlist             | F knownvariables          |
| F expandedpathlistfromvariable | F load                    |
| F expandpath                   | F loadbinfile             |
| F expandpathfromvariable       | D loaders                 |
| F expansion                    | F loadlualib              |
| F extendtexmfvariable          | F loadtexfile             |
| F filtered_from_content        | F loadtime                |
| F find_file                    | F locateformat            |
| F find_files                   | D locators                |
| F findbinfile                  | S luacnffallback          |
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| F findfiles                    | N maxreadlevel            |
| F findgivenfile                | F methodhandler           |
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| F findpk                       | F openfile                |
| F findtexfile                  | F opentexfile             |
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| F findwildcardfiles            | F popextrapath            |
| F formatofsuffix               | F popinputname            |
| F formatofvariable             | F poppath                 |
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| F foundintree                  | F prependhash             |
| F foundintrees                 | F pushextrapath           |
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| F getdetails                   | F registerextrapath       |
| F getenv                       | F registerfile            |
| F getextrapaths                | F registerfilehash        |
| F getfilehashes                | F registerintrees         |

- F registermethod
- F registerzipfile
- D relations
- F renew
- F renewcache
- F repath
- F reset
- F resetextrapaths
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- D savers
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- F scanfiles
- F setdynamic
- F setenv
- F showdetails
- F showpath
- F simplescanfiles
- F splitconfigurationpath
- F splitexpansions
- F splitmethod
- F splitpath
- F stackpath
- F starttiming
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- D suffixes
- D suffixmap
- F suffixofformat
- F texdatablob
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- F uselibrary
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- F usezipfile
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- F disablelibraries
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- F enable
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- F finalizer
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- F logcalls
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- D colors
- F decomposehangul
- F getdata
- F getinjector
- F getsplitter
- F getstatus
- D handlers
- D helpers
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- F reset
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| D splitters              |          | F readcardinal1le   |       |
| <b>select</b>            | function | F readcardinal2     |       |
| <b>serial</b>            | table    | F readcardinal2le   |       |
| F __gc                   |          | F readcardinal3     |       |
| S __name                 |          | F readcardinal3le   |       |
| F __tostring             |          | F readcardinal4     |       |
| F close                  |          | F readcardinal4le   |       |
| F open                   |          | F readcardinaltable |       |
| F send                   |          | F readcline         |       |
| F tostring               |          | F readcstring       |       |
| F write                  |          | F readdouble        |       |
| <b>set</b>               | table    | F readdoublele      |       |
| F contains               |          | F readfixed2        |       |
| F create                 |          | F readfixed4        |       |
| F tolist                 |          | F readfloat         |       |
| F tonumber               |          | F readfloatle       |       |
| F totable                |          | F readfloatle12     |       |
| <b>setinspector</b>      | function | F readfloatle6      |       |
| <b>setmathparameters</b> | function | F readinteger1      |       |
| <b>setmetatable</b>      | function | F readinteger1le    |       |
| <b>sha2</b>              | table    | F readinteger2      |       |
| F HASH256                |          | F readinteger2le    |       |
| F HASH384                |          | F readinteger3      |       |
| F HASH512                |          | F readinteger3le    |       |
| F HEX256                 |          | F readinteger4      |       |
| F HEX384                 |          | F readinteger4le    |       |
| F HEX512                 |          | F readintegertable  |       |
| F digest256              |          | F tocardinal1       |       |
| F digest384              |          | F tocardinal1le     |       |
| F digest512              |          | F tocardinal2       |       |
| F hash256                |          | F tocardinal2le     |       |
| F hash384                |          | F tocardinal3       |       |
| F hash512                |          | F tocardinal3le     |       |
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| F sum256                 |          | N BLOCKSIZE         |       |
| F sum384                 |          | N _SETSIZE          |       |
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| <b>sio</b>               | table    | F bind              |       |
| F read2dot14             |          | F choose            |       |
| F readbytes              |          | F connect           |       |
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| F connect6        |              | F elapsedtime           |              |
| F gettime         |              | B enable                |              |
| F newtry          |              | F formatruntime         |              |
| F protect         |              | F hastiming             |              |
| F report          |              | F memused               |              |
| F select          |              | F register              |              |
| F sink            |              | F resettiming           |              |
| F skip            |              | F runtime               |              |
| F sleep           |              | F savefmtstatus         |              |
| F source          |              | F show                  |              |
| F tcp             |              | F showmemory            |              |
| F tcp4            |              | F showusage             |              |
| F tcp6            |              | F starttiming           |              |
| F try             |              | F stoptiming            |              |
| <b>sorters</b>    | <b>table</b> | N threshold             |              |
| F basicsorter     |              | F timed                 |              |
| D definitions     |              | F tracefunction         |              |
| F firstofsplit    |              | <b>status</b>           | <b>table</b> |
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| F replacementlist |              | F getcallbackstate      |              |
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| F sort            |              | F getdumpstate          |              |
| F strip           |              | F geterrorlinestate     |              |
| D tracers         |              | F geterrorstate         |              |
| F update          |              | F getexpandstate        |              |
| <b>sparse</b>     | <b>table</b> | F getfilestate          |              |
| F concat          |              | F getfontstate          |              |
| F get             |              | F gethalferrorlinestate |              |
| F max             |              | F gethashstate          |              |
| F min             |              | F gethyphenationstate   |              |
| F new             |              | F getinputstate         |              |
| F range           |              | F getinsertstate        |              |
| F restore         |              | F getlanguagestate      |              |
| F set             |              | F getlinebreakstate     |              |
| F traverse        |              | F getlookupstate        |              |
| F wipe            |              | F getluastate           |              |
| <b>statistics</b> | <b>table</b> | F getmarkstate          |              |
| F benchmarktimer  |              | F getmvlstate           |              |
| F checkfmtstatus  |              | F getneststate          |              |
| F currenttime     |              | F getnodestate          |              |
| F elapsed         |              | F getparameterstate     |              |
| F elapsedindeed   |              | F getpoolstate          |              |
| F elapsedseconds  |              | F getreadstate          |              |

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| F getsavestate    | F chrtointeger    |
| F getsparsestate  | F collapsespaces  |
| F getstringstate  | F containsws      |
| F gettexstate     | F count           |
| F gettokenstate   | F dectointeger    |
| F getwarningstate | F dump            |
| F list            | F escapedpattern  |
| F resetmessages   | F escapedquotes   |
| <b>storage</b>    | F f6              |
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| F mark            | F format          |
| F marked          | F formatter       |
| N max             | D formatters      |
| N min             | F fullstrip       |
| N nofmodules      | F gmatch          |
| N noftables       | F gsub            |
| F register        | F hextocharacters |
| D shared          | F hextointeger    |
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| N tofmodules      | F itself          |
| N toftables       | F len             |
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| F disable         | F longtostring    |
| F enable          | F lower           |
| F flush           | F match           |
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| F push            | F octtointeger    |
| F start           | F optionalquoted  |
| F stop            | F pack            |
| F synchronize     | F packrowscolumns |
| <b>string</b>     | F packsize        |
| F autodouble      | F quote           |
| F autosingle      | F quoted          |
| F booleanstring   | F rep             |
| F byte            | F replacenewlines |
| F bytepairs       | F reverse         |
| F bytes           | F split           |
| F bytetable       | F splitintolines  |
| F char            | F splitlines      |
| F characterpairs  | F splitup         |
| F characters      | F strip           |
| F checkedsplit    | F sub             |

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| F texnewlines          | F concat               |
| F tformat              | F contains             |
| F toHEX                | F copy                 |
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| F todec                | F deserialize          |
| F todimen              | F fastcopy             |
| F tohex                | F fastserialize        |
| F tonodes              | F filtered             |
| F topattern            | F flattened            |
| F totable              | F fromhash             |
| F tounicode16          | F getn                 |
| F toutf                | F has_one_entry        |
| F toutf16              | F hashed               |
| F toutf32              | F identical            |
| F toutf8               | F imerge               |
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| F unpack               | F is_empty             |
| F unquote              | F is_simple_table      |
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| F utfcharacter         | F makeweak             |
| F utfcharacters        | F merge                |
| F utfcharacterstable   | F merged               |
| F utflength            | F mirrored             |
| F utfpadd              | F move                 |
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| F utfvalue             | F print                |
| F utfvalues            | F remove               |
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| F currentsectionnumber | F setmetatableindex    |
| <b>table</b>           | F setmetatableindices  |
|                        | F setmetatablenewindex |
| F append               | F sort                 |
| F are_equal            |                        |
| F combine              |                        |

|                   |                           |
|-------------------|---------------------------|
| F sorted          | D discoptioncodes         |
| F sortedhash      | D discpartcodes           |
| F sortedhashkeys  | F dontcomplain            |
| F sortedhashonly  | D doublescriptoptioncodes |
| F sortedindexonly | F enableprimitives        |
| F sortedkeys      | F equal                   |
| F sortedpairs     | F error                   |
| F sparse          | F expandasvalue           |
| F strip           | D experiments             |
| F sub             | F extraprimitives         |
| F swapped         | F fatalerror              |
| F tocontext       | D fillcodes               |
| F tocsv           | D flagcodes               |
| F tofile          | F fontidentifier          |
| F tohandle        | F fontname                |
| F tohash          | F forcehmode              |
| F toxml           | F forceinputstatefile     |
| F unique          | F forceinputstateline     |
| F unpack          | F fprint                  |

## tex table

|                         |                                 |
|-------------------------|---------------------------------|
| F attributetracking     | F get                           |
| D automigrationcodes    | F getadjustoptionvalues         |
| D autoparagraphcodes    | F getalignmentcontextvalues     |
| F badness               | F getamcode                     |
| D badnessmodecodes      | F getappendadjustcontextvalues  |
| F balance               | F getappendmigratecontextvalues |
| D balancecallbackcodes  | F getattribute                  |
| D boxcontinuationcodes  | F getautomigrationvalues        |
| D boxoptioncodes        | F getautoparagraphvalues        |
| D breakcodes            | F getbadnessmodevalues          |
| D buildcodes            | F getbalancecallbackvalues      |
| D charactercontrolcodes | F getbalanceparameterfields     |
| D charactertagcodes     | F getbalanceresultfields        |
| F chardef               | F getbalancestepoptionvalues    |
| D classoptioncodes      | F getbox                        |
| D conditionals          | F getboxcontinuationvalues      |
| D constants             | F getboxdir                     |
| F cprint                | F getboxoptionvalues            |
| N ctdcatcodes           | F getboxresourcebox             |
| N ctxcatcodes           | F getboxresourcedimensions      |
| F definefont            | F getbreakcontextvalues         |
| F dimensiondef          | F getbuildcontextvalues         |
| D directioncodes        | F getcardinalvalue              |
| D directives            | F getcatcode                    |



|                               |                               |
|-------------------------------|-------------------------------|
| F getcccode                   | F getinsertheight             |
| F getcharactercontrolvalues   | F getinsertlimit              |
| F getcharacteritagvalues      | F getinsertmaxplaced          |
| F getcount                    | F getinsertmultiplier         |
| F getcurrentmvl               | F getinsertoptionvalues       |
| F getdelcode                  | F getinsertsplitvalues        |
| F getdelcodes                 | F getinsertwidth              |
| F getdimen                    | F getintegervalue             |
| F getdimensionvalue           | F getinteraction              |
| F getdirectionvalues          | F getinteractionmodes         |
| F getdiscoptionvalues         | F getiovalues                 |
| F getdiscpartvalues           | F getkerneloptionvalues       |
| F getdoublescriptoptionvalues | F getkernoptionvalues         |
| F getemptyparagraphmodevalues | F getlargestusedmark          |
| F geterrorrecoverymodevalues  | F getlccode                   |
| F geterrorvalues              | F getlinebreakparameterfields |
| F getfillvalues               | F getlinebreakresultfields    |
| F getflagvalues               | F getlinebreakstatevalues     |
| F getfloat                    | F getlinedir                  |
| F getfontdimension            | F getlist                     |
| F getfontoffamily             | F getlistanchorvalues         |
| F getfrozenparvalues          | F getlistfields               |
| F getglue                     | F getlistgeometryvalues       |
| F getglueoptionvalues         | F getlistsignvalues           |
| F getglyphdata                | F getlocalbox                 |
| F getglyphdiscvalues          | F getlocalboxlocations        |
| F getglyphoptionvalues        | F getlocallevel               |
| F getglyphprotectionvalues    | F getmark                     |
| F getglyphscales              | F getmarknames                |
| F getglyphscript              | F getmath                     |
| F getglyphstate               | F getmathclassoptionvalues    |
| F getgroupvalues              | F getmathcode                 |
| F gethccode                   | F getmathcodes                |
| F gethelptext                 | F getmathcontrolvalues        |
| F gethmcode                   | F getmathdir                  |
| F gethyphenationvalues        | F getmathgluevalues           |
| F getiftypes                  | F getmathoptionvalues         |
| F getinputstatefile           | F getmathparametervalues      |
| F getinputstateline           | F getmathscriptordervalue     |
| F getinputstatemode           | F getmathscriptsmodevalues    |
| F getinsertcategoryvalues     | F getmathstylenamevalues      |
| F getinsertcontent            | F getmathstylevalues          |
| F getinsertdepth              | F getmathstylevariant         |
| F getinsertdistance           | F getmathsurroundvalues       |

|                               |                                |
|-------------------------------|--------------------------------|
| F getmathvariantpresets       | F getskip                      |
| F getmathvariantvalues        | F getsnappingvalues            |
| F getmeasure                  | F getspacefactorvalues         |
| F getmissingcharactervalues   | F getspacevalues               |
| F getmode                     | F getspacevalues               |
| F getmodevalues               | F getspecialmathclassvalues    |
| F getmuglue                   | F getspecification             |
| F getmuskip                   | F getspecificationoptionvalues |
| F getmvloptionvalues          | F gettextcontrolvalues         |
| F getnest                     | F gettextdir                   |
| F getnestfields               | F gettoks                      |
| F getnestlevel                | F getuccode                    |
| F getnoadoptoptionvalues      | F getuleaderlocationvalues     |
| F getnormalizelinevalues      | F getunitclassvalues           |
| F getnormalizeparvalues       | F getusedmarks                 |
| F getnospacesmodevalues       | F getvsplitcontextvalues       |
| F getoutputactive             | D glueoptioncodes              |
| F getoverloadstate            | D glyphdiscodes                |
| F getpacktypevalues           | D glyphoptioncodes             |
| F getpagecontextvalues        | D groupcodes                   |
| F getpagestate                | F hashtokens                   |
| F getpagestatevalues          | N hshcatcodes                  |
| F getparagraphfillmodevalues  | D hyphenationcodes             |
| F getparagraphoptionvalues    | D ifs                          |
| F getparametermodevalues      | N inicatcodes                  |
| F getparcontextvalues         | D insertcategorycodes          |
| F getpardataspecifications    | D insertoptioncodes            |
| F getpardir                   | D insertsplitletters           |
| F getparmodevalues            | F integerdef                   |
| F getparstate                 | D interactioncodes             |
| F getparstatefields           | F isattribute                  |
| F getpartriggervalue          | F isbox                        |
| F getpenaltyoptionvalues      | F iscount                      |
| F getpositvalue               | F isdefined                    |
| F getprepoststatevalues       | F isdimen                      |
| F getprimitivelegacies        | F isfloat                      |
| F getprimitiveorigins         | F isglue                       |
| F getprotrusionboundaryvalues | F ismuglue                     |
| F getregisterindex            | F ismuskip                     |
| F getruleoptionvalues         | F isprintable                  |
| F getrunstate                 | F isskip                       |
| F getrunstatevalues           | F istoks                       |
| F getsfcode                   | F lastnodetype                 |
| F getshapingpenaltiesvalues   | F less                         |

|                           |                       |
|---------------------------|-----------------------|
| F lessequal               | D ruleoptioncodes     |
| F linebreak               | F runlocal            |
| D linebreakstatecodes     | D runstatecodes       |
| D listanchorcodes         | F runstring           |
| D listgeometrycodes       | F runtoks             |
| D listsigncodes           | F saveboxresource     |
| N luacatcodes             | F scale               |
| F mathchardef             | F scantoks            |
| D mathcontrolcodes        | F set                 |
| D mathoptioncodes         | F setamcode           |
| D mathparametercodes      | F setattribute        |
| D mathscriptordercodes    | F setbalanceshape     |
| D mathscriptsmodecodes    | F setbox              |
| D modes                   | F setboxdir           |
| F mprint                  | F setcardinalvalue    |
| D mvloptioncodes          | F setcatcode          |
| N nilcatcodes             | F setcccode           |
| D noadoptioncodes         | F setconditional      |
| D normalizelinecodes      | F setcount            |
| D normalizeparcodes       | F setdelcode          |
| D nospacesmodecodes       | F setdimen            |
| N notcatcodes             | F setdimensionvalue   |
| F number                  | F setfloat            |
| D paragraphfillmodecodes  | F setglue             |
| D paragraphoptioncodes    | F setglyphdata        |
| D parcontextcodes         | F setglyphscript      |
| D partriggercodes         | F setglyphstate       |
| D penaltyoptioncodes      | F sethccode           |
| F poplocal                | F sethmcodes          |
| F popsavelevel            | F setinputstatefile   |
| F positdef                | F setinputstateline   |
| F preparelinebreak        | F setinputstatemode   |
| F primitives              | F setinsertcontent    |
| F print                   | F setinsertdistance   |
| D protrusionboundarycodes | F setinsertlimit      |
| N prtcatcodes             | F setinsertmaxplaced  |
| F pushlocal               | F setinsertmultiplier |
| F pushsavelevel           | F setintegervalue     |
| F quitlocal               | F setinteraction      |
| F quittoks                | F setlccode           |
| F resetparagraph          | F setlinedir          |
| N rlnatcodes              | F setlist             |
| F romannumeral            | F setlocalbox         |
| F round                   | F setmath             |

F setmathcode  
 F setmathdir  
 F setmuglue  
 F setmuskip  
 F setnest  
 F setoverloadstate  
 F setpardir  
 F setpositvalue  
 F setrunstate  
 F setsfcode  
 F setskip  
 F setspcode  
 F settextrdir  
 F settoks  
 F setuccode  
 D shapingpenaltiescodes  
 F shiftparshape  
 F shipout  
 F showcontext  
 D snappingcodes  
 F snapshotpar  
 F sp  
 D spacefactorcodes  
 D spaceskipcodes  
 D specialmathclasscodes  
 D specificationoptioncodes  
 F splitbox  
 D splitcodes  
 F sprint  
 F stringtocodesbitmap  
 D systemmodes  
 F takebox  
 N texcatcodes  
 D textcontrolcodes  
 F tonumber  
 F toscaled  
 N tpatcatcodes  
 N tpbcatcodes  
 F tprint  
 F trackedattribute  
 D trackers  
 F triggerbuildpage  
 N txtcatcodes  
 F uprint

F useboxresource  
 D userdiscoptioncodes  
 D userglyphoptioncodes  
 N vrbcatcodes  
 F write

## **texconditionals**

data table

## **texconfig**

table

S firstline  
 S formatname  
 N functionsize  
 F init  
 B kpse\_init  
 N propertiessize  
 S shell\_escape

## **texconstants**

data table

## **texifs**

data table

## **texio**

table

F closeinput  
 F forceendoffile  
 F getinputindex  
 F getselectorvalues  
 F getsourcefilename  
 F setlogfile  
 F terminal  
 F write  
 F write\_nl  
 F writenl  
 F writeselector  
 F writeselectorlf  
 F writeselectornl

## **texisdefined**

data table

## **texmodes**

data table

## **texsystemmodes**

data table

## **thirddata**

table

## **toboolean**

function

## **todecimal**

function

## **token**

table

F create  
 F expand  
 F expand\_macro  
 F expandmacro  
 F future\_expand  
 F futureexpand  
 F get\_active

|                     |                      |
|---------------------|----------------------|
| F get_cmdchracs     | F getinteger         |
| F get_cmdname       | F getlength          |
| F get_command       | F getmacro           |
| F get_csname        | F getmeaning         |
| F get_cstoken       | F getnoaligned       |
| F get_dimension     | F getparameters      |
| F get_expandable    | F getpermanent       |
| F get_fields        | F getprimitive       |
| F get_flags         | F getprimitives      |
| F get_frozen        | F getprotected       |
| F get_id            | F getrange           |
| F get_immutable     | F gettok             |
| F get_index         | F gettolerant        |
| F get_instance      | F gobble             |
| F get_integer       | F gobble_dimension   |
| F get_macro         | F gobble_integer     |
| F get_meaning       | F gobbledimension    |
| F get_noaligned     | F gobbleinteger      |
| F get_parameters    | F grab               |
| F get_permanent     | F grabstring         |
| F get_primitive     | F inhibit            |
| F get_protected     | F is_defined         |
| F get_range         | F is_next_char       |
| F get_tok           | F is_token           |
| F get_tolerant      | F isactivestring     |
| F getactive         | F isdefined          |
| F getcmdchracs      | F isnextchar         |
| F getcmdname        | F istoken            |
| F getcommand        | F locatemacro        |
| F getcommandvalues  | F new                |
| F getconstant       | F peek_next          |
| F getcsname         | F peek_next_char     |
| F getcstoken        | F peek_next_expanded |
| F getdimension      | F peeknext           |
| F getexpandable     | F peeknextchar       |
| F getexpansion      | F peeknextexpanded   |
| F getfields         | F pop_macro          |
| F getflags          | F popmacro           |
| F getfrozen         | F push_macro         |
| F getfunctionvalues | F pushmacro          |
| F getid             | F put_back           |
| F getimmutable      | F put_next           |
| F getindex          | F putback            |
| F getinstance       | F putnext            |

|                        |                         |
|------------------------|-------------------------|
| F save_lua             | F scandelimited         |
| F savelua              | F scandetokened         |
| F scan_argument        | F scandimen             |
| F scan_box             | F scandimenargument     |
| F scan_cardinal        | F scandimension         |
| F scan_char            | F scandimensionargument |
| F scan_cmdchr          | F scanfloat             |
| F scan_cmdchr_expanded | F scanglue              |
| F scan_code            | F scanint               |
| F scan_csname          | F scaninteger           |
| F scan_delimited       | F scanintegerargument   |
| F scan_dimen           | F scankey               |
| F scan_float           | F scankeyword           |
| F scan_glue            | F scankeywordcs         |
| F scan_integer         | F scanletters           |
| F scan_key             | F scanluacardinal       |
| F scan_keyword         | F scanluainteger        |
| F scan_keyword_cs      | F scanluanumber         |
| F scan_letters         | F scannext              |
| F scan_luacardinal     | F scannextchar          |
| F scan_luainteger      | F scannextexpanded      |
| F scan_luanumber       | F scanposit             |
| F scan_next            | F scanreal              |
| F scan_next_char       | F scanscale             |
| F scan_next_expanded   | F scanskip              |
| F scan_real            | F scanstring            |
| F scan_scale           | F scantoken             |
| F scan_skip            | F scantokencode         |
| F scan_string          | F scantokenlist         |
| F scan_token           | F scantokenstring       |
| F scan_token_code      | F scantoks              |
| F scan_tokenlist       | F scanvalue             |
| F scan_toks            | F scanword              |
| F scan_value           | F serialize             |
| F scan_word            | F set_dimension         |
| F scanargument         | F set_integer           |
| F scanbox              | F set_lua               |
| F scancardinal         | F set_macro             |
| F scanchar             | F setdimension          |
| F scancmdchr           | F setinteger            |
| F scancmdchrexpanded   | F setlua                |
| F scancode             | F setmacro              |
| F scancsname           | F setmacrofrommark      |
| F scancstoken          | F setmacrotonothing     |

|                      |          |                    |          |
|----------------------|----------|--------------------|----------|
| F skip_next          |          | F savehash         |          |
| F skip_next_expanded |          | F show             |          |
| F skipnext           |          | F value            |          |
| F skipnextexpanded   |          | F values           |          |
| F type               |          | <b>type</b>        | function |
| F undefine_macro     |          | <b>typesetters</b> | table    |
| F undefinemacro      |          | F marksuspects     |          |
| <b>tokens</b>        | table    | F showinvisibles   |          |
| F astable            |          | F showsuspects     |          |
| D cache              |          | <b>unpack</b>      | function |
| D commands           |          | <b>updaters</b>    | table    |
| F compile            |          | F apply            |          |
| F constant           |          | F register         |          |
| F create             |          | <b>url</b>         | table    |
| F defined            |          | F addscheme        |          |
| F flags              |          | F barepath         |          |
| F getinteger         |          | F construct        |          |
| F gobble             |          | F decode           |          |
| F isactivestring     |          | F encode           |          |
| F isdefined          |          | F escape           |          |
| F istoken            |          | F filename         |          |
| F new                |          | F hashed           |          |
| F pickup             |          | F hasscheme        |          |
| F popmacro           |          | F query            |          |
| F pushmacro          |          | F split            |          |
| F setinteger         |          | F toquery          |          |
| D values             |          | F unescape         |          |
| <b>tonumber</b>      | function | F unescapeget      |          |
| <b>tostring</b>      | function | <b>userdata</b>    | table    |
| <b>tracers</b>       | table    | <b>utf</b>         | table    |
| F printererror       |          | F byte             |          |
| F printwarning       |          | F bytepairs        |          |
| F showlines          |          | F char             |          |
| <b>trackers</b>      | table    | F characters       |          |
| D data               |          | F chrln            |          |
| F default            |          | F count            |          |
| F disable            |          | F filetype         |          |
| F dumphashtofile     |          | F is_valid         |          |
| F enable             |          | F len              |          |
| F list               |          | F length           |          |
| S name               |          | F lower            |          |
| F register           |          | F remapper         |          |
| F report             |          | F replacer         |          |
| F reset              |          | F split            |          |

F splitlines  
F sub  
F substituter  
F tocodes  
F toeight  
F toentities  
F toString  
F totable  
F upper  
F ustring  
F utf16\_to\_utf8\_be  
F utf16\_to\_utf8\_be\_t  
F utf16\_to\_utf8\_le  
F utf16\_to\_utf8\_le\_t  
F utf16\_to\_utf8\_t  
F utf32\_to\_utf8\_be  
F utf32\_to\_utf8\_be\_t  
F utf32\_to\_utf8\_le  
F utf32\_to\_utf8\_le\_t  
F utf32\_to\_utf8\_t  
F utf8\_to\_utf16  
F utf8\_to\_utf16\_be  
F utf8\_to\_utf16\_le  
F utf8\_to\_utf8\_t  
F values  
F xstring

## **utf8** table

F char  
S charpattern  
F codepoint  
F codes  
F len  
F offset

## **utilities** table

## **vector** table

F add  
F append  
F average  
F ceiling  
F checkoverlap  
F collectoverlap  
F concat  
F copy  
F crossproduct

F delete  
F determinant  
F div  
F equal  
F exchange  
F floor  
F get  
F getdimensions  
F getepsilon  
F getmemory  
F getmesh  
F getmeshtypes  
F getpointtypes  
F getrc  
F getrow  
F getsizes  
F getstacking  
F gettype  
F homogenize  
F identity  
F inner  
F insert  
F inverse  
F ismesh  
F ispoints  
F issingular  
F isvector  
F iszero  
F meshname  
F mul  
F negate  
F new  
F normalize  
F onecolumn  
F onerow  
F product  
F remove  
F replace  
F round  
F rowechelon  
F set  
F setepsilon  
F setnext  
F setrc



|                         |                 |                 |              |
|-------------------------|-----------------|-----------------|--------------|
| F setstacking           |                 | F __eq          |              |
| F slice                 |                 | F __mul         |              |
| F sort                  |                 | S __name        |              |
| F sub                   |                 | F __pow         |              |
| F swap                  |                 | F __sub         |              |
| F tostring              |                 | F __tostring    |              |
| F totable               |                 | F __unm         |              |
| F transpose             |                 | F abs           |              |
| F triangulate           |                 | F acos          |              |
| F truncate              |                 | F acosh         |              |
| F type                  |                 | F arg           |              |
| <b>visualizers</b>      | <b>table</b>    | F asin          |              |
| D escapepatterns        |                 | F asinh         |              |
| F getspecification      |                 | F atan          |              |
| F getvisualizer         |                 | F atanh         |              |
| F load                  |                 | F conj          |              |
| F makenested            |                 | F cos           |              |
| F makepattern           |                 | F cosh          |              |
| F newgrammar            |                 | F dawson        |              |
| F newhandler            |                 | F erf           |              |
| F parsedxml             |                 | F erfc          |              |
| F pattern               |                 | F erfcx         |              |
| F register              |                 | F erfi          |              |
| F registerescapecommand |                 | F exp           |              |
| F registerescapeline    |                 | F i             |              |
| F registerescapepattern |                 | F imag          |              |
| S signal                |                 | F log           |              |
| U signalpattern         |                 | F new           |              |
| D specifications        |                 | F pow           |              |
| F visualize             |                 | F proj          |              |
| F visualizebuffer       |                 | F real          |              |
| F visualizefile         |                 | F sin           |              |
| F visualizestring       |                 | F sinh          |              |
| F write                 |                 | F sqrt          |              |
| F writeargument         |                 | F tan           |              |
| F writedefault          |                 | F tanh          |              |
| F writeemptyline        |                 | F topair        |              |
| F writenewline          |                 | F tostring      |              |
| F writespace            |                 | F totable       |              |
| <b>warn</b>             | <b>function</b> | F voigt         |              |
| <b>xcomplex</b>         | <b>table</b>    | F voigt_hwhm    |              |
| F I                     |                 | <b>xdecimal</b> | <b>table</b> |
| F __add                 |                 | F __add         |              |
| F __div                 |                 | F __band        |              |

F \_\_bor  
 F \_\_bxor  
 F \_\_div  
 F \_\_eq  
 F \_\_idiv  
 F \_\_le  
 F \_\_lt  
 F \_\_mod  
 F \_\_mul  
 S \_\_name  
 F \_\_pow  
 F \_\_shl  
 F \_\_shr  
 F \_\_sub  
 F \_\_tostring  
 F \_\_unm  
 F abs  
 F band  
 F bor  
 F bxor  
 F conj  
 F copy  
 F exp  
 F getprecision  
 F ln  
 F log  
 F max  
 F min  
 F minus  
 F new  
 F plus  
 F pow  
 F rotate  
 F setprecision  
 F shift  
 F sqrt  
 F toengstring  
 F tonumber  
 F tostring  
 F trim

## xinterval

F \_\_add  
 F \_\_div  
 F \_\_eq

## table

F \_\_le  
 F \_\_lt  
 F \_\_mod  
 F \_\_mul  
 S \_\_name  
 F \_\_pow  
 F \_\_sub  
 F \_\_tostring  
 F \_\_unm  
 F abs  
 F acos  
 F acosh  
 F asin  
 F asinh  
 F asstring  
 F atan  
 F atanh  
 F cos  
 F cosh  
 F diameter  
 F equal  
 F erf  
 F erfc  
 F exp  
 F exp10  
 F exp2  
 F expm1  
 F greater  
 F greaterequal  
 F hull  
 F intersect  
 F less  
 F lessequal  
 F log  
 F log10  
 F log1p  
 F log2  
 F mid  
 F mod  
 F new  
 F neweps  
 F pow  
 F seteq  
 F setge

|                |                      |
|----------------|----------------------|
| F setgt        | F hypot              |
| F setle        | N inf                |
| F setlt        | F isfinite           |
| F sin          | F isinf              |
| F sinh         | F isnan              |
| F sqr          | F isnormal           |
| F sqrt         | F j0                 |
| F tan          | F j1                 |
| F tanh         | F jn                 |
| F topair       | F ldexp              |
| F tostring     | F lgamma             |
| F totable      | F log                |
| F usedendian   | F log10              |
| F within       | F loglp              |
| <b>xmath</b>   | F log2               |
|                | F logb               |
| F acos         | F modf               |
| F acosh        | N nan                |
| F asin         | F nearbyint          |
| F asinh        | F nextafter          |
| F atan         | F normalize          |
| F atan2        | N pi                 |
| F atanh        | F pow                |
| F cbrt         | F rad                |
| F ceil         | F remainder          |
| F copysign     | F remquo             |
| F cos          | F round              |
| F cosh         | F scalbn             |
| F crossproduct | F sin                |
| F deg          | F sinh               |
| F dotproduct   | F sqrt               |
| F erf          | F tan                |
| F erfc         | F tanh               |
| F exp          | F tgamma             |
| F exp2         | F trunc              |
| F expm1        | F y0                 |
| F fabs         | F y1                 |
| F fdim         | F yn                 |
| F floor        | <b>xml</b>           |
| F fma          | <b>table</b>         |
| F fmax         | F addentitiesdoctype |
| F fmin         | F addspecialcode     |
| F fmod         | F after              |
| F frexp        | F all                |
| F gamma        | F applylpath         |

|                    |                         |
|--------------------|-------------------------|
| F applyselector    | F getbuffer             |
| F assign           | F gethandlers           |
| F attribute        | F gethandlersfunction   |
| F badinclusions    | F include               |
| F before           | F inclusion             |
| F body             | F inclusions            |
| F cdata            | F index                 |
| F cdatatotext      | F inheritattributes     |
| F checkbom         | F inheritedconvert      |
| F checkedempty     | F inject                |
| F checkerror       | F inject_element        |
| F checknamespace   | F inject_element_after  |
| F checkns          | F inject_element_before |
| F cleansed         | F injectafter           |
| F collect          | F injectbefore          |
| F collect_elements | F insert                |
| F collect_tags     | F insert_element        |
| F collect_texts    | F insert_element_after  |
| F collected        | F insert_element_before |
| F collecttexts     | F insertafter           |
| F compactelement   | F insertbefore          |
| F content          | F insertcdata           |
| F convert          | F insertcomment         |
| F copy             | F inspect               |
| F count            | F is_valid              |
| F cprint           | F kind                  |
| S defaultprotocol  | F last                  |
| F delete           | F lastmatch             |
| F delete_element   | F load                  |
| F despecialized    | F lpath                 |
| F each             | F lpathcached           |
| F each_element     | F lpathcalls            |
| F elements         | F lshow                 |
| F elementtocdata   | F makestandalone        |
| F empty            | F match                 |
| D entities         | F name                  |
| F erase            | F newhandlers           |
| F errorhandler     | F nodesettostring       |
| F escaped          | F originalload          |
| F expand           | F package               |
| F fillin           | F parent                |
| F filter           | U parsedentitylpeg      |
| F first            | F popmatch              |
| F found            | F position              |

D privatecodes  
 F privatetoken  
 F process  
 F process\_attributes  
 F process\_elements  
 F processattributes  
 F pure  
 F pushmatch  
 F raw  
 F registerentity  
 F registersns  
 F reindex  
 F remap\_name  
 F remap\_namespace  
 F remap\_tag  
 F remapname  
 F remapnamespace  
 F remaptag  
 F rename  
 F renamespace  
 U reparsedentitylpeg  
 F replace  
 F replace\_element  
 F resetinjectors  
 F resolvns  
 F root  
 F save  
 F selection  
 F separate  
 F serialize  
 F serializetotext  
 F setcdata  
 F setcomment  
 F sethandlersfunction  
 F setinjectors  
 F setproperty  
 F settings  
 D specialcodes  
 F sprint  
 F string  
 F strip  
 F strip\_whitespace  
 F stripelement  
 F stripleadingspaces

F table  
 F text  
 F textonly  
 F texttocdata  
 F tocdata  
 F toelement  
 F toString  
 F totable  
 F toxml  
 F tprint  
 F traverse  
 F unescaped  
 U unescapedentitylpeg  
 F unprivatized  
 F unspecialized  
 F wipe  
 F withelement  
 F withelements  
 F wrap

## xpcall

function

## xzip

table

F Adler32  
 F compress  
 F compresssize  
 F crc32  
 F decompress  
 F decompresssize

## zbuffer

table

F bounds  
 F composite  
 F crop  
 F edgepoint  
 F flatten  
 F get  
 F getcolor  
 F getconstants  
 F getdepth  
 F getnormal  
 F getnormalmodes  
 F getopacity  
 F getsizes  
 F implicit  
 F iszbuffer  
 F meshmesh

F new  
F points  
F process  
F project  
F resolve  
F set  
F setcolor  
F setdepth  
F setnormal  
F setopacity  
F setup  
F smoothen  
F stipple\_0  
F stipple\_1  
F stipple\_2  
F stipple\_3  
F stipple\_4  
F stipple\_setup

F tobytes  
F tostring  
F totable  
F transform  
F trianglebounds  
F triangled  
F triangles  
F type

## zip

table

F closearchive  
F openarchive

## zlib

table

F adler32  
F compress  
F compresssize  
F crc32  
F decompress  
F decompresssize

# globals (context)

|                            |          |                    |       |
|----------------------------|----------|--------------------|-------|
| <b>CONTEXT</b>             | number   | D values           |       |
| <b>LUAFORMAT</b>           | number   | <b>backends</b>    | table |
| <b>LUAMAJORVERSION</b>     | number   | F closeout         |       |
| <b>LUAMINORVERSION</b>     | number   | S current          |       |
| <b>LUATEXENGINE</b>        | string   | F getcallbackstate |       |
| <b>LUATEXFORMATID</b>      | number   | F initialize       |       |
| <b>LUATEXFUNCTIONALITY</b> | number   | F latelua          |       |
| <b>LUATEXVERBOSE</b>       | string   | F nothing          |       |
| <b>LUATEXVERSION</b>       | number   | F openout          |       |
| <b>LUAVERSION</b>          | number   | F setencryption    |       |
| <b>MP</b>                  | function | F writeout         |       |
| <b>SAVEDREQUIRE</b>        | function | <b>basexx</b>      | table |
| <b>aes</b>                 | table    | F decode16         |       |
| F decode                   |          | F decode64         |       |
| F encode                   |          | F decode85         |       |
| F random                   |          | F decodeLZW        |       |
| <b>arg</b>                 | table    | F decodeRL         |       |
| <b>assert</b>              | function | F encode16         |       |
| <b>attributes</b>          | table    | F encode64         |       |
| F applyoverloads           |          | F encode85         |       |
| F checkvalues              |          | F encodeLZW        |       |
| D counts                   |          | F encodeRL         |       |
| F define                   |          | <b>bibtex</b>      | table |
| F getvalue                 |          | <b>bit32</b>       | table |
| F getvalues                |          | F arshift          |       |
| F hasvalues                |          | F band             |       |
| D list                     |          | F bnot             |       |
| F markasoverload           |          | F bor              |       |
| D names                    |          | F btest            |       |
| D numbers                  |          | F bxor             |       |
| F ofnode                   |          | F extract          |       |
| F private                  |          | F lrotate          |       |
| F public                   |          | F lshift           |       |
| F registervalue            |          | F replace          |       |
| F restore                  |          | F rrotate          |       |
| F save                     |          | F rshift           |       |
| F setcleaner               |          | <b>bitset</b>      | table |
| F showcurren               |          | F asstring         |       |
| F system                   |          | F get              |       |
| F tooverloads              |          | F new              |       |
| N unsetvalue               |          | F reset            |       |

|                   |              |                            |              |
|-------------------|--------------|----------------------------|--------------|
| F set             |              | F parpassidentifier        |              |
| F totable         |              | D passnames                |              |
| F traverse        |              | F registerpasspasscallback |              |
| F wipe            |              | <b>bytemap</b>             | <b>table</b> |
| <b>blobs</b>      | <b>table</b> | F bounds                   |              |
| F append          |              | F clip                     |              |
| F dimensions      |              | F copy                     |              |
| F dispose         |              | F dimensions               |              |
| F new             |              | F domain                   |              |
| F pack            |              | F downgrade                |              |
| F strdp           |              | F downsample               |              |
| F strhd           |              | F fillgray                 |              |
| F strht           |              | F fillrgb                  |              |
| F strwd           |              | F filter                   |              |
| F tohpack         |              | F getbyte                  |              |
| F tonodes         |              | F getbytes                 |              |
| F tovpack         |              | F getluminance             |              |
| F write           |              | F getvalue                 |              |
| <b>boolean</b>    | <b>table</b> | F hasbytegray              |              |
| F tocontext       |              | F hasbyterange             |              |
| F tonumber        |              | F hasbytergb               |              |
| <b>buffers</b>    | <b>table</b> | F new                      |              |
| F append          |              | F newdomain                |              |
| F assign          |              | F overlay                  |              |
| F collectcontent  |              | F process                  |              |
| F empty           |              | F reduce                   |              |
| F erase           |              | F reset                    |              |
| F exists          |              | F setgray                  |              |
| F get             |              | F setgrayadd               |              |
| F getcontent      |              | F setgraymin               |              |
| F getctxlua       |              | F setrgb                   |              |
| F getlines        |              | F slicegray                |              |
| F getmkvi         |              | F slicerange               |              |
| F gettex          |              | F slicergb                 |              |
| F loadcontent     |              | F totable                  |              |
| F prepend         |              | <b>caches</b>              | <b>table</b> |
| F raw             |              | S base                     |              |
| F run             |              | F collapsecontent          |              |
| F samplefile      |              | F configfiles              |              |
| F splitsamplefile |              | F contentstate             |              |
| F undent          |              | F getfirstreadablefile     |              |
| <b>builders</b>   | <b>table</b> | F getreadablepaths         |              |
| F checkparcontext |              | F getwritablepath          |              |
| F checksidefloat  |              | F hashed                   |              |



|                        |                 |
|------------------------|-----------------|
| F is_writable          |                 |
| F loadcontent          |                 |
| F loaddata             |                 |
| S more                 |                 |
| F savecontent          |                 |
| F savedata             |                 |
| F setfirstwritablefile |                 |
| F setluanames          |                 |
| F treehash             |                 |
| F usedpaths            |                 |
| <b>callback</b>        | <b>table</b>    |
| F find                 |                 |
| F getcontracts         |                 |
| F getcontractvalues    |                 |
| F getindex             |                 |
| F getoptions           |                 |
| F getoptionvalues      |                 |
| F getstate             |                 |
| F getstatevalues       |                 |
| F known                |                 |
| F list                 |                 |
| F names                |                 |
| F register             |                 |
| F setoptions           |                 |
| F setstate             |                 |
| F testonly             |                 |
| F usage                |                 |
| <b>callbacks</b>       | <b>table</b>    |
| F freeze               |                 |
| F getcomment           |                 |
| F getstates            |                 |
| F known                |                 |
| F optimizer            |                 |
| F pop                  |                 |
| F push                 |                 |
| F register             |                 |
| F report               |                 |
| <b>catcodes</b>        | <b>table</b>    |
| D names                |                 |
| D numbers              |                 |
| F register             |                 |
| <b>characterkern</b>   | <b>function</b> |
| <b>characters</b>      | <b>table</b>    |
| F addcharacternames    |                 |
| F adobename            |                 |
| D bidi                 |                 |
| F blockrange           |                 |
| D blocks               |                 |
| D can_have_space       |                 |
| D categories           |                 |
| F category             |                 |
| D categorytags         |                 |
| D collapsed            |                 |
| D combined             |                 |
| F contextname          |                 |
| D csletters            |                 |
| D data                 |                 |
| D decomposed           |                 |
| F description          |                 |
| D descriptions         |                 |
| D detailtags           |                 |
| F emojiized            |                 |
| D encodes              |                 |
| D entities             |                 |
| D fallbacks            |                 |
| D filters              |                 |
| F fromnumber           |                 |
| D fschars              |                 |
| D fscodes              |                 |
| F getrange             |                 |
| D graphemes            |                 |
| D hangul               |                 |
| F initialize           |                 |
| D is_character         |                 |
| D is_command           |                 |
| D is_hyphenator        |                 |
| D is_letter            |                 |
| D is_mark              |                 |
| D is_nothing           |                 |
| D is_punctuation       |                 |
| D is_spacing           |                 |
| D is_symbol            |                 |
| D lcchars              |                 |
| F lccode               |                 |
| D lccodes              |                 |
| F lettered             |                 |
| D lhash                |                 |
| D linebreaks           |                 |

- F lower
- D mathlists
- D oftfscripts
- D ranges
- F registerentities
- F remap\_hangul\_syllabe
- F setactivecatcodes
- F setlettercatcodes
- F setothercatcodes
- F shape
- F shaped
- D shash
- D shchars
- D shcodes
- F showstring
- D specialchars
- D specials
- D synonyms
- F tonumber
- D ucchars
- F uccode
- D uccodes
- D uhash
- F unicodechar
- F upper
- D verbosemarks

## chemistry

table

- F component
- F define
- D definitions
- S format
- F inlinechemical
- S instance
- S method
- F molecule
- U moleculeparser
- F start
- F stop
- F undefine

## collectgarbage

function

## commands

table

- F Alphabetic
- F MPLIBconvertfollowtext
- F MPLIBconvertoutlinetext

- F MPc
- F MPd
- F MPfromhash
- F MPH
- F MPll
- F MPLr
- F MPls
- F MPn
- F MPp
- F MPpardata
- F MPplus
- F MPpos
- F MPr
- F MPrest
- F MPrs
- F MPul
- F MPur
- F MPw
- F MPwhd
- F MPx
- F MPxy
- F MPxywhd
- F MPy
- F Romannumerals
- F actualday
- F actualmonth
- F actualyear
- F adaptfontfeature
- F adaptivecheckbox
- F addbtxentry
- F addcounter
- F adddefaultsuffix
- F addfeature
- F addfontpath
- F addgrapheme
- F addlinenumbers
- F addlinklistelement
- F addreplacements
- F addreplacementslist
- F addrivers
- F addtocolumnset
- F addtocolumnsetmvl
- F addtojavascriptpreamble
- F addtolist

|                             |                         |
|-----------------------------|-------------------------|
| F align_point_delta         | F bitwisenot            |
| F allinputpaths             | F bitwiseor             |
| F alphabetic                | F bitwiseset            |
| F analysefloatmethod        | F bitwiseshift          |
| F analyzeitemgroup          | F bitwisexor            |
| F analyzelist               | F blockcolumnset        |
| F analyzelocalanchors       | F blockligatures        |
| F analyzeregister           | F boolcase              |
| F analyzerotate             | F boxdir                |
| F analyzeupcomingitemgroup  | F boxline               |
| F applypattern              | F boxlinedp             |
| F applytobox                | F boxlineht             |
| F applytofirstcharacter     | F boxlineil             |
| F arraycompare              | F boxlinein             |
| F arrayequals               | F boxlineir             |
| F arraymaxvalue             | F boxlinelh             |
| F arrayminvalue             | F boxlinelp             |
| F arraysizes                | F boxlinels             |
| F arrayvalue                | F boxlinemaxdp          |
| F ascender                  | F boxlinemaxht          |
| F askedreference            | F boxlinemaxwd          |
| F assignbuffer              | F boxlinend             |
| F async                     | F boxlinenh             |
| F attachinsertion           | F boxlinew              |
| F attachinsertions          | F boxlinerh             |
| F atwordboundary            | F boxlinerp             |
| F autofinishstructurelevels | F boxliners             |
| F autoinsertnextspace       | F boxlines              |
| F automigrationcodestring   | F boxlinewd             |
| F autonextstructurelevel    | F boxoptioncodestring   |
| F autorule                  | F boxrangedp            |
| F autosetups                | F boxrangeht            |
| F autostarttext             | F boxrangewd            |
| F autostoptext              | F boxtostring           |
| F badnessmodecodestring     | F bposcolumn            |
| F balanceshapeextra         | F bposcolumnregistered  |
| F benchmarktimer            | F bs                    |
| F bf                        | F btxabbreviatedjournal |
| F bi                        | F btxaddentry           |
| F bitmapimage               | F btxaddjournal         |
| F bitwise                   | F btxauthor             |
| F bitwiseand                | F btxauthorfield        |
| F bitwiseflip               | F btxauthortoregister   |
| F bitwisenil                | F btxcmdstring          |

|                             |                              |
|-----------------------------|------------------------------|
| F btxcollectlistentries     | F btxsetspecification        |
| F btxcurrentfirstnames      | F btxshortcut                |
| F btxcurrentinitials        | F btxshowauthorremapping     |
| F btxcurrentjuniors         | F btxsingularorplural        |
| F btxcurrentsurnames        | F btxtoregister              |
| F btxcurrentvons            | F btxusedataset              |
| F btxdetail                 | F btxuservariable            |
| F btxdirect                 | F build_page_after_shipout   |
| F btxdoif                   | F build_page_before_shipout  |
| F btxdoifelse               | F build_page_setup           |
| F btxdoifelsecitedone       | F capheight                  |
| F btxdoifelsecombiinlist    | F charactercontrolcodestring |
| F btxdoifelsematches        | F chardescription            |
| F btxdoifelsesameasprevious | F checkcolumnset             |
| F btxdoifelseuservariable   | F checkcolumnsetmvl          |
| F btxdoifnot                | F checkcountersetup          |
| F btxentrytobuffer          | F checkedconversion          |
| F btxentrytostring          | F checkedemoji               |
| F btxexpandedjournal        | F checkedpagefloat           |
| F btxfetchlistentries       | F checkinjector              |
| F btxfield                  | F checkinterfacechain        |
| F btxfirstofrange           | F checkmarkedpages           |
| F btxflush                  | F checknesting               |
| F btxflushlistcombi         | F checkspacingavailable      |
| F btxflushlistentry         | F checkstrutdepth            |
| F btxflushlisttag           | F chemicalcomponent          |
| F btxflushmarked            | F chinesenumerals            |
| F btxflushpages             | F chr                        |
| F btxhandlecite             | F classoptioncodestring      |
| F btxhandlenocite           | F cleancolumnset             |
| F btxlastofrange            | F cleanfontname              |
| F btxlistvariant            | F cleanupattributes          |
| F btxloaddefinitionfile     | F cleanupbox                 |
| F btxloadjournalist         | F cleanuplayers              |
| F btxloadreplacementfile    | F clonefield                 |
| F btxmissing                | F closein                    |
| F btxoneorrange             | F closeout                   |
| F btxpreparelistentries     | F codedef                    |
| F btxremapauthor            | F collectedmath              |
| F btxsavedataset            | F collectfloat               |
| F btxsavejournalist         | F collecthelp                |
| F btxsetdataset             | F columnsetbalancefirst      |
| F btxsetentry               | F columnsetbalancelast       |
| F btxsetregister            | F columnsetbalancepage       |

|                                  |                            |
|----------------------------------|----------------------------|
| F columnsetcolumnwidth           | F cosd                     |
| F columnsetdisablewide           | F ctxfunction              |
| F columnsetenablewide            | F ctxluamatch              |
| F columnsetgoto                  | F ctxluamatchfile          |
| F columnsethascontent            | F ctxresetter              |
| F columnsethspan                 | F ctxscanner               |
| F columnsetlastfuture            | F currentcomponent         |
| F columnsetlastmvl               | F currentdate              |
| F columnsetlimit                 | F currentdesignsize        |
| F columnsetnoto                  | F currentenvironment       |
| F columnsetresetdirect           | F currentfontinstancespec  |
| F columnsetresetshapeextra       | F currentposthyphenchar    |
| F columnsetreshape               | F currentprehyphenchar     |
| F columnsetsblockrows            | F currentproduct           |
| F columnsetsetshapeextra         | F currentproject           |
| F columnsetshape                 | F currentsectiontolist     |
| F columnsetshapebalancecheck     | F datasetsize              |
| F columnsetshapebalanceextend    | F datasetvariable          |
| F columnsetshapecolumn           | F day                      |
| F columnsetshapecount            | F daynmem                  |
| F columnsetshapeindex            | F dayname                  |
| F columnsetshapepage             | F decimals                 |
| F columnsetshapresetbalancerange | F decrementcounter         |
| F columnsetshapslots             | F decrementsubcounter      |
| F columnsetshapspread            | F deferredenhancelist      |
| F columnsetshapevsize            | F deferredenhancereference |
| F columnsetshapeworst            | F deferredenhanceregister  |
| F columnsetshowgrid              | F defineaccents            |
| F columnsetshowshape             | F defineattribute          |
| F columnsetshowtracedgrid        | F defineboundary           |
| F columnsetstate                 | F definebreakpoint         |
| F columnsetstatus                | F definebreakpoints        |
| F combineglyphs                  | F definecharacterspacing   |
| F component                      | F definechemical           |
| F concatcommalist                | F definecolumnset          |
| F consultfloat                   | F defineconversion         |
| F continueifinputfile            | F definecounter            |
| F convertbtxdatasettoxml         | F definedlayoutcomponent   |
| F converteddimen                 | F definefallbackfamily     |
| F converttmpstopdf               | F definefield              |
| F copyboxfromcache               | F definefieldset           |
| F copyboxline                    | F definefilefallback       |
| F copyposition                   | F definefilesynonym        |
| F cos                            | F definefont_one           |

|                              |                              |
|------------------------------|------------------------------|
| F definefont_two             | F directboxfromcache         |
| F definefontcolorpalette     | F disableautofontscript      |
| F definefontfamily           | F disablecollectingmath      |
| F definefontfamilypreset     | F disabledirectives          |
| F definefontfeature          | F disableexperiments         |
| F definefontscriptset        | F disablejobsave             |
| F definefontsolution         | F disableparbuilder          |
| F definehyphenationfeatures  | F disableregime              |
| F defineinsertion            | F disablespellchecking       |
| F defineintermediatecolor    | F disablestatetriggering     |
| F definelabcolorglobal       | F disablestream              |
| F definelabcolorlocal        | F disabletrackers            |
| F definelabels               | F discardfromlist            |
| F definelanguage             | F disoptioncodestring        |
| F definelinkedlist           | F disposeboxesincache        |
| F definemathpreset           | F disposeluatable            |
| F definemultitonecolorglobal | F doanalyzeframedbox         |
| F definenamespace            | F doif                       |
| F definenote                 | F doifcounter                |
| F defineparallel             | F doifelse                   |
| F defineparbuilder           | F doifelseblack              |
| F defineprocesscolordummy    | F doifelseboxincache         |
| F defineprocesscolorglobal   | F doifelsebuffer             |
| F defineprocesscolorlocal    | F doifelsebufferempty        |
| F defineprogram              | F doifelsecharinfo           |
| F definereference            | F doifelseconverter          |
| F defineregister             | F doifelsecounter            |
| F definesectionlevels        | F doifelsecounteractive      |
| F definespotcolorglobal      | F doifelsecurrentfontfeature |
| F definestructureblock       | F doifelseedimenstring       |
| F definestructureset         | F doifelseedrawingblack      |
| F definesubcolumnset         | F doifelsefieldcategory      |
| F definetransparency         | F doifelsefieldset           |
| F definetransparencyglobal   | F doifelsefiledefined        |
| F definetypeface             | F doifelsefileexist          |
| F defineviewerlayer          | F doifelsefirstchar          |
| F delayedcolumnsetsheet      | F doifelsefirstsectionpage   |
| F depthnumber                | F doifelsefontfeature        |
| F depthofstring              | F doifelsefontpresent        |
| F descender                  | F doifelseindented           |
| F detachinsertions           | F doifelseinelement          |
| F diagnostic                 | F doifelseinputfile          |
| F digits_normal              | F doifelseleapyear           |
| F digits_reverse             | F doifselisthasentry         |

|                                 |                              |
|---------------------------------|------------------------------|
| F doifselisthasnumber           | F doifnothing                |
| F doifselisthaspage             | F doifposition               |
| F doifselisthastitle            | F doifsame                   |
| F doifselmarkedcontent          | F doifsomething              |
| F doifselmarkedpage             | F doifunknownfontfeature     |
| F doifselmessage                | F domainattribute            |
| F doifselnoteonsamepage         | F doreshapeframedbox         |
| F doifselobjectfound            | F doresyncframedbox          |
| F doifselobjectreferencefound   | F dosaveposition             |
| F doifselolderversion           | F dosavepositionplus         |
| F doifseloverlapping            | F dosavepositionwhd          |
| F doifselparallel               | F dosetposition              |
| F doifselparentfile             | F dosetpositionbox           |
| F doifselpathexist              | F dosetpositionplus          |
| F doifselpendingpagecontent     | F dosetpositionstrut         |
| F doifselposition               | F dosetpositionstrutkind     |
| F doifselpositiononpage         | F dosetpositionwhd           |
| F doifselpositionsonsamepage    | F dostarttextfile            |
| F doifselpositionsonthispage    | F dostoptextfile             |
| F doifselpositionsused          | F doublescriptmodecodestring |
| F doifselrangeonpage            | F droptablegroup             |
| F doifselreference              | F elapsedsteptime            |
| F doifselreferencechecked       | F elapsedtime                |
| F doifselreferenced             | F embedimagefromstore        |
| F doifselrighttoleftinbox       | F enablealignmentcharacter   |
| F doifselesame                  | F enablealignskips           |
| F doifselomething               | F enableasciimode            |
| F doifselspaces                 | F enableautofences           |
| F doifselstackedfloats          | F enableautofontscript       |
| F doifselstructureblockempty    | F enablebackgroundalign      |
| F doifselssynonymshown          | F enablebackgroundboxes      |
| F doifselssynonymused           | F enablecollectingmath       |
| F doifselsetaggednoteonsamepage | F enablecolor                |
| F doifselseurldefined           | F enablecolorintents         |
| F doifselseutfmathaccent        | F enablecrlf                 |
| F doifselseutfmathfiller        | F enabledirectives           |
| F doifselseutfmathlimop         | F enabledriver               |
| F doifselsevalidhashedfiles     | F enableexperiments          |
| F doifselsevisualizer           | F enablefontcolorsschemes    |
| F doifmarkedpage                | F enablehidecontent          |
| F doifnot                       | F enablemathalignrelocate    |
| F doifnotcounter                | F enablematrixornaments      |
| F doifnotecontent               | F enablematrixrules          |
| F doifnotsame                   | F enableparbuilder           |

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|---------------------------------|-----------------------------|
| F enableregime                  | F filterdefaultreference    |
| F enablescriptunstacking        | F filterreference           |
| F enablesidefloatchecker        | F finalizeobjectbox         |
| F enablespellchecking           | F finalizeshipoutbox        |
| F enablestacking                | F findregisterinternal      |
| F enablestatetriggering         | F finishcolumnsetflush      |
| F enablestream                  | F finishexport              |
| F enableswapping                | F firstinset                |
| F enabletighten                 | F firstsectionpage          |
| F enabletrackers                | F firstsubcountervalue      |
| F enabletransparency            | F flagcodestring            |
| F enableutf                     | F flushcolumnsetareas       |
| F enhancelinkedlist             | F flushcolumnsetcolumn      |
| F enhancelist                   | F flushfloat                |
| F enhanceregister               | F flushlabeledfloat         |
| F enhancesynonym                | F flushnoterange            |
| F environment                   | F flushnotes                |
| F eposcolumn                    | F flushpageblocks           |
| F eposcolumnregistered          | F flushpageinjectionsafter  |
| F erasebuffer                   | F flushpageinjectionsbefore |
| F execute                       | F flushpostponedblocks      |
| F exportbox                     | F flushpostponednotes       |
| F exportformdata                | F flushstream               |
| F exportreferences              | F flushstreamcopy           |
| F expression                    | F fontchar                  |
| F extendregister                | F fontcharbyindex           |
| F extendregisterlate            | F fontcollectionclone       |
| F extensiblecode                | F fontcollectiondefine      |
| F fakeword                      | F fontcollectionprepare     |
| F featureattribute              | F fontcollectionregister    |
| F feedback                      | F fontcollectionreport      |
| F fetchlanguagegettext          | F fontcollectionreset       |
| F fetchmultipar                 | F fontlookupgetkey          |
| F fetchposboxes                 | F fontlookupgetkeyofindex   |
| F fetchterminaldata             | F fontlookupinitialize      |
| F figure_flush_registered_pages | F fontlookupnooffound       |
| F figure_mergegoodies           | F forceendjob               |
| F figure_nof_registered_pages   | F forcepageflush            |
| F figure_register_page          | F forcequitjob              |
| F figuresetdimensions           | F forcestrutdepth           |
| F filebasename                  | F format                    |
| F filedirname                   | F formatcolor               |
| F filenameonly                  | F formatgray                |
| F filesuffix                    | F formatluminance           |



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|----------------------------------|------------------------------|
| F formatluminanceratio           | F getmathtextblob            |
| F formatone                      | F getmathvariant             |
| F framedoutline                  | F getmessage                 |
| F from                           | F getmpextensions            |
| F frozenparcodestring            | F getnaturaldimensions       |
| F frule                          | F getnaturalwd               |
| F getMPdata                      | F getnofpreviouslines        |
| F getbidimode                    | F getnotestate               |
| F getbodyfontsizes               | F getnotestorecount          |
| F getboxcountfromcache           | F getnotestoreflushed        |
| F getboxdpfromcache              | F getobjectdimensions        |
| F getboxfromcache                | F getobjectreference         |
| F getboxhtdpfromcache            | F getobjectreferencepage     |
| F getboxhtfromcache              | F getparshape                |
| F getboxwdfromcache              | F getparwrapper              |
| F getbuffer                      | F getprivatechar             |
| F getbuffercontent               | F getprivateglyphslot        |
| F getbufferctxlua                | F getprivatemathchar         |
| F getbuffermkvi                  | F getprivateslot             |
| F getbuffertex                   | F getprogram                 |
| F getcharactercasingcategory     | F getrandomdimen             |
| F getclubpenalties               | F getrandomfloat             |
| F getcurrentprefixspec           | F getrandomnumber            |
| F getcurrentreferencelabel       | F getrandomseed              |
| F getcurrentreferencemetadata    | F getreadfilename            |
| F getdefaultfieldvalue           | F getreferencesectionnumber  |
| F getdisplaywidowpenalties       | F getrivers                  |
| F getdocumentargument            | F getsetupstring             |
| F getdocumentfilename            | F getshapeparameter          |
| F getfieldcategory               | F getsomefullstructurenumber |
| F getfile                        | F getsomestructurenumber     |
| F getfirstcharacter              | F getspecificstructuretitle  |
| F getfloatvariable               | F getstructurelevel          |
| F getfromluatable                | F getstructurenumber         |
| F getfullstructurenumber         | F getsubstring               |
| F getinterlinepenalties          | F getsynchronizemarking      |
| F getinternalcachedlistreference | F geturl                     |
| F getinternallistreference       | F getvisual                  |
| F getinternalorderreference      | F getwidowpenalties          |
| F getmathblobindex               | F globalpopmacro             |
| F getmathblobmapping             | F globalpushmacro            |
| F getmathcharone                 | F gluetoinesinbox            |
| F getmathchartwo                 | F glyphoptioncodestring      |
| F getmathmathblob                | F gobblenested               |

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|-------------------------|------------------------------|
| F gotocolumnsetslot     | F ifxmlltext                 |
| F grabbuffer            | F ignorefile                 |
| F groupcodestring       | F ignoretagsinexport         |
| F groupsetgroup         | F immediatemessage           |
| F grph_nodes_initialize | F includesvgbuffer           |
| F grph_nodes_register   | F includesvgfile             |
| F grph_nodes_reset      | F incrementcounter           |
| F grph_nodes_resolve    | F incrementsubcounter        |
| F handlemathhang        | F inhibitprimitive           |
| F hboxtovbox            | F initializeexport           |
| F heightofstring        | F initializejob              |
| F helpaction            | F initializemathcollapsing   |
| F helpreference         | F initializemathfencing      |
| F hour                  | F initializemathitalics      |
| F hpackbox              | F initializemathkernpairs    |
| F hpackedbox            | F initializemathnumbers      |
| F hsplit                | F initializemathspacing      |
| F hyphenateddigits      | F initializestream           |
| F hyphenatedlist        | F initializetexdirectives    |
| F hyphenationcodestring | F initializetexexperiments   |
| F ic                    | F initializetextrackers      |
| F idxfromluatable       | F injectcurrentreference     |
| F if_common             | F injectcurrentreferencehtdp |
| F ifallmodes            | F injectdisable              |
| F ifbitwiseand          | F injectnowhite              |
| F ifblobindexdone       | F injectreference            |
| F ifboxhaslist          | F injectvpenalty             |
| F ifcommon              | F injectvskip                |
| F ifeof                 | F injectvspacing             |
| F ifgluelineinbox       | F injectzerobaselineskip     |
| F ifinset               | F inlinechemical             |
| F iflua                 | F inputfilebarename          |
| F ifmode                | F inputfilename              |
| F ifmodes               | F inputfilerealsuffix        |
| F ifmvl                 | F inputfilesuffix            |
| F ifpositunum           | F insertattachment           |
| F ifsetups              | F insertcategorycodestring   |
| F ifsystemmode          | F insertcomment              |
| F iftext                | F insertfield                |
| F ifxml                 | F insertionnumber            |
| F ifxmllatt             | F insertoptioncodestring     |
| F ifxmllattempty        | F insertrenderingwindow      |
| F ifxmlempty            | F inspectluatable            |
| F ifxmlselfempty        | F installedlanguages         |

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|-------------------------------|-------------------------|
| F installtexdirective         | F listexternal          |
| F installtexexperiment        | F listgroupindex        |
| F installtextracker           | F listlabel             |
| F interceptsamepagecontent    | F listlocation          |
| F it                          | F listnamespaces        |
| F jobfilefullname             | F listprefixednumber    |
| F jobfilename                 | F listprefixedpage      |
| F jobfilesuffix               | F listrawnnumber        |
| F jobposd                     | F listrawrealpage       |
| F jobposh                     | F listrawuserpage       |
| F jobposp                     | F listsigncodestring    |
| F jobposw                     | F listsize              |
| F jobposx                     | F listtitle             |
| F jobposy                     | F listuserdata          |
| F languagecharwidth           | F lmt_parameters_define |
| F languagenumber              | F lmt_parameters_preset |
| F lastblobindex               | F lmt_parameters_reset  |
| F lastnodeequals              | F loadcldfile           |
| F lastnodeid                  | F loadcldfileonce       |
| F lastnodeidstring            | F loadexamodes          |
| F lastnodesubtypestring       | F loadfontgoodies       |
| F lastparwrapper              | F loadlibrary           |
| F lastrpos                    | F loadluafile           |
| F lastsavedboxresourceindex   | F loadluafileonce       |
| F lastsavedimageresourceindex | F loadluamodule         |
| F lastsavedimageresourcepages | F loadmapfile           |
| F lastsectionpage             | F loadmapline           |
| F lastsubcountervalue         | F loadspellchecklist    |
| F lastunitstring              | F loadtexfile           |
| F lastxpos                    | F loadtexfileonce       |
| F lastypos                    | F loadtypescriptfile    |
| F lateindent                  | F loadvisualizer        |
| F latelua                     | F loadxmlfile           |
| F lateundent                  | F loadxmlfileonce       |
| F lchexnumber                 | F localanchorx          |
| F lchexnumbers                | F localanchory          |
| F leapyear                    | F localboxindent        |
| F limited                     | F localboxindex         |
| F linebuffering               | F localboxlefthang      |
| F linedir                     | F localboxleftoffset    |
| F lineprofilebox              | F localboxleftskip      |
| F linesegment                 | F localboxlinenumber    |
| F listanchorcodestring        | F localboxlinewidth     |
| F listbookmark                | F localboxlocalwidth    |

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|----------------------------|------------------------|
| F localboxmarkonce         | F mathhorizontalcode   |
| F localboxovershoot        | F mathoptioncodestring |
| F localboxparfillleftskip  | F mathrenderset        |
| F localboxparfillrightskip | F mathstrutcode        |
| F localboxparinitleftskip  | F mathvariantcode      |
| F localboxparinitrightskip | F mathvariantcount     |
| F localboxprogress         | F mathvariantslot      |
| F localboxrighthang        | F maxbufferwidth       |
| F localboxrightoffset      | F message              |
| F localboxrightskip        | F metapostescaped      |
| F localpopmacro            | F metapostformatted    |
| F localpushmacro           | F metapostgraphformat  |
| F locatefilepath           | F minute               |
| F loopcs                   | F mixcleanup           |
| F loopcsn                  | F mixfinalize          |
| F lousiness                | F mixflushlist         |
| F luacardinal              | F mixflushrest         |
| F luacount                 | F mixgetsplit          |
| F luadimen                 | F mixsetsplit          |
| F luaexpression            | F mixstate             |
| F luafloat                 | F modecodestring       |
| F luainteger               | F molecule             |
| F luanumber                | F month                |
| F luarandom                | F monthmnem            |
| F luascript                | F monthname            |
| F luathread                | F mpblobtext           |
| F lxmlid                   | F mpcolor              |
| F manipulated              | F mpgetblob            |
| F markblobindexdone        | F mpgettext            |
| F markcontent              | F mpgraphic            |
| F markedpageparameter      | F mpgraphictexttoks    |
| F markedpages              | F mpoptions            |
| F markingnumber            | F mpoutlinetoks        |
| F markingtitle             | F mppopvariables       |
| F markinjector             | F mppushvariables      |
| F markpage                 | F mprunset             |
| F markregionbox            | F mpruntab             |
| F markregionboxcorrected   | F mprunvar             |
| F markregionboxtagged      | F mpsetclippath        |
| F markregionboxtaggedkind  | F mpsetoutercolor      |
| F markregionboxtaggedn     | F mpsetsxsy            |
| F markvisualfonts          | F mpsettext            |
| F mathclassvalue           | F mptexget             |
| F mathcontrolcodestring    | F mptexreset           |

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|------------------------------|-------------------------------|
| F mptexset                   | F octnumber                   |
| F mptexsetfrombuffer         | F openin                      |
| F mptexttoks                 | F openout                     |
| F mvloptioncodestring        | F operatingsystem             |
| F namedstructureuservariable | F ordinal                     |
| F namedstructurevariable     | F outputfilename              |
| F naturalwd                  | F overloadbookmark            |
| F negatedcolorcomponent      | F overloadlowercase           |
| F newarray                   | F overloadsattribute          |
| F newhashedtable             | F overloaduppercase           |
| F newindexedtable            | F pageblockrealpage           |
| F newluatable                | F pageofinternal              |
| F newparwrapper              | F pageoverflow                |
| F newrepeatablerandom        | F pagestaterealpage           |
| F newstacking                | F pagestaterealpageorder      |
| F nextparallel               | F paragraphfillmodecodestring |
| F nextsubcountervalue        | F paragraphoptioncodestring   |
| F nodimen                    | F parcontextcodestring        |
| F nofbufferlines             | F pardir                      |
| F nofconversions             | F parpasscallback             |
| F nofdays                    | F parpassidentifier           |
| F nofdetachedinsertions      | F parpos                      |
| F noflinesinbox              | F partriggercodestring        |
| F nofmathvariants            | F pdfbackendcurrentresources  |
| F nofstackedfloats           | F pdfbackendsetcatalog        |
| F nologbuffering             | F pdfbackendsetcolorspace     |
| F normalizecodestring        | F pdfbackendsettextgstate     |
| F normalizedbodyfontsize     | F pdfbackendsetinfo           |
| F normalizeparcodestring     | F pdfbackendsetname           |
| F nospacesmodecodestring     | F pdfbackendsetpageattribute  |
| F notedeltapage              | F pdfbackendsetpageresource   |
| F noteinternal               | F pdfbackendsetpagesattribute |
| F notelistindex              | F pdfbackendsetpattern        |
| F noteprefixednumber         | F pdfbackendsetshade          |
| F notetitle                  | F pdfcolor                    |
| F ntb_split_section_check    | F pdfextension                |
| F ntb_split_section_fetch    | F pdffeedback                 |
| F ntb_split_section_locate   | F pdflastobj                  |
| F ntb_split_section_reset    | F pdfliteral                  |
| F ntimes                     | F pdfobj                      |
| F number                     | F pdfrefobj                   |
| F numbers                    | F pdfsetuserdata              |
| F objectreference            | F pdfvariable                 |
| F objectreferencepage        | F percentageof                |

F pfabs  
F pfacos  
F pfacosh  
F pfasin  
F pfasinh  
F pfatan  
F pfatanh  
F pfatantwo  
F pfceil  
F pfcos  
F pfcosh  
F pfdeg  
F pfexp  
F pffloor  
F pflog  
F pfmod  
F pfpow  
F pfrad  
F pfrem  
F pfround  
F pfsin  
F pfsinh  
F pfsqrt  
F pftan  
F pftanh  
F pgfxcolorspec  
F pickupbuffer  
F pickupmarkedcontent  
F pickuppunctuation  
F placeparallel  
F popfloat  
F pophyphenation  
F poplist  
F poplocalanchors  
F poplogfile  
F popparagraphtweak  
F poppath  
F poprandomseed  
F popreferenceprefix  
F popregime  
F poprunstate  
F popsectionblock  
F positunum  
F positunumdef

F postlistbox  
F postlistcopy  
F postponenotes  
F prefixedconverted  
F prelistbox  
F prelistcopy  
F preparecolumnsetflush  
F prerolledpages  
F presetcolumnsetsheets  
F presetcolumnsetspreadsheets  
F presetmathematics  
F presetmathfontalternate  
F presetsymbollist  
F previoussubcountervalue  
F privatecharacter  
F privatecharactercode  
F processbodyfontsizes  
F processbuffer  
F processcollectedmath  
F processedfile  
F processedfiles  
F processfile  
F processfilemany  
F processfilenone  
F processfileonce  
F processjob  
F processlist  
F processregister  
F processsplit  
F processsynonyms  
F product  
F profilebox  
F profilemacro  
F project  
F pruneboxesincache  
F purefontname  
F pushfloat  
F pushhyphenation  
F pushlist  
F pushlocalanchors  
F pushlogfile  
F pushparagraphtweak  
F pushpath  
F pushrandomseed

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|---------------------------------|--------------------------------|
| F pushreferenceprefix           | F registerfilescheme           |
| F pushregime                    | F registerfontalternative      |
| F pushrunstate                  | F registerfontfamily           |
| F pushsectionblock              | F registerforcedbreak          |
| F pushstream                    | F registergroupset             |
| F putboxincache                 | F registerhashedfiles          |
| F putincolumnset                | F registerhelp                 |
| F putincolumnsetdirect          | F registerhyphenationexception |
| F putnextboxincache             | F registerhyphenationpattern   |
| F qrcodebuffer                  | F registeritemgroup            |
| F qrcodestring                  | F registerlabel                |
| F rawMPdata                     | F registerlanguagefeatures     |
| F rawcharacter                  | F registerlinenumbering        |
| F rawsetupstring                | F registermaintextcolor        |
| F rawsubcountervalue            | F registermpextensible         |
| F read                          | F registernamedlua             |
| F readfromterminal              | F registernamespace            |
| F readline                      | F registerobject               |
| F readlinedirect                | F registerpattern              |
| F recolorbox                    | F registerpostponedblock       |
| F referencecolumn               | F registerreferencedobject     |
| F referencepagedetail           | F registerrendering            |
| F referencepagestate            | F registersection              |
| F referenceposx                 | F registershownsynonym         |
| F referenceposy                 | F registersnaprange            |
| F referencerealpage             | F registerstructureprocessor   |
| F registeranchorbox             | F registersynchronize          |
| F registerattachment            | F registersynonym              |
| F registerbackendsymbol         | F registertransliteration      |
| F registerbackground            | F registerunit                 |
| F registerbodyfontsize          | F registerunitshortcut         |
| F registerbookmark              | F registerunknownbodysize      |
| F registerboxglyph              | F registerusedsynonym          |
| F registerbuttons               | F registeruserdata             |
| F registercolorintent           | F registerverbosemathlabel     |
| F registercolumnsetarea         | F removelastline               |
| F registercolumnsetsheet        | F renderingvar                 |
| F registercolumnsetspreadsheets | F reorderfloats                |
| F registerengineclass           | F replacefeature               |
| F registerfallbackfont          | F replacepospxpywhd            |
| F registerfiguremask            | F replacewildcard              |
| F registerfigurepattern         | F reportstructure              |
| F registerfigurerange           | F resavefloat                  |
| F registerfiguresuffix          | F reservedautoregiontag        |

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|----------------------------|---------------------------|
| F resetMPinstance          | F ruby                    |
| F resetalternate           | F ruleoptioncodestring    |
| F resetbackgroundrowdata   | F runcontextjob           |
| F resetboxesincache        | F safechar                |
| F resetcharacteralign      | F samplefile              |
| F resetcolumnset           | F saveboxresource         |
| F resetcomparingtext       | F savebuffer              |
| F resetcounter             | F savecounter             |
| F reseteffect              | F savecurrentattributes   |
| F resetfeature             | F savecurrentvalue        |
| F resetfontfeature         | F savedlistnumber         |
| F resetfontsolution        | F savedlistprefixednumber |
| F resetforcedbreak         | F savedlisttitle          |
| F resetinjector            | F savefloat               |
| F resetlogfile             | F saveimageresource       |
| F resetmapfiles            | F savemargindata          |
| F resetmathattributes      | F savenote                |
| F resetnullfont            | F savepagedata            |
| F resetparallel            | F savepageinjections      |
| F resetpath                | F saveparallel            |
| F resetreference           | F savepos                 |
| F resetscript              | F savespecificfloat       |
| F resetstructureprocessor  | F savestructureblock      |
| F resetsubcounter          | F scangivendimensions     |
| F resetsynchronizemarking  | F scriptsmodecodestring   |
| F resetsynonyms            | F second                  |
| F resettimer               | F selectstructureblock    |
| F resettrackers            | F set_extensible_data     |
| F resolvedemoji            | F setadaptive             |
| F restartcounter           | F setalternate            |
| F restartsubcounter        | F setautoprefix           |
| F restorecounter           | F setbackgroundrowdata    |
| F restorecurrentattributes | F setboxglyph             |
| F restorenote              | F setboxline              |
| F restoreobject            | F setbreakpoints          |
| F reuserandomseed          | F setcharacteralign       |
| F reversevboxcontent       | F setcharacteraligndetail |
| F revivefeature            | F setcharactercasing      |
| F rightmathkern            | F setcharactercleaning    |
| F romannumerals            | F setcharacterkerning     |
| F rotateparshape           | F setcharacterspacing     |
| F rounded                  | F setcharacterstripping   |
| F roundedfill              | F setcloseddocumentaction |
| F roundedoutline           | F setclosepageaction      |



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|------------------------------|-----------------------------|
| F setcolormodel              | F setinitialsectionblock    |
| F setcolumnsetarea           | F setinjector               |
| F setcolumnsetlines          | F setinsertionlocation      |
| F setcolumnsetmvl            | F setinsertmigration        |
| F setcolumnsetpresent        | F setinterfaceconstant      |
| F setcolumnsetproperties     | F setinterfaceelement       |
| F setcolumnsetstart          | F setinterfacemessage       |
| F setcomparetext             | F setinterfacemessages      |
| F setcomparingtext           | F setinterfacevariable      |
| F setcounter                 | F setinternalreference      |
| F setdataset                 | F setitaliccorrection       |
| F setdestinationattribute    | F setjavascriptpreamble     |
| F setdigitsmanipulation      | F setjobcomment             |
| F setdirection               | F setlanguageexceptions     |
| F setdocumentargument        | F setlanguageoptions        |
| F setdocumentcommandline     | F setlanguagepatterns       |
| F setdocumentctxfile         | F setlanguagesynonym        |
| F setdocumentdefaultargument | F setlinefiller             |
| F setdocumentenvironments    | F setlineprofile            |
| F setdocumentfilename        | F setlocalalignanchor       |
| F setdocumentfilenames       | F setlocalmarkanchor        |
| F setdocumentmodes           | F setlocalsyncanchor        |
| F setdocumentmodules         | F setluatable               |
| F seteffect                  | F setmathalphabet           |
| F setelementbackentag        | F setmathattribute          |
| F setelementnature           | F setmathfontalternate      |
| F setextrafontkerns          | F setmathfontstyle          |
| F setfieldcalculationset     | F setmathsnapping           |
| F setfigureconversion        | F setmathvariant            |
| F setfigurelookuporder       | F setmetaglyph              |
| F setfigurepaths             | F setmpextensions           |
| F setfirstline               | F setnaturalwd              |
| F setfontfeature             | F setnegative               |
| F setfontofid                | F setnextinternalreferences |
| F setfontsolution            | F setnextreferenceorder     |
| F setformat                  | F setnotestate              |
| F setformsmethod             | F setopendocumentaction     |
| F sethsizecolumnset          | F setopenpageaction         |
| F sethsizecolumnspan         | F setowncounter             |
| F sethyphenatedurlcharacters | F setownsubcounter          |
| F sethyphenationfeatures     | F setpagecolormodel         |
| F sethyphenationmethod       | F setpagestate              |
| F setidentityvariable        | F setpagestaterealpago      |
| F setinitial                 | F setpagetransition         |

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|--------------------------|---------------------------------|
| F setparagraphmetashape  | F settaghighlight               |
| F setparagraphshape      | F settagitem                    |
| F setparbuilder          | F settagitemgroup               |
| F setparwrapper          | F settaglist                    |
| F setpdfcompression      | F settagmarginanchor            |
| F setperiodkerning       | F settagmarginext               |
| F setpostlistbox         | F settagmpgraphic               |
| F setprelistbox          | F settagnotation                |
| F setprofile             | F settagnotationsymbol          |
| F setrandomseed          | F settagparagraph               |
| F setrealign             | F settagproperty                |
| F setrealspaces          | F settagpublication             |
| F setreferencelinkmethod | F settagregister                |
| F setregionbox           | F settagsetreferencedescription |
| F setregionboxtagged     | F settagsorting                 |
| F setregistermethod      | F settagssubsentencesymbol      |
| F setreplacements        | F settagssynonym                |
| F setruby                | F settagtablecell               |
| F setrule                | F settagtabulatecell            |
| F setscript              | F settransliteration            |
| F setsectionblock        | F setupbookmarks                |
| F setsectionentry        | F setupcanvas                   |
| F setshadow              | F setupcharacterspacing         |
| F setshift               | F setupcomparingtext            |
| F setsofthyphens         | F setupexport                   |
| F setsplitlisthtdp       | F setupfontexpansion            |
| F setstacking            | F setupfontprotrusion           |
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## floats table

- F analysemethod
- F checkedpagefloat
- F collect
- F consult
- F flush
- F getvariable
- F hasstacked
- F nofstacked
- F pop
- F push
- F resave
- F save
- F stacked

## font table

- F addcharacters
- F addquality
- F current
- F define
- F getfont
- F getfontdimen
- F getfontfields
- F getfontspec
- F getmathcharacterfields
- F getmathindex
- F getmathspec
- F getparameterfields
- F gettextcharacterfields
- F id
- F max
- F nextid
- F setfont
- F setfontdimen
- F xscaled
- F yscaled

## fonts table

- F current
- F currentid
- F definitypeface
- F enhanceid
- D expansions
- D formats
- F getalternate
- F getindices
- F getkern
- F getmultiple
- F getsubstitution
- F infofont
- F mode
- D names
- D nulldata
- D protrusions
- F registersanitizer
- F reserveid
- F toeffect

## functions table

- F dummy

## getmetatable function

## globaldata table

- N CONTEXTLMTXMODE
- N LUAFORMAT
- N LUAMAJORVERSION
- N LUAMINORVERSION
- S LUATEXENGINE
- N LUATEXFORMATID
- N LUATEXFUNCTIONALITY
- S LUATEXVERBOSE
- N LUATEXVERSION
- N LUAVERSION
- F SAVEDREQUIRE
- F assert
- F characterkern
- F collectgarbage
- F dimen
- F dofile
- F error
- F getmetatable
- F inspect
- F ipairs

F load  
 D loaders  
 F loadfile  
 F loadstring  
 F next  
 F optionalrequire  
 F pairs  
 F pcall  
 F print  
 F rawequal  
 F rawget  
 F rawlen  
 F rawset  
 F require  
 F select  
 F setinspector  
 F setmathparameters  
 F setmetatable  
 D texconditionals  
 D texconstants  
 D texifs  
 D texisdefined  
 D texmodes  
 D texsystemmodes  
 F toboolean  
 F todecimal  
 F tonumber  
 F tostring  
 F type  
 F unpack  
 F warn  
 F xpcall

## globals

N CONTEXTLMTXMODE  
 N LUAFORMAT  
 N LUAMAJORVERSION  
 N LUAMINORVERSION  
 S LUAEXENGINE  
 N LUAEXFORMATID  
 N LUAEXFUNCTIONALITY  
 S LUAEXVERBOSE  
 N LUAEXVERSION  
 N LUAVERSION  
 F SAVEDREQUIRE

## table

F assert  
 F characterkern  
 F collectgarbage  
 F dimen  
 F dofile  
 F error  
 F getmetatable  
 F inspect  
 F ipairs  
 F load  
 D loaders  
 F loadfile  
 F loadstring  
 F next  
 F optionalrequire  
 F pairs  
 F pcall  
 F print  
 F rawequal  
 F rawget  
 F rawlen  
 F rawset  
 F require  
 F select  
 F setinspector  
 F setmathparameters  
 F setmetatable  
 D texconditionals  
 D texconstants  
 D texifs  
 D texisdefined  
 D texmodes  
 D texsystemmodes  
 F toboolean  
 F todecimal  
 F tonumber  
 F tostring  
 F type  
 F unpack  
 F warn  
 F xpcall

## graphics

F bpsize  
 F identify

## table

|                        |          |                        |       |
|------------------------|----------|------------------------|-------|
| F registerpdfcompactor |          | F checkedspecification |       |
| F registerpdffont      |          | D complete             |       |
| <b>gzip</b>            | table    | D constants            |       |
| F compress             |          | D corenamespaces       |       |
| F compressed           |          | S currentinterface     |       |
| F decompress           |          | S currentresponse      |       |
| F load                 |          | F definecommand        |       |
| F save                 |          | F defined              |       |
| F suffix               |          | F doifelsemessage      |       |
| <b>helper</b>          | table    | D elements             |       |
| F getprofile           |          | D formats              |       |
| F mathnested           |          | F getmessage           |       |
| F mathprocess          |          | F getnamespace         |       |
| F mathstep             |          | F getsetupstring       |       |
| F snapindeed           |          | D hashedsetups         |       |
| F snaplist             |          | F implement            |       |
| F traversecharmethod   |          | F interfacedcommand    |       |
| F traverseglyphmethod  |          | F knownscanner         |       |
| F traversehorizontal   |          | F message              |       |
| F traversevertical     |          | F nameofscanner        |       |
| <b>images</b>          | table    | F registernamespace    |       |
| F check                |          | F registerscanner      |       |
| F checksize            |          | D scanners             |       |
| F copy                 |          | F setconstant          |       |
| F create               |          | F setcount             |       |
| F embed                |          | F setdimen             |       |
| F embedfromstore       |          | F setelement           |       |
| F fetchdata            |          | F setformat            |       |
| F print                |          | F setmacro             |       |
| F resetdata            |          | F setmessage           |       |
| F resetstore           |          | F setmessages          |       |
| F scan                 |          | F settranslation       |       |
| F serialize            |          | D setupstrings         |       |
| F storedata            |          | F setvariable          |       |
| F totable              |          | F showmessage          |       |
| F wrap                 |          | D storedscanners       |       |
| <b>inspect</b>         | function | F tolist               |       |
| <b>interactions</b>    | table    | D translations         |       |
| D attachments          |          | D usednamespaces       |       |
| D linkedlists          |          | D variables            |       |
| D renderings           |          | F writestatus          |       |
| <b>interfaces</b>      | table    | <b>io</b>              | table |
| D cachedsetups         |          | F ask                  |       |
| F cachesetup           |          | F bytes                |       |

|                 |          |                    |       |
|-----------------|----------|--------------------|-------|
| F characters    |          | F delete           |       |
| F close         |          | F found            |       |
| F copydata      |          | F insert           |       |
| F exists        |          | F nearest          |       |
| S fileseparator |          | F nearestindex     |       |
| F flush         |          | F new              |       |
| F gobble        |          | F newweighted      |       |
| F input         |          | F reset            |       |
| F lines         |          | F totable          |       |
| F loadchunk     |          | F traverse         |       |
| F loaddata      |          | <b>kpse</b>        | table |
| F loadlines     |          | <b>language</b>    | table |
| F newreader     |          | F clean            |       |
| F noflines      |          | F clearhyphenation |       |
| F open          |          | F clearpatterns    |       |
| F output        |          | F current          |       |
| S pathseparator |          | F gethjcode        |       |
| F popen         |          | F haslanguage      |       |
| F read          |          | F hyphenate        |       |
| F readall       |          | F hyphenation      |       |
| F readnumber    |          | F hyphenationmin   |       |
| F readstring    |          | F id               |       |
| F savedata      |          | F new              |       |
| F size          |          | F patterns         |       |
| U stderr        |          | F postexhyphenchar |       |
| U stdin         |          | F posthyphenchar   |       |
| U stdout        |          | F preexhyphenchar  |       |
| F tmpfile       |          | F prehyphenchar    |       |
| F type          |          | F sethjcode        |       |
| F write         |          | F setwordhandler   |       |
| <b>ipairs</b>   | function | <b>languages</b>   | table |
| <b>job</b>      | table    | F addgoodiesdata   |       |
| F comment       |          | F addgoodiesfile   |       |
| F disablesave   |          | F associate        |       |
| F getrandomseed |          | D associated       |       |
| F initialize    |          | F association      |       |
| F load          |          | D counters         |       |
| F loadother     |          | F current          |       |
| B pack          |          | D data             |       |
| N packversion   |          | D decimals         |       |
| F register      |          | F define           |       |
| F save          |          | F explicithyphen   |       |
| N version       |          | F exportlanguage   |       |
| <b>kdtree</b>   | table    | F flatten          |       |



- F getdata
- F getmetadata
- F getnumber
- F goodiefiles
- F hyphenate
- F installed
- F installhandler
- F loadable
- F loadwords
- F noflattened
- D numbers
- F pdflanguage
- F posttexhyphenchar
- F posthyphenchar
- F preexhyphenchar
- F prehyphenchar
- D registered
- F serializediscretionary
- F setexceptions
- F setgoodieshandler
- F setoptions
- F setpatterns
- F setsynonym
- F setwordhandler
- F showdiscretionaries
- F strippedgoodiewords
- F tolang
- F unload
- N version
- F visualizediscretionaries

## layouts table

- F getcanvas
- F getpagedimensions
- F setupcanvas

## lfs table

- F attributes
- F canonicalize
- F chdir
- F currentdir
- F dir
- F expandpath
- F isdir
- F isfile
- F isfound

- F isreadabledir
- F isreadablefile
- F iswriteabledir
- F iswriteablefile
- F link
- F mkdir
- F mkdirs
- F modification
- F readlink
- F rmdir
- F setexecutable
- F symlink
- F symlinktarget
- F touch
- F walkdir

## libraries table

- D basiclua
- D basictex
- D builtin
- D extralua
- D extratex
- D functions
- D globals
- D obsolete
- D optional

## lmx table

- F color
- F convert
- F convertfile
- F convertstring
- F einject
- F eprint
- F escape
- F finalized
- F finject
- F fprint
- F get
- F getvariables
- F htmfile
- F include
- F initialize
- F inject
- F lmxfile
- F loadedfile

|                        |            |                           |       |
|------------------------|------------|---------------------------|-------|
| F make                 |            | F settarget               |       |
| F new                  |            | F settimedlog             |       |
| F popupfile            |            | F settranslations         |       |
| F print                |            | F show                    |       |
| F pv                   |            | F start_page_number       |       |
| F report               |            | F start_run               |       |
| F reset                |            | F startererrorlogging     |       |
| F result               |            | F startfilelogging        |       |
| F set                  |            | F status                  |       |
| F show                 |            | F stop_page_number        |       |
| F stripcss             |            | F stoperrorlogging        |       |
| F tv                   |            | F stopfilelogging         |       |
| F type                 |            | F subdirect               |       |
| F urlescape            |            | F subreport               |       |
| F variable             |            | F system                  |       |
| <b>load</b>            | function   | F texerrormessage         |       |
| <b>loaders</b>         | data table | F writer                  |       |
| <b>loadfile</b>        | function   | <b>lpdf</b>               | table |
| <b>loadstring</b>      | function   | F action                  |       |
| <b>logs</b>            | table      | F adddocumentcolorspace   |       |
| F application          |            | F adddocumenttextgstate   |       |
| F categories           |            | F adddocumentpattern      |       |
| F direct               |            | F adddocumentshade        |       |
| F disable              |            | F addtocatalog            |       |
| F enable               |            | F addtoinfo               |       |
| F finalactions         |            | F addtonames              |       |
| F flush                |            | F addtopageattributes     |       |
| F loggingerrors        |            | F addtopageresources      |       |
| F messenger            |            | F addtopagesattributes    |       |
| F new                  |            | F addxmpinfo              |       |
| F newline              |            | F annotationspecification |       |
| F poplogfile           |            | F array                   |       |
| F poptarget            |            | F attachmentsymbols       |       |
| F pushlogfile          |            | F boolean                 |       |
| F pushtarget           |            | F border                  |       |
| F registerfinalactions |            | F checkedkey              |       |
| F report               |            | F checkedresources        |       |
| F reporter             |            | F checkedvalue            |       |
| F resetlogfile         |            | F circularshade           |       |
| F setformats           |            | F close                   |       |
| F setformatters        |            | F cmykcode                |       |
| F setlogfile           |            | F collectedresources      |       |
| F setmessenger         |            | F color                   |       |
| F setprocessor         |            | F colorspace              |       |

|                            |                             |
|----------------------------|-----------------------------|
| D colorspaceconstants      | F getnofpages               |
| F colorspec                | F getpageproperties         |
| F colorvalue               | F getpagereference          |
| F colorvalues              | F getpagetags               |
| F commentsymbols           | F getpos                    |
| F compressdata             | F gettaggingstatus          |
| F compresslevel            | F gettrailerid              |
| F constant                 | F getversion                |
| F copyarray                | F getxformname              |
| F copyboolean              | F graycode                  |
| F copyconstant             | F hasmatrix                 |
| F copydictionary           | F immediateobject           |
| F copyfloat                | F includeimage              |
| F copyinteger              | F initialize                |
| F copynumber               | F injectxmpinfo             |
| F copystring               | F insertxmpinfo             |
| F copyunicode              | F lastreferredpage          |
| F deferredobject           | F layerreference            |
| F delayedobject            | F limited                   |
| F destinationspecification | F linearshade               |
| F dictionary               | F literal                   |
| F docstring                | F majorversion              |
| F escaped                  | F makenametree              |
| F finalizedocument         | F minorversion              |
| F finalizepage             | F noffontregistries         |
| F finalizesignature        | F nopages                   |
| F finishtransparencycode   | F null                      |
| F flushfonts               | F number                    |
| F flushgroup               | F object                    |
| F flushobject              | F objectcompresslevel       |
| F flushstreamfileobject    | F pagereference             |
| F flushstreamobject        | F patternstream             |
| D formats                  | F pdfcolor                  |
| F frombytes                | F pdftimestamp              |
| F fromeight                | F preparesignature          |
| F frompdfdoc               | F print                     |
| F fromsixteen              | F procset                   |
| F getcatalog               | B protectresources          |
| F getcompression           | F quads                     |
| F getfontname              | F rectangle                 |
| F getfontobjectnumber      | F reference                 |
| F getformatoption          | F registerannotation        |
| F getinfo                  | F registerdestination       |
| F getmatrix                | F registerdocumentfinalizer |

F registerfontmethod  
 F registerinitializer  
 F registerpagefinalizer  
 F registerpattern  
 F registersignature  
 F replacexmpinfo  
 F reserveobject  
 F rgbcode  
 F rotationcm  
 F setcompression  
 F setencryption  
 F setpageorder  
 F setpagestate  
 F setpagetag  
 F setutfstringmode  
 F setversion  
 F sharedobject  
 F shareobject  
 F shareobjectreference  
 F splitsixteen  
 F spotcode  
 F stream  
 F string  
 F supportedformats  
 F tocidset  
 F toeight  
 F topdfdoc  
 F topermissions  
 F tosixteen  
 F transparency  
 F transparencycode  
 F transparencyvalue  
 F unicode  
 D usedfontstreams  
 F verbose

## lpeg table

F B  
 F C  
 F Carg  
 F Cb  
 F Cc  
 F Cf  
 F Cg  
 F Cmt

F Cp  
 F Cs  
 F Ct  
 F P  
 F R  
 F S  
 F UP  
 F UR  
 F US  
 F V  
 F afterprefix  
 F anywhere  
 F append  
 F balancer  
 F beforesuffix  
 F checkedsplit  
 F containsws  
 F counter  
 F endstripper  
 F finder  
 F firstofsplit  
 F frontstripper  
 F instringchecker  
 F is\_lpeg  
 F keeper  
 F locale  
 F match  
 F oneof  
 D patterns  
 F pcode  
 F print  
 F ptree  
 F replacer  
 F secondofsplit  
 F setmaxstack  
 F setutfcasers  
 F split  
 F splitat  
 F splitter  
 F stripper  
 F times  
 F tsplitat  
 F tsplitter  
 F type

|                           |              |                        |              |
|---------------------------|--------------|------------------------|--------------|
| F utfR                    |              | S startupfile          |              |
| F utfchartabletopattern   |              | F subnibble            |              |
| F utfreplacer             |              | F usestringcall        |              |
| S version                 |              | S version              |              |
| <b>lua</b>                | <b>table</b> | <b>luac</b>            | <b>table</b> |
| F addnibble               |              | F inspect              |              |
| D bytedata                |              | F print                |              |
| F callbytecode            |              | <b>luatex</b>          | <b>table</b> |
| F comparekeys             |              | F abort                |              |
| F finalizeinitex          |              | F cleanup              |              |
| N firstbytecode           |              | F cleanuptempfiles     |              |
| F get_functions_table     |              | F currentfile          |              |
| F getallkeys              |              | F currentlevel         |              |
| F getbytecode             |              | F currenttotal         |              |
| F getcheckedkeys          |              | F poprunstate          |              |
| F getcurrenttime          |              | F pushrunstate         |              |
| F getdebuginfo            |              | F registerdumpactions  |              |
| F getexitcode             |              | F registerpageactions  |              |
| F getfunctionstable       |              | F registerstartactions |              |
| F getnibble               |              | F registerstopactions  |              |
| F getnumerickeys          |              | F registertempfile     |              |
| F getpreciseseconds       |              | N starttime            |              |
| F getpreciseticks         |              | F wrapup               |              |
| F getruntime              |              | <b>lxml</b>            | <b>table</b> |
| F getstacktop             |              | F addindex             |              |
| F getstartupfile          |              | F all                  |              |
| F getstringkeys           |              | F applyfunction        |              |
| F getversion              |              | F applyselectors       |              |
| N lastbytecode            |              | F att                  |              |
| F luastringcall           |              | F attribute            |              |
| D messages                |              | F badinclusions        |              |
| F mpsstringcall           |              | F chainattribute       |              |
| F newindex                |              | F checkindex           |              |
| F newtable                |              | F collected            |              |
| D numbers                 |              | F command              |              |
| F protectglobal           |              | F concat               |              |
| F registercode            |              | F concatrange          |              |
| F registeredcodes         |              | F content              |              |
| F registerglobal          |              | F context              |              |
| F registerinitexfinalizer |              | F convert              |              |
| F registername            |              | F count                |              |
| F setbytecode             |              | F delete               |              |
| F setexitcode             |              | F depth                |              |
| F setnibble               |              | F direct               |              |

|                   |                    |
|-------------------|--------------------|
| F displayverbatim | F name             |
| F doif            | F namespace        |
| F doifelse        | F nonSPACE         |
| F doifelseempty   | F par              |
| F doifelsetext    | F param            |
| F doifempty       | F parameter        |
| F doifnot         | F path             |
| F doifnotempty    | F popmatch         |
| F doifnottext     | F pos              |
| F doiftext        | F position         |
| F element         | F pure             |
| F elements        | F pushmatch        |
| D entities        | F raw              |
| F fileline        | F rawroot          |
| F filename        | F rawtex           |
| F filter          | F refatt           |
| F filterlist      | F register         |
| F first           | F registerentity   |
| F flush           | F removesetup      |
| F flushlinewise   | F replace          |
| F flushsetups     | F resetcdata       |
| F flushspacewise  | F resetsetups      |
| F get_id          | F resolvedentity   |
| F getid           | F root             |
| F getindex        | F save             |
| F id              | F serialize        |
| U idsplitter      | F setaction        |
| F ifatt           | F setatt           |
| F ifattempty      | F setattribute     |
| F include         | F setcdata         |
| F inclusion       | F setcommandtonone |
| F inclusions      | F setcommandtotext |
| F index           | F setpar           |
| F inlineverbatim  | F setparam         |
| F installsetup    | F setparameter     |
| F last            | F setsetup         |
| F lastatt         | F snippet          |
| F lastmatch       | F splitid          |
| F lastpar         | F startraw         |
| F load            | F startunesCAPED   |
| F loadbuffer      | F stopraw          |
| F loaddata        | F stopunesCAPED    |
| F main            | F store            |
| F match           | F strip            |

- F stripped
- F tag
- F texatt
- F text
- F tobuffer
- F tofile
- F toparameters
- F toverbatim
- F verbatim
- F withinindex
- D xmltexhandler

## math

## table

- F abs
- F acos
- F asin
- F atan
- F atan2
- F ceil
- F ceiling
- F cos
- F cosd
- F cosh
- F deg
- F div
- F even
- F exp
- F floor
- F fmod
- F frexp
- N huge
- N initialseed
- F ldexp
- F log
- F log10
- F max
- N maxinteger
- F min
- N mininteger
- F mod
- F modf
- F odd
- N pi
- F pow
- F rad

- F random
- F randomseed
- F round
- F setrandomseedi
- F sin
- F sind
- F sinh
- F sqrt
- F tan
- F tand
- F tanh
- F tointeger
- F type
- F ult

## mathematics

## table

- F addfallbacks
- F addvariant
- D alphabets
- D alternatives
- F beforepassingfonttotex
- F big
- D boldmap
- D charactersets
- F checkaftercopyingfont
- D classes
- D classnames
- D codes
- F define
- F dimensions
- F disablecollecting
- F enablecollecting
- F extensiblecode
- D extensibles
- N extrabase
- F fallbackstyleattr
- F familyname
- F functiontype
- D gaps
- F getblobindex
- F getboth
- F getdomainname
- F gettextensibledata
- F getmathblob
- F getoptionallabel

F getstyle  
 F gettextblob  
 F getverboselabel  
 F horizontalcode  
 F initializeparameters  
 F makefont  
 D mapremap  
 F markblobindexdone  
 F overloadparameters  
 F preset  
 N privatebase  
 F registerextensible  
 F registersimplematrix  
 F registerverboselabel  
 F remapalphabets  
 F renderextensible  
 F renderset  
 F resetattributes  
 F scaleparameters  
 F setextensibledata  
 F setmathsymbol  
 D sets  
 F setsnapping  
 F setvariant  
 F strutcode  
 D styles  
 F syncboth  
 F syncname  
 F syncstyle  
 F toengineclass  
 F toname  
 F tostyle  
 F tweakaftercopyingfont  
 F tweakbeforecopyingfont  
 F utfmathaccent  
 F utfmathclass  
 F utfmathcommand  
 F utfmathfiller  
 F utfmathstretch  
 F variantcode  
 F variantcount  
 F virtualize  
 D virtualized

md5

table

|                      |              |
|----------------------|--------------|
| F HEX                |              |
| F dec                |              |
| F hex                |              |
| F sum                |              |
| F sumHEXA            |              |
| F sumhexa            |              |
| <b>metapost</b>      | <b>table</b> |
| F checkformat        |              |
| F checktracingonline |              |
| F cm                 |              |
| F colorinitializer   |              |
| F convert            |              |
| F currentmpx         |              |
| F currentmpxstatus   |              |
| S defaultformat      |              |
| S defaultinstance    |              |
| S defaultmethod      |              |
| F directrun          |              |
| F escaped            |              |
| D extensiondata      |              |
| F filter             |              |
| F filterclippath     |              |
| F flush              |              |
| F flushnormalpath    |              |
| F format_number      |              |
| F format_string      |              |
| F formatnumber       |              |
| F getbackendoption   |              |
| F getbendtolerance   |              |
| F getbuffer          |              |
| F getclippath        |              |
| F getextensions      |              |
| F getextradata       |              |
| F gethashentries     |              |
| F getinstancebyname  |              |
| F getinstanceinfo    |              |
| F getinstancenames   |              |
| F getjobdata         |              |
| F getmovetolerance   |              |
| F getparameter       |              |
| F getparameterpreset |              |
| F getparameterset    |              |
| F getrecord          |              |
| F getstatistics      |              |



|                        |                       |
|------------------------|-----------------------|
| F gettext              | F registertokens      |
| F gettolerance         | F remappedtext        |
| F graphic              | F remaptext           |
| F graphic_base_pass    | F reporterror         |
| F hascurvature         | F reportsvgfonts      |
| F includesvgbuffer     | F reset               |
| F includesvgfile       | F resetbackendoptions |
| F installplugin        | F resetlastlog        |
| F isobject             | F resetplugins        |
| S lastlog              | F run                 |
| F load                 | F runinternal         |
| F maketext             | F runscript           |
| F mesh                 | F scanparameters      |
| F mesher               | F scriptindex         |
| F meshupper            | F setbendtolerance    |
| N n                    | F setbuffer           |
| F newlogger            | F setextensions       |
| N nofruns              | F setlogging          |
| F nofscriptruns        | F setmovetolerance    |
| F nvformat             | F setoutercolor       |
| F pdfliterals          | F setparameter        |
| F pluginactions        | F setparameterset     |
| F points               | F setrecord           |
| F pointset             | F settext             |
| F popformat            | F settolerance        |
| F popscriptrunner      | F setvariable         |
| F popvariable          | B showlog             |
| F popvariables         | F showsvgpage         |
| F preparetexttextsdata | F simple              |
| F process              | F splitprescript      |
| F processing           | F startgraphic        |
| F processplugins       | F startsvghashing     |
| F pushformat           | F stopgraphic         |
| F pushscriptrunner     | F stopsvghashing      |
| F pushvariable         | F svformat            |
| F pushvariables        | B svgcachefile        |
| B quitonerror          | F svgcolorremapper    |
| B reducetogray         | F svgglyphomp         |
| F registerdirect       | F svgshapestomp       |
| F registerfilter       | F svgshapestopdf      |
| F registermesher       | F svgsplitcolor       |
| F registermeshupper    | F svgtomp             |
| F registerpointset     | F svgtopdf            |
| F registerscript       | F sxsy                |

|                       |              |                       |
|-----------------------|--------------|-----------------------|
| B texerrors           |              | D buff-imp-tex        |
| F texexp              |              | D buff-imp-xml        |
| F theclippath         |              | D buff-ini            |
| F tographic           |              | D buff-par            |
| F totable             |              | D buff-ver            |
| F typesvgpage         |              | D char-enc            |
| F unload              |              | D chem-ini            |
| F untagvariable       |              | D chem-str            |
| <b>mime</b>           | <b>table</b> | D cldf-com            |
| F b64                 |              | D cldf-ver            |
| F decode              |              | D colo-icc            |
| F dot                 |              | D colo-ini            |
| F encode              |              | D cont-run            |
| F eol                 |              | D core-con            |
| F normalize           |              | D core-ctx            |
| F qp                  |              | D core-dat            |
| F qpwrp               |              | D core-sys            |
| F report              |              | D driv-shp            |
| F stuff               |              | D file-mod            |
| F unb64               |              | D file-syn            |
| F unqp                |              | D font-afk            |
| F wrap                |              | D font-agl            |
| F wrp                 |              | D font-aux            |
| <b>moduledata</b>     | <b>table</b> | D font-cff            |
| <b>modules</b>        | <b>table</b> | D font-chk            |
| D anch-loc            |              | D font-cid            |
| D anch-pgr            |              | D font-col            |
| D anch-snc            |              | D font-con            |
| D back-exp            |              | D font-ctx            |
| D back-exp-imp-fnt    |              | D font-def            |
| D back-exp-imp-mth    |              | D font-dsp            |
| D back-exp-imp-ref    |              | D font-enc            |
| D back-exp-imp-tag    |              | D font-enh            |
| D back-imp-lua        |              | D font-fbk            |
| D back-imp-mps        |              | D font-fmp            |
| D back-imp-pdf        |              | D font-hsh            |
| D back-imp-pdp        |              | D font-imp-checks     |
| D blob-ini            |              | D font-imp-combining  |
| D buff-imp-default    |              | D font-imp-compound   |
| D buff-imp-escaped    |              | D font-imp-digits     |
| D buff-imp-lua        |              | D font-imp-dimensions |
| D buff-imp-mp         |              | D font-imp-effects    |
| D buff-imp-nested     |              | D font-imp-italics    |
| D buff-imp-parsed-xml |              | D font-imp-ligatures  |

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|-----------------------|---------------------------|
| D font-imp-math       | D font-tpk                |
| D font-imp-notused    | D font-trt                |
| D font-imp-properties | D font-ttf                |
| D font-imp-quality    | D font-txt                |
| D font-imp-reorder    | D font-vfc                |
| D font-imp-scripts    | D font-vir                |
| D font-imp-spacekerns | D good-ctx                |
| D font-imp-tex        | D good-gen                |
| D font-imp-text       | D good-ini                |
| D font-imp-tracing    | D good-mth                |
| D font-imp-tweaks     | D grph-bmp                |
| D font-imp-unicode    | D grph-con                |
| D font-ini            | D grph-epd                |
| D font-lig            | D grph-fil                |
| D font-log            | D grph-img                |
| D font-lua            | D grph-inc                |
| D font-map            | D grph-pat                |
| D font-mpf            | D grph-raw                |
| D font-mps            | D grph-rul                |
| D font-nod            | D grph-trf                |
| D font-ogr            | D grph-u3d                |
| D font-one            | D java-ini                |
| D font-onr            | D lang-cnt                |
| D font-osd            | D lang-def                |
| D font-ota            | D lang-dis                |
| D font-otc            | D lang-exc                |
| D font-otd            | D lang-hup                |
| D font-oth            | D lang-hyp                |
| D font-oti            | D lang-ini                |
| D font-otj            | D lang-lab                |
| D font-otl            | D lang-rep                |
| D font-oto            | D lang-tra                |
| D font-otr            | D lang-txt                |
| D font-ots            | D lang-url                |
| D font-ott            | D lang-wrd                |
| D font-oup            | D libs-imp-curl           |
| D font-pat            | D libs-imp-ghostscript    |
| D font-prv            | D libs-imp-graphicsmagick |
| D font-san            | D libs-ini                |
| D font-sel            | D lpdf-ano                |
| D font-shp            | D lpdf-col                |
| D font-sol            | D lpdf-epa                |
| D font-syn            | D lpdf-epd                |
| D font-tfm            | D lpdf-fix                |

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|-------------------------|----------------|
| D lpdf-fix-imp-contents | D meta-fun     |
| D lpdf-fix-imp-fonts    | D meta-imp-txt |
| D lpdf-fld              | D meta-ini     |
| D lpdf-fmt              | D meta-lua     |
| D lpdf-grp              | D meta-nod     |
| D lpdf-img              | D meta-pdf     |
| D lpdf-ini              | D meta-tex     |
| D lpdf-lmt              | D mlib-ctx     |
| D lpdf-mis              | D mlib-fio     |
| D lpdf-ren              | D mlib-int     |
| D lpdf-rul              | D mlib-lmp     |
| D lpdf-sig              | D mlib-lmt     |
| D lpdf-tag              | D mlib-lua     |
| D lpdf-u3d              | D mlib-mat     |
| D lpdf-wid              | D mlib-mpf     |
| D lpdf-xmp              | D mlib-msh     |
| D lxml-css              | D mlib-pdf     |
| D lxml-dir              | D mlib-pos     |
| D lxml-ent              | D mlib-pps     |
| D lxml-ini              | D mlib-ptr     |
| D lxml-mms              | D mlib-pts     |
| D lxml-sor              | D mlib-ran     |
| D lxml-tex              | D mlib-run     |
| D math-act              | D mlib-scen    |
| D math-ali              | D mlib-snc     |
| D math-dim              | D mlib-svg     |
| D math-ext              | D node-ali     |
| D math-fbk              | D node-bck     |
| D math-fnt              | D node-fnt     |
| D math-frc              | D node-par     |
| D math-inc              | D node-ref     |
| D math-ini              | D node-rul     |
| D math-map              | D node-tra     |
| D math-mps              | D node-typ     |
| D math-noa              | D pack-obj     |
| D math-pre              | D pack-ori     |
| D math-rad              | D pack-rul     |
| D math-ren              | D page-blk     |
| D math-spa              | D page-cst     |
| D math-tag              | D page-flt     |
| D math-txt              | D page-ini     |
| D math-vfu              | D page-inj     |
| D meta-blb              | D page-ins     |
| D meta-fnt              | D page-lin     |

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|------------|------------|
| D page-mix | D strc-con |
| D page-otr | D strc-doc |
| D page-pst | D strc-flt |
| D page-sel | D strc-ini |
| D page-str | D strc-itm |
| D phys-dim | D strc-lev |
| D publ-aut | D strc-lst |
| D publ-dat | D strc-mar |
| D publ-fnd | D strc-mat |
| D publ-inc | D strc-not |
| D publ-ini | D strc-num |
| D publ-jrn | D strc-pag |
| D publ-oth | D strc-ref |
| D publ-reg | D strc-reg |
| D publ-sor | D strc-rsc |
| D publ-tra | D strc-syn |
| D publ-usr | D strc-tag |
| D regi-ini | D strc-usr |
| D scrn-but | D symb-emj |
| D scrn-fld | D symb-ini |
| D scrn-hlp | D syst-cmp |
| D scrn-ini | D tabl-ali |
| D scrn-pag | D tabl-ntb |
| D scrn-ref | D tabl-tbl |
| D scrn-wid | D tabl-xtb |
| D scrp-cjk | D task-ini |
| D scrp-eth | D trac-bld |
| D scrp-ini | D trac-jus |
| D scrp-tha | D trac-par |
| D scrp-tib | D trac-riv |
| D sort-ini | D trac-sig |
| D sort-lan | D trac-vis |
| D spac-ali | D type-ini |
| D spac-brk | D typo-adj |
| D spac-chr | D typo-bld |
| D spac-hor | D typo-brk |
| D spac-ovl | D typo-cap |
| D spac-par | D typo-chr |
| D spac-pas | D typo-cln |
| D spac-prf | D typo-dha |
| D spac-twi | D typo-dif |
| D spac-ver | D typo-dig |
| D strc-bkm | D typo-dir |
| D strc-blk | D typo-drp |

|            |                              |
|------------|------------------------------|
| D typo-duc | F getbytemap                 |
| D typo-fkr | F getbytemapdata             |
| D typo-fln | F getbytemapdirect           |
| D typo-hid | F getlastaddtype             |
| D typo-inj | F graphformat                |
| D typo-itc | F integer                    |
| D typo-krn | F lmt_function_x             |
| D typo-lan | F lmt_glyphshape_accent      |
| D typo-lbx | F lmt_glyphshape_baseline    |
| D typo-man | F lmt_glyphshape_boundingbox |
| D typo-mar | F lmt_glyphshape_depth       |
| D typo-ovl | F lmt_glyphshape_height      |
| D typo-pag | F lmt_glyphshape_italic      |
| D typo-par | F lmt_glyphshape_llx         |
| D typo-pnc | F lmt_glyphshape_n           |
| D typo-prc | F lmt_glyphshape_path        |
| D typo-rep | F lmt_glyphshape_start       |
| D typo-rub | F lmt_glyphshape_stop        |
| D typo-spa | F lmt_glyphshape_usedbox     |
| D typo-stc | F lmt_glyphshape_usedline    |
| D typo-sus | F lmt_glyphshape_width       |
| D typo-syn | F lmt_mesh_set               |
| D typo-tal | F lmt_mesh_update            |
| D typo-wrp | F lmt_poisson_generate       |
| D unic-ini | F lmt_start_potrace          |

## mp

## function

|                          |                          |
|--------------------------|--------------------------|
| F autoinject             | F lmt_stop_potrace       |
| F boolean                | F lmt_svg_include        |
| F bytemap_load_from_file | F loadbytemap            |
| F cleaned                | F loadbytemapfile        |
| F color                  | F mesh_register          |
| F dataset                | F mf_blob_add            |
| F direct                 | F mf_blob_dimensions     |
| F downgradebytemap       | F mf_blob_index          |
| F downsamplebytemap      | F mf_blob_new            |
| F draw                   | F mf_blob_raw_content    |
| F expandtex              | F mf_blob_raw_dimensions |
| F fill                   | F mf_blob_raw_reset      |
| F fillbytemap            | F mf_blob_raw_toutf      |
| F flatten                | F mf_blob_raw_wipe       |
| F flush                  | F mf_blob_size           |
| F format                 | F mf_blob_width          |
| F formatted              | F mf_external_figure     |
| F fprint                 | F mf_finish_saving_data  |

- F mf\_formatted\_text
- F mf\_get\_outline\_text
- F mf\_graphic\_text
- F mf\_inject\_blob
- F mf\_map\_move
- F mf\_map\_text
- F mf\_named\_color
- F mf\_outline\_text
- F mf\_save\_data
- F mf\_script\_index
- F mf\_start\_saving\_data
- F mf\_stop\_saving\_data
- F mode
- F n
- F newbytemap
- F number
- F numeric
- F pair
- F pairpoints
- F path
- F pathpoints
- F points
- F points\_register
- F processbytemap
- F processingmode
- F quadruple
- F quadruplepoints
- F quoted
- F report
- F setbytemap
- F setbytemapoctave
- F size
- F solve
- F string
- F systemmode
- F transform
- F triplet
- F tripletpoints
- F vprint

## **mplib** table

- F downgradebytemap
- F downsamplebytemap
- F execute
- F expandtex

- F fillbytemap
- F finder
- F finish
- F getbytemap
- F getbytemapdata
- F getbytemapdirect
- F getcallbackstate
- F getcodes
- F getcolormodels
- F getfields
- F gethashentries
- F gethashentry
- F getinternalactions
- F getknotstates
- F getlastaddtype
- F getlogtargets
- F getobjectdata
- F getobjectpeninfo
- F getobjectstacking
- F getobjecttypes
- F getprimitivelegacies
- F getprimitiveorigins
- F getprimitives
- F getpruneoptions
- F getresultstates
- F getscaledbits
- F getscantypes
- F getstates
- F getstatistics
- F getstatus
- F gettolerance
- F gettype
- F gettypes
- F injectboolean
- F injectcmymkcolor
- F injectcolor
- F injectinteger
- F injectnumber
- F injectnumeric
- F injectpair
- F injectpath
- F injectpoints
- F injectstring
- F injecttokens

|                        |          |                          |  |
|------------------------|----------|--------------------------|--|
| F injecttransform      |          | F copylist               |  |
| F injectvector         |          | F count                  |  |
| F injectvectors        |          | F currentattributes      |  |
| F injectwhatever       |          | F dimensions             |  |
| F new                  |          | F effectiveglue          |  |
| F newbytemap           |          | F endofmath              |  |
| F peninfo              |          | F family_font            |  |
| F processbytemap       |          | F fields                 |  |
| B realltimelogging     |          | F findattribute          |  |
| F scanboolean          |          | F flattendiscretionaries |  |
| F scanmykcolor         |          | F flushlist              |  |
| F scancolor            |          | F flushnode              |  |
| F scanexpression       |          | F free                   |  |
| F scaninteger          |          | F getattribute           |  |
| F scannext             |          | F getcachestate          |  |
| F scannumber           |          | F getfield               |  |
| F scannumeric          |          | F getfielderror          |  |
| F scanpair             |          | F getglue                |  |
| F scanpath             |          | F getnodeerrorvalues     |  |
| F scanpen              |          | F getpropertyestable     |  |
| F scanproperty         |          | F getproperty            |  |
| F scanstring           |          | F gluetostring           |  |
| F scansymbol           |          | F hasattribute           |  |
| F scantoken            |          | F hasfield               |  |
| F scantransform        |          | F hasglyph               |  |
| F setbytemap           |          | F hpack                  |  |
| F setbytemapoctave     |          | F hyphenating            |  |
| F settolerance         |          | F id                     |  |
| F showcontext          |          | F insertafter            |  |
| F skiptoken            |          | F insertbefore           |  |
| F solvepath            |          | F instock                |  |
| F version              |          | F intracking             |  |
| <b>nanojpeg</b>        | table    | F inuse                  |  |
| F decode               |          | F isnode                 |  |
| <b>next</b>            | function | F iszeroglue             |  |
| <b>noads</b>           | table    | F kerning                |  |
| F getmathmode          |          | F lastnode               |  |
| F process              |          | F length                 |  |
| F processnested        |          | F ligaturing             |  |
| F processouter         |          | F makeextensible         |  |
| <b>node</b>            | table    | F mlisttohlist           |  |
| F checkdiscretionaries |          | F new                    |  |
| F checkdiscretionary   |          | F protectglyph           |  |
| F copy                 |          | F protectglyphs          |  |



- F protrusionskipable
- F rangedimensions
- F remove
- F serialized
- F setattribute
- F setfield
- F setfielderror
- F setglue
- F setproperty
- F show
- F size
- F slide
- F subtypes
- F tail
- F todirect
- F tonode
- F toString
- F traverse
- F traverseid
- F type
- F types
- F unprotectglyph
- F unprotectglyphs
- F unsetattribute
- F usedlist
- F usesfont
- F vpack
- F write

## nodes

- D accentcodes
- F aligned
- F append
- F apply
- F applyvisuals
- F astable
- F basepoints
- D boundarycodes
- D choicecodes
- F concat
- F copy
- F copy\_node
- F copylist
- F countall
- F currentattributes

## table

- F delete
- D dircodes
- D directioncodes
- D directionvalues
- D dirvalues
- D disccodes
- D discretionarycodes
- D fencecodes
- F fields
- D fillcodes
- D fillvalues
- F firstdirinbox
- F flush
- F flushlist
- F flushnode
- D fractioncodes
- F fullhpack
- F getattr
- F getattribute
- F getboth
- F getbox
- F getchar
- F getfield
- F getfont
- F getid
- F getleader
- F getlist
- F getnext
- F getprev
- F getproperty
- F getsubtype
- D gluecodes
- D glyphcodes
- F hasattribute
- F hasfield
- F hpack
- F idsandsubtypes
- F idstousing
- F insertafter
- F insertbefore
- F insertlistafter
- F installattributehandler
- F is\_display\_math
- F isdirect

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|---------------------|------------------------|
| F isnode            | F setfield             |
| F isnut             | F setfont              |
| D kerncodes         | F setleader            |
| F leftmarginwidth   | F setlink              |
| F link              | F setlist              |
| F linked            | F setnext              |
| F list              | F setprev              |
| D listcodes         | F setproperty          |
| F listtutf          | F showboxes            |
| D literalvalues     | F showlist             |
| F locate            | F showsimplelist       |
| D markcodes         | D skipcodes            |
| D mathcodes         | F somepenalty          |
| D mathvalues        | F somespace            |
| F maxboxwidth       | F splitbox             |
| F new               | F startofpar           |
| D noadcodes         | F stripdiscretionaries |
| D nodecodes         | D subtypes             |
| F nopts             | F tail                 |
| F packlist          | F takeattr             |
| D parcodes          | F takebox              |
| D penaltycodes      | F tobasepoints         |
| F points            | F tocentimeters        |
| F prepend           | F tociceros            |
| F print             | F todidots             |
| F pts               | F todimen              |
| D radicalcodes      | F toinches             |
| F rehpak            | F tomillimeters        |
| F remove            | F tonode               |
| F repackhlist       | F tonodes              |
| F replace           | F tonut                |
| F report            | F topics               |
| F rightmarginwidth  | F topoints             |
| D rulecodes         | F toscaledpoints       |
| F serialize         | F tosequence           |
| F serializebox      | F toString             |
| F serialized        | F totable              |
| F setattr           | F totree               |
| F setattrtribute    | F toutf                |
| F setattrrlist      | F traverse             |
| F setboth           | F traverseid           |
| F setbox            | F unsetattribute       |
| F setboxtonaturalwd | F upcomingproperties   |
| F setchar           | F usedlist             |

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| F vianodes         |       | F topoints             |          |
| F vianuts          |       | F toscaled             |          |
| F visualizebox     |       | F toscaledpoints       |          |
| F vpack            |       | F totoves              |          |
| D whatsitcodes     |       | F valid                |          |
| F write            |       | <b>objects</b>         | table    |
| <b>number</b>      | table | F dimensions           |          |
| F %                |       | F found                |          |
| F basepoints       |       | N n                    |          |
| F bytetodecimal    |       | F page                 |          |
| F clamp            |       | F reference            |          |
| F compactfloat     |       | F register             |          |
| F decimaltobyte    |       | F restore              |          |
| D dimenfactors     |       | <b>optional</b>        | table    |
| F formatted        |       | <b>optionalrequire</b> | function |
| F formattedfloat   |       | <b>os</b>              | table    |
| F formattedinteger |       | F __getenv__           |          |
| F fractiontobyte   |       | F __setenv__           |          |
| F idiv             |       | S binsuffix            |          |
| N maxdimen         |       | N bits                 |          |
| F nopts            |       | F clock                |          |
| F nubasepoints     |       | F converttime          |          |
| F nupoints         |       | F date                 |          |
| F percent          |       | F difftime             |          |
| F points           |       | F enableansi           |          |
| F pts              |       | D env                  |          |
| F signed           |       | F execute              |          |
| F sparseexponent   |       | F exit                 |          |
| F splitdimen       |       | F fulltime             |          |
| F to16dot16        |       | F getcodepage          |          |
| F tobasepoints     |       | F getenv               |          |
| F tobitstring      |       | F getnamevalues        |          |
| F tocentimeters    |       | F gettimeofday         |          |
| F tociceros        |       | F gettypevalues        |          |
| F tocontext        |       | F getunamefields       |          |
| F todidots         |       | F isleapyear           |          |
| F todimen          |       | F launch               |          |
| F toediths         |       | S libsuffix            |          |
| F toevenhex        |       | F localtime            |          |
| F toinches         |       | S name                 |          |
| F tomillimeters    |       | S newline              |          |
| F tonumberf        |       | F nofdays              |          |
| F tonumberg        |       | F now                  |          |
| F topicas          |       | F pipeto               |          |

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|-----------------|-------|----------------------|----------|
| S platform      |       | <b>pagebuilders</b>  | table    |
| F remove        |       | <b>pairs</b>         | function |
| F rename        |       | <b>parametersets</b> | table    |
| F resultof      |       | <b>pcall</b>         | function |
| F runtime       |       | <b>pdf</b>           | table    |
| S selfbin       |       | F immediateobj       |          |
| S selfcore      |       | <b>pdfdecode</b>     | table    |
| S selfdir       |       | <b>pdfe</b>          | table    |
| S selflink      |       | F arraytotable       |          |
| S selfname      |       | F close              |          |
| S selfpath      |       | F closestream        |          |
| F serialwrite   |       | F dictionarytotable  |          |
| F setenv        |       | F getarray           |          |
| F setexitcode   |       | F getboolean         |          |
| F setlocale     |       | F getbox             |          |
| F sleep         |       | F getcatalog         |          |
| F startuptime   |       | F getdictionary      |          |
| F time          |       | F getencodingvalues  |          |
| F timezone      |       | F getfieldtypes      |          |
| F tmpname       |       | F getfromarray       |          |
| F today         |       | F getfromdictionary  |          |
| S type          |       | F getfromobject      |          |
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| F validate      |       | F getinfo            |          |
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| F openfile        |       | F atan         |       |
| F openstream      |       | F band         |       |
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| F __bxor          |       | F rounded      |       |
| F __div           |       | F shift        |       |
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| D authormap           |          | F bitmap           |          |
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| F found               |          | F disable          |          |
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| F findwildcardfiles            | F popextrapath            |
| F formatofsuffix               | F popinputname            |
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 F pushextrapath  
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| F concat          |              | F getfontstate          |              |
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| F getboxoptionvalues          | F getinputstatefile           |
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| F getpagestatevalues         | N hshcatcodes                  |
| F getparagraphfillmodevalues | D hyphenationcodes             |
| F getparagraphoptionvalues   | D ifs                          |
| F getparametermodevalues     | N inicatcodes                  |
| F getparcontextvalues        | D insertcategorycodes          |
| F getpardataspecifications   | D insertoptioncodes            |
| F getpardir                  | D insertsplitcodes             |
| F getparmodevalues           | F integerdef                   |
| F getparstate                | D interactioncodes             |
| F getparstatefields          | F isattribute                  |
| F getpartriggervalue         | F isbox                        |
| F getpenaltyoptionvalues     | F iscount                      |
| F getpositvalue              | F isdefined                    |
| F getprepoststatevalues      | F isdimen                      |

F isfloat  
 F isglue  
 F ismuglue  
 F ismuskip  
 F isprintable  
 F isskip  
 F istoks  
 F lastnodetype  
 F less  
 F lessequal  
 F linebreak  
 D linebreakstatecodes  
 D listanchorcodes  
 D listgeometrycodes  
 D listsigncodes  
 N luacatcodes  
 F mathchardef  
 D mathcontrolcodes  
 D mathoptioncodes  
 D mathparametercodes  
 D mathscriptordercodes  
 D mathscriptsmodecodes  
 D modes  
 F mprint  
 D mvloptioncodes  
 N nilcatcodes  
 D noadoptoptioncodes  
 D normalizelinecodes  
 D normalizeparcodes  
 D nospacesmodecodes  
 N notcatcodes  
 F number  
 D paragraphfillmodecodes  
 D paragraphoptioncodes  
 D parcontextcodes  
 D partriggercodes  
 D penaltyoptioncodes  
 F poplocal  
 F popsavelevel  
 F positdef  
 F preparelinebreak  
 F primitives  
 F print  
 D protrusionboundarycodes

N prtcategories  
 F pushlocal  
 F pushsavelevel  
 F quitlocal  
 F quittoks  
 F resetparagraph  
 N rlncategories  
 F romannumeral  
 F round  
 D ruleoptioncodes  
 F runlocal  
 D runstatecodes  
 F runstring  
 F runtoks  
 F saveboxresource  
 F scale  
 F scantoks  
 F set  
 F setamcode  
 F setattribute  
 F setbalanceshape  
 F setbox  
 F setboxdir  
 F setcardinalvalue  
 F setcatcode  
 F setcccode  
 F setconditional  
 F setcount  
 F setdelcode  
 F setdimen  
 F setdimensionvalue  
 F setfloat  
 F setglue  
 F setglyphdata  
 F setglyphscript  
 F setglyphstate  
 F sethccode  
 F sethmcodes  
 F setinputstatefile  
 F setinputstateline  
 F setinputstatemode  
 F setinsertcontent  
 F setinsertdistance  
 F setinsertlimit



F setinsertmaxplaced  
 F setinsertmultiplier  
 F setintegervalue  
 F setinteraction  
 F setlccode  
 F setlinedir  
 F setlist  
 F setlocalbox  
 F setmath  
 F setmathcode  
 F setmathdir  
 F setmuglue  
 F setmuskip  
 F setnest  
 F setoverloadstate  
 F setpardir  
 F setpositvalue  
 F setrunstate  
 F setsfcode  
 F setskip  
 F setspcode  
 F settextrdir  
 F settoks  
 F setuccode  
 D shapingpenaltiescodes  
 F shiftparshape  
 F shipout  
 F showcontext  
 D snappingcodes  
 F snapshotpar  
 F sp  
 D spacefactorcodes  
 D spaceskipcodes  
 D specialmathclasscodes  
 D specificationoptioncodes  
 F splitbox  
 D splitcodes  
 F sprint  
 F stringtocodesbitmap  
 D systemmodes  
 F takebox  
 N texcatcodes  
 D textcontrolcodes  
 F tonumber

F toscaled  
 N tpcatcodes  
 N tpbcatcodes  
 F tprint  
 F trackedattribute  
 D trackers  
 F triggerbuildpage  
 N txtcatcodes  
 F uprint  
 F useboxresource  
 D userdiscoptioncodes  
 D userglyphoptioncodes  
 N vrbcatcodes  
 F write

## **texconditionals**

data table

## **texconfig**

table

S firstline  
 S formatname  
 N functionsize  
 F init  
 B kpse\_init  
 N propertiessize  
 S shell\_escape

## **texconstants**

data table

## **texifs**

data table

## **texio**

table

F closeinput  
 F forceendoffile  
 F getinputindex  
 F getselectorvalues  
 F getsourcefilename  
 F setlogfile  
 F terminal  
 F write  
 F write\_nl  
 F writenl  
 F writeselector  
 F writeselectorlf  
 F writeselectornl

## **texisdefined**

data table

## **texmodes**

data table

## **texsystemmodes**

data table

## **thirddata**

table

## **toboolean**

function

**todecimal****token**

F create  
F expand  
F expand\_macro  
F expandmacro  
F future\_expand  
F futureexpand  
F get\_active  
F get\_cmdchrscs  
F get\_cmdname  
F get\_command  
F get\_csname  
F get\_cstoken  
F get\_dimension  
F get\_expandable  
F get\_fields  
F get\_flags  
F get\_frozen  
F get\_id  
F get\_immutable  
F get\_index  
F get\_instance  
F get\_integer  
F get\_macro  
F get\_meaning  
F get\_noaligned  
F get\_parameters  
F get\_permanent  
F get\_primitive  
F get\_protected  
F get\_range  
F get\_tok  
F get\_tolerant  
F getactive  
F getcmdchrscs  
F getcmdname  
F getcommand  
F getcommandvalues  
F getconstant  
F getcsname  
F getcstoken  
F getdimension  
F getexpandable

**function****table**

F getexpansion  
F getfields  
F getflags  
F getfrozen  
F getfunctionvalues  
F getid  
F getimmutable  
F getindex  
F getinstance  
F getinteger  
F getlength  
F getmacro  
F getmeaning  
F getnoaligned  
F getparameters  
F getpermanent  
F getprimitive  
F getprimitives  
F getprotected  
F getrange  
F gettok  
F gettolerant  
F gobble  
F gobble\_dimension  
F gobble\_integer  
F gobbledimension  
F gobbleinteger  
F grab  
F grabstring  
F inhibit  
F is\_defined  
F is\_next\_char  
F is\_token  
F isactivestring  
F isdefined  
F isnextchar  
F istoken  
F locatemacro  
F new  
F peek\_next  
F peek\_next\_char  
F peek\_next\_expanded  
F peeknext  
F peeknextchar

|                        |                         |
|------------------------|-------------------------|
| F peeknextexpanded     | F scanargument          |
| F pop_macro            | F scanbox               |
| F popmacro             | F scancardinal          |
| F push_macro           | F scanchar              |
| F pushmacro            | F scancmdchr            |
| F put_back             | F scancmdchrexploded    |
| F put_next             | F scancode              |
| F putback              | F scanscname            |
| F putnext              | F scancstoken           |
| F save_lua             | F scandelimited         |
| F savelua              | F scandetokened         |
| F scan_argument        | F scandimen             |
| F scan_box             | F scandimenargument     |
| F scan_cardinal        | F scandimension         |
| F scan_char            | F scandimensionargument |
| F scan_cmdchr          | F scanfloat             |
| F scan_cmdchr_expanded | F scanglue              |
| F scan_code            | F scanint               |
| F scan_csname          | F scaninteger           |
| F scan_delimited       | F scanintegerargument   |
| F scan_dimen           | F scankey               |
| F scan_float           | F scankeyword           |
| F scan_glue            | F scankeywordcs         |
| F scan_integer         | F scanletters           |
| F scan_key             | F scanluacardinal       |
| F scan_keyword         | F scanluainteger        |
| F scan_keyword_cs      | F scanluanumber         |
| F scan_letters         | F scannext              |
| F scan_luacardinal     | F scannextchar          |
| F scan_luainteger      | F scannextexpanded      |
| F scan_luanumber       | F scanposit             |
| F scan_next            | F scanreal              |
| F scan_next_char       | F scanscale             |
| F scan_next_expanded   | F scanskip              |
| F scan_real            | F scanstring            |
| F scan_scale           | F scantoken             |
| F scan_skip            | F scantokencode         |
| F scan_string          | F scantokenlist         |
| F scan_token           | F scantokenstring       |
| F scan_token_code      | F scantoks              |
| F scan_tokenlist       | F scanvalue             |
| F scan_toks            | F scanword              |
| F scan_value           | F serialize             |
| F scan_word            | F set_dimension         |

|                      |          |                    |          |
|----------------------|----------|--------------------|----------|
| F set_integer        |          | F default          |          |
| F set_lua            |          | F disable          |          |
| F set_macro          |          | F dumphashtofile   |          |
| F setdimension       |          | F enable           |          |
| F setinteger         |          | F list             |          |
| F setlua             |          | S name             |          |
| F setmacro           |          | F register         |          |
| F setmacrofrommark   |          | F report           |          |
| F setmacrotonothing  |          | F reset            |          |
| F skip_next          |          | F savehash         |          |
| F skip_next_expanded |          | F show             |          |
| F skipnext           |          | F value            |          |
| F skipnextexpanded   |          | F values           |          |
| F type               |          | <b>type</b>        | function |
| F undefine_macro     |          | <b>typesetters</b> | table    |
| F undefinemacro      |          | F marksuspects     |          |
| <b>tokens</b>        | table    | F showinvisibles   |          |
| F astable            |          | F showsuspects     |          |
| D cache              |          | <b>unpack</b>      | function |
| D commands           |          | <b>updaters</b>    | table    |
| F compile            |          | F apply            |          |
| F constant           |          | F register         |          |
| F create             |          | <b>url</b>         | table    |
| F defined            |          | F addscheme        |          |
| F flags              |          | F barepath         |          |
| F getinteger         |          | F construct        |          |
| F gobble             |          | F decode           |          |
| F isactivestring     |          | F encode           |          |
| F isdefined          |          | F escape           |          |
| F istoken            |          | F filename         |          |
| F new                |          | F hashed           |          |
| F pickup             |          | F hasscheme        |          |
| F popmacro           |          | F query            |          |
| F pushmacro          |          | F split            |          |
| F setinteger         |          | F toquery          |          |
| D values             |          | F unescape         |          |
| <b>tonumber</b>      | function | F unescapeget      |          |
| <b>tostring</b>      | function | <b>userdata</b>    | table    |
| <b>tracers</b>       | table    | <b>utf</b>         | table    |
| F printererror       |          | F byte             |          |
| F printwarning       |          | F bytepairs        |          |
| F showlines          |          | F char             |          |
| <b>trackers</b>      | table    | F characters       |          |
| D data               |          | F chrln            |          |

|                      |              |                  |  |
|----------------------|--------------|------------------|--|
| F count              |              | F add            |  |
| F filetype           |              | F append         |  |
| F is_valid           |              | F average        |  |
| F len                |              | F ceiling        |  |
| F length             |              | F checkoverlap   |  |
| F lower              |              | F collectoverlap |  |
| F remapper           |              | F concat         |  |
| F replacer           |              | F copy           |  |
| F split              |              | F crossproduct   |  |
| F splitlines         |              | F delete         |  |
| F sub                |              | F determinant    |  |
| F substituter        |              | F div            |  |
| F tocodes            |              | F equal          |  |
| F toeight            |              | F exchange       |  |
| F toentities         |              | F floor          |  |
| F tostring           |              | F get            |  |
| F totable            |              | F getdimensions  |  |
| F upper              |              | F getepsilon     |  |
| F ustring            |              | F getmemory      |  |
| F utf16_to_utf8_be   |              | F getmesh        |  |
| F utf16_to_utf8_be_t |              | F getmeshtypes   |  |
| F utf16_to_utf8_le   |              | F getpointtypes  |  |
| F utf16_to_utf8_le_t |              | F getrc          |  |
| F utf16_to_utf8_t    |              | F getrow         |  |
| F utf32_to_utf8_be   |              | F getsizes       |  |
| F utf32_to_utf8_be_t |              | F getstacking    |  |
| F utf32_to_utf8_le   |              | F gettype        |  |
| F utf32_to_utf8_le_t |              | F homogenize     |  |
| F utf32_to_utf8_t    |              | F identity       |  |
| F utf8_to_utf16      |              | F inner          |  |
| F utf8_to_utf16_be   |              | F insert         |  |
| F utf8_to_utf16_le   |              | F inverse        |  |
| F utf8_to_utf8_t     |              | F ismesh         |  |
| F values             |              | F ispoints       |  |
| F xstring            |              | F issingular     |  |
| <b>utf8</b>          | <b>table</b> | F isvector       |  |
| F char               |              | F iszero         |  |
| S charpattern        |              | F meshname       |  |
| F codepoint          |              | F mul            |  |
| F codes              |              | F negate         |  |
| F len                |              | F new            |  |
| F offset             |              | F normalize      |  |
| <b>utilities</b>     | <b>table</b> | F onecolumn      |  |
| <b>vector</b>        | <b>table</b> | F onerow         |  |

F product  
 F remove  
 F replace  
 F round  
 F rowechelon  
 F set  
 F setepsilon  
 F setnext  
 F setrc  
 F setstacking  
 F slice  
 F sort  
 F sub  
 F swap  
 F tostring  
 F totable  
 F transpose  
 F triangulate  
 F truncate  
 F type

## visualizers

table

D escapepatterns  
 F getspecification  
 F getvisualizer  
 F load  
 F makenested  
 F makepattern  
 F newgrammar  
 F newhandler  
 F parsedxml  
 F pattern  
 F register  
 F registerescapecommand  
 F registerescapepline  
 F registerescapepattern  
 S signal  
 U signalpattern  
 D specifications  
 F visualize  
 F visualizebuffer  
 F visualizefile  
 F visualizestring  
 F write  
 F writeargument

F writedefault  
 F writeemptyline  
 F writenewline  
 F writespace

## warn

## xcomplex

F I  
 F \_\_add  
 F \_\_div  
 F \_\_eq  
 F \_\_mul  
 S \_\_name  
 F \_\_pow  
 F \_\_sub  
 F \_\_tostring  
 F \_\_unm  
 F abs  
 F acos  
 F acosh  
 F arg  
 F asin  
 F asinh  
 F atan  
 F atanh  
 F conj  
 F cos  
 F cosh  
 F dawson  
 F erf  
 F erfc  
 F erfcx  
 F erfi  
 F exp  
 F i  
 F imag  
 F log  
 F new  
 F pow  
 F proj  
 F real  
 F sin  
 F sinh  
 F sqrt  
 F tan

## function

table

|                 |              |                  |              |
|-----------------|--------------|------------------|--------------|
| F tanh          |              | F sqrt           |              |
| F topair        |              | F toengstring    |              |
| F toString      |              | F tonumber       |              |
| F totable       |              | F toString       |              |
| F voigt         |              | F trim           |              |
| F voigt_hwhm    |              | <b>xinterval</b> | <b>table</b> |
| <b>xdecimal</b> | <b>table</b> | F __add          |              |
| F __add         |              | F __div          |              |
| F __band        |              | F __eq           |              |
| F __bor         |              | F __le           |              |
| F __bxor        |              | F __lt           |              |
| F __div         |              | F __mod          |              |
| F __eq          |              | F __mul          |              |
| F __idiv        |              | S __name         |              |
| F __le          |              | F __pow          |              |
| F __lt          |              | F __sub          |              |
| F __mod         |              | F __toString     |              |
| F __mul         |              | F __unm          |              |
| S __name        |              | F abs            |              |
| F __pow         |              | F acos           |              |
| F __shl         |              | F acosh          |              |
| F __shr         |              | F asin           |              |
| F __sub         |              | F asinh          |              |
| F __toString    |              | F asstring       |              |
| F __unm         |              | F atan           |              |
| F abs           |              | F atanh          |              |
| F band          |              | F cos            |              |
| F bor           |              | F cosh           |              |
| F bxor          |              | F diameter       |              |
| F conj          |              | F equal          |              |
| F copy          |              | F erf            |              |
| F exp           |              | F erfc           |              |
| F getprecision  |              | F exp            |              |
| F ln            |              | F exp10          |              |
| F log           |              | F exp2           |              |
| F max           |              | F expm1          |              |
| F min           |              | F greater        |              |
| F minus         |              | F greaterequal   |              |
| F new           |              | F hull           |              |
| F plus          |              | F intersect      |              |
| F pow           |              | F less           |              |
| F rotate        |              | F lessequal      |              |
| F setprecision  |              | F log            |              |
| F shift         |              | F log10          |              |

F log1p  
F log2  
F mid  
F mod  
F new  
F neweps  
F pow  
F seteq  
F setge  
F setgt  
F settle  
F setlt  
F sin  
F sinh  
F sqr  
F sqrt  
F tan  
F tanh  
F topair  
F toString  
F totable  
F usedendian  
F within

## xmath

F acos  
F acosh  
F asin  
F asinh  
F atan  
F atan2  
F atanh  
F cbrt  
F ceil  
F copysign  
F cos  
F cosh  
F crossproduct  
F deg  
F dotproduct  
F erf  
F erfc  
F exp  
F exp2  
F expm1

## table

F fabs  
F fdim  
F floor  
F fma  
F fmax  
F fmin  
F fmod  
F frexp  
F gamma  
F hypot  
N inf  
F isfinite  
F isinf  
F isnan  
F isnormal  
F j0  
F j1  
F jn  
F ldexp  
F lgamma  
F log  
F log10  
F log1p  
F log2  
F logb  
F modf  
N nan  
F nearbyint  
F nextafter  
F normalize  
N pi  
F pow  
F rad  
F remainder  
F remquo  
F round  
F scalbn  
F sin  
F sinh  
F sqrt  
F tan  
F tanh  
F tgamma  
F trunc



|                      |              |                         |
|----------------------|--------------|-------------------------|
| F y0                 |              | D entities              |
| F y1                 |              | F erase                 |
| F yn                 |              | F errorhandler          |
| <b>xml</b>           | <b>table</b> | F escaped               |
| F addentitiesdoctype |              | F expand                |
| F addspecialcode     |              | F fillin                |
| F after              |              | F filter                |
| F all                |              | F first                 |
| F applylpath         |              | F found                 |
| F applyselector      |              | F getbuffer             |
| F assign             |              | F gethandlers           |
| F attribute          |              | F gethandlersfunction   |
| F badinclusions      |              | F include               |
| F before             |              | F inclusion             |
| F body               |              | F inclusions            |
| F cdata              |              | F index                 |
| F cdatatotext        |              | F inheritattributes     |
| F checkbom           |              | F inheritedconvert      |
| F checkedempty       |              | F inject                |
| F checkerror         |              | F inject_element        |
| F checknamespace     |              | F inject_element_after  |
| F checkns            |              | F inject_element_before |
| F cleansed           |              | F injectafter           |
| F collect            |              | F injectbefore          |
| F collect_elements   |              | F insert                |
| F collect_tags       |              | F insert_element        |
| F collect_texts      |              | F insert_element_after  |
| F collected          |              | F insert_element_before |
| F collecttexts       |              | F insertafter           |
| F compactelement     |              | F insertbefore          |
| F content            |              | F insertcdata           |
| F convert            |              | F insertcomment         |
| F copy               |              | F inspect               |
| F count              |              | F is_valid              |
| F cprint             |              | F kind                  |
| S defaultprotocol    |              | F last                  |
| F delete             |              | F lastmatch             |
| F delete_element     |              | F load                  |
| F despecialized      |              | F lpath                 |
| F each               |              | F lpathcached           |
| F each_element       |              | F lpathcalls            |
| F elements           |              | F lshow                 |
| F elementtocdata     |              | F makestandalone        |
| F empty              |              | F match                 |

F name  
 F newhandlers  
 F nodesettostring  
 F originalload  
 F package  
 F parent  
 U parsedentitylpeg  
 F popmatch  
 F position  
 D privatecodes  
 F privatetoken  
 F process  
 F process\_attributes  
 F process\_elements  
 F processattributes  
 F pure  
 F pushmatch  
 F raw  
 F registerentity  
 F registers  
 F reindex  
 F remap\_name  
 F remap\_namespace  
 F remap\_tag  
 F remapname  
 F remapnamespace  
 F remaptag  
 F rename  
 F renamespace  
 U reparsedentitylpeg  
 F replace  
 F replace\_element  
 F resetinjectors  
 F resolvens  
 F root  
 F save  
 F selection  
 F separate  
 F serialize  
 F serializetotext  
 F setcdata  
 F setcomment  
 F sethandlersfunction  
 F setinjectors

F setproperty  
 F settings  
 D specialcodes  
 F sprint  
 F string  
 F strip  
 F strip\_whitespace  
 F stripelement  
 F stripleadingspaces  
 F table  
 F text  
 F textonly  
 F texttocdata  
 F tocdata  
 F toelement  
 F tostring  
 F totable  
 F toxml  
 F tprint  
 F traverse  
 F unescaped  
 U unescapedentitylpeg  
 F unprivatized  
 F unspecialized  
 F wipe  
 F withelement  
 F withelements  
 F wrap

## xpcall

function

## xzip

table

F adler32  
 F compress  
 F compresssize  
 F crc32  
 F decompress  
 F decompresssize

## zbuffer

table

F bounds  
 F composite  
 F crop  
 F edgepoint  
 F flatten  
 F get  
 F getcolor

F getconstants  
F getdepth  
F getnormal  
F getnormalmodes  
F getopacity  
F getsize  
F implicit  
F iszbuffer  
F meshmesh  
F new  
F points  
F process  
F project  
F resolve  
F set  
F setcolor  
F setdepth  
F setnormal  
F setopacity  
F setup  
F smoothen  
F stipple\_0  
F stipple\_1

F stipple\_2  
F stipple\_3  
F stipple\_4  
F stipple\_setup  
F tobytes  
F tostring  
F totable  
F transform  
F trianglebounds  
F triangled  
F triangles  
F type

## zip

table

F closearchive  
F openarchive

## zlib

table

F adler32  
F compress  
F compresssize  
F crc32  
F decompress  
F decompresssize

# graphics (context)

|                    |          |                             |          |
|--------------------|----------|-----------------------------|----------|
| <b>backgrounds</b> | table    | <b>identify</b>             | function |
| F fetchmultipar    |          | <b>registerpdfcompactor</b> | function |
| F pair             |          | <b>registerpdffont</b>      | function |
| F path             |          | <b>synchronizers</b>        | table    |
| F point            |          | F collapse                  |          |
| <b>bitmaps</b>     | table    | F collect                   |          |
| F flush            |          | F extend                    |          |
| F load             |          | F getbottom                 |          |
| F new              |          | F getd                      |          |
| F placeholder      |          | F geth                      |          |
| F tocontext        |          | F getkind                   |          |
| <b>bpsize</b>      | function | F getsizes                  |          |
| <b>colorspaces</b> | table    | F gettask                   |          |
| <b>identifiers</b> | table    | F gettop                    |          |
| F bitmap           |          | F getw                      |          |
| F jp2              |          | F getx                      |          |
| F jpeg             |          | F gety                      |          |
| F jpg              |          | F prune                     |          |
| F png              |          |                             |          |

## gzip (context)

**compress**    function  
**compressed** function  
**decompress** function

**load**        function  
**save**        function  
**suffix**      function

## helper (context)

|                    |          |
|--------------------|----------|
| <b>getprofile</b>  | function |
| <b>mathnested</b>  | function |
| <b>mathprocess</b> | function |
| <b>mathstep</b>    | function |
| <b>snapindeed</b>  | function |

|                            |          |
|----------------------------|----------|
| <b>snplist</b>             | function |
| <b>traversecharmmethod</b> | function |
| <b>traverseglyphmethod</b> | function |
| <b>traversehorizontal</b>  | function |
| <b>traversevertical</b>    | function |

# images (context)

|                       |          |                  |          |
|-----------------------|----------|------------------|----------|
| <b>check</b>          | function | <b>B crop</b>    |          |
| <b>checksize</b>      | function | <b>B media</b>   |          |
| <b>copy</b>           | function | <b>B none</b>    |          |
| <b>create</b>         | function | <b>B trim</b>    |          |
| <b>embed</b>          | function | <b>storedata</b> | function |
| <b>embedfromstore</b> | function | <b>totable</b>   | function |
| <b>fetchdata</b>      | function | <b>types</b>     | table    |
| <b>keys</b>           | table    | <b>N jbig2</b>   |          |
| <b>print</b>          | function | <b>N jp2</b>     |          |
| <b>resetdata</b>      | function | <b>N jpg</b>     |          |
| <b>resetstore</b>     | function | <b>N none</b>    |          |
| <b>scan</b>           | function | <b>N pdf</b>     |          |
| <b>serialize</b>      | function | <b>N png</b>     |          |
| <b>sizes</b>          | table    | <b>N stream</b>  |          |
| <b>B art</b>          |          | <b>wrap</b>      | function |
| <b>B bleed</b>        |          |                  |          |

# interactions (context)

**attachments** data table

**fields** table

F clone

F define

F defineset

F insert

**general** table

F getidentity

F getidentityvariable

**help** table

F collect

F reference

F register

**javascripts** table

F addtopreamble

F code

D codes

F flushpreambles

D functions

D preambles

F setpreamble

F storecode

F storepreamble

F usepreamblenow

F usescripts

**linkedList** data table

**pages** table

F setpagetransition

F setupcanvas

**references** table

F setcloseddocumentaction

F setclosepageaction

F setopendocumentaction

F setopenpageaction

**renderings** data table



# interfaces (context)

|                             |            |                          |            |
|-----------------------------|------------|--------------------------|------------|
| <b>arrays</b>               | table      | <b>namesofscanners</b>   | table      |
| <b>cachedsetups</b>         | data table | <b>namespaces</b>        | table      |
| <b>cachesetup</b>           | function   | F define                 |            |
| <b>cardinals</b>            | table      | F list                   |            |
| <b>checkedspecification</b> | function   | <b>numbers</b>           | table      |
| <b>complete</b>             | data table | <b>registernamespace</b> | function   |
| <b>constants</b>            | data table | <b>registerscanner</b>   | function   |
| <b>corenamespaces</b>       | data table | <b>scanners</b>          | data table |
| <b>currentinterface</b>     | string     | <b>setconstant</b>       | function   |
| <b>currentresponse</b>      | string     | <b>setcount</b>          | function   |
| <b>definecommand</b>        | function   | <b>setdimen</b>          | function   |
| <b>defined</b>              | function   | <b>setelement</b>        | function   |
| <b>doifelsemessage</b>      | function   | <b>setformat</b>         | function   |
| <b>elements</b>             | data table | <b>setmacro</b>          | function   |
| <b>floats</b>               | table      | <b>setmessage</b>        | function   |
| <b>formats</b>              | data table | <b>setmessages</b>       | function   |
| <b>getmessage</b>           | function   | <b>settranslation</b>    | function   |
| <b>getnamespace</b>         | function   | <b>setupstrings</b>      | data table |
| <b>getsetupstring</b>       | function   | <b>setvariable</b>       | function   |
| <b>hashedsetups</b>         | data table | <b>showmessage</b>       | function   |
| <b>implement</b>            | function   | <b>storedscanners</b>    | data table |
| <b>integers</b>             | table      | <b>tolist</b>            | function   |
| <b>interfacedcommand</b>    | function   | <b>translations</b>      | data table |
| <b>interfaces</b>           | table      | <b>usednamespaces</b>    | data table |
| <b>knownscanner</b>         | function   | <b>variables</b>         | data table |
| <b>message</b>              | function   | <b>writestatus</b>       | function   |
| <b>nameofscanner</b>        | function   |                          |            |

## io (basic lua)

|                      |                     |                      |                     |
|----------------------|---------------------|----------------------|---------------------|
| <b>ask</b>           | function            | <b>open</b>          | overloaded function |
| <b>bytes</b>         | function            | <b>output</b>        | overloaded function |
| <b>characters</b>    | function            | <b>pathseparator</b> | string              |
| <b>close</b>         | function            | <b>popen</b>         | overloaded function |
| <b>copydata</b>      | function            | <b>read</b>          | function            |
| <b>exists</b>        | function            | <b>readall</b>       | function            |
| <b>fileseparator</b> | string              | <b>readnumber</b>    | function            |
| <b>flush</b>         | function            | <b>readstring</b>    | function            |
| <b>gobble</b>        | function            | <b>savadata</b>      | function            |
| <b>input</b>         | overloaded function | <b>size</b>          | function            |
| <b>lines</b>         | overloaded function | <b>stderr</b>        | userdata            |
| <b>loadchunk</b>     | function            | <b>stdin</b>         | userdata            |
| <b>loaddata</b>      | function            | <b>stdout</b>        | userdata            |
| <b>loadlines</b>     | function            | <b>tmpfile</b>       | function            |
| <b>newreader</b>     | function            | <b>type</b>          | function            |
| <b>noflines</b>      | function            | <b>write</b>         | function            |

# job (context)

|                          |          |                    |       |
|--------------------------|----------|--------------------|-------|
| <b>comment</b>           | function | D collected        |       |
| <b>datasets</b>          | table    | F countervalue     |       |
| D collected              |          | F extend           |       |
| F extend                 |          | F realpage         |       |
| F getdata                |          | F realpageorder    |       |
| F setdata                |          | F setstate         |       |
| D tobesaved              |          | D tobesaved        |       |
| <b>disablesave</b>       | function | <b>parwrappers</b> | table |
| <b>embedded</b>          | table    | D collected        |       |
| D collected              |          | D tobesaved        |       |
| D tobesaved              |          | <b>positions</b>   | table |
| <b>fileobjreferences</b> | table    | F b_column         |       |
| D collected              |          | F b_region         |       |
| D tobesaved              |          | F c                |       |
| <b>files</b>             | table    | D collected        |       |
| D collected              |          | F column           |       |
| F context                |          | F columnofpos      |       |
| B forcerun               |          | F copy             |       |
| F run                    |          | F depth            |       |
| D tobesaved              |          | F e_column         |       |
| <b>getrandomseed</b>     | function | F e_region         |       |
| <b>initialize</b>        | function | F enhance          |       |
| <b>linkedlists</b>       | table    | F extra            |       |
| D collected              |          | F get              |       |
| D tobesaved              |          | F getcolumndata    |       |
| <b>load</b>              | function | F getfree          |       |
| <b>loadother</b>         | function | F gethpos          |       |
| <b>meshedup</b>          | table    | F gethvrpos        |       |
| D collected              |          | F getpos           |       |
| D tobesaved              |          | F getreserved      |       |
| <b>objects</b>           | table    | F getrpos          |       |
| D collected              |          | F getsync          |       |
| F get                    |          | F gettobesaved     |       |
| F number                 |          | F getvpos          |       |
| F page                   |          | F hangafter        |       |
| F save                   |          | F hangindent       |       |
| F set                    |          | F height           |       |
| D tobesaved              |          | F hsize            |       |
| <b>pack</b>              | boolean  | N lastr            |       |
| <b>packversion</b>       | number   | N lastx            |       |
| <b>pagestates</b>        | table    | N lasty            |       |

|                    |                  |          |
|--------------------|------------------|----------|
| F leftskip         | F upperleft      |          |
| F lowerleft        | F upperright     |          |
| F lowerright       | F used           |          |
| F markregionbox    | F whd            |          |
| F n                | F width          |          |
| F onsamepage       | F x              |          |
| F overlapping      | F xy             |          |
| F p                | F y              |          |
| F page             | <b>register</b>  | function |
| F paragraph        | <b>save</b>      | function |
| F parindent        | <b>structure</b> | table    |
| F position         | D collected      |          |
| F r                | <b>variables</b> | table    |
| F region           | D checksums      |          |
| F registerhandlers | D collected      |          |
| F replace          | F getchecksum    |          |
| F rightskip        | F makechecksum   |          |
| F set              | F restore        |          |
| F setall           | F save           |          |
| F setregionbox     | F setchecksum    |          |
| F setspec          | D tobesaved      |          |
| F settobesaved     | <b>version</b>   | number   |
| D tobesaved        |                  |          |

## kdtree (context)

|                     |          |
|---------------------|----------|
| <b>delete</b>       | function |
| <b>found</b>        | function |
| <b>insert</b>       | function |
| <b>nearest</b>      | function |
| <b>nearestindex</b> | function |

|                    |          |
|--------------------|----------|
| <b>new</b>         | function |
| <b>newweighted</b> | function |
| <b>reset</b>       | function |
| <b>totable</b>     | function |
| <b>traverse</b>    | function |

## language (basic tex)

|                         |          |
|-------------------------|----------|
| <b>clean</b>            | function |
| <b>clearhyphenation</b> | function |
| <b>clearpatterns</b>    | function |
| <b>current</b>          | function |
| <b>gethjcode</b>        | function |
| <b>haslanguage</b>      | function |
| <b>hyphenate</b>        | function |
| <b>hyphenation</b>      | function |
| <b>hyphenationmin</b>   | function |

|                         |          |
|-------------------------|----------|
| <b>id</b>               | function |
| <b>new</b>              | function |
| <b>patterns</b>         | function |
| <b>postexhyphenchar</b> | function |
| <b>posthyphenchar</b>   | function |
| <b>preexhyphenchar</b>  | function |
| <b>prehyphenchar</b>    | function |
| <b>sethjcode</b>        | function |
| <b>setwordhandler</b>   | function |

# languages (context)

|                        |            |                                 |            |
|------------------------|------------|---------------------------------|------------|
| <b>addgoodiesdata</b>  | function   | <b>numbers</b>                  | data table |
| <b>addgoodiesfile</b>  | function   | <b>pdflanguage</b>              | function   |
| <b>appliedoptions</b>  | table      | <b>postexhyphenchar</b>         | function   |
| <b>associate</b>       | function   | <b>posthyphenchar</b>           | function   |
| <b>associated</b>      | data table | <b>preexhyphenchar</b>          | function   |
| <b>association</b>     | function   | <b>prehyphenchar</b>            | function   |
| <b>counters</b>        | data table | <b>registered</b>               | data table |
| <b>current</b>         | function   | <b>registeredactions</b>        | table      |
| <b>data</b>            | data table | <b>replacements</b>             | table      |
| <b>decimals</b>        | data table | F add                           |            |
| <b>define</b>          | function   | F addlist                       |            |
| <b>explicitthyphen</b> | function   | F handler                       |            |
| <b>exportlanguage</b>  | function   | F set                           |            |
| <b>flatten</b>         | function   | <b>serializediscretionary</b>   | function   |
| <b>frequencies</b>     | table      | <b>setexceptions</b>            | function   |
| F averagecharwidth     |            | <b>setgoodieshandler</b>        | function   |
| F getdata              |            | <b>setoptions</b>               | function   |
| <b>getdata</b>         | function   | <b>setpatterns</b>              | function   |
| <b>getmetadata</b>     | function   | <b>setsynonym</b>               | function   |
| <b>getnumber</b>       | function   | <b>setwordhandler</b>           | function   |
| <b>goodiefiles</b>     | function   | <b>showdiscretionaries</b>      | function   |
| <b>hjcounts</b>        | table      | <b>strippedgoodiewords</b>      | function   |
| <b>hyphenate</b>       | function   | <b>tolang</b>                   | function   |
| <b>hyphenators</b>     | table      | <b>transliteration</b>          | table      |
| D featuresets          |            | F exception                     |            |
| F handler              |            | F handler                       |            |
| F popmethod            |            | F register                      |            |
| F pushmethod           |            | F set                           |            |
| F setmethod            |            | F use                           |            |
| F setup                |            | <b>unload</b>                   | function   |
| F showhyphenationtrace |            | <b>urls</b>                     | table      |
| <b>installed</b>       | function   | D characters                    |            |
| <b>installhandler</b>  | function   | N lefthyphenmin                 |            |
| <b>labels</b>          | table      | D mapping                       |            |
| F define               |            | B packslashes                   |            |
| F split                |            | N righthyphenmin                |            |
| <b>loadable</b>        | function   | F setcharacters                 |            |
| <b>loadwords</b>       | function   | <b>version</b>                  | number     |
| <b>logger</b>          | table      | <b>visualizediscretionaries</b> | function   |
| F report               |            | <b>words</b>                    | table      |
| <b>nofflattened</b>    | function   | F check                         |            |

F disable

F found

F enable

F load

B enables

N threshold



# layouts (context)

|                          |          |                          |       |
|--------------------------|----------|--------------------------|-------|
| <b>getcanvas</b>         | function | <b>status</b>            | table |
| <b>getpagedimensions</b> | function | F isleftpage             |       |
| <b>setupcanvas</b>       | function | F lefttorrightpageaction |       |

## lfs (extra lua)

|                       |                     |                        |                     |
|-----------------------|---------------------|------------------------|---------------------|
| <b>attributes</b>     | overloaded function | <b>iswriteablefile</b> | function            |
| <b>canonicalize</b>   | function            | <b>link</b>            | overloaded function |
| <b>chdir</b>          | overloaded function | <b>mkdir</b>           | overloaded function |
| <b>currentdir</b>     | function            | <b>makedirs</b>        | function            |
| <b>dir</b>            | overloaded function | <b>modification</b>    | function            |
| <b>expandpath</b>     | function            | <b>readlink</b>        | function            |
| <b>isdir</b>          | overloaded function | <b>rmdir</b>           | overloaded function |
| <b>isfile</b>         | overloaded function | <b>setexecutable</b>   | function            |
| <b>isfound</b>        | function            | <b>symlink</b>         | function            |
| <b>isreadabledir</b>  | function            | <b>symlinktarget</b>   | function            |
| <b>isreadablefile</b> | function            | <b>touch</b>           | overloaded function |
| <b>iswriteabledir</b> | function            | <b>walkdir</b>         | function            |

## libraries (context)

**basiclua** data table

**basictex** data table

**builtin** data table

**extralua** data table

**extratex** data table

**functions** data table

**globals** data table

**optional** data table

# lmx (context)

|                          |          |
|--------------------------|----------|
| <b>color</b>             | function |
| <b>convert</b>           | function |
| <b>convertfile</b>       | function |
| <b>convertstring</b>     | function |
| <b>einject</b>           | function |
| <b>eprint</b>            | function |
| <b>escape</b>            | function |
| <b>finalized</b>         | function |
| <b>finject</b>           | function |
| <b>fprint</b>            | function |
| <b>get</b>               | function |
| <b>getvariables</b>      | function |
| <b>htmfile</b>           | function |
| <b>html</b>              | table    |
| F a                      |          |
| F td                     |          |
| F th                     |          |
| <b>include</b>           | function |
| <b>inherited</b>         | table    |
| S color-background-four  |          |
| S color-background-one   |          |
| S color-background-three |          |
| S color-background-two   |          |
| S title                  |          |
| <b>initialize</b>        | function |
| <b>inject</b>            | function |

|                           |          |
|---------------------------|----------|
| <b>lmxfile</b>            | function |
| <b>loadedfile</b>         | function |
| <b>make</b>               | function |
| <b>new</b>                | function |
| <b>popupfile</b>          | function |
| <b>print</b>              | function |
| <b>pv</b>                 | function |
| <b>report</b>             | function |
| <b>reset</b>              | function |
| <b>result</b>             | function |
| <b>set</b>                | function |
| <b>show</b>               | function |
| <b>stripcss</b>           | function |
| <b>tv</b>                 | function |
| <b>type</b>               | function |
| <b>urlescape</b>          | function |
| <b>variable</b>           | function |
| <b>variables</b>          | table    |
| S color-background-blue   |          |
| S color-background-body   |          |
| S color-background-green  |          |
| S color-background-main   |          |
| S color-background-purple |          |
| S color-background-yellow |          |
| S title-default           |          |

# logs (context)

|                             |          |                            |          |
|-----------------------------|----------|----------------------------|----------|
| <b>application</b>          | function | <b>setformats</b>          | function |
| <b>categories</b>           | function | <b>setformatters</b>       | function |
| <b>direct</b>               | function | <b>setlogfile</b>          | function |
| <b>disable</b>              | function | <b>setmessenger</b>        | function |
| <b>enable</b>               | function | <b>setprocessor</b>        | function |
| <b>finalactions</b>         | function | <b>settarget</b>           | function |
| <b>flush</b>                | function | <b>settimedlog</b>         | function |
| <b>loggingerrors</b>        | function | <b>settranslations</b>     | function |
| <b>messenger</b>            | function | <b>show</b>                | function |
| <b>new</b>                  | function | <b>start_page_number</b>   | function |
| <b>newline</b>              | function | <b>start_run</b>           | function |
| <b>poplogfile</b>           | function | <b>startererrorlogging</b> | function |
| <b>poptarget</b>            | function | <b>startfilelogging</b>    | function |
| <b>private</b>              | table    | <b>status</b>              | function |
| F enablepagetiming          |          | <b>stop_page_number</b>    | function |
| F getpagetiming             |          | <b>stoperrorlogging</b>    | function |
| <b>pushlogfile</b>          | function | <b>stopfilelogging</b>     | function |
| <b>pushtarget</b>           | function | <b>subdirect</b>           | function |
| <b>registerfinalactions</b> | function | <b>subreport</b>           | function |
| <b>report</b>               | function | <b>system</b>              | function |
| <b>reporter</b>             | function | <b>texerrormessage</b>     | function |
| <b>resetlogfile</b>         | function | <b>writer</b>              | function |

# lpdf (context)

|                                |            |                                 |          |
|--------------------------------|------------|---------------------------------|----------|
| <b>action</b>                  | function   | <b>copystring</b>               | function |
| <b>adddocumentcolorspace</b>   | function   | <b>copyunicode</b>              | function |
| <b>adddocumenttextgstate</b>   | function   | <b>deferredobject</b>           | function |
| <b>adddocumentpattern</b>      | function   | <b>delayedobject</b>            | function |
| <b>adddocumentshade</b>        | function   | <b>destinationspecification</b> | function |
| <b>addtocatalog</b>            | function   | <b>dictionary</b>               | function |
| <b>addtoinfo</b>               | function   | <b>docstring</b>                | function |
| <b>addtonames</b>              | function   | <b>encodingvalues</b>           | table    |
| <b>addtopageattributes</b>     | function   | N base16                        |          |
| <b>addtopageresources</b>      | function   | N base85                        |          |
| <b>addtopagesattributes</b>    | function   | N decoded                       |          |
| <b>addxmpinfo</b>              | function   | N encoded                       |          |
| <b>annotationspecification</b> | function   | N plain                         |          |
| <b>array</b>                   | function   | N utf16be                       |          |
| <b>attachmentsymbols</b>       | function   | N utf16le                       |          |
| <b>boolean</b>                 | function   | <b>endpoints</b>                | table    |
| <b>border</b>                  | function   | S Div                           |          |
| <b>checkedkey</b>              | function   | S Part                          |          |
| <b>checkedresources</b>        | function   | S Sect                          |          |
| <b>checkedvalue</b>            | function   | <b>epdf</b>                     | table    |
| <b>circularshade</b>           | function   | F allcontent                    |          |
| <b>close</b>                   | function   | F analyzefonts                  |          |
| <b>cmykcode</b>                | function   | F contentplugin                 |          |
| <b>collectedresources</b>      | function   | F contenttostring               |          |
| <b>color</b>                   | function   | F contenttotext                 |          |
| <b>colorspace</b>              | function   | D encryptioncodes               |          |
| <b>colorspaceconstants</b>     | data table | F expanded                      |          |
| <b>colorspec</b>               | function   | F expandwidths                  |          |
| <b>colorvalue</b>              | function   | F fontplugin                    |          |
| <b>colorvalues</b>             | function   | F getpagecontent                |          |
| <b>commentsymbols</b>          | function   | F getstructure                  |          |
| <b>compressdata</b>            | function   | F load                          |          |
| <b>compresslevel</b>           | function   | F mergewidths                   |          |
| <b>constant</b>                | function   | D objectcodes                   |          |
| <b>copyarray</b>               | function   | F objectrange                   |          |
| <b>copyboolean</b>             | function   | F objecttype                    |          |
| <b>copyconstant</b>            | function   | F pageplugin                    |          |
| <b>copydictionary</b>          | function   | F parsecontent                  |          |
| <b>copyfloat</b>               | function   | F producer                      |          |
| <b>copyinteger</b>             | function   | F unload                        |          |
| <b>copynumber</b>              | function   | F verboseobject                 |          |

|                               |            |                            |          |
|-------------------------------|------------|----------------------------|----------|
| <b>escaped</b>                | function   | S \                        |          |
| <b>finalizedocument</b>       | function   | <b>h_hex_4</b>             | table    |
| <b>finalizepage</b>           | function   | <b>hasmatrix</b>           | function |
| <b>finalizesignature</b>      | function   | <b>immediateobject</b>     | function |
| <b>finishtransparencycode</b> | function   | <b>includeimage</b>        | function |
| <b>flags</b>                  | table      | <b>initialize</b>          | function |
| <b>flushfonts</b>             | function   | <b>injectors</b>           | table    |
| <b>flushgroup</b>             | function   | F bitmap                   |          |
| <b>flushobject</b>            | function   | F jp2                      |          |
| <b>flushstreamfileobject</b>  | function   | F jpg                      |          |
| <b>flushstreamobject</b>      | function   | F png                      |          |
| <b>fonts</b>                  | table      | <b>injectxmpinfo</b>       | function |
| F color_direct                |            | <b>insertxmpinfo</b>       | function |
| F color_indirect              |            | <b>lastreferredpage</b>    | function |
| <b>formats</b>                | data table | <b>layerreference</b>      | function |
| <b>frombytes</b>              | function   | <b>limited</b>             | function |
| <b>fromeight</b>              | function   | <b>linearshade</b>         | function |
| <b>frompdfdoc</b>             | function   | <b>literal</b>             | function |
| <b>fromsixteen</b>            | function   | <b>majorversion</b>        | function |
| <b>getcatalag</b>             | function   | <b>makenametree</b>        | function |
| <b>getcompression</b>         | function   | <b>minorversion</b>        | function |
| <b>getfontname</b>            | function   | <b>noffontregistries</b>   | function |
| <b>getfontobjectnumber</b>    | function   | <b>nofpages</b>            | function |
| <b>getformatoption</b>        | function   | <b>null</b>                | function |
| <b>getinfo</b>                | function   | <b>number</b>              | function |
| <b>getmatrix</b>              | function   | <b>object</b>              | function |
| <b>getnofpages</b>            | function   | <b>objectcompresslevel</b> | function |
| <b>getpageproperties</b>      | function   | <b>pagereference</b>       | function |
| <b>getpagereference</b>       | function   | <b>patternstream</b>       | function |
| <b>getpagetags</b>            | function   | <b>pdfcolor</b>            | function |
| <b>getpos</b>                 | function   | <b>pdftimestamp</b>        | function |
| <b>gettaggingstatus</b>       | function   | <b>permissions</b>         | table    |
| <b>gettrailerid</b>           | function   | N access                   |          |
| <b>getversion</b>             | function   | N add                      |          |
| <b>getxformname</b>           | function   | N assemble                 |          |
| <b>graycode</b>               | function   | N extract                  |          |
| <b>h_hex_2</b>                | table      | N fillin                   |          |
| S                             |            | N modify                   |          |
| S                             |            | N print                    |          |
| S                             |            | N quality                  |          |
| S                             |            | <b>preparesignature</b>    | function |
| S                             |            | <b>print</b>               | function |
| S (                           |            | <b>procset</b>             | function |
| S )                           |            | <b>protectresources</b>    | boolean  |

|                                  |          |                               |            |
|----------------------------------|----------|-------------------------------|------------|
| <b>quads</b>                     | function | <b>splitsixteen</b>           | function   |
| <b>rectangle</b>                 | function | <b>spotcode</b>               | function   |
| <b>reference</b>                 | function | <b>stream</b>                 | function   |
| <b>registerannotation</b>        | function | <b>string</b>                 | function   |
| <b>registerdestination</b>       | function | <b>supportedformats</b>       | function   |
| <b>registerdocumentfinalizer</b> | function | <b>tocidset</b>               | function   |
| <b>registerfontmethod</b>        | function | <b>toeight</b>                | function   |
| <b>registerinitializer</b>       | function | <b>topdfdoc</b>               | function   |
| <b>registerpagefinalizer</b>     | function | <b>topermissions</b>          | function   |
| <b>registerpattern</b>           | function | <b>tosixteen</b>              | function   |
| <b>registersignature</b>         | function | <b>transparency</b>           | function   |
| <b>replacexmpinfo</b>            | function | <b>transparencycode</b>       | function   |
| <b>reserveobject</b>             | function | <b>transparencygroups</b>     | table      |
| <b>rgbcode</b>                   | function | B none                        |            |
| <b>rotationcm</b>                | function | <b>transparencyvalue</b>      | function   |
| <b>setcompression</b>            | function | <b>unicode</b>                | function   |
| <b>setencryption</b>             | function | <b>usedcharacters</b>         | table      |
| <b>setpageorder</b>              | function | <b>usedfontnames</b>          | table      |
| <b>setpagestate</b>              | function | N dejavusansmonoboldttf-1-0-0 |            |
| <b>setpagetag</b>                | function | <b>usedfontobjects</b>        | table      |
| <b>setutfstringmode</b>          | function | N dejavusansmonoboldttf-1-0-0 |            |
| <b>setversion</b>                | function | <b>usedfontstreams</b>        | data table |
| <b>sharedobject</b>              | function | <b>usedindices</b>            | table      |
| <b>shareobject</b>               | function | <b>verbose</b>                | function   |
| <b>shareobjectreference</b>      | function |                               |            |



## lpeg (basic lua)

|                     |          |                              |            |
|---------------------|----------|------------------------------|------------|
| <b>B</b>            | function | <b>firstofsplit</b>          | function   |
| <b>C</b>            | function | <b>frontstripper</b>         | function   |
| <b>Carg</b>         | function | <b>instringchecker</b>       | function   |
| <b>Cb</b>           | function | <b>is_lpeg</b>               | function   |
| <b>Cc</b>           | function | <b>keeper</b>                | function   |
| <b>Cf</b>           | function | <b>locale</b>                | function   |
| <b>Cg</b>           | function | <b>match</b>                 | function   |
| <b>Cmt</b>          | function | <b>oneof</b>                 | function   |
| <b>Cp</b>           | function | <b>patterns</b>              | data table |
| <b>Cs</b>           | function | <b>pcode</b>                 | function   |
| <b>Ct</b>           | function | <b>print</b>                 | function   |
| <b>P</b>            | function | <b>ptree</b>                 | function   |
| <b>R</b>            | function | <b>replacer</b>              | function   |
| <b>S</b>            | function | <b>secondofsplit</b>         | function   |
| <b>UP</b>           | function | <b>setmaxstack</b>           | function   |
| <b>UR</b>           | function | <b>setutfcasers</b>          | function   |
| <b>US</b>           | function | <b>split</b>                 | function   |
| <b>V</b>            | function | <b>splitat</b>               | function   |
| <b>afterprefix</b>  | function | <b>splitter</b>              | function   |
| <b>anywhere</b>     | function | <b>stripper</b>              | function   |
| <b>append</b>       | function | <b>times</b>                 | function   |
| <b>balancer</b>     | function | <b>tsplitat</b>              | function   |
| <b>beforesuffix</b> | function | <b>tsplitter</b>             | function   |
| <b>checkedsplit</b> | function | <b>type</b>                  | function   |
| <b>containsws</b>   | function | <b>utfR</b>                  | function   |
| <b>counter</b>      | function | <b>utfchartabletopattern</b> | function   |
| <b>endstripper</b>  | function | <b>utfreplacer</b>           | function   |
| <b>finder</b>       | function | <b>version</b>               | string     |

## lua (basic tex)

|                            |                     |                                |            |
|----------------------------|---------------------|--------------------------------|------------|
| <b>addnibble</b>           | function            | <b>getstringkeys</b>           | function   |
| <b>bytecode</b>            | table               | <b>getversion</b>              | function   |
| <b>bytedata</b>            | data table          | <b>lastbytecode</b>            | number     |
| <b>bytdone</b>             | table               | <b>luastringcall</b>           | function   |
| <b>callbytecode</b>        | function            | <b>messages</b>                | data table |
| <b>comparekeys</b>         | function            | <b>mpsstringcall</b>           | function   |
| <b>finalizeinitex</b>      | function            | <b>newindex</b>                | function   |
| <b>firstbytecode</b>       | number              | <b>newtable</b>                | function   |
| <b>get_functions_table</b> | function            | <b>numbers</b>                 | data table |
| <b>getallkeys</b>          | function            | <b>protectglobal</b>           | function   |
| <b>getbytecode</b>         | function            | <b>registercode</b>            | function   |
| <b>getcheckedkeys</b>      | function            | <b>registeredcodes</b>         | function   |
| <b>getcurrenttime</b>      | function            | <b>registerglobal</b>          | function   |
| <b>getdebuginfo</b>        | function            | <b>registerinitexfinalizer</b> | function   |
| <b>getexitcode</b>         | function            | <b>registername</b>            | function   |
| <b>getfunctionstable</b>   | overloaded function | <b>setbytecode</b>             | function   |
| <b>getnibble</b>           | function            | <b>setexitcode</b>             | function   |
| <b>getnumerickeys</b>      | function            | <b>setnibble</b>               | function   |
| <b>getpreciseseconds</b>   | function            | <b>startupfile</b>             | string     |
| <b>getpreciseticks</b>     | function            | <b>subnibble</b>               | function   |
| <b>getruntime</b>          | function            | <b>usestringcall</b>           | function   |
| <b>getstacktop</b>         | function            | <b>version</b>                 | string     |
| <b>getstartupfile</b>      | function            |                                |            |

## luac (extra lua)

```
inspect function  
print    function
```

# luatex (context)

|                             |          |                    |          |
|-----------------------------|----------|--------------------|----------|
| <b>abort</b>                | function | F disable          |          |
| <b>cleanup</b>              | function | F enable           |          |
| <b>cleanuptempfiles</b>     | function | F finish           |          |
| <b>currentfile</b>          | function | F isenabled        |          |
| <b>currentlevel</b>         | function | F pause            |          |
| <b>currenttotal</b>         | function | F popline          |          |
| <b>poprunstate</b>          | function | F pushline         |          |
| <b>pushrunstate</b>         | function | F registerdisabler |          |
| <b>registerdumpactions</b>  | function | F registerenabler  |          |
| <b>registerpageactions</b>  | function | F resetfilename    |          |
| <b>registerstartactions</b> | function | F resume           |          |
| <b>registerstopactions</b>  | function | F setfilename      |          |
| <b>registertempfile</b>     | function | F setup            |          |
| <b>starttime</b>            | number   | F start            |          |
| <b>synctex</b>              | table    | F stop             |          |
| F blockfilename             |          | F wrapup           |          |
| F collect                   |          | <b>wrapup</b>      | function |

# lxml (context)

|                          |            |                         |                     |
|--------------------------|------------|-------------------------|---------------------|
| <b>addindex</b>          | function   | <b>filter</b>           | function            |
| <b>all</b>               | function   | <b>filterlist</b>       | function            |
| <b>applyfunction</b>     | function   | <b>first</b>            | function            |
| <b>applyselectors</b>    | function   | <b>flush</b>            | function            |
| <b>att</b>               | function   | <b>flushlinewise</b>    | function            |
| <b>attribute</b>         | function   | <b>flushsetups</b>      | function            |
| <b>badinclusions</b>     | function   | <b>flushspacewise</b>   | function            |
| <b>chainattribute</b>    | function   | <b>get_id</b>           | <del>function</del> |
| <b>checkindex</b>        | function   | <b>getid</b>            | function            |
| <b>collected</b>         | function   | <b>getindex</b>         | function            |
| <b>command</b>           | function   | <b>id</b>               | function            |
| <b>concat</b>            | function   | <b>idsplitter</b>       | userdata            |
| <b>concatrange</b>       | function   | <b>ifatt</b>            | function            |
| <b>content</b>           | function   | <b>ifattempty</b>       | function            |
| <b>context</b>           | function   | <b>include</b>          | function            |
| <b>convert</b>           | function   | <b>inclusion</b>        | function            |
| <b>count</b>             | function   | <b>inclusions</b>       | function            |
| <b>delete</b>            | function   | <b>index</b>            | function            |
| <b>depth</b>             | function   | <b>inlineverbatimim</b> | function            |
| <b>direct</b>            | function   | <b>installsetup</b>     | function            |
| <b>directives</b>        | table      | <b>last</b>             | function            |
| F after                  |            | <b>lastatt</b>          | function            |
| F before                 |            | <b>lastmatch</b>        | function            |
| F handle                 |            | <b>lastpar</b>          | function            |
| F load                   |            | <b>load</b>             | function            |
| F setup                  |            | <b>loadbuffer</b>       | function            |
| <b>displayverbatimim</b> | function   | <b>loaddata</b>         | function            |
| <b>doif</b>              | function   | <b>loaded</b>           | table               |
| <b>doifelse</b>          | function   | <b>main</b>             | function            |
| <b>doifelseempty</b>     | function   | <b>match</b>            | function            |
| <b>doifelsetext</b>      | function   | <b>name</b>             | function            |
| <b>doifempty</b>         | function   | <b>namespace</b>        | function            |
| <b>doifnot</b>           | function   | <b>nonspace</b>         | function            |
| <b>doifnotempty</b>      | function   | <b>par</b>              | function            |
| <b>doifnottext</b>       | function   | <b>param</b>            | function            |
| <b>doiftext</b>          | function   | <b>parameter</b>        | function            |
| <b>element</b>           | function   | <b>path</b>             | function            |
| <b>elements</b>          | function   | <b>popmatch</b>         | function            |
| <b>entities</b>          | data table | <b>pos</b>              | function            |
| <b>fileline</b>          | function   | <b>position</b>         | function            |
| <b>filename</b>          | function   | <b>pure</b>             | function            |

|                         |          |
|-------------------------|----------|
| <b>pushmatch</b>        | function |
| <b>raw</b>              | function |
| <b>rawroot</b>          | function |
| <b>rawtex</b>           | function |
| <b>refatt</b>           | function |
| <b>register</b>         | function |
| <b>registerentity</b>   | function |
| <b>removesetup</b>      | function |
| <b>replace</b>          | function |
| <b>resetcdata</b>       | function |
| <b>resetsetups</b>      | function |
| <b>resolvedentity</b>   | function |
| <b>root</b>             | function |
| <b>save</b>             | function |
| <b>serialize</b>        | function |
| <b>setaction</b>        | function |
| <b>setatt</b>           | function |
| <b>setattribute</b>     | function |
| <b>setcdata</b>         | function |
| <b>setcommandtonone</b> | function |
| <b>setcommandtotext</b> | function |
| <b>setpar</b>           | function |
| <b>setparam</b>         | function |
| <b>setparameter</b>     | function |
| <b>setsetup</b>         | function |
| <b>snippet</b>          | function |

|                         |            |
|-------------------------|------------|
| <b>sorters</b>          | table      |
| F add                   |            |
| F compare               |            |
| F flush                 |            |
| F reset                 |            |
| F show                  |            |
| F sort                  |            |
| <b>splitid</b>          | function   |
| <b>startraw</b>         | function   |
| <b>startunesescaped</b> | function   |
| <b>stopraw</b>          | function   |
| <b>stopunesescaped</b>  | function   |
| <b>store</b>            | function   |
| <b>strip</b>            | function   |
| <b>stripped</b>         | function   |
| <b>tag</b>              | function   |
| <b>texatt</b>           | function   |
| <b>text</b>             | function   |
| <b>tobuffer</b>         | function   |
| <b>tofile</b>           | function   |
| <b>toparameters</b>     | function   |
| <b>toverbatim</b>       | function   |
| <b>verbatim</b>         | function   |
| <b>withindex</b>        | function   |
| <b>xmltexhandler</b>    | data table |

## math (basic lua)

|                    |          |                       |                     |
|--------------------|----------|-----------------------|---------------------|
| <b>abs</b>         | function | <b>maxinteger</b>     | number              |
| <b>acos</b>        | function | <b>min</b>            | function            |
| <b>asin</b>        | function | <b>mininteger</b>     | number              |
| <b>atan</b>        | function | <b>mod</b>            | function            |
| <b>atan2</b>       | function | <b>modf</b>           | function            |
| <b>ceil</b>        | function | <b>odd</b>            | function            |
| <b>ceiling</b>     | function | <b>pi</b>             | number              |
| <b>cos</b>         | function | <b>pow</b>            | function            |
| <b>cosd</b>        | function | <b>rad</b>            | function            |
| <b>cosh</b>        | function | <b>random</b>         | overloaded function |
| <b>deg</b>         | function | <b>randomseed</b>     | function            |
| <b>div</b>         | function | <b>round</b>          | function            |
| <b>even</b>        | function | <b>setrandomseedi</b> | function            |
| <b>exp</b>         | function | <b>sin</b>            | function            |
| <b>floor</b>       | function | <b>sind</b>           | function            |
| <b>fmod</b>        | function | <b>sinh</b>           | function            |
| <b>frexp</b>       | function | <b>sqrt</b>           | function            |
| <b>huge</b>        | number   | <b>tan</b>            | function            |
| <b>initialseed</b> | number   | <b>tand</b>           | function            |
| <b>ldexp</b>       | function | <b>tanh</b>           | function            |
| <b>log</b>         | function | <b>tointeger</b>      | function            |
| <b>log10</b>       | function | <b>type</b>           | function            |
| <b>max</b>         | function | <b>ult</b>            | function            |

# mathematics (context)

|                               |            |                             |            |
|-------------------------------|------------|-----------------------------|------------|
| <b>addfallbacks</b>           | function   | <b>familyname</b>           | function   |
| <b>addvariant</b>             | function   | <b>functiontype</b>         | function   |
| <b>alphabets</b>              | data table | <b>gaps</b>                 | data table |
| <b>alternatives</b>           | data table | <b>getblobindex</b>         | function   |
| <b>beforepassingfonttotex</b> | function   | <b>getboth</b>              | function   |
| <b>big</b>                    | function   | <b>getdomainname</b>        | function   |
| <b>boldmap</b>                | data table | <b>getextensibledata</b>    | function   |
| <b>categories</b>             | table      | <b>getmathblob</b>          | function   |
| D functions                   |            | <b>getoptionallabel</b>     | function   |
| <b>charactersets</b>          | data table | <b>getstyle</b>             | function   |
| <b>checkaftercopyingfont</b>  | function   | <b>gettextblob</b>          | function   |
| <b>classes</b>                | data table | <b>getverboselabel</b>      | function   |
| <b>classnames</b>             | data table | <b>horizontalcode</b>       | function   |
| <b>codes</b>                  | data table | <b>initializeparameters</b> | function   |
| <b>define</b>                 | function   | <b>makefont</b>             | function   |
| <b>dictionaries</b>           | table      | <b>mapremap</b>             | data table |
| D data                        |            | <b>markblobindexdone</b>    | function   |
| D defaults                    |            | <b>overloadparameters</b>   | function   |
| D groups                      |            | <b>preset</b>               | function   |
| F groupset                    |            | <b>presets</b>              | table      |
| F groupsetgroup               |            | <b>privatebase</b>          | number     |
| F name                        |            | <b>registerextensible</b>   | function   |
| D names                       |            | <b>registersimplematrix</b> | function   |
| F registercharacter           |            | <b>registerverboselabel</b> | function   |
| F registergroup               |            | <b>remapalphabets</b>       | function   |
| F registergroupset            |            | <b>renderextensible</b>     | function   |
| D sets                        |            | <b>renderings</b>           | table      |
| D variants                    |            | D mappings                  |            |
| <b>dimensions</b>             | function   | D numbers                   |            |
| <b>disablecollecting</b>      | function   | D sets                      |            |
| <b>domainlabels</b>           | table      | <b>renderset</b>            | function   |
| <b>domainnames</b>            | table      | <b>resetattributes</b>      | function   |
| N default                     |            | <b>scaleparameters</b>      | function   |
| <b>enablecollecting</b>       | function   | <b>setextensibledata</b>    | function   |
| <b>extensiblecode</b>         | function   | <b>setmathsymbol</b>        | function   |
| <b>extensibles</b>            | data table | <b>sets</b>                 | data table |
| <b>extrabase</b>              | number     | <b>setsnapping</b>          | function   |
| <b>fallbacks</b>              | table      | <b>setvariant</b>           | function   |
| F apply                       |            | <b>strutcode</b>            | function   |
| F install                     |            | <b>styles</b>               | data table |
| <b>fallbackstyleattr</b>      | function   | <b>syncboth</b>             | function   |



|                               |          |                        |            |
|-------------------------------|----------|------------------------|------------|
| <b>syncname</b>               | function | F fontname             |            |
| <b>syncstyle</b>              | function | F inspect              |            |
| <b>tagging</b>                | table    | F keepbases            |            |
| B mfenced                     |          | F kernpairs            |            |
| <b>toengineclass</b>          | function | F kerns                |            |
| <b>toname</b>                 | function | F margins              |            |
| <b>tostyle</b>                | function | F message              |            |
| <b>tweakaftercopyingfont</b>  | function | F moveitalics          |            |
| <b>tweakbeforecopyingfont</b> | function | F movelimits           |            |
| <b>tweaks</b>                 | table    | F parameters           |            |
| F accentdimensions            |          | F radicalbodymargins   |            |
| F action                      |          | F radicaldegreeanchors |            |
| F addactuarian                |          | F replace              |            |
| F addarrows                   |          | F replacealphabets     |            |
| F addbars                     |          | F replacerules         |            |
| F addcomposites               |          | F setoptions           |            |
| F addequals                   |          | F setovershoots        |            |
| F addfourier                  |          | F showinfo             |            |
| F addmirrors                  |          | F simplifykerns        |            |
| F addparts                    |          | F sortvariants         |            |
| F addprivates                 |          | F staircase            |            |
| F addrules                    |          | F strut                |            |
| F addscripts                  |          | F substitute           |            |
| F bigslots                    |          | F topanchors           |            |
| F checkaccents                |          | F tweaks               |            |
| F checkspacing                |          | F version              |            |
| F copyaccents                 |          | F wipeanchors          |            |
| F dimensions                  |          | F wipecues             |            |
| F extendaccents               |          | F wipeitalics          |            |
| F fallbacks                   |          | F wipevariants         |            |
| F fixaccents                  |          | <b>utfmathaccent</b>   | function   |
| F fixellipses                 |          | <b>utfmathclass</b>    | function   |
| F fixintegrals                |          | <b>utfmathcommand</b>  | function   |
| F fixoldschool                |          | <b>utfmathfiller</b>   | function   |
| F fixprimes                   |          | <b>utfmathstretch</b>  | function   |
| F fixradicals                 |          | <b>variantcode</b>     | function   |
| F fixslashes                  |          | <b>variantcount</b>    | function   |
| F fixvariants                 |          | <b>virtualize</b>      | function   |
| F flattenaccents              |          | <b>virtualized</b>     | data table |

## md5 (extra lua)

|            |          |
|------------|----------|
| <b>HEX</b> | function |
| <b>dec</b> | function |
| <b>hex</b> | function |

|                |          |
|----------------|----------|
| <b>sum</b>     | function |
| <b>sumHEXA</b> | function |
| <b>sumhexa</b> | function |

# metapost (context)

|                           |          |                   |
|---------------------------|----------|-------------------|
| <b>checkformat</b>        | function | N newinternal     |
| <b>checktracingonline</b> | function | N nullary         |
| <b>cm</b>                 | function | N numeric         |
| <b>codes</b>              | table    | N of              |
| N addto                   |          | N ofbinary        |
| N ampersand               |          | N onlyset         |
| N and                     |          | N parametertype   |
| N assignment              |          | N pathconnect     |
| N atleast                 |          | N pathjoin        |
| N begingroup              |          | N plusorminus     |
| N btex                    |          | N primarybinary   |
| N capsule                 |          | N primarydef      |
| N colon                   |          | N property        |
| N comma                   |          | N protection      |
| N controls                |          | N relax           |
| N curl                    |          | N repeatloop      |
| N cycle                   |          | N rightbrace      |
| N definedmacro            |          | N rightbracket    |
| N delimiters              |          | N rightdelimiter  |
| N endgroup                |          | N runscript       |
| N equals                  |          | N save            |
| N etex                    |          | N scantokens      |
| N everyjob                |          | N secondarybinary |
| N exittest                |          | N secondarydef    |
| N expandafter             |          | N semicolon       |
| N fiorelse                |          | N setbounds       |
| N if                      |          | N shipout         |
| N input                   |          | N show            |
| N interim                 |          | N slash           |
| N internal                |          | N step            |
| N iteration               |          | N stop            |
| N leftbrace               |          | N str             |
| N leftbracket             |          | N string          |
| N leftdelimiter           |          | N tag             |
| N let                     |          | N tension         |
| N macrodef                |          | N tertiarybinary  |
| N macrospecial            |          | N tertiarydef     |
| N maketext                |          | N thingstoadd     |
| N message                 |          | N to              |
| N mode                    |          | N typename        |
| N newbytemap              |          | N unary           |

|                           |            |                             |          |
|---------------------------|------------|-----------------------------|----------|
| N undefinedcs             |            | <b>gettext</b>              | function |
| N until                   |            | <b>gettolerance</b>         | function |
| N void                    |            | <b>graphic</b>              | function |
| N with                    |            | <b>graphic_base_pass</b>    | function |
| N within                  |            | <b>hascurvature</b>         | function |
| N write                   |            | <b>includesvgbuffer</b>     | function |
| <b>colorinitializer</b>   | function   | <b>includesvgfile</b>       | function |
| <b>convert</b>            | function   | <b>installplugin</b>        | function |
| <b>currentmpx</b>         | function   | <b>isobject</b>             | function |
| <b>currentmpxstatus</b>   | function   | <b>knownparameters</b>      | table    |
| <b>defaultformat</b>      | string     | <b>lastlog</b>              | string   |
| <b>defaultinstance</b>    | string     | <b>load</b>                 | function |
| <b>defaultmethod</b>      | string     | <b>maketext</b>             | function |
| <b>directrun</b>          | function   | <b>mesh</b>                 | function |
| <b>escaped</b>            | function   | <b>mesher</b>               | function |
| <b>exectime</b>           | table      | <b>meshupper</b>            | function |
| <b>extensiondata</b>      | data table | <b>metafonts</b>            | table    |
| <b>filter</b>             | function   | <b>metafun</b>              | table    |
| <b>filterclippath</b>     | function   | F interpolate               |          |
| <b>flush</b>              | function   | F topath                    |          |
| <b>flushers</b>           | table      | <b>mptopdf</b>              | table    |
| <b>flushnormalpath</b>    | function   | F convertmpstopdf           |          |
| <b>fonts</b>              | table      | N nofconverted              |          |
| F define                  |            | <b>n</b>                    | number   |
| <b>format_number</b>      | function   | <b>newlogger</b>            | function |
| <b>format_string</b>      | function   | <b>nofruns</b>              | number   |
| <b>formatnumber</b>       | function   | <b>nofscriptruns</b>        | function |
| <b>getbackendoption</b>   | function   | <b>nvformat</b>             | function |
| <b>getbendtolerance</b>   | function   | <b>pdfliterals</b>          | function |
| <b>getbuffer</b>          | function   | <b>pluginactions</b>        | function |
| <b>getclippath</b>        | function   | <b>points</b>               | function |
| <b>getextensions</b>      | function   | <b>pointset</b>             | function |
| <b>getextradata</b>       | function   | <b>popformat</b>            | function |
| <b>gethashentries</b>     | function   | <b>popscriptrunner</b>      | function |
| <b>getinstancebyname</b>  | function   | <b>popvariable</b>          | function |
| <b>getinstanceinfo</b>    | function   | <b>popvariables</b>         | function |
| <b>getinstancenames</b>   | function   | <b>preparetexttextsdata</b> | function |
| <b>getjobdata</b>         | function   | <b>process</b>              | function |
| <b>getmovetolerance</b>   | function   | <b>processing</b>           | function |
| <b>getparameter</b>       | function   | <b>processplugins</b>       | function |
| <b>getparameterpreset</b> | function   | <b>properties</b>           | table    |
| <b>getparameterset</b>    | function   | <b>pushformat</b>           | function |
| <b>getrecord</b>          | function   | <b>pushscriptrunner</b>     | function |
| <b>getstatistics</b>      | function   | <b>pushvariable</b>         | function |

|                            |          |                         |          |
|----------------------------|----------|-------------------------|----------|
| <b>pushvariables</b>       | function | <b>svgcachefile</b>     | boolean  |
| <b>quitonerror</b>         | boolean  | <b>svgcolorremapper</b> | function |
| <b>reducetogray</b>        | boolean  | <b>svgfactors</b>       | table    |
| <b>registerdirect</b>      | function | N %                     |          |
| <b>registerfilter</b>      | function | N bp                    |          |
| <b>registermesher</b>      | function | N cm                    |          |
| <b>registermeshupper</b>   | function | N em                    |          |
| <b>registerpointset</b>    | function | N ex                    |          |
| <b>registerscript</b>      | function | N in                    |          |
| <b>registertokens</b>      | function | N mm                    |          |
| <b>remappedtext</b>        | function | N pc                    |          |
| <b>remaptext</b>           | function | N pt                    |          |
| <b>reporterror</b>         | function | N px                    |          |
| <b>reportsvgfonts</b>      | function | <b>svgglyphtomp</b>     | function |
| <b>reset</b>               | function | <b>svgshapestomp</b>    | function |
| <b>resetbackendoptions</b> | function | <b>svgshapestopdf</b>   | function |
| <b>resetlastlog</b>        | function | <b>svgsplitcolor</b>    | function |
| <b>resetplugins</b>        | function | <b>svgtomp</b>          | function |
| <b>run</b>                 | function | <b>svgtopdf</b>         | function |
| <b>runinternal</b>         | function | <b>sxsy</b>             | function |
| <b>runscript</b>           | function | <b>tex</b>              | table    |
| <b>scanparameters</b>      | function | F get                   |          |
| <b>scriptindex</b>         | function | F reset                 |          |
| <b>setbendtolerance</b>    | function | F set                   |          |
| <b>setbuffer</b>           | function | F setfrombuffer         |          |
| <b>setextensions</b>       | function | <b>texerrors</b>        | boolean  |
| <b>setlogging</b>          | function | <b>texexp</b>           | function |
| <b>setmovetolerance</b>    | function | <b>theclippath</b>      | function |
| <b>setoutercolor</b>       | function | <b>tographic</b>        | function |
| <b>setparameter</b>        | function | <b>totable</b>          | function |
| <b>setparameterset</b>     | function | <b>types</b>            | table    |
| <b>setrecord</b>           | function | N boolean               |          |
| <b>settext</b>             | function | N cmykcolor             |          |
| <b>settolerance</b>        | function | N color                 |          |
| <b>setvariable</b>         | function | N dependent             |          |
| <b>showlog</b>             | boolean  | N independent           |          |
| <b>showsvgpage</b>         | function | N known                 |          |
| <b>simple</b>              | function | N nep                   |          |
| <b>splitprescript</b>      | function | N numeric               |          |
| <b>startgraphic</b>        | function | N pair                  |          |
| <b>startsvghashing</b>     | function | N path                  |          |
| <b>stopgraphic</b>         | function | N pen                   |          |
| <b>stopsvghashing</b>      | function | N picture               |          |
| <b>svformat</b>            | function | N protodependent        |          |

|                  |                      |          |
|------------------|----------------------|----------|
| N string         | N unknownpicture     |          |
| N structured     | N unknownstring      |          |
| N suffixedmacro  | N unsuffixedmacro    |          |
| N tokenlist      | N vacuous            |          |
| N transform      | <b>typesvgpage</b>   | function |
| N unknownboolean | <b>unload</b>        | function |
| N unknownnep     | <b>untagvariable</b> | function |
| N unknownpath    | <b>variables</b>     | table    |
| N unknownpen     |                      |          |

## mime (context)

**b64** function

**decode** function

**decodet** table

F base64

F quoted-printable

**dot** function

**encode** function

**encodet** table

F base64

F quoted-printable

**eol** function

**normalize** function

**qp** function

**qpwrp** function

**report** function

**stuff** function

**unb64** function

**unqp** function

**wrap** function

**wrapt** table

F base64

F default

F quoted-printable

F text

**wrp** function

# mp (context)

|                               |          |                                   |          |
|-------------------------------|----------|-----------------------------------|----------|
| <b>autoinject</b>             | function | <b>flatten</b>                    | function |
| <b>aux</b>                    | function | <b>flush</b>                      | function |
| F boolean                     |          | <b>format</b>                     | function |
| F color                       |          | <b>formatted</b>                  | function |
| F direct                      |          | <b>fprint</b>                     | function |
| F draw                        |          | <b>get</b>                        | table    |
| F fill                        |          | F hashentry                       |          |
| F flush                       |          | <b>getbytemap</b>                 | function |
| F fprint                      |          | <b>getbytemapdata</b>             | function |
| F integer                     |          | <b>getbytemapdirect</b>           | function |
| F number                      |          | <b>getlastaddtype</b>             | function |
| F numeric                     |          | <b>graphformat</b>                | function |
| F pair                        |          | <b>inject</b>                     | function |
| F pairpoints                  |          | F boolean                         |          |
| F path                        |          | F cmykcolor                       |          |
| F pathpoints                  |          | F color                           |          |
| F points                      |          | F integer                         |          |
| F print                       |          | F number                          |          |
| F quadruple                   |          | F numeric                         |          |
| F quadruplepoints             |          | F pair                            |          |
| F quoted                      |          | F path                            |          |
| F size                        |          | F points                          |          |
| F string                      |          | F pt                              |          |
| F transform                   |          | F quadruplet                      |          |
| F triplet                     |          | F string                          |          |
| F tripletpoints               |          | F tokens                          |          |
| F vprint                      |          | F transform                       |          |
| <b>boolean</b>                | function | F triplet                         |          |
| <b>bytemap_load_from_file</b> | function | F vector                          |          |
| <b>cleaned</b>                | function | F vectors                         |          |
| <b>color</b>                  | function | F whatever                        |          |
| <b>dataset</b>                | function | F whd                             |          |
| <b>datasets</b>               | function | F xy                              |          |
| F load                        |          | <b>integer</b>                    | function |
| <b>direct</b>                 | function | <b>lmt_function_x</b>             | function |
| <b>downgradebytemap</b>       | function | <b>lmt_glyphshape_accent</b>      | function |
| <b>downsamplebytemap</b>      | function | <b>lmt_glyphshape_baseline</b>    | function |
| <b>draw</b>                   | function | <b>lmt_glyphshape_boundingbox</b> | function |
| <b>expandtex</b>              | function | <b>lmt_glyphshape_depth</b>       | function |
| <b>fill</b>                   | function | <b>lmt_glyphshape_height</b>      | function |
| <b>fillbytemap</b>            | function | <b>lmt_glyphshape_italic</b>      | function |



|                                |          |                        |          |
|--------------------------------|----------|------------------------|----------|
| <b>lmt_glyphshape_llx</b>      | function | <b>n</b>               | function |
| <b>lmt_glyphshape_n</b>        | function | <b>newbytemap</b>      | function |
| <b>lmt_glyphshape_path</b>     | function | <b>number</b>          | function |
| <b>lmt_glyphshape_start</b>    | function | <b>numeric</b>         | function |
| <b>lmt_glyphshape_stop</b>     | function | <b>pair</b>            | function |
| <b>lmt_glyphshape_usedbox</b>  | function | <b>pairpoints</b>      | function |
| <b>lmt_glyphshape_usedline</b> | function | <b>path</b>            | function |
| <b>lmt_glyphshape_width</b>    | function | <b>pathpoints</b>      | function |
| <b>lmt_mesh_set</b>            | function | <b>points</b>          | function |
| <b>lmt_mesh_update</b>         | function | <b>points_register</b> | function |
| <b>lmt_poisson_generate</b>    | function | <b>print</b>           | function |
| <b>lmt_start_potrace</b>       | function | F boolean              |          |
| <b>lmt_step_potrace</b>        | function | F color                |          |
| <b>lmt_stop_potrace</b>        | function | F direct               |          |
| <b>lmt_svg_include</b>         | function | F draw                 |          |
| <b>loadbytemap</b>             | function | F fill                 |          |
| <b>loadbytemapfile</b>         | function | F flush                |          |
| <b>mesh_register</b>           | function | F fprint               |          |
| <b>mf_blob_add</b>             | function | F integer              |          |
| <b>mf_blob_dimensions</b>      | function | F number               |          |
| <b>mf_blob_index</b>           | function | F numeric              |          |
| <b>mf_blob_new</b>             | function | F pair                 |          |
| <b>mf_blob_raw_content</b>     | function | F pairpoints           |          |
| <b>mf_blob_raw_dimensions</b>  | function | F path                 |          |
| <b>mf_blob_raw_reset</b>       | function | F pathpoints           |          |
| <b>mf_blob_raw_toutf</b>       | function | F points               |          |
| <b>mf_blob_raw_wipe</b>        | function | F print                |          |
| <b>mf_blob_size</b>            | function | F quadruple            |          |
| <b>mf_blob_width</b>           | function | F quadruplepoints      |          |
| <b>mf_external_figure</b>      | function | F quoted               |          |
| <b>mf_finish_saving_data</b>   | function | F size                 |          |
| <b>mf_formatted_text</b>       | function | F string               |          |
| <b>mf_get_outline_text</b>     | function | F transform            |          |
| <b>mf_graphic_text</b>         | function | F triplet              |          |
| <b>mf_inject_blob</b>          | function | F tripletpoints        |          |
| <b>mf_map_move</b>             | function | F vprint               |          |
| <b>mf_map_text</b>             | function | <b>processbytemap</b>  | function |
| <b>mf_named_color</b>          | function | <b>processingmode</b>  | function |
| <b>mf_outline_text</b>         | function | <b>quadruple</b>       | function |
| <b>mf_save_data</b>            | function | <b>quadruplepoints</b> | function |
| <b>mf_script_index</b>         | function | <b>quoted</b>          | function |
| <b>mf_start_saving_data</b>    | function | <b>report</b>          | function |
| <b>mf_stop_saving_data</b>     | function | <b>scan</b>            | table    |
| <b>mode</b>                    | function | F boolean              |          |

F cmykcolor  
F color  
F expression  
F integer  
F next  
F number  
F numeric  
F pair  
F path  
F pen  
F property  
F string  
F symbol  
F token  
F transform

F whatever  
**set**  
**setbytemap**  
**setbytemapoctave**  
**size**  
**skip**  
F token  
**solve**  
**string**  
**systemmode**  
**transform**  
**triplet**  
**tripletpoints**  
**vprint**

table  
function  
function  
function  
table  
  
function  
function  
function  
function  
function  
function

## mplib (extra tex)

|                             |          |                        |                     |
|-----------------------------|----------|------------------------|---------------------|
| <b>downgradebyte</b>        | function | <b>injectcmymcolor</b> | function            |
| <b>downsamplebyte</b>       | function | <b>injectcolor</b>     | function            |
| <b>execute</b>              | function | <b>injectinteger</b>   | function            |
| <b>expandtex</b>            | function | <b>injectnumber</b>    | function            |
| <b>fillbyte</b>             | function | <b>injectnumeric</b>   | function            |
| <b>finder</b>               | function | <b>injectpair</b>      | function            |
| <b>finders</b>              | table    | <b>injectpath</b>      | function            |
| F file                      |          | <b>injectpoints</b>    | function            |
| F mpstemplate               |          | <b>injectstring</b>    | function            |
| <b>finish</b>               | function | <b>injecttokens</b>    | function            |
| <b>getbyte</b>              | function | <b>injecttransform</b> | function            |
| <b>getbyte</b>              | function | <b>injectvector</b>    | function            |
| <b>getbyte</b>              | function | <b>injectvectors</b>   | function            |
| <b>getcallbackstate</b>     | function | <b>injectwhatever</b>  | function            |
| <b>getcodes</b>             | function | <b>new</b>             | overloaded function |
| <b>getcolormodels</b>       | function | <b>newbyte</b>         | function            |
| <b>getfields</b>            | function | <b>peninfo</b>         | function            |
| <b>gethashentries</b>       | function | <b>processbyte</b>     | function            |
| <b>gethashentry</b>         | function | <b>propertycodes</b>   | table               |
| <b>getinternalactions</b>   | function | <b>realtime</b>        | boolean             |
| <b>getknotstates</b>        | function | <b>scanboolean</b>     | function            |
| <b>getlastadddtype</b>      | function | <b>scancmymcolor</b>   | function            |
| <b>getlogtargets</b>        | function | <b>scancolor</b>       | function            |
| <b>getobjectdata</b>        | function | <b>scanexpression</b>  | function            |
| <b>getobjectpeninfo</b>     | function | <b>scaninteger</b>     | function            |
| <b>getobjectstacking</b>    | function | <b>scannext</b>        | function            |
| <b>getobjecttypes</b>       | function | <b>scannumber</b>      | function            |
| <b>getprimitivelegacies</b> | function | <b>scannumeric</b>     | function            |
| <b>getprimitiveorigins</b>  | function | <b>scanpair</b>        | function            |
| <b>getprimitives</b>        | function | <b>scanpath</b>        | function            |
| <b>getpruneoptions</b>      | function | <b>scanpen</b>         | function            |
| <b>getresultstates</b>      | function | <b>scanproperty</b>    | function            |
| <b>getscaledbits</b>        | function | <b>scanstring</b>      | function            |
| <b>getscantypes</b>         | function | <b>scansymbol</b>      | function            |
| <b>getstates</b>            | function | <b>scantoken</b>       | function            |
| <b>getstatistics</b>        | function | <b>scantransform</b>   | function            |
| <b>getstatus</b>            | function | <b>setbyte</b>         | function            |
| <b>gettolerance</b>         | function | <b>setbyte</b>         | function            |
| <b>gettype</b>              | function | <b>settolerance</b>    | function            |
| <b>gettypes</b>             | function | <b>showcontext</b>     | function            |
| <b>injectboolean</b>        | function | <b>skiptoken</b>       | function            |

**solvepath**  
**version**

function  
function

# nanojpeg (context)

**decode** function

# noads (context)

|                    |          |                      |              |
|--------------------|----------|----------------------|--------------|
| <b>getmathmode</b> | function |                      | F relocate   |
| <b>handlers</b>    | table    |                      | F render     |
| F alternates       |          |                      | F showtree   |
| F autofences       |          |                      | F snap       |
| F classes          |          |                      | F spacing    |
| F collapse         |          |                      | F suspicious |
| F dictionaries     |          |                      | F tags       |
| F export           |          |                      | F unscript   |
| F families         |          |                      | F unstack    |
| F fencing          |          |                      | F variants   |
| F intervals        |          | <b>process</b>       | function     |
| F kernpairs        |          | <b>processnested</b> | function     |
| F makeup           |          | <b>processors</b>    | table        |
| F normalize        |          | <b>processouter</b>  | function     |
| F numbers          |          |                      |              |

# node (basic tex)

|                             |          |                           |
|-----------------------------|----------|---------------------------|
| <b>checkdiscretionaries</b> | function | F getanchors              |
| <b>checkdiscretionary</b>   | function | F getattribute            |
| <b>copy</b>                 | function | F getattributelist        |
| <b>copylist</b>             | function | F getattributes           |
| <b>count</b>                | function | F getboth                 |
| <b>currentattributes</b>    | function | F getbottom               |
| <b>dimensions</b>           | function | F getbottomdelimiter      |
| <b>direct</b>               | table    | F getbox                  |
| F addmargins                |          | F getcachestate           |
| F addxoffset                |          | F getchar                 |
| F addxymargins              |          | F getchardict             |
| F addyoffset                |          | F getcharspec             |
| F appendaftertail           |          | F getchoice               |
| F appendcurrenttail         |          | F getclass                |
| F beginofmath               |          | F getcontinuation         |
| F checkdiscretionaries      |          | F getcontrol              |
| F checkdiscretionary        |          | F getcornerkerns          |
| F collapsing                |          | F getcurrenttail          |
| F copy                      |          | F getdata                 |
| F copylist                  |          | F getdegree               |
| F copyonly                  |          | F getdelimiter            |
| F count                     |          | F getdenominator          |
| F currentattributes         |          | F getdepth                |
| F dimensions                |          | F getdirection            |
| F effectiveglue             |          | F getdisc                 |
| F endofmath                 |          | F getdiscpart             |
| F exchange                  |          | F getexcept               |
| F fields                    |          | F getexpansion            |
| F findattribute             |          | F getfam                  |
| F findattributerange        |          | F getfield                |
| F findnode                  |          | F getfielderror           |
| F firstchar                 |          | F getfirstdirectioninlist |
| F firstglyph                |          | F getfont                 |
| F firstglyphnode            |          | F getfontcheck            |
| F firstitalicglyph          |          | F getgeometry             |
| F flattendiscretionaries    |          | F getglue                 |
| F flattenleaders            |          | F getglyphdata            |
| F flushlist                 |          | F getglyphdimensions      |
| F flushnode                 |          | F getheight               |
| F free                      |          | F getid                   |
| F freeze                    |          | F getidsubtype            |

F getidsubtypenext  
F getindex  
F getinputfields  
F getkern  
F getkerndimension  
F getlanguage  
F getleader  
F getleftdelimiter  
F getlist  
F getlistdimensions  
F getlocalparboxstate  
F getmvlolist  
F getnaturalwhd  
F getnext  
F getnodeerrorvalues  
F getnodes  
F getnormalizedline  
F getnucleus  
F getnumerator  
F getoffsets  
F getoptions  
F getorientation  
F getparstate  
F getpenalty  
F getpost  
F getpre  
F getprev  
F getprime  
F getpropertystable  
F getproperty  
F getreplace  
F getrightdelimiter  
F getruledimensions  
F getscale  
F getscales  
F getscrip  
F getscripts  
F getshift  
F getshort  
F getslant  
F getspeciallist  
F getstate  
F getsub  
F getsubpre

F getsubtype  
F getsup  
F getsuppre  
F gettop  
F gettopdelimiter  
F gettotal  
F getusage  
F getusedattributes  
F getweight  
F getwhd  
F getwidth  
F getwordrange  
F getxscale  
F getxyscales  
F getsyscale  
F gluetostring  
F hasattribute  
F hasdimensions  
F hasdiscoption  
F hasfield  
F hasgeometry  
F hasglyph  
F hasglyphoption  
F hasidsubtype  
F hasoptions  
F hasusage  
F hpack  
F hyphenating  
F id  
F ignoremathskip  
F insertafter  
F insertbefore  
F insertcurrentthead  
F isboth  
F ischar  
F ischarcheck  
F ischardisc  
F isdirect  
F isfontcheck  
F isglyph  
F isitalicglyph  
F isloop  
F isnext  
F isnextchar



|                       |                       |
|-----------------------|-----------------------|
| F isnextcharcheck     | F setattribute        |
| F isnextglyph         | F setattributelist    |
| F isnode              | F setattributes       |
| F isprev              | F setattributesinlist |
| F isprevchar          | F setboth             |
| F isprevcharcheck     | F setbottom           |
| F isprevglyph         | F setbottomdelimiter  |
| F issimilarglyph      | F setbox              |
| F issnapped           | F setchar             |
| F isspeciallist       | F setchardict         |
| F isvalid             | F setchoice           |
| F iszeroglue          | F setclass            |
| F kerning             | F setcontinuation     |
| F lastnode            | F setcontrol          |
| F length              | F setcurrenttail      |
| F ligaturing          | F setdata             |
| F makeextensible      | F setdegree           |
| F markglyphprocessing | F setdelimiter        |
| F migrate             | F setdenominator      |
| F mlisttohlist        | F setdepth            |
| F naturalhsize        | F setdirection        |
| F naturalwidth        | F setdisc             |
| F new                 | F setdiscpart         |
| F newcontinuationatom | F setexcept           |
| F newfontcheck        | F setexpansion        |
| F newmathglyph        | F setfam              |
| F newtextglyph        | F setfield            |
| F patchattributes     | F setfielderror       |
| F patchparshape       | F setfont             |
| F prependbeforehead   | F setfontcheck        |
| F prependcurrenttail  | F setgeometry         |
| F protectglyph        | F setglue             |
| F protectglyphs       | F setglyphdata        |
| F protectglyphsbase   | F setheight           |
| F protectglyphsnone   | F setindex            |
| F protrusionskipable  | F setinputfields      |
| F rangedimensions     | F setkern             |
| F remove              | F setlanguage         |
| F removefromlist      | F setleader           |
| F repack              | F setleftdelimiter    |
| F reverse             | F setlink             |
| F samefontcheck       | F setlist             |
| F serialized          | F setnext             |
| F setanchors          | F setnucleus          |

|                     |                               |          |
|---------------------|-------------------------------|----------|
| F setnumerator      | F tovaliddirect               |          |
| F setoffsets        | F traverse                    |          |
| F setoptions        | F traversesechar              |          |
| F setorientation    | F traversesecontent           |          |
| F setpenalty        | F traverseseglyph             |          |
| F setpost           | F traverseseid                |          |
| F setpre            | F traverseseitalic            |          |
| F setprev           | F traverseseleader            |          |
| F setprime          | F traverselist                |          |
| F setproperty       | F traversesepossible          |          |
| F setreplace        | F type                        |          |
| F setrightdelimiter | F types                       |          |
| F setruledimensions | F unprotectglyph              |          |
| F setscale          | F unprotectglyphs             |          |
| F setscales         | F unsetattribute              |          |
| F setscript         | F unsetattributes             |          |
| F setscripts        | F updatefirstandbotmark       |          |
| F setshift          | F updatefirstmarks            |          |
| F setslant          | F updatemarks                 |          |
| F setsnapped        | F updatetopmarks              |          |
| F setspeciallist    | F usedlist                    |          |
| F setsplit          | F usesfont                    |          |
| F setstate          | F vbalance                    |          |
| F setsub            | F verticalbreak               |          |
| F setsubpre         | F vpack                       |          |
| F setsubtype        | F write                       |          |
| F setup             | F xscaled                     |          |
| F setsuppre         | F yscaled                     |          |
| F settop            | <b>effectiveglue</b>          | function |
| F settopdelimiter   | <b>endofmath</b>              | function |
| F settotal          | <b>family_font</b>            | function |
| F setweight         | <b>fields</b>                 | function |
| F setwhd            | <b>findattribute</b>          | function |
| F setwidth          | <b>flattendiscretionaries</b> | function |
| F show              | <b>flushlist</b>              | function |
| F size              | <b>flushnode</b>              | function |
| F slide             | <b>free</b>                   | function |
| F softenhyphens     | <b>getattribute</b>           | function |
| F startofpar        | <b>getcachestate</b>          | function |
| F subtypes          | <b>getfield</b>               | function |
| F tail              | <b>getfielderror</b>          | function |
| F todirect          | <b>getglue</b>                | function |
| F tonode            | <b>getnodeerrorvalues</b>     | function |
| F tostring          | <b>getpropertystable</b>      | function |

|                             |          |                        |          |
|-----------------------------|----------|------------------------|----------|
| <b>getproperty</b>          | function | <b>remove</b>          | function |
| <b>gluetostring</b>         | function | <b>serialized</b>      | function |
| <b>hasattribute</b>         | function | <b>setattribute</b>    | function |
| <b>hasfield</b>             | function | <b>setfield</b>        | function |
| <b>hasglyph</b>             | function | <b>setfielderror</b>   | function |
| <b>hpack</b>                | function | <b>setglue</b>         | function |
| <b>hyphenating</b>          | function | <b>setproperty</b>     | function |
| <b>id</b>                   | function | <b>show</b>            | function |
| <b>insertafter</b>          | function | <b>size</b>            | function |
| <b>insertbefore</b>         | function | <b>slide</b>           | function |
| <b>instock</b>              | function | <b>subtypes</b>        | function |
| <b>intracking</b>           | function | <b>tail</b>            | function |
| <b>inuse</b>                | function | <b>todirect</b>        | function |
| <b>isnode</b>               | function | <b>tonode</b>          | function |
| <b>iszeroglue</b>           | function | <b>tostring</b>        | function |
| <b>kerning</b>              | function | <b>traverse</b>        | function |
| <b>lastnode</b>             | function | <b>traverseid</b>      | function |
| <b>length</b>               | function | <b>type</b>            | function |
| <b>ligaturing</b>           | function | <b>types</b>           | function |
| <b>makeextensible</b>       | function | <b>unprotectglyph</b>  | function |
| <b>mlisttohlist</b>         | function | <b>unprotectglyphs</b> | function |
| <b>new</b>                  | function | <b>unsetattribute</b>  | function |
| <b>protectglyph</b>         | function | <b>usedlist</b>        | function |
| <b>protectglyphs</b>        | function | <b>usesfont</b>        | function |
| <b>protrusionskipppable</b> | function | <b>vpack</b>           | function |
| <b>rangedimensions</b>      | function | <b>write</b>           | function |

# nodes (context)

|                           |            |                      |            |
|---------------------------|------------|----------------------|------------|
| <b>accentcodes</b>        | data table | <b>fencecodes</b>    | data table |
| <b>adaptive</b>           | table      | <b>fields</b>        | function   |
| F handlehorizontal        |            | <b>fieldtypes</b>    | table      |
| F handlevertical          |            | <b>fillcodes</b>     | data table |
| F set                     |            | <b>fillvalues</b>    | data table |
| <b>aligned</b>            | function   | <b>firstdirinbox</b> | function   |
| <b>append</b>             | function   | <b>flush</b>         | function   |
| <b>apply</b>              | function   | <b>flushlist</b>     | function   |
| <b>applyvisuals</b>       | function   | <b>flushnode</b>     | function   |
| <b>astable</b>            | function   | <b>fractioncodes</b> | data table |
| <b>basepoints</b>         | function   | <b>fullhpack</b>     | function   |
| <b>boundarycodes</b>      | data table | <b>getattr</b>       | function   |
| <b>boxes</b>              | table      | <b>getattribute</b>  | function   |
| F count                   |            | <b>getboth</b>       | function   |
| F dimensions              |            | <b>getbox</b>        | function   |
| F direct                  |            | <b>getchar</b>       | function   |
| F dispose                 |            | <b>getfield</b>      | function   |
| F found                   |            | <b>getfont</b>       | function   |
| F prune                   |            | <b>getid</b>         | function   |
| F reset                   |            | <b>getleader</b>     | function   |
| F restore                 |            | <b>getlist</b>       | function   |
| F save                    |            | <b>getnext</b>       | function   |
| F savenode                |            | <b>getprev</b>       | function   |
| <b>builders</b>           | table      | <b>getproperty</b>   | function   |
| <b>choicecodes</b>        | data table | <b>getsubtype</b>    | function   |
| <b>concat</b>             | function   | <b>gluecodes</b>     | data table |
| <b>copy</b>               | function   | <b>glyphcodes</b>    | data table |
| <b>copy_node</b>          | function   | <b>handlers</b>      | table      |
| <b>copyleft</b>           | function   | F adjusters          |            |
| <b>countall</b>           | function   | F aligncharacter     |            |
| <b>currentattributes</b>  | function   | F backgrounds        |            |
| <b>delete</b>             | function   | F backgroundspace    |            |
| <b>destinations</b>       | table      | F backgroundsvbox    |            |
| N attribute               |            | F characters         |            |
| F handler                 |            | F checkglyphs        |            |
| <b>dircodes</b>           | data table | F checkparcounter    |            |
| <b>directioncodes</b>     | data table | F cleanupbox         |            |
| <b>directionvalues</b>    | data table | F cleanuppage        |            |
| <b>dirvalues</b>          | data table | F export             |            |
| <b>disccodes</b>          | data table | F finalizebox        |            |
| <b>discretionarycodes</b> | data table | F finalizelist       |            |

|                                |          |                        |            |
|--------------------------------|----------|------------------------|------------|
| F flatten                      |          | <b>is_display_math</b> | function   |
| F flattenline                  |          | <b>isdirect</b>        | function   |
| F mathalignrelocate            |          | <b>isnode</b>          | function   |
| F mathmatrixornaments          |          | <b>isnut</b>           | function   |
| F mathmatrixrules              |          | <b>kerncodes</b>       | data table |
| F migrate                      |          | <b>leftmarginwidth</b> | function   |
| F protectglyphs                |          | <b>linefillers</b>     | table      |
| F recolor                      |          | F filler               |            |
| F show                         |          | F handler              |            |
| F showhyphenation              |          | F set                  |            |
| F showpreamble                 |          | <b>lines</b>           | table      |
| F stripping                    |          | D data                 |            |
| F tabulateskip                 |          | F finalize             |            |
| F textbackgrounds              |          | F number               |            |
| F unprotectglyphs              |          | <b>link</b>            | function   |
| F visualizehyphenation         |          | <b>linked</b>          | function   |
| F wipe                         |          | <b>list</b>            | function   |
| <b>hasattribute</b>            | function | <b>listcodes</b>       | data table |
| <b>hasfield</b>                | function | <b>listtoutf</b>       | function   |
| <b>hpack</b>                   | function | <b>literalvalues</b>   | data table |
| <b>idsandsubtypes</b>          | function | <b>locate</b>          | function   |
| <b>idstostring</b>             | function | <b>markcodes</b>       | data table |
| <b>injections</b>              | table    | <b>markers</b>         | table      |
| F copy                         |          | F get                  |            |
| F getligaindex                 |          | F set                  |            |
| F getthreshold                 |          | <b>mathcodes</b>       | data table |
| F handler                      |          | <b>mathvalues</b>      | data table |
| F installgetspaceboth          |          | <b>maxboxwidth</b>     | function   |
| F installnewkern               |          | <b>migrations</b>      | table      |
| F isspace                      |          | F setinserts           |            |
| F keepcounts                   |          | F setmarks             |            |
| F reset                        |          | <b>new</b>             | function   |
| F resetcounts                  |          | <b>noadcodes</b>       | data table |
| F setcursive                   |          | <b>nodecodes</b>       | data table |
| F setkern                      |          | <b>nopts</b>           | function   |
| F setligaindex                 |          | <b>nuts</b>            | table      |
| F setmark                      |          | F addmargins           |            |
| F setmove                      |          | F addxoffset           |            |
| F setposition                  |          | F addxymargins         |            |
| F setspacekerns                |          | F addyoffset           |            |
| <b>insertafter</b>             | function | F append               |            |
| <b>insertbefore</b>            | function | F appendaftertail      |            |
| <b>insertlistafter</b>         | function | F apply                |            |
| <b>installattributehandler</b> | function | F applyvisuals         |            |

|                          |                           |
|--------------------------|---------------------------|
| F beginofmath            | F getboth                 |
| F checkdiscretionaries   | F getbottom               |
| F checkdiscretionary     | F getbottomdelimiter      |
| F checkslide             | F getbox                  |
| F collapsing             | F getboxglue              |
| F concat                 | F getchar                 |
| F copy                   | F getchardict             |
| F copy_properties        | F getcharspec             |
| F copylist               | F getchoice               |
| F copynode               | F getclass                |
| F copyonly               | F getcomponents           |
| F copyvisual             | F getcontinuation         |
| F count                  | F getcontrol              |
| F countall               | F getcornerkerns          |
| F currentattributes      | F getdata                 |
| F delete                 | F getdegree               |
| F dimensions             | F getdelimiter            |
| F dirdimensions          | F getdenominator          |
| F effectiveglue          | F getdepth                |
| F endofmath              | F getdirection            |
| F exchange               | F getdisc                 |
| F findattribute          | F getdiscpart             |
| F findattributerange     | F getdiscretionary        |
| F findnode               | F getexcept               |
| F firstchar              | F getexpansion            |
| F firstglyph             | F getfam                  |
| F firstglyphnode         | F getfield                |
| F firstitalicglyph       | F getfirstdirectioninlist |
| F flattendiscretionaries | F getfont                 |
| F flattenleaders         | F getfontcheck            |
| F flush                  | F getgeometry             |
| F flushlist              | F getglue                 |
| F flushnode              | F getglyphdata            |
| F free                   | F getglyphdimensions      |
| F freeze                 | F getheight               |
| F fullhpack              | F getid                   |
| F fullhpackrl            | F getidsubtype            |
| F getanchors             | F getidsubtypenext        |
| F getattr                | F getindex                |
| F getattribute           | F getinputfields          |
| F getattributelist       | F getkern                 |
| F getattributes          | F getkerndimension        |
| F getattrlist            | F getlang                 |
| F getattrts              | F getlanguage             |

|                       |                     |
|-----------------------|---------------------|
| F getleader           | F gettotal          |
| F getleftdelimiter    | F getusedattributes |
| F getlist             | F getvalue          |
| F getlistdimensions   | F getweight         |
| F getlocalparboxstate | F getwhd            |
| F getmiddle           | F getwidth          |
| F getmiddledelimiter  | F getwordrange      |
| F getmvlolist         | F getxscale         |
| F getnaturalwhd       | F getxyscales       |
| F getnext             | F getyscale         |
| F getnormalizedline   | F gluetostring      |
| F getnucleus          | F hasattribute      |
| F getnumerator        | F hasdimensions     |
| F getoffsets          | F hasdiscoption     |
| F getoptions          | F hasfield          |
| F getorientation      | F hasgeometry       |
| F getparstate         | F hasglyph          |
| F getpenalty          | F hasglyphoption    |
| F getpost             | F hasidsubtype      |
| F getpre              | F hasoptions        |
| F getprev             | F hpack             |
| F getprime            | F hyphenating       |
| F getprop             | F ignoremathskip    |
| F getreplace          | F insertafter       |
| F getrightdelimiter   | F insertbefore      |
| F getruledata         | F insertlistafter   |
| F getruledimensions   | F is_display_math   |
| F getscale            | F isboth            |
| F getscales           | F ischar            |
| F getscrip            | F ischarcheck       |
| F getscripts          | F isdirect          |
| F getshift            | F isdone            |
| F getshort            | F isglyph           |
| F getslant            | F isitalicglyph     |
| F getspeciallist      | F isnext            |
| F getstate            | F isnextchar        |
| F getsub              | F isnextcharcheck   |
| F getsubpre           | F isnextglyph       |
| F getsubtype          | F isnode            |
| F getsup              | F isnut             |
| F getsuppre           | F isprev            |
| F getsurround         | F isprevchar        |
| F gettop              | F isprevcharcheck   |
| F gettopdelimiter     | F isprevglyph       |

|                       |                         |
|-----------------------|-------------------------|
| F issimilarglyph      | F setattr               |
| F issnapped           | F setattrtribute        |
| F iszeroglue          | F setattrtributelist    |
| F kerning             | F setattrtributes       |
| F lastnode            | F setattrtributesinlist |
| F length              | F setattrrlist          |
| F ligaturing          | F setattrrs             |
| F link                | F setattrrsinlist       |
| F linked              | F setboth               |
| F locate              | F setbottom             |
| F migrate             | F setbox                |
| F migratebox          | F setboxglue            |
| F mlisttohlist        | F setchar               |
| F naturalhsize        | F setchardict           |
| F naturalwidth        | F setchoice             |
| F nestedtracedslide   | F setclass              |
| F new                 | F setcomponents         |
| F newcontinuationatom | F setcontinuation       |
| F newfontcheck        | F setcontrol            |
| F newmathglyph        | F setdata               |
| F openup              | F setdegree             |
| F patchattributes     | F setdelimiter          |
| F patchparshape       | F setdenominator        |
| F prepend             | F setdepth              |
| F prependbeforehead   | F setdirection          |
| F processranges       | F setdisc               |
| F processwords        | F setdiscpart           |
| F protectglyph        | F setdiscretionary      |
| F protectglyphs       | F setexcept             |
| F protectglyphsbase   | F setexpansion          |
| F protectglyphsnone   | F setfam                |
| F protrusionskipable  | F setfield              |
| F rangedimensions     | F setfont               |
| F rawprop             | F setfontcheck          |
| F reference           | F setgeometry           |
| F rehpick             | F setglue               |
| F remove              | F setglyphdata          |
| F removefromlist      | F setheight             |
| F repack              | F setinputfields        |
| F repackhlist         | F setkern               |
| F replace             | F setlang               |
| F reverse             | F setlanguage           |
| F serialized          | F setleader             |
| F setanchors          | F setleftdelimiter      |



|                      |                         |
|----------------------|-------------------------|
| F setlink            | F slide                 |
| F setlist            | F softenhyphens         |
| F setmiddledelimiter | F somepenalty           |
| F setnext            | F somespace             |
| F setnucleus         | F splitbox              |
| F setnumerator       | F startofpar            |
| F setoffsets         | F striprange            |
| F setoptions         | F tail                  |
| F setorientation     | F takeattr              |
| F setparproperty     | F takebox               |
| F setpenalty         | F takelist              |
| F setpost            | F theprop               |
| F setpre             | F tonode                |
| F setprev            | F tonodes               |
| F setprime           | F tonut                 |
| F setprop            | F tosequence            |
| F setreplace         | F toString              |
| F setrightdelimiter  | F tracedslide           |
| F setruledata        | F traverse              |
| F setruledimensions  | F traversesechar        |
| F setscale           | F traversesecontent     |
| F setscales          | F traverseseglyph       |
| F setscript          | F traverseseid          |
| F setscripts         | F traverseseitalic      |
| F setshift           | F traverseseleader      |
| F setslant           | F traverselist          |
| F setsnapped         | F traversepossible      |
| F setspeciallist     | F unprotectglyph        |
| F setsplit           | F unprotectglyphs       |
| F setstate           | F unsetattribute        |
| F setsub             | F unsetattributes       |
| F setsubpre          | F untracedslide         |
| F setsubtype         | F updatefirstandbotmark |
| F setup              | F updatefirstmarks      |
| F setsuppre          | F updatemarks           |
| F setsurround        | F updatetopmarks        |
| F settop             | F usedlist              |
| F setvalue           | F usesfont              |
| F setvisual          | F verticalbreak         |
| F setvisuals         | F vianodes              |
| F setweight          | F vianuts               |
| F setwhd             | F vpack                 |
| F setwidth           | F write                 |
| F show               | F xscaled               |

|                     |            |                          |            |
|---------------------|------------|--------------------------|------------|
| F yscaled           |            | replace                  | function   |
| <b>packlist</b>     | function   | <b>report</b>            | function   |
| <b>parcodes</b>     | data table | <b>rightmarginwidth</b>  | function   |
| <b>penaltycodes</b> | data table | <b>rulecodes</b>         | data table |
| <b>plugindata</b>   | table      | <b>rules</b>             | table      |
| F finalizer         |            | F handler                |            |
| F initializer       |            | F process                |            |
| S name              |            | F set                    |            |
| F processor         |            | F userrule               |            |
| <b>points</b>       | function   | <b>serialize</b>         | function   |
| <b>pool</b>         | table      | <b>serializebox</b>      | function   |
| F baselineskip      |            | <b>serialized</b>        | function   |
| F cleanup           |            | <b>setattr</b>           | function   |
| F directliteral     |            | <b>setattribute</b>      | function   |
| F glyph             |            | <b>setattrlist</b>       | function   |
| F hlist             |            | <b>setboth</b>           | function   |
| F latelua           |            | <b>setbox</b>            | function   |
| F leader            |            | <b>setboxtonaturalwd</b> | function   |
| F lineskip          |            | <b>setchar</b>           | function   |
| F literal           |            | <b>setfield</b>          | function   |
| F originliteral     |            | <b>setfont</b>           | function   |
| F register          |            | <b>setleader</b>         | function   |
| F restore           |            | <b>setlink</b>           | function   |
| F save              |            | <b>setlist</b>           | function   |
| F savepos           |            | <b>setnext</b>           | function   |
| F setmatrix         |            | <b>setprev</b>           | function   |
| F stock             |            | <b>setProperty</b>       | function   |
| F usage             |            | <b>shadows</b>           | table      |
| F usernode          |            | F handler                |            |
| F vlist             |            | F set                    |            |
| <b>prepend</b>      | function   | <b>shifts</b>            | table      |
| <b>print</b>        | function   | F handler                |            |
| <b>processors</b>   | table      | F set                    |            |
| F trace_glyph_run_a |            | <b>showboxes</b>         | function   |
| F trace_glyph_run_b |            | <b>showlist</b>          | function   |
| <b>properties</b>   | table      | <b>showsimplelist</b>    | function   |
| <b>pts</b>          | function   | <b>skipcodes</b>         | data table |
| <b>radicalcodes</b> | data table | <b>snapshots</b>         | table      |
| <b>references</b>   | table      | F getsamples             |            |
| N attribute         |            | F resetsamples           |            |
| F handler           |            | F takesample             |            |
| <b>rehpack</b>      | function   | <b>somepenalty</b>       | function   |
| <b>remove</b>       | function   | <b>somespace</b>         | function   |
| <b>repackhlist</b>  | function   | <b>specialskipcodes</b>  | table      |

|                             |            |                           |            |
|-----------------------------|------------|---------------------------|------------|
| <b>splitbox</b>             | function   | <b>tracers</b>            | table      |
| <b>startofpar</b>           | function   | F directions              |            |
| <b>stripdiscretionaries</b> | function   | F fontchar                |            |
| <b>stripping</b>            | table      | F resetproperties         |            |
| F set                       |            | F rule                    |            |
| <b>subtypes</b>             | data table | F setlist                 |            |
| <b>tail</b>                 | function   | F setproperties           |            |
| <b>takeattr</b>             | function   | F start                   |            |
| <b>takebox</b>              | function   | F stop                    |            |
| <b>tasks</b>                | table      | <b>traverse</b>           | function   |
| F actions                   |            | <b>traverseid</b>         | function   |
| F appendaction              |            | <b>typesetters</b>        | table      |
| F disableaction             |            | F hbox                    |            |
| F disablegroup              |            | F hpack                   |            |
| F enableaction              |            | F tohbox                  |            |
| F enablegroup               |            | F tohpack                 |            |
| F freeze                    |            | F tonodes                 |            |
| F freezegroup               |            | F tovbox                  |            |
| F new                       |            | F tovpack                 |            |
| F prependaction             |            | F vpack                   |            |
| F removeaction              |            | <b>unsetattribute</b>     | function   |
| F replaceaction             |            | <b>upcomingproperties</b> | function   |
| F restart                   |            | <b>usedlist</b>           | function   |
| F setaction                 |            | <b>vianodes</b>           | function   |
| F showactions               |            | <b>vianuts</b>            | function   |
| F table                     |            | <b>visualizebox</b>       | function   |
| <b>tobasepoints</b>         | function   | <b>visualizers</b>        | table      |
| <b>tocentimeters</b>        | function   | F attribute               |            |
| <b>tociceros</b>            | function   | F box                     |            |
| <b>todidots</b>             | function   | F definelayer             |            |
| <b>todimen</b>              | function   | F enable                  |            |
| <b>toinches</b>             | function   | F getusedfont             |            |
| <b>tomillimeters</b>        | function   | F handler                 |            |
| <b>tonode</b>               | function   | F markfonts               |            |
| <b>tonodes</b>              | function   | F register                |            |
| <b>tonut</b>                | function   | F registercache           |            |
| <b>topicas</b>              | function   | F setfont                 |            |
| <b>topoints</b>             | function   | F setlayer                |            |
| <b>toscaledpoints</b>       | function   | F setruledglyph           |            |
| <b>tosequence</b>           | function   | F setvisual               |            |
| <b>tostring</b>             | function   | <b>vpack</b>              | function   |
| <b>totable</b>              | function   | <b>whatsitcodes</b>       | data table |
| <b>totree</b>               | function   | <b>write</b>              | function   |
| <b>toutf</b>                | function   |                           |            |

## number (context)

|                         |            |
|-------------------------|------------|
|                         | function   |
| <b>basepoints</b>       | function   |
| <b>bytetodecimal</b>    | function   |
| <b>clamp</b>            | function   |
| <b>compactfloat</b>     | function   |
| <b>decimaltobyte</b>    | function   |
| <b>dimenfactors</b>     | data table |
| <b>formatted</b>        | function   |
| <b>formattedfloat</b>   | function   |
| <b>formattedinteger</b> | function   |
| <b>fractiontobyte</b>   | function   |
| <b>idiv</b>             | function   |
| <b>maxdimen</b>         | number     |
| <b>nopts</b>            | function   |
| <b>nubasepoints</b>     | function   |
| <b>nupoints</b>         | function   |
| <b>percent</b>          | function   |
| <b>points</b>           | function   |
| <b>pts</b>              | function   |
| <b>signed</b>           | function   |
| <b>sparseexponent</b>   | function   |

|                       |          |
|-----------------------|----------|
| <b>splitdimen</b>     | function |
| <b>tol6dot16</b>      | function |
| <b>tobasepoints</b>   | function |
| <b>tobitstring</b>    | function |
| <b>tocentimeters</b>  | function |
| <b>toxiceros</b>      | function |
| <b>tocontext</b>      | function |
| <b>todidots</b>       | function |
| <b>todimen</b>        | function |
| <b>toediths</b>       | function |
| <b>toevenhex</b>      | function |
| <b>toinches</b>       | function |
| <b>tomillimeters</b>  | function |
| <b>tonumberf</b>      | function |
| <b>tonumberg</b>      | function |
| <b>topicas</b>        | function |
| <b>topoints</b>       | function |
| <b>toscaled</b>       | function |
| <b>toscaledpoints</b> | function |
| <b>totoves</b>        | function |
| <b>valid</b>          | function |

## objects (context)

|                   |          |                  |          |
|-------------------|----------|------------------|----------|
| <b>data</b>       | table    | <b>page</b>      | function |
| <b>dimensions</b> | function | <b>reference</b> | function |
| <b>found</b>      | function | <b>register</b>  | function |
| <b>n</b>          | number   | <b>restore</b>   | function |

# optional (context)

## **curl** table

F escape  
F fetch  
F getversion  
F initialize  
F unescape

## **foreign** table

F abivalues  
F getbuffer  
F initialize  
F load  
F newbuffer  
F totable  
F types

## **ghostscript** table

F execute  
F initialize

## **graphicsmagick** table

F execute  
F getnoisetypes  
F initialize

## **hb** table

F getshapers  
F getversion  
F initialize  
F loadfont  
F shapestring

## **imagemagick** table

F execute  
F initialize

## **kpse** table

F expand\_braces  
F expand\_path  
F expand\_var  
F find\_file  
F find\_files  
F get\_file\_types  
F initialize  
F readable\_file  
F set\_program\_name  
F var\_value

## **library** table

F load

## **loaded** table

## **lz4** table

F compress  
F decompresssize  
F framecompress  
F framedecompress  
F initialize

## **lzma** table

F compress  
F decompress  
F initialize

## **lzo** table

F compress  
F decompresssize  
F initialize

## **mujs** table

F dofile  
F execute  
F initialize  
F reset  
F setclosefile  
F setconsole  
F setfindfile  
F setopenfile  
F setreadfile  
F setseekfile

## **mysql** table

F close  
F execute  
F getmessage  
F initialize  
F open

## **openssl** table

F getversion  
F initialize  
F sign  
F verify

## **postgres** table

F close

|               |       |              |       |
|---------------|-------|--------------|-------|
| F execute     |       | F open       |       |
| F getmessage  |       | <b>zint</b>  | table |
| F initialize  |       | F execute    |       |
| F open        |       | F initialize |       |
| <b>sqlite</b> | table | <b>zstd</b>  | table |
| F close       |       | F compress   |       |
| F execute     |       | F decompress |       |
| F getmessage  |       | F initialize |       |
| F initialize  |       |              |       |

## os (basic lua)

|                       |                     |                      |                     |
|-----------------------|---------------------|----------------------|---------------------|
| <b>__getenv__</b>     | function            | <b>platform</b>      | string              |
| <b>__setenv__</b>     | function            | <b>remove</b>        | overloaded function |
| <b>binsuffix</b>      | string              | <b>rename</b>        | overloaded function |
| <b>binsuffixes</b>    | table               | <b>resultof</b>      | function            |
| <b>bits</b>           | number              | <b>runtime</b>       | function            |
| <b>clock</b>          | function            | <b>selfarg</b>       | table               |
| <b>converttime</b>    | function            | <b>selfbin</b>       | string              |
| <b>date</b>           | overloaded function | <b>selfcore</b>      | string              |
| <b>difftime</b>       | function            | <b>selfdir</b>       | string              |
| <b>enableansi</b>     | function            | <b>selflink</b>      | string              |
| <b>env</b>            | data table          | <b>selfname</b>      | string              |
| <b>execute</b>        | overloaded function | <b>selfpath</b>      | string              |
| <b>exit</b>           | overloaded function | <b>serialwrite</b>   | function            |
| <b>fulltime</b>       | function            | <b>setenv</b>        | overloaded function |
| <b>getcodepage</b>    | function            | <b>setexitcode</b>   | function            |
| <b>getenv</b>         | overloaded function | <b>setlocale</b>     | overloaded function |
| <b>getnamevalues</b>  | function            | <b>sleep</b>         | function            |
| <b>gettimeofday</b>   | function            | <b>startuptime</b>   | function            |
| <b>gettypevalues</b>  | function            | <b>time</b>          | function            |
| <b>getunamefields</b> | function            | <b>timezone</b>      | function            |
| <b>isleapyear</b>     | function            | <b>tmpname</b>       | function            |
| <b>launch</b>         | function            | <b>today</b>         | function            |
| <b>libsuffix</b>      | string              | <b>type</b>          | string              |
| <b>localtime</b>      | function            | <b>uname</b>         | function            |
| <b>name</b>           | string              | <b>uuid</b>          | function            |
| <b>newline</b>        | string              | <b>validdatetime</b> | function            |
| <b>nofdays</b>        | function            | <b>weekday</b>       | function            |
| <b>now</b>            | function            | <b>where</b>         | function            |
| <b>pipeto</b>         | function            | <b>which</b>         | function            |



## package (basic lua)

|                      |          |                    |          |
|----------------------|----------|--------------------|----------|
| <b>config</b>        | string   | <b>loadlib</b>     | function |
| <b>cpath</b>         | string   | <b>luapaths</b>    | function |
| <b>extralibpath</b>  | function | <b>path</b>        | string   |
| <b>extralibpaths</b> | function | <b>poplibpath</b>  | function |
| <b>extralupath</b>   | function | <b>popluapath</b>  | function |
| <b>extralupaths</b>  | function | <b>preload</b>     | table    |
| <b>helpers</b>       | table    | <b>pushlibpath</b> | function |
| <b>libpaths</b>      | function | <b>pushluapath</b> | function |
| <b>loaded</b>        | table    | <b>searchers</b>   | table    |
| <b>loaders</b>       | table    | <b>searchpath</b>  | function |

# pagebuilders (context)

## **columnsets**      table

F add  
F addmvl  
F block  
F blockrows  
F check  
F checkmvl  
F clean  
F define  
F definesub  
F delayedsheet  
F finishflush  
F flushareas  
F flushcolumn  
F goto  
F gotoslot  
F limit  
F noto  
F prepareflush  
F presetsheets  
F presetspreadsheets  
F put  
F putdirect  
F registerarea  
F registersheet  
F registerspreadsheets  
F reset  
F resetdirect

F reshape  
F setarea  
F setextra  
F sethsize  
F sethspan  
F setlines  
F setmvl  
F setpresent  
F setproperties  
F setstart  
F shape  
F showshape  
F state  
F status  
F submvl  
F submvlbalance

## **injections**      table

F flushafter  
F flushbefore  
F save

## **mixedcolumns**      table

F cleanup  
F finalize  
F getlist  
F getrest  
F getsplit  
F setsplit

pdf (context)

**immediateobj** function

## pdf (extra tex)

|                          |          |
|--------------------------|----------|
| <b>arraytotable</b>      | function |
| <b>close</b>             | function |
| <b>closestream</b>       | function |
| <b>dictionarytotable</b> | function |
| <b>getarray</b>          | function |
| <b>getboolean</b>        | function |
| <b>getbox</b>            | function |
| <b>getcatalag</b>        | function |
| <b>getdictionary</b>     | function |
| <b>getencodingvalues</b> | function |
| <b>getfieldtypes</b>     | function |
| <b>getfromarray</b>      | function |
| <b>getfromdictionary</b> | function |
| <b>getfromobject</b>     | function |
| <b>getfromreference</b>  | function |
| <b>getfromstream</b>     | function |
| <b>getinfo</b>           | function |
| <b>getinteger</b>        | function |
| <b>getmemoryusage</b>    | function |
| <b>getname</b>           | function |
| <b>getnofobjects</b>     | function |
| <b>getnofpages</b>       | function |

|                        |          |
|------------------------|----------|
| <b>getnumber</b>       | function |
| <b>getobjectrange</b>  | function |
| <b>getpage</b>         | function |
| <b>getpages</b>        | function |
| <b>getpermissions</b>  | function |
| <b>getsize</b>         | function |
| <b>getstatus</b>       | function |
| <b>getstatusvalues</b> | function |
| <b>getstream</b>       | function |
| <b>getstring</b>       | function |
| <b>gettrailer</b>      | function |
| <b>getversion</b>      | function |
| <b>new</b>             | function |
| <b>open</b>            | function |
| <b>openfile</b>        | function |
| <b>openstream</b>      | function |
| <b>pagestotable</b>    | function |
| <b>readfromstream</b>  | function |
| <b>readwholestream</b> | function |
| <b>type</b>            | function |
| <b>unencrypt</b>       | function |

# physics (context)

**units** table

F registershortcut

F registerunit

D tables

## pngdecode (extra tex)

|                    |          |                        |          |
|--------------------|----------|------------------------|----------|
| <b>applyfilter</b> | function | <b>setexperiment</b>   | function |
| <b>expand</b>      | function | <b>splitmask</b>       | function |
| <b>frompalette</b> | function | <b>tocmyk</b>          | function |
| <b>interlace</b>   | function | <b>togray</b>          | function |
| <b>makemask</b>    | function | <b>tomask</b>          | function |
| <b>palettemask</b> | function | <b>transparentmask</b> | function |
| <b>reduce</b>      | function |                        |          |

## posit (extra lua)

|                   |          |                  |          |
|-------------------|----------|------------------|----------|
| <b>NaN</b>        | function | <b>exp2</b>      | function |
| <b>NaR</b>        | function | <b>expand</b>    | function |
| <b>__add</b>      | function | <b>floor</b>     | function |
| <b>__band</b>     | function | <b>frombyte</b>  | function |
| <b>__bor</b>      | function | <b>fromchar</b>  | function |
| <b>__bxor</b>     | function | <b>fromposit</b> | function |
| <b>__div</b>      | function | <b>integer</b>   | function |
| <b>__eq</b>       | function | <b>log</b>       | function |
| <b>__le</b>       | function | <b>log10</b>     | function |
| <b>__lt</b>       | function | <b>log1p</b>     | function |
| <b>__mul</b>      | function | <b>log2</b>      | function |
| <b>__name</b>     | string   | <b>logb</b>      | function |
| <b>__pow</b>      | function | <b>max</b>       | function |
| <b>__shl</b>      | function | <b>min</b>       | function |
| <b>__shr</b>      | function | <b>modf</b>      | function |
| <b>__sub</b>      | function | <b>new</b>       | function |
| <b>__tostring</b> | function | <b>pow</b>       | function |
| <b>__unm</b>      | function | <b>rotate</b>    | function |
| <b>abs</b>        | function | <b>round</b>     | function |
| <b>acos</b>       | function | <b>rounded</b>   | function |
| <b>asin</b>       | function | <b>shift</b>     | function |
| <b>atan</b>       | function | <b>sin</b>       | function |
| <b>band</b>       | function | <b>sqrt</b>      | function |
| <b>bor</b>        | function | <b>tan</b>       | function |
| <b>bxor</b>       | function | <b>tobyte</b>    | function |
| <b>ceil</b>       | function | <b>tochar</b>    | function |
| <b>conj</b>       | function | <b>tonumber</b>  | function |
| <b>copy</b>       | function | <b>toposit</b>   | function |
| <b>cos</b>        | function | <b>tostring</b>  | function |
| <b>exp</b>        | function |                  |          |

## potrace (extra tex)

|                    |          |
|--------------------|----------|
| <b>checked</b>     | function |
| <b>concat</b>      | function |
| <b>contourplot</b> | function |
| <b>convert</b>     | function |
| <b>count</b>       | function |
| <b>flush</b>       | function |
| <b>free</b>        | function |
| <b>fromfile</b>    | function |
| <b>getbitmap</b>   | function |

|                         |          |
|-------------------------|----------|
| <b>getnewfields</b>     | function |
| <b>getpolicyvalues</b>  | function |
| <b>getprocessfields</b> | function |
| <b>new</b>              | function |
| <b>process</b>          | function |
| <b>setbitmap</b>        | function |
| <b>stripped</b>         | function |
| <b>totable</b>          | function |
| <b>trace</b>            | function |



# publications (context)

|                         |            |                             |            |
|-------------------------|------------|-----------------------------|------------|
| <b>addtexentry</b>      | function   | <b>concatstate</b>          | function   |
| <b>analyze</b>          | function   | <b>converttoxml</b>         | function   |
| <b>anywhere</b>         | function   | <b>currentspecification</b> | table      |
| <b>authorcache</b>      | data table | S author                    |            |
| <b>authorcollapsers</b> | data table | S comment                   |            |
| <b>authorhashers</b>    | table      | S copyright                 |            |
| F invertedshort         |            | S name                      |            |
| F normal                |            | S version                   |            |
| F normalinverted        |            | D virtual                   |            |
| F normalshort           |            | <b>datasets</b>             | table      |
| F sequential            |            | <b>detailed</b>             | table      |
| F sequentialshort       |            | <b>endnotes_to_btx</b>      | function   |
| F short                 |            | <b>enhancer</b>             | table      |
| <b>authormap</b>        | data table | S arguments                 |            |
| <b>authortoregister</b> | function   | B dirty                     |            |
| <b>btx</b>              | table      | N steps                     |            |
| <b>casters</b>          | table      | <b>enhancers</b>            | table      |
| F author                |            | F details                   |            |
| F keyword               |            | F order                     |            |
| F pagenumber            |            | F suffixes                  |            |
| F range                 |            | <b>finder</b>               | function   |
| <b>checkeddataset</b>   | function   | <b>firstofrange</b>         | function   |
| <b>citevariants</b>     | table      | <b>found</b>                | function   |
| F author                |            | <b>get</b>                  | function   |
| F authornum             |            | <b>getcasted</b>            | function   |
| F authoryear            |            | <b>getdetail</b>            | function   |
| F authoryears           |            | <b>getdirect</b>            | function   |
| F category              |            | <b>getfaster</b>            | function   |
| F default               |            | <b>getfield</b>             | function   |
| F entry                 |            | <b>getfuzzy</b>             | function   |
| F index                 |            | <b>getindex</b>             | function   |
| F keywords              |            | <b>inspectors</b>           | data table |
| F num                   |            | <b>journals</b>             | table      |
| F page                  |            | F abbreviated               |            |
| F short                 |            | F add                       |            |
| F tag                   |            | F expanded                  |            |
| F textnum               |            | F load                      |            |
| F year                  |            | F save                      |            |
| <b>collected</b>        | data table | <b>lastofrange</b>          | function   |
| <b>components</b>       | table      | <b>lists</b>                | table      |
| F author                |            | F collectentries            |            |

|                            |          |                           |          |
|----------------------------|----------|---------------------------|----------|
| F combiinlist              |          | F xml                     |          |
| F fetchentries             |          | <b>search</b>             | function |
| F filterall                |          | <b>setcategory</b>        | function |
| F flushcombi               |          | <b>setspecification</b>   | function |
| F flushentry               |          | <b>singularorplural</b>   | function |
| F flushtag                 |          | <b>sortedentries</b>      | function |
| F nofregistered            |          | <b>sortmethods</b>        | table    |
| F prepareentries           |          | <b>specifications</b>     | table    |
| F register                 |          | <b>splitauthor</b>        | function |
| F sameasprevious           |          | <b>splitoneauthor</b>     | function |
| F uservariable             |          | <b>statistics</b>         | table    |
| <b>listvariants</b>        | table    | N nofbytes                |          |
| F default                  |          | N nofdatasets             |          |
| F index                    |          | N nofdefinitions          |          |
| F num                      |          | N nofshortcuts            |          |
| F short                    |          | <b>tables</b>             | table    |
| F tag                      |          | D implicits               |          |
| <b>load</b>                | function | D origins                 |          |
| <b>loadbibdata</b>         | function | D privates                |          |
| <b>loaddefinitionfile</b>  | function | D sets                    |          |
| <b>loaders</b>             | table    | D specials                |          |
| F bbl                      |          | D types                   |          |
| F bib                      |          | D virtuals                |          |
| F buffer                   |          | <b>tags</b>               | function |
| F endnote                  |          | <b>tracers</b>            | table    |
| F json                     |          | F showdatasetauthors      |          |
| F lua                      |          | F showdatasetcompleteness |          |
| F refman                   |          | F showdatasetfields       |          |
| F registercleaner          |          | F showentry               |          |
| F xml                      |          | F showfields              |          |
| <b>loadreplacementfile</b> | function | F showhashedauthors       |          |
| <b>meanings</b>            | table    | F showtables              |          |
| <b>new</b>                 | function | F showvariants            |          |
| <b>okay</b>                | function | <b>typesetters</b>        | table    |
| <b>oneorange</b>           | function | F author                  |          |
| <b>parenttag</b>           | function | F string                  |          |
| <b>permitted</b>           | function | <b>usedataset</b>         | function |
| <b>prerollcmdstring</b>    | function | <b>usedentries</b>        | function |
| <b>refman_to_btx</b>       | function | <b>writers</b>            | table    |
| <b>resolvedname</b>        | function | F author                  |          |
| <b>save</b>                | function | F keyword                 |          |
| <b>savers</b>              | table    | F pagenumber              |          |
| F bib                      |          | F range                   |          |
| F lua                      |          |                           |          |

## qrcodegen (context)

**bitmap**    function

**generate**   function

## regimes (context)

|                    |            |
|--------------------|------------|
| <b>backmapping</b> | data table |
| <b>cleanup</b>     | function   |
| <b>disable</b>     | function   |
| <b>enable</b>      | function   |
| <b>fromregime</b>  | function   |
| <b>list</b>        | function   |

|                  |            |
|------------------|------------|
| <b>mapping</b>   | data table |
| <b>pop</b>       | function   |
| <b>process</b>   | function   |
| <b>push</b>      | function   |
| <b>toregime</b>  | function   |
| <b>translate</b> | function   |

# resolvers (context)

|                                     |            |                                |            |
|-------------------------------------|------------|--------------------------------|------------|
| <b>allprefixes</b>                  | function   | <b>findfiles</b>               | function   |
| <b>appendhash</b>                   | function   | <b>findgivenfile</b>           | function   |
| <b>automount</b>                    | function   | <b>findgivenfiles</b>          | function   |
| <b>booleanvariable</b>              | function   | <b>findlib</b>                 | function   |
| <b>cacheversion</b>                 | string     | <b>findpath</b>                | function   |
| <b>checkedvariable</b>              | function   | <b>findpk</b>                  | function   |
| <b>cleanedpathlist</b>              | function   | <b>findtexfile</b>             | function   |
| <b>cleaners</b>                     | data table | <b>findwildcardfile</b>        | function   |
| <b>cleanpath</b>                    | function   | <b>findwildcardfiles</b>       | function   |
| <b>cleanpathlist</b>                | function   | <b>formatofsuffix</b>          | function   |
| <b>cleanupbinfile</b>               | function   | <b>formatofvariable</b>        | function   |
| <b>concatinators</b>                | data table | <b>formats</b>                 | data table |
| <b>configbanner</b>                 | string     | <b>foundintree</b>             | function   |
| <b>configurationfiles</b>           | function   | <b>foundintrees</b>            | function   |
| <b>dangerous</b>                    | data table | <b>generators</b>              | data table |
| <b>datastate</b>                    | function   | <b>get_from_content</b>        | function   |
| <b>decompressors</b>                | table      | <b>getdetails</b>              | function   |
| F register                          |            | <b>getenv</b>                  | function   |
| F unregister                        |            | <b>getextrapaths</b>           | function   |
| <b>defaultsuffixes</b>              | table      | <b>getfilehashes</b>           | function   |
| <b>dirstatus</b>                    | function   | <b>gethashes</b>               | function   |
| <b>dowithfilesandreport</b>         | function   | <b>getownpath</b>              | function   |
| <b>dowithfilesintree</b>            | function   | <b>getreadfilename</b>         | function   |
| <b>dowithpath</b>                   | function   | <b>hashers</b>                 | data table |
| <b>dowithvariable</b>               | function   | <b>homedir</b>                 | string     |
| <b>executers</b>                    | table      | <b>ignorelibrary</b>           | function   |
| <b>expandbraces</b>                 | function   | <b>initialized</b>             | function   |
| <b>expandedpathfromlist</b>         | function   | <b>installinputfilehandler</b> | function   |
| <b>expandedpathlist</b>             | function   | <b>installinputlinehandler</b> | function   |
| <b>expandedpathlistfromvariable</b> | function   | <b>jobpath</b>                 | function   |
| <b>expandpath</b>                   | function   | <b>jobs</b>                    | table      |
| <b>expandpathfromvariable</b>       | function   | F collectstructure             |            |
| <b>expansion</b>                    | function   | F currentcomponent             |            |
| <b>extendtexmfvariable</b>          | function   | F currentenvironment           |            |
| <b>filtered_from_content</b>        | function   | F currentproduct               |            |
| <b>find_file</b>                    | function   | F currentproject               |            |
| <b>find_files</b>                   | function   | F currentstructure             |            |
| <b>findbinfile</b>                  | function   | F justacomponent               |            |
| <b>findctxfile</b>                  | function   | F namelist                     |            |
| <b>finders</b>                      | data table | D processors                   |            |
| <b>findfile</b>                     | function   | F productcomponent             |            |

|                  |            |                        |            |
|------------------|------------|------------------------|------------|
| F usefile        |            | registerextrapath      | function   |
| joinpath         | function   | registerfile           | function   |
| knownvariables   | function   | registerfilehash       | function   |
| libraries        | table      | registerintrees        | function   |
| F optionalloaded |            | registermethod         | function   |
| F validoptional  |            | registerzipfile        | function   |
| load             | function   | relations              | data table |
| loadbinfile      | function   | renew                  | function   |
| loaders          | data table | renewcache             | function   |
| loadlualib       | function   | repath                 | function   |
| loadtexfile      | function   | reset                  | function   |
| loadtime         | function   | resetextrapaths        | function   |
| locateformat     | function   | resetresolve           | function   |
| locators         | data table | resolve                | function   |
| luacnffallback   | string     | savers                 | data table |
| luacnfname       | string     | scandata               | function   |
| luacnfspec       | string     | scanfiles              | function   |
| luacnfstate      | string     | schemes                | table      |
| macros           | table      | F cleanname            |            |
| F convertfile    |            | F fetchstring          |            |
| F decodecomment  |            | F fetchtable           |            |
| F enablecomment  |            | F install              |            |
| F encodecomment  |            | setdynamic             | function   |
| F preprocessed   |            | setenv                 | function   |
| F processmk      |            | showdetails            | function   |
| F processmklx    |            | showpath               | function   |
| F processmkvi    |            | simplescanfiles        | function   |
| F version        |            | splitconfigurationpath | function   |
| maxreadlevel     | number     | splitexpansions        | function   |
| methodhandler    | function   | splitmethod            | function   |
| missinglibrary   | function   | splitpath              | function   |
| openbinfile      | function   | stackpath              | function   |
| openers          | data table | starttiming            | function   |
| openfile         | function   | stoptiming             | function   |
| opentexfile      | function   | suffices               | data table |
| ownpath          | string     | suffixmap              | data table |
| popextrapath     | function   | suffixofformat         | function   |
| popinputname     | function   | texdatablob            | function   |
| poppath          | function   | toppath                | function   |
| prefixes         | data table | tracers                | data table |
| prependhash      | function   | unexpandedpath         | function   |
| pushextrapath    | function   | unexpandedpathlist     | function   |
| pushinputname    | function   | unresolve              | function   |
| pushpath         | function   | updaterelations        | function   |

|                      |            |                                 |          |
|----------------------|------------|---------------------------------|----------|
| <b>updatescript</b>  | function   | <b>variable</b>                 | function |
| <b>uselibrary</b>    | function   | <b>variableofformat</b>         | function |
| <b>usertypes</b>     | data table | <b>variableofformatorsuffix</b> | function |
| <b>usezipfile</b>    | function   | <b>wildcardpattern</b>          | function |
| <b>validatedpath</b> | function   |                                 |          |

## sandbox (context)

|                         |          |
|-------------------------|----------|
| <b>blockrequire</b>     | function |
| <b>disablelibraries</b> | function |
| <b>disablerunners</b>   | function |
| <b>enable</b>           | function |
| <b>filehandlerone</b>   | function |
| <b>filehandlertwo</b>   | function |
| <b>finalizer</b>        | function |
| <b>getrunner</b>        | function |
| <b>initializer</b>      | function |
| <b>iohandler</b>        | function |
| <b>logcalls</b>         | function |
| <b>logfiles</b>         | function |
| <b>original</b>         | function |
| <b>overload</b>         | function |

|                          |          |
|--------------------------|----------|
| <b>possiblepath</b>      | function |
| <b>redefine</b>          | function |
| <b>register</b>          | function |
| <b>registerbinary</b>    | function |
| <b>registerlibrary</b>   | function |
| <b>registerroot</b>      | function |
| <b>registerrunner</b>    | function |
| <b>report</b>            | function |
| <b>setfilenamelogger</b> | function |
| <b>setlogger</b>         | function |
| <b>setreporter</b>       | function |
| <b>settrace</b>          | function |
| <b>validfilename</b>     | function |



# scripts (context)

**autofontfeature** table

F disable  
F enable  
F handler

**colors** data table

**decomposehangul** function

**defaults** table

N inter\_char\_half\_shrink\_factor  
N inter\_char\_half\_stretch\_factor  
N inter\_char\_hangul\_penalty  
N inter\_char\_quarter\_shrink\_factor  
N inter\_char\_quarter\_stretch\_factor  
N inter\_char\_shrink\_factor  
N inter\_char\_stretch\_factor  
N inter\_word\_stretch\_factor

**getdata** function

**getinjector** function

**getsplitter** function

**getstatus** function

**handlers** data table

**helpers** data table

**injectors** data table

**inserters** table

F nobreakspace\_after

F nobreakspace\_before

F space\_after

F space\_before

F zerowidthspace\_after

F zerowidthspace\_before

**installdataset** function

**installmethod** function

**numbertodataset** data table

**numbertohandler** data table

**reset** function

**set** function

**setstatus** function

**splitters** data table

## serial (context)

|                   |          |
|-------------------|----------|
| <b>__gc</b>       | function |
| <b>__name</b>     | string   |
| <b>__tostring</b> | function |
| <b>close</b>      | function |

|                 |          |
|-----------------|----------|
| <b>open</b>     | function |
| <b>send</b>     | function |
| <b>tostring</b> | function |
| <b>write</b>    | function |

## set (context)

**contains**    function

**create**     function

**tolist**     function

**tonumber**   function

**totable**    function

## sha2 (extra lua)

|                  |          |
|------------------|----------|
| <b>HASH256</b>   | function |
| <b>HASH384</b>   | function |
| <b>HASH512</b>   | function |
| <b>HEX256</b>    | function |
| <b>HEX384</b>    | function |
| <b>HEX512</b>    | function |
| <b>digest256</b> | function |
| <b>digest384</b> | function |
| <b>digest512</b> | function |

|                |          |
|----------------|----------|
| <b>hash256</b> | function |
| <b>hash384</b> | function |
| <b>hash512</b> | function |
| <b>hex256</b>  | function |
| <b>hex384</b>  | function |
| <b>hex512</b>  | function |
| <b>sum256</b>  | function |
| <b>sum384</b>  | function |
| <b>sum512</b>  | function |

## sio (extra lua)

**read2dot14**            function  
**readbytes**            function  
**readbytetable**        function  
**readcardinal1**        function  
**readcardinal1le**      function  
**readcardinal2**        function  
**readcardinal2le**      function  
**readcardinal3**        function  
**readcardinal3le**      function  
**readcardinal4**        function  
**readcardinal4le**      function  
**readcardinaltable**    function  
**readcline**            function  
**readcstring**          function  
**readdouble**            function  
**readdoublele**         function  
**readfixed2**            function  
**readfixed4**            function  
**readfloat**            function  
**readfloatle**          function

**readfloatle12**        function  
**readfloatle6**        function  
**readinteger1**        function  
**readinteger1le**      function  
**readinteger2**        function  
**readinteger2le**      function  
**readinteger3**        function  
**readinteger3le**      function  
**readinteger4**        function  
**readinteger4le**      function  
**readintegertable**    function  
**tocardinal1**          function  
**tocardinal1le**        function  
**tocardinal2**          function  
**tocardinal2le**        function  
**tocardinal3**          function  
**tocardinal3le**        function  
**tocardinal4**          function  
**tocardinal4le**        function

# socket (extra lua)

|                       |          |                   |          |
|-----------------------|----------|-------------------|----------|
| <b>BLOCKSIZE</b>      | number   | <b>sinkt</b>      | table    |
| <b>_SETSIZE</b>       | number   | F close-when-done |          |
| <b>_SOCKETINVALID</b> | number   | F default         |          |
| <b>bind</b>           | function | F http-chunked    |          |
| <b>choose</b>         | function | F keep-open       |          |
| <b>connect</b>        | function | <b>skip</b>       | function |
| <b>connect4</b>       | function | <b>sleep</b>      | function |
| <b>connect6</b>       | function | <b>source</b>     | function |
| <b>dns</b>            | table    | <b>sourcet</b>    | table    |
| F getaddrinfo         |          | F by-length       |          |
| F gethostname         |          | F default         |          |
| F getnameinfo         |          | F http-chunked    |          |
| F tohostname          |          | F until-closed    |          |
| F toip                |          | <b>tcp</b>        | function |
| <b>gettime</b>        | function | <b>tcp4</b>       | function |
| <b>headers</b>        | table    | <b>tcp6</b>       | function |
| F lower               |          | <b>tp</b>         | table    |
| F normalize           |          | N TIMEOUT         |          |
| <b>http</b>           | table    | F connect         |          |
| N TIMEOUT             |          | <b>try</b>        | function |
| S USERAGENT           |          | <b>url</b>        | table    |
| F genericform         |          | F absolute        |          |
| F open                |          | F build           |          |
| F request             |          | F build_path      |          |
| <b>newtry</b>         | function | F escape          |          |
| <b>protect</b>        | function | F parse           |          |
| <b>report</b>         | function | F parse_path      |          |
| <b>select</b>         | function | F unescape        |          |
| <b>sink</b>           | function |                   |          |

## sorters (context)

**basicsorter** function

**comparers** table

F basic

**constants** table

S defaultdigits

S defaultlanguage

S defaultmethod

N digitmaximum

N digitoffset

N ignoredoffset

N replacementoffset

**definitions** data table

**firstofsplit** function

**packch** function

**packuc** function

**replacementlist** function

**setlanguage** function

**sort** function

**splitters** table

F utf

**strip** function

**tracers** data table

**update** function

## sparse (extra lua)

|               |          |
|---------------|----------|
| <b>concat</b> | function |
| <b>get</b>    | function |
| <b>max</b>    | function |
| <b>min</b>    | function |
| <b>new</b>    | function |

|                 |          |
|-----------------|----------|
| <b>range</b>    | function |
| <b>restore</b>  | function |
| <b>set</b>      | function |
| <b>traverse</b> | function |
| <b>wipe</b>     | function |



## statistics (context)

|                       |          |                      |          |
|-----------------------|----------|----------------------|----------|
| <b>benchmarktimer</b> | function | <b>hastiming</b>     | function |
| <b>checkfmtstatus</b> | function | <b>memused</b>       | function |
| <b>currenttime</b>    | function | <b>register</b>      | function |
| <b>elapsed</b>        | function | <b>resettiming</b>   | function |
| <b>elapsedindeed</b>  | function | <b>runtime</b>       | function |
| <b>elapsedseconds</b> | function | <b>savefmtstatus</b> | function |
| <b>elapseddtime</b>   | function | <b>show</b>          | function |
| <b>enable</b>         | boolean  | <b>showmemory</b>    | function |
| <b>feedback</b>       | table    | <b>showusage</b>     | function |
| F info                |          | <b>starttiming</b>   | function |
| F processstates       |          | <b>stoptiming</b>    | function |
| F register            |          | <b>threshold</b>     | number   |
| F reportstates        |          | <b>timed</b>         | function |
| F setstate            |          | <b>tracefunction</b> | function |
| <b>formatruntime</b>  | function |                      |          |

## status (basic tex)

|                              |          |                          |          |
|------------------------------|----------|--------------------------|----------|
| <b>getbalancestate</b>       | function | <b>getluastate</b>       | function |
| <b>getbufferstate</b>        | function | <b>getmarkstate</b>      | function |
| <b>getcallbackstate</b>      | function | <b>getmvlstate</b>       | function |
| <b>getconstants</b>          | function | <b>getneststate</b>      | function |
| <b>getdumpstate</b>          | function | <b>getnodestate</b>      | function |
| <b>geterrorlinestate</b>     | function | <b>getparameterstate</b> | function |
| <b>geterrorstate</b>         | function | <b>getpoolstate</b>      | function |
| <b>getexpandstate</b>        | function | <b>getreadstate</b>      | function |
| <b>getfilestate</b>          | function | <b>getsavestate</b>      | function |
| <b>getfontstate</b>          | function | <b>getsparsestate</b>    | function |
| <b>gethalferrorlinestate</b> | function | <b>getstringstate</b>    | function |
| <b>gethashstate</b>          | function | <b>gettexstate</b>       | function |
| <b>gethyphenationstate</b>   | function | <b>gettokenstate</b>     | function |
| <b>getinputstate</b>         | function | <b>getwarningstate</b>   | function |
| <b>getinsertstate</b>        | function | <b>iocodes</b>           | table    |
| <b>getlanguagestate</b>      | function | <b>list</b>              | function |
| <b>getlinebreakstate</b>     | function | <b>resetmessages</b>     | function |
| <b>getlookupstate</b>        | function |                          |          |

## storage (context)

|                   |          |                        |            |
|-------------------|----------|------------------------|------------|
| <b>allocate</b>   | function | <b>noftables</b>       | number     |
| <b>data</b>       | table    | <b>register</b>        | function   |
| <b>mark</b>       | function | <b>shared</b>          | data table |
| <b>marked</b>     | function | <b>storedfunctions</b> | data table |
| <b>max</b>        | number   | <b>strip</b>           | boolean    |
| <b>min</b>        | number   | <b>tofmodules</b>      | number     |
| <b>nofmodules</b> | number   | <b>toftables</b>       | number     |

## streams (context)

|                   |          |
|-------------------|----------|
| <b>collect</b>    | function |
| <b>disable</b>    | function |
| <b>enable</b>     | function |
| <b>flush</b>      | function |
| <b>initialize</b> | function |

|                    |          |
|--------------------|----------|
| <b>push</b>        | function |
| <b>start</b>       | function |
| <b>stop</b>        | function |
| <b>synchronize</b> | function |

## string (basic lua)

|                        |          |                        |          |
|------------------------|----------|------------------------|----------|
| <b>autodouble</b>      | function | <b>optionalquoted</b>  | function |
| <b>autosingle</b>      | function | <b>pack</b>            | function |
| <b>booleanstring</b>   | function | <b>packrowscolumns</b> | function |
| <b>byte</b>            | function | <b>packsize</b>        | function |
| <b>bytepairs</b>       | function | <b>quote</b>           | function |
| <b>bytes</b>           | function | <b>quoted</b>          | function |
| <b>bytetable</b>       | function | <b>rep</b>             | function |
| <b>char</b>            | function | <b>replacenewlines</b> | function |
| <b>characterpairs</b>  | function | <b>reverse</b>         | function |
| <b>characters</b>      | function | <b>split</b>           | function |
| <b>checkedsplit</b>    | function | <b>splitintolines</b>  | function |
| <b>chrtointeger</b>    | function | <b>splitlines</b>      | function |
| <b>collapsespaces</b>  | function | <b>splitup</b>         | function |
| <b>containsws</b>      | function | <b>strip</b>           | function |
| <b>count</b>           | function | <b>sub</b>             | function |
| <b>dectointeger</b>    | function | <b>texhashed</b>       | function |
| <b>dump</b>            | function | <b>texnewlines</b>     | function |
| <b>escapedpattern</b>  | function | <b>tformat</b>         | function |
| <b>escapedquotes</b>   | function | <b>toHEX</b>           | function |
| <b>f6</b>              | function | <b>toboolean</b>       | function |
| <b>find</b>            | function | <b>tobytes</b>         | function |
| <b>format</b>          | function | <b>tocontext</b>       | function |
| <b>formatter</b>       | function | <b>todec</b>           | function |
| <b>formatters</b>      | function | <b>todimen</b>         | function |
| <b>fullstrip</b>       | function | <b>tohex</b>           | function |
| <b>gmatch</b>          | function | <b>tonodes</b>         | function |
| <b>gsub</b>            | function | <b>topattern</b>       | function |
| <b>hashes</b>          | table    | <b>totable</b>         | function |
| <b>hextocharacters</b> | function | <b>tounicode16</b>     | function |
| <b>hextointeger</b>    | function | <b>toutf</b>           | function |
| <b>is_boolean</b>      | function | <b>toutf16</b>         | function |
| <b>is_empty</b>        | function | <b>toutf32</b>         | function |
| <b>itself</b>          | function | <b>toutf8</b>          | function |
| <b>len</b>             | function | <b>tracedchar</b>      | function |
| <b>limit</b>           | function | <b>unescapedquotes</b> | function |
| <b>linetable</b>       | function | <b>unpack</b>          | function |
| <b>longtostring</b>    | function | <b>unquote</b>         | function |
| <b>lower</b>           | function | <b>unquoted</b>        | function |
| <b>match</b>           | function | <b>upper</b>           | function |
| <b>nospaces</b>        | function | <b>utf16toutf8</b>     | function |
| <b>octtointeger</b>    | function | <b>utfcharacter</b>    | function |

|                           |          |
|---------------------------|----------|
| <b>utfcharacters</b>      | function |
| <b>utfcharacteretable</b> | function |
| <b>utflength</b>          | function |
| <b>utfpadd</b>            | function |
| <b>utfpadding</b>         | function |
| <b>utftabletostring</b>   | function |

|                      |          |
|----------------------|----------|
| <b>utfvalue</b>      | function |
| <b>utfvalues</b>     | function |
| <b>utfvaluetable</b> | function |
| <b>valid</b>         | function |
| <b>wordsplitter</b>  | function |

# structures (context)

|                   |          |                             |          |
|-------------------|----------|-----------------------------|----------|
| <b>atlocation</b> | function | F record                    |          |
| <b>blocks</b>     | table    | F reset                     |          |
| D collected       |          | F restart                   |          |
| F define          |          | F restore                   |          |
| F empty           |          | F save                      |          |
| F print           |          | F set                       |          |
| F save            |          | F setlevel                  |          |
| F select          |          | F setoffset                 |          |
| F setstate        |          | F setown                    |          |
| D states          |          | F setstate                  |          |
| D tobesaved       |          | F setvalue                  |          |
| F uservariable    |          | F subs                      |          |
| <b>bookmarks</b>  | table    | D tobesaved                 |          |
| F finalize        |          | F value                     |          |
| F flatten         |          | F way                       |          |
| F installhandler  |          | <b>currentsectionnumber</b> | function |
| F merge           |          | <b>descriptions</b>         | table    |
| S method          |          | <b>documents</b>            | table    |
| F overload        |          | D data                      |          |
| F place           |          | F initialize                |          |
| F register        |          | F preset                    |          |
| F setopened       |          | F reset                     |          |
| F setup           |          | <b>floats</b>               | table    |
| <b>counters</b>   | table    | <b>formulas</b>             | table    |
| F add             |          | <b>helpers</b>              | table    |
| F analyze         |          | F analyze                   |          |
| F check           |          | F formulanumber             |          |
| D collected       |          | F merged                    |          |
| F compact         |          | F page                      |          |
| F converted       |          | F pageofinternal            |          |
| F currentvalue    |          | F prefix                    |          |
| D data            |          | F prefixlastpage            |          |
| F define          |          | F prefixpage                |          |
| F first           |          | F simplify                  |          |
| F get             |          | F title                     |          |
| F isset           |          | F touserdata                |          |
| F last            |          | <b>inserts</b>              | table    |
| F next            |          | F attach                    |          |
| F prev            |          | D data                      |          |
| F previous        |          | F define                    |          |
| F raw             |          | F detach                    |          |

F getdata  
F getlocation  
F getname  
F getnumber  
F nofdetached  
F setlocation  
F setup  
D stored

**itemgroups** table

D collected  
D tobesaved

**lists** table

F addto  
F analyze  
F bookmark  
D cached  
D collected  
F discard  
F enhance  
F external  
F filter  
F formulainumber  
F getresult  
F groupindex  
F hasnumberdata  
F haspagedata  
F hastitledata  
F integrate  
F iscached  
D kinds  
F label  
F location  
D names  
F number  
D ordered  
F page  
F popnesting  
F prefix  
F prefixednumber  
F prefixedpage  
F process  
D pushed  
F pushnesting  
F rawnumber

F rawrealpage  
F rawuserpage  
F realpage  
F reorder  
F reordered  
F savedlistnumber  
F savedlistprefixednumber  
F savedlisttitle  
D sectionblocks  
F size  
F title  
D tobesaved  
F use  
F userdata  
F userpage  
F uservalue

**marks** table

F number  
F title

**notes** table

F checkpagechange  
F define  
F flushrange  
F get  
F getdeltapage  
F getinternal  
F getn  
F getstate  
F getstorecount  
F getstoreflushed  
F hascontent  
F internal  
F internalid  
F listindex  
D numbers  
F ordered  
F postpone  
F resetpostponed  
F restore  
F save  
F setstate  
D states  
F store

**pages** table



F allmarked  
F analyze  
D collected  
F fraction  
F getlabels  
F in\_body  
F is\_odd  
F mark  
F marked  
F markedlist  
N nopages  
F number  
F on\_right  
F save  
D tobesaved  
F toranges

## references

F analyze  
F checked  
F checkedfile  
F checkedfileorurl  
F checkedprogram  
F checkedrealpage  
F checkedurl  
D collected  
U componentsplitter  
F currentmetadata  
F currentorder  
D currentset  
F define  
D defined  
D derived  
F enhance  
D executers  
F export  
F filter  
F filterdefault  
D filters  
D flaginternals  
F forceinnermode  
F from  
D functions  
F get  
F getcurrentlabel

## table

F getcurrentmetadata  
F getcurrentprefixspec  
F getinnermethod  
F getinternalcachedlistreference  
F getinternallistreference  
F getinternalreference  
F getoutermethod  
D handlers  
F identify  
F import  
F inject  
F injectcurrentset  
S innermethod  
F integrate  
D internals  
F load  
F loadpresets  
F mark  
S outermethod  
F pagefromhash  
D pages  
U prefixsplitter  
D productdata  
F realpageofpage  
F referenced  
U referencesplitter  
D referred  
F referredpage  
F registerfinalizer  
F registerinitializer  
F registerpage  
F reset  
D runners  
F set  
F setandgetattribute  
F setinternalreference  
F setlinkmethod  
F setnextinternal  
F setnextorder  
F setreferencerealpage  
F setstructure  
D specials  
F splitcomponent  
F splitprefix

|                       |              |                     |              |
|-----------------------|--------------|---------------------|--------------|
| F splitreference      |              | F fullnumber        |              |
| F synchronizepage     |              | F getcurrentlevel   |              |
| D tobereferred        |              | F getlevel          |              |
| D tobesaved           |              | F getnumber         |              |
| D tracedpages         |              | F internalreference |              |
| D usedinternals       |              | D levelmap          |              |
| D usedviews           |              | D levels            |              |
| F valid               |              | F matchingtilldepth |              |
| <b>registers</b>      | <b>table</b> | F numberatdepth     |              |
| F analyze             |              | F numbers           |              |
| D collected           |              | F popblock          |              |
| F compare             |              | F prefixedconverted |              |
| F define              |              | F pushblock         |              |
| F enhance             |              | F register          |              |
| F extend              |              | D registered        |              |
| F filter              |              | F reportstructure   |              |
| F filtercollected     |              | F save              |              |
| F finalize            |              | F set               |              |
| F findinternal        |              | F setblock          |              |
| F flush               |              | F setchecker        |              |
| F get                 |              | F setentry          |              |
| F integrate           |              | F setinitialblock   |              |
| F prepare             |              | F setlevel          |              |
| F process             |              | F setnumber         |              |
| F setmethod           |              | F showstructure     |              |
| F sort                |              | F structuredata     |              |
| F store               |              | F title             |              |
| D tobesaved           |              | D tobesaved         |              |
| F tweak               |              | F typesetnumber     |              |
| F unique              |              | F userdata          |              |
| F use                 |              | B verbose           |              |
| F userdata            |              | <b>sets</b>         | <b>table</b> |
| <b>sections</b>       | <b>table</b> | F define            |              |
| F autodepth           |              | F get               |              |
| F cct                 |              | F getall            |              |
| D collected           |              | D setlist           |              |
| F current             |              | <b>specials</b>     | <b>table</b> |
| F currentblock        |              | D collected         |              |
| F currentid           |              | F retrieve          |              |
| F currentlevel        |              | F store             |              |
| F currentsectionindex |              | D tobesaved         |              |
| F depthnumber         |              | <b>synonyms</b>     | <b>table</b> |
| F finddata            |              | F analyzed          |              |
| F findnumber          |              | D collected         |              |

F compare  
F define  
F enhance  
F filter  
F finalize  
F firstuserpage  
F flush  
F isshown  
F isused  
F meaning  
F pages  
F prepare  
F process  
F register  
F registershown  
F registerused  
F reset  
F resetall  
F resetlist  
F resetshown  
F resetused  
F sort  
F synonym  
D tobesaved  
F userpages

## tags

F analyze  
F checknesting  
F current  
F elementtag  
F enabled  
F exportbox  
F finishexport  
F getdescriptionindex  
F getextradata  
F getid  
F getitemgroup  
F getmetadata  
F getoption  
F getsorting  
F getsynonym  
F gettablecell  
F gettabulatecell  
F gettagreferencedescription

## table

F handler  
F initializeexport  
D labels  
F last  
F lastinchain  
F localexport  
F locatedtag  
D overloads  
F pop  
D properties  
F push  
F registerextradata  
F registermetadata  
F restart  
F setaspect  
F setattributefield  
F setcodepoint  
F setcombination  
F setcombinationpair  
F setconstruct  
F setdescriptionindex  
F setfigure  
F setfloat  
F setformulacontent  
F sethighlight  
F setitem  
F setitemgroup  
F setmpgraphic  
F setnotation  
F setnotationsymbol  
F setparagraph  
F setproperty  
F setpublication  
F setregister  
F setsorting  
F setsynonym  
F settablecell  
F settabulatecell  
F settagdelimitedsymbol  
F settagsubsentencesymbol  
F setupexport  
F setuserproperties  
D specifications  
F start

F stop

D taglist

F strip

## table (basic lua)

|                        |          |                             |          |
|------------------------|----------|-----------------------------|----------|
| <b>append</b>          | function | <b>pack</b>                 | function |
| <b>are_equal</b>       | function | <b>prepend</b>              | function |
| <b>combine</b>         | function | <b>print</b>                | function |
| <b>compact</b>         | function | <b>remove</b>               | function |
| <b>concat</b>          | function | <b>reverse</b>              | function |
| <b>contains</b>        | function | <b>reversed</b>             | function |
| <b>copy</b>            | function | <b>save</b>                 | function |
| <b>count</b>           | function | <b>sequenced</b>            | function |
| <b>create</b>          | function | <b>serialize</b>            | function |
| <b>derive</b>          | function | <b>setmetatablecall</b>     | function |
| <b>deserialize</b>     | function | <b>setmetatableindex</b>    | function |
| <b>fastcopy</b>        | function | <b>setmetatableindices</b>  | function |
| <b>fastserialize</b>   | function | <b>setmetatablenewindex</b> | function |
| <b>filtered</b>        | function | <b>sort</b>                 | function |
| <b>flattened</b>       | function | <b>sorted</b>               | function |
| <b>fromhash</b>        | function | <b>sortedhash</b>           | function |
| <b>getn</b>            | function | <b>sortedhashkeys</b>       | function |
| <b>has_one_entry</b>   | function | <b>sortedhashonly</b>       | function |
| <b>hashed</b>          | function | <b>sortedindexonly</b>      | function |
| <b>identical</b>       | function | <b>sortedkeys</b>           | function |
| <b>imerge</b>          | function | <b>sortedpairs</b>          | function |
| <b>imerged</b>         | function | <b>sparse</b>               | function |
| <b>insert</b>          | function | <b>strip</b>                | function |
| <b>is_empty</b>        | function | <b>sub</b>                  | function |
| <b>is_simple_table</b> | function | <b>swapped</b>              | function |
| <b>keys</b>            | function | <b>tocontext</b>            | function |
| <b>load</b>            | function | <b>tocsv</b>                | function |
| <b>loweredkeys</b>     | function | <b>tofile</b>               | function |
| <b>makewweak</b>       | function | <b>tohandle</b>             | function |
| <b>merge</b>           | function | <b>tohash</b>               | function |
| <b>merged</b>          | function | <b>toxml</b>                | function |
| <b>mirrored</b>        | function | <b>unique</b>               | function |
| <b>move</b>            | function | <b>unpack</b>               | function |

# tex (basic tex)

|                              |            |                                      |            |
|------------------------------|------------|--------------------------------------|------------|
| <b>amcode</b>                | table      | <b>directives</b>                    | data table |
| <b>attribute</b>             | table      | <b>discoptioncodes</b>               | data table |
| <b>attributetracking</b>     | function   | <b>discpartcodes</b>                 | data table |
| <b>automigrationcodes</b>    | data table | <b>dontcomplain</b>                  | function   |
| <b>autoparagraphcodes</b>    | data table | <b>doublescriptoptioncodes</b>       | data table |
| <b>badness</b>               | function   | <b>enableprimitives</b>              | function   |
| <b>badnessmodecodes</b>      | data table | <b>equal</b>                         | function   |
| <b>balance</b>               | function   | <b>error</b>                         | function   |
| <b>balancecallbackcodes</b>  | data table | <b>expandasvalue</b>                 | function   |
| <b>boundaries</b>            | table      | <b>experiments</b>                   | data table |
| D names                      |            | <b>extraprimitives</b>               | function   |
| D numbers                    |            | <b>fatalerror</b>                    | function   |
| F system                     |            | <b>fillcodes</b>                     | data table |
| <b>box</b>                   | table      | <b>flagcodes</b>                     | data table |
| <b>boxcontinuationcodes</b>  | data table | <b>fontidentifier</b>                | function   |
| <b>boxoptioncodes</b>        | data table | <b>fontname</b>                      | function   |
| <b>boxresources</b>          | table      | <b>forcehmode</b>                    | function   |
| F getbox                     |            | <b>forceinputstatefile</b>           | function   |
| F getdimensions              |            | <b>forceinputstateline</b>           | function   |
| F save                       |            | <b>fprint</b>                        | function   |
| F use                        |            | <b>frozenparcodes</b>                | data table |
| <b>breakcodes</b>            | data table | <b>get</b>                           | function   |
| <b>buildcodes</b>            | data table | <b>getadjustoptionvalues</b>         | function   |
| <b>catcode</b>               | table      | <b>getalignmentcontextvalues</b>     | function   |
| <b>cccode</b>                | table      | <b>getamcode</b>                     | function   |
| <b>charactercontrolcodes</b> | data table | <b>getappendadjustcontextvalues</b>  | function   |
| <b>charactertagcodes</b>     | data table | <b>getappendmigratecontextvalues</b> | function   |
| <b>chardef</b>               | function   | <b>getattribute</b>                  | function   |
| <b>classoptioncodes</b>      | data table | <b>getautomigrationvalues</b>        | function   |
| <b>codes</b>                 | table      | <b>getautoparagraphvalues</b>        | function   |
| <b>conditionals</b>          | data table | <b>getbadnessmodevalues</b>          | function   |
| <b>constants</b>             | data table | <b>getbalancecallbackvalues</b>      | function   |
| <b>count</b>                 | table      | <b>getbalanceparameterfields</b>     | function   |
| <b>cprint</b>                | function   | <b>getbalanceresultfields</b>        | function   |
| <b>ctdcatcodes</b>           | number     | <b>getbalancestepoptionvalues</b>    | function   |
| <b>ctxcatcodes</b>           | number     | <b>getbox</b>                        | function   |
| <b>definefont</b>            | function   | <b>getboxcontinuationvalues</b>      | function   |
| <b>delcode</b>               | table      | <b>getboxdir</b>                     | function   |
| <b>dimen</b>                 | table      | <b>getboxoptionvalues</b>            | function   |
| <b>dimensiondef</b>          | function   | <b>getboxresourcebox</b>             | function   |
| <b>directioncodes</b>        | data table | <b>getboxresourcedimensions</b>      | function   |

|                                    |          |                                    |          |
|------------------------------------|----------|------------------------------------|----------|
| <b>getbreakcontextvalues</b>       | function | <b>getinsertcategoryvalues</b>     | function |
| <b>getbuildcontextvalues</b>       | function | <b>getinsertcontent</b>            | function |
| <b>getcardinalvalue</b>            | function | <b>getinsertdepth</b>              | function |
| <b>getcatcode</b>                  | function | <b>getinsertdistance</b>           | function |
| <b>getcccode</b>                   | function | <b>getinsertheight</b>             | function |
| <b>getcharactercontrolvalues</b>   | function | <b>getinsertlimit</b>              | function |
| <b>getcharactertagvalues</b>       | function | <b>getinsertmaxplaced</b>          | function |
| <b>getcount</b>                    | function | <b>getinsertmultiplier</b>         | function |
| <b>getcurrentmvl</b>               | function | <b>getinsertoptionvalues</b>       | function |
| <b>getdelcode</b>                  | function | <b>getinsertsplitvalues</b>        | function |
| <b>getdelcodes</b>                 | function | <b>getinsertwidth</b>              | function |
| <b>getdimen</b>                    | function | <b>getintegervalue</b>             | function |
| <b>getdimensionvalue</b>           | function | <b>getinteraction</b>              | function |
| <b>getdirectionvalues</b>          | function | <b>getinteractionmodes</b>         | function |
| <b>getdiscoptionvalues</b>         | function | <b>getiovalues</b>                 | function |
| <b>getdiscpartvalues</b>           | function | <b>getkerneloptionvalues</b>       | function |
| <b>getdoublescriptoptionvalues</b> | function | <b>getkernoptionvalues</b>         | function |
| <b>getemptyparagraphmodevalues</b> | function | <b>getlargestusedmark</b>          | function |
| <b>geterrorrecoverymodevalues</b>  | function | <b>getlccode</b>                   | function |
| <b>geterrorvalues</b>              | function | <b>getlinebreakparameterfields</b> | function |
| <b>getfillvalues</b>               | function | <b>getlinebreakresultfields</b>    | function |
| <b>getflagvalues</b>               | function | <b>getlinebreakstatevalues</b>     | function |
| <b>getfloat</b>                    | function | <b>getlinedir</b>                  | function |
| <b>getfontdimension</b>            | function | <b>getlist</b>                     | function |
| <b>getfontofffamily</b>            | function | <b>getlistanchorvalues</b>         | function |
| <b>getfrozenparvalues</b>          | function | <b>getlistfields</b>               | function |
| <b>getglue</b>                     | function | <b>getlistgeometryvalues</b>       | function |
| <b>getglueoptionvalues</b>         | function | <b>getlistsignvalues</b>           | function |
| <b>getglyphdata</b>                | function | <b>getlocalbox</b>                 | function |
| <b>getglyphdiscvalues</b>          | function | <b>getlocalboxlocations</b>        | function |
| <b>getglyphoptionvalues</b>        | function | <b>getlocallevel</b>               | function |
| <b>getglyphprotectionvalues</b>    | function | <b>getmark</b>                     | function |
| <b>getglyphscales</b>              | function | <b>getmarknames</b>                | function |
| <b>getglyphscript</b>              | function | <b>getmath</b>                     | function |
| <b>getglyphstate</b>               | function | <b>getmathclassoptionvalues</b>    | function |
| <b>getgroupvalues</b>              | function | <b>getmathcode</b>                 | function |
| <b>gethccode</b>                   | function | <b>getmathcodes</b>                | function |
| <b>gethelptext</b>                 | function | <b>getmathcontrolvalues</b>        | function |
| <b>gethmcodes</b>                  | function | <b>getmathdir</b>                  | function |
| <b>gethyphenationvalues</b>        | function | <b>getmathgluevalues</b>           | function |
| <b>getitypes</b>                   | function | <b>getmathoptionvalues</b>         | function |
| <b>getinputstatefile</b>           | function | <b>getmathparametervalue</b>       | function |
| <b>getinputstateline</b>           | function | <b>getmathscriptordervalue</b>     | function |
| <b>getinputstatemode</b>           | function | <b>getmathscriptsmodevalue</b>     | function |

|                                    |          |                                     |            |
|------------------------------------|----------|-------------------------------------|------------|
| <b>getmathstylenamevalues</b>      | function | <b>getrunstate</b>                  | function   |
| <b>getmathstylevalues</b>          | function | <b>getrunstatevalues</b>            | function   |
| <b>getmathstylevariant</b>         | function | <b>getsfcode</b>                    | function   |
| <b>getmathsurroundvalues</b>       | function | <b>getshapingpenaltiesvalues</b>    | function   |
| <b>getmathvariantpresets</b>       | function | <b>getskip</b>                      | function   |
| <b>getmathvariantvalues</b>        | function | <b>getsnappingvalues</b>            | function   |
| <b>getmeasure</b>                  | function | <b>getspacefactorvalues</b>         | function   |
| <b>getmissingcharactervalues</b>   | function | <b>getspaceskipvalues</b>           | function   |
| <b>getmode</b>                     | function | <b>getspcode</b>                    | function   |
| <b>getmodevalues</b>               | function | <b>getspecialmathclassvalues</b>    | function   |
| <b>getmuglue</b>                   | function | <b>getspecification</b>             | function   |
| <b>getmuskip</b>                   | function | <b>getspecificationoptionvalues</b> | function   |
| <b>getmvloptionvalues</b>          | function | <b>gettextcontrolvalues</b>         | function   |
| <b>getnest</b>                     | function | <b>gettextdir</b>                   | function   |
| <b>getnestfields</b>               | function | <b>gettoks</b>                      | function   |
| <b>getnestlevel</b>                | function | <b>getuccode</b>                    | function   |
| <b>getnoadoptoptionvalues</b>      | function | <b>getuleaderlocationvalues</b>     | function   |
| <b>getnormalizelinevalues</b>      | function | <b>getunitclassvalues</b>           | function   |
| <b>getnormalizeparvalues</b>       | function | <b>getusedmarks</b>                 | function   |
| <b>getnospacesmodevalues</b>       | function | <b>getvsplitcontextvalues</b>       | function   |
| <b>getoutputactive</b>             | function | <b>glue</b>                         | table      |
| <b>getoverloadstate</b>            | function | <b>glueoptioncodes</b>              | data table |
| <b>getpacktypevalues</b>           | function | <b>glyphdisccodes</b>               | data table |
| <b>getpagecontextvalues</b>        | function | <b>glyphoptioncodes</b>             | data table |
| <b>getpagestate</b>                | function | <b>groupcodes</b>                   | data table |
| <b>getpagestatevalues</b>          | function | <b>hashtokens</b>                   | function   |
| <b>getparagraphfillmodevalues</b>  | function | <b>hccode</b>                       | table      |
| <b>getparagraphoptionvalues</b>    | function | <b>hmcode</b>                       | table      |
| <b>getparametermodevalues</b>      | function | <b>hshcatcodes</b>                  | number     |
| <b>getparcontextvalues</b>         | function | <b>hyphenationcodes</b>             | data table |
| <b>getpardataspecifications</b>    | function | <b>ifs</b>                          | data table |
| <b>getpardir</b>                   | function | <b>inimatcodes</b>                  | number     |
| <b>getparmodevalues</b>            | function | <b>insertcategorycodes</b>          | data table |
| <b>getparstate</b>                 | function | <b>insertoptioncodes</b>            | data table |
| <b>getparstatefields</b>           | function | <b>insertsplitcodes</b>             | data table |
| <b>getpartriggervalue</b>          | function | <b>integerdef</b>                   | function   |
| <b>getpenaltyoptionvalues</b>      | function | <b>interactioncodes</b>             | data table |
| <b>getpositvalue</b>               | function | <b>isattribute</b>                  | function   |
| <b>getprepoststatevalues</b>       | function | <b>isbox</b>                        | function   |
| <b>getprimitivelegacies</b>        | function | <b>iscount</b>                      | function   |
| <b>getprimitiveorigins</b>         | function | <b>isdefined</b>                    | function   |
| <b>getprotrusionboundaryvalues</b> | function | <b>isdimen</b>                      | function   |
| <b>getregisterindex</b>            | function | <b>isfloat</b>                      | function   |
| <b>getruleoptionvalues</b>         | function | <b>isglue</b>                       | function   |



|                            |            |                              |
|----------------------------|------------|------------------------------|
| <b>ismuglue</b>            | function   | N mathfirstuserclass         |
| <b>ismuskip</b>            | function   | N mathlastuserclass          |
| <b>isprintable</b>         | function   | N maxattributeregisterindex  |
| <b>isskip</b>              | function   | N maxboxregisterindex        |
| <b>istoks</b>              | function   | N maxbytecodeindex           |
| <b>lastnodetype</b>        | function   | N maxcalculatedbadness       |
| <b>lccode</b>              | table      | N maxcardinal                |
| <b>less</b>                | function   | N maxcharactercode           |
| <b>lessequal</b>           | function   | N maxdatavalue               |
| <b>linebreak</b>           | function   | N maxdimen                   |
| <b>linebreakstatecodes</b> | data table | N maxdimenregisterindex      |
| <b>listanchorcodes</b>     | data table | N maxdimension               |
| <b>listgeometrycodes</b>   | data table | N maxdimensionregisterindex  |
| <b>lists</b>               | table      | N maxendlinecharacter        |
| <b>listsigncodes</b>       | data table | N maxfloatregisterindex      |
| <b>luacatcodes</b>         | number     | N maxfontadjustshrinkfactor  |
| <b>magicconstants</b>      | table      | N maxfontadjuststep          |
| S activecharacternamespace |            | N maxfontadjuststretchfactor |
| N allfitnessvalues         |            | N maxfunctionreference       |
| N assumedmathcontrol       |            | N maxglueregisterindex       |
| N awfulbad                 |            | N maxhalfvalue               |
| N decentcriterion          |            | N maxhalfword                |
| N defaultcatcodetable      |            | N maxinteger                 |
| N defaultcharactercontrol  |            | N maxintegerregisterindex    |
| N defaultdeadcycles        |            | N maxintregisterindex        |
| N defaulteqnogapstep       |            | N maxlimitedscale            |
| N defaultthangafter        |            | N maxmarkindex               |
| N defaultoutputbox         |            | N maxmathclasscode           |
| N defaultpredisplaygap     |            | N maxmathfamilyindex         |
| N defaultrule              |            | N maxmathscalingfactor       |
| N defaultspacefactor       |            | N maxmathstylescale          |
| N defaulttolerance         |            | N maxmuglueregisterindex     |
| N deplorable               |            | N maxmvlindex                |
| N ejectpenalty             |            | N maxnewlinecharacter        |
| N ignoreddepth             |            | N maxnofbytecodes            |
| N infinitebad              |            | N maxnofcatcodetables        |
| N infinitepenalty          |            | N maxnoffitnessvalues        |
| N infinity                 |            | N maxnoffonts                |
| N largewidthexcess         |            | N maxnoflanguages            |
| N loosecriterion           |            | N maxnofmarks                |
| N mathallclass             |            | N maxnofmathfamilies         |
| N mathbeginclass           |            | N maxquarterword             |
| N mathdefaultpenalty       |            | N maxscalefactor             |
| N mathendclass             |            | N maxsizeofword              |

|                          |            |                                |            |
|--------------------------|------------|--------------------------------|------------|
| N maxspacefactor         |            | <b>mathparametercodes</b>      | data table |
| N maxtoksregisterindex   |            | <b>mathscriptordercodes</b>    | data table |
| N maxtwinlength          |            | <b>mathscriptsmodecodes</b>    | data table |
| N mincardinal            |            | <b>modelevels</b>              | table      |
| N mindatavalue           |            | N horizontal                   |            |
| N mindimen               |            | N math                         |            |
| N mindimension           |            | N unset                        |            |
| N minhalfword            |            | N vertical                     |            |
| N mininfinity            |            | <b>modes</b>                   | data table |
| N mininteger             |            | <b>mprint</b>                  | function   |
| N minmvlindex            |            | <b>muglue</b>                  | table      |
| N minnofitnessvalues     |            | <b>muskip</b>                  | table      |
| N minquarterword         |            | <b>mvloptioncodes</b>          | data table |
| N minscalefactor         |            | <b>nest</b>                    | table      |
| N minspacefactor         |            | <b>nilcatcodes</b>             | number     |
| N nocatcodetable         |            | <b>noadoptioncodes</b>         | data table |
| N null                   |            | <b>normalizelinecodes</b>      | data table |
| N nullflag               |            | <b>normalizeparcodes</b>       | data table |
| N nullfont               |            | <b>nospacesmodecodes</b>       | data table |
| N onebp                  |            | <b>notcatcodes</b>             | number     |
| N presetrulethickness    |            | <b>number</b>                  | function   |
| N runningrule            |            | <b>paragraphfillmodecodes</b>  | data table |
| N saparthigh             |            | <b>paragraphoptioncodes</b>    | data table |
| N sapartlow              |            | <b>parcontextcodes</b>         | data table |
| N sapartmiddle           |            | <b>partriggercodes</b>         | data table |
| N smallstretchability    |            | <b>penaltyoptioncodes</b>      | data table |
| N specialspacefactor     |            | <b>poplocal</b>                | function   |
| N texeqtbsize            |            | <b>popsavelevel</b>            | function   |
| N texhashprime           |            | <b>posit</b>                   | table      |
| N texhashsize            |            | <b>positdef</b>                | function   |
| N texmemorymode          |            | <b>preparelinebreak</b>        | function   |
| N two                    |            | <b>primitives</b>              | function   |
| N undefinedmathparameter |            | <b>print</b>                   | function   |
| N unity                  |            | <b>protrusionboundarycodes</b> | data table |
| N unusedattributevalue   |            | <b>prtcacodes</b>              | number     |
| N unusedmathfamily       |            | <b>pushlocal</b>               | function   |
| N unusedmathstyle        |            | <b>pushsavelevel</b>           | function   |
| N unusedscriptvalue      |            | <b>quitlocal</b>               | function   |
| N unusedstatevalue       |            | <b>quittoks</b>                | function   |
| N zeroglue               |            | <b>resetparagraph</b>          | function   |
| <b>mathchardef</b>       | function   | <b>rlncatcodes</b>             | number     |
| <b>mathcode</b>          | table      | <b>romannumeral</b>            | function   |
| <b>mathcontrolcodes</b>  | data table | <b>round</b>                   | function   |
| <b>mathoptioncodes</b>   | data table | <b>ruleoptioncodes</b>         | data table |

|                            |            |                                 |                     |
|----------------------------|------------|---------------------------------|---------------------|
| <b>runlocal</b>            | function   | <b>setmathdir</b>               | function            |
| <b>runstatecodes</b>       | data table | <b>setmuglue</b>                | function            |
| <b>runstring</b>           | function   | <b>setmuskip</b>                | function            |
| <b>runtoks</b>             | function   | <b>setnest</b>                  | function            |
| <b>saveboxresource</b>     | function   | <b>setoverloadstate</b>         | function            |
| <b>scale</b>               | function   | <b>setpardir</b>                | function            |
| <b>scantoks</b>            | function   | <b>setpositvalue</b>            | function            |
| <b>set</b>                 | function   | <b>setrunstate</b>              | overloaded function |
| <b>setamcode</b>           | function   | <b>setsfcode</b>                | function            |
| <b>setattribute</b>        | function   | <b>setskip</b>                  | function            |
| <b>setbalanceshape</b>     | function   | <b>setspcode</b>                | function            |
| <b>setbox</b>              | function   | <b>settetmdir</b>               | function            |
| <b>setboxdir</b>           | function   | <b>settoks</b>                  | function            |
| <b>setcardinalvalue</b>    | function   | <b>setuccode</b>                | function            |
| <b>setcatcode</b>          | function   | <b>sfcode</b>                   | table               |
| <b>setcccode</b>           | function   | <b>shapingpenaltiescodes</b>    | data table          |
| <b>setconditional</b>      | function   | <b>shiftparshape</b>            | function            |
| <b>setcount</b>            | function   | <b>shipout</b>                  | function            |
| <b>setdelcode</b>          | function   | <b>showcontext</b>              | function            |
| <b>setdimen</b>            | function   | <b>skip</b>                     | table               |
| <b>setdimensionvalue</b>   | function   | <b>snappingcodes</b>            | data table          |
| <b>setfloat</b>            | function   | <b>snapshotpar</b>              | function            |
| <b>setglue</b>             | function   | <b>sp</b>                       | function            |
| <b>setglyphdata</b>        | function   | <b>spacefactorcodes</b>         | data table          |
| <b>setglyphscript</b>      | function   | <b>spaceskipcodes</b>           | data table          |
| <b>setglyphstate</b>       | function   | <b>specialmathclasscodes</b>    | data table          |
| <b>sethcode</b>            | function   | <b>specificationoptioncodes</b> | data table          |
| <b>sethmcode</b>           | function   | <b>splitbox</b>                 | function            |
| <b>setinputstatefile</b>   | function   | <b>splitcodes</b>               | data table          |
| <b>setinputstateline</b>   | function   | <b>sprint</b>                   | function            |
| <b>setinputstatemode</b>   | function   | <b>stringtocodesbitmap</b>      | function            |
| <b>setinsertcontent</b>    | function   | <b>systemmodes</b>              | data table          |
| <b>setinsertdistance</b>   | function   | <b>takebox</b>                  | function            |
| <b>setinsertlimit</b>      | function   | <b>texcatcodes</b>              | number              |
| <b>setinsertmaxplaced</b>  | function   | <b>textcontrolcodes</b>         | data table          |
| <b>setinsertmultiplier</b> | function   | <b>toks</b>                     | table               |
| <b>setintegervalue</b>     | function   | <b>tonumber</b>                 | function            |
| <b>setinteraction</b>      | function   | <b>toscaled</b>                 | function            |
| <b>setlcode</b>            | function   | <b>tpacatcodes</b>              | number              |
| <b>setlinedir</b>          | function   | <b>tpbcacodes</b>               | number              |
| <b>setlist</b>             | function   | <b>tprint</b>                   | function            |
| <b>setlocalbox</b>         | function   | <b>trackedattribute</b>         | function            |
| <b>setmath</b>             | function   | <b>trackers</b>                 | data table          |
| <b>setmathcode</b>         | function   | <b>triggerbuildpage</b>         | function            |

**txtcatcodes**  
**uccode**  
**uprint**  
**useboxresource**

number  
**table**  
function  
function

**userdiscoptioncodes**  
**userglyphoptioncodes**  
**vrbcacodes**  
**write**

data table  
data table  
number  
**function**

# texconfig (basic tex)

|                                                  |                     |                                               |        |
|--------------------------------------------------|---------------------|-----------------------------------------------|--------|
| <b>buffer</b> <b>size</b>                        | table               | <b>language</b> <b>size</b>                   | table  |
| N size                                           |                     | N size                                        |        |
| N step                                           |                     | N step                                        |        |
| <b>error</b> <b>line</b> <b>size</b>             | table               | <b>mark</b> <b>size</b>                       | table  |
| N size                                           |                     | N size                                        |        |
| <b>expand</b> <b>size</b>                        | table               | N step                                        |        |
| N size                                           |                     | <b>nest</b> <b>size</b>                       | table  |
| <b>file</b> <b>size</b>                          | table               | N size                                        |        |
| N size                                           |                     | N step                                        |        |
| N step                                           |                     | <b>node</b> <b>size</b>                       | table  |
| <b>first</b> <b>line</b>                         | string              | N size                                        |        |
| <b>font</b> <b>size</b>                          | table               | N step                                        |        |
| N size                                           |                     | <b>param</b> <b>eter</b> <b>size</b>          | table  |
| N step                                           |                     | N size                                        |        |
| <b>format</b> <b>name</b>                        | string              | N step                                        |        |
| <b>function</b> <b>size</b>                      | number              | <b>pool</b> <b>size</b>                       | table  |
| <b>half</b> <b>error</b> <b>line</b> <b>size</b> | table               | N size                                        |        |
| N size                                           |                     | N step                                        |        |
| <b>hash</b> <b>size</b>                          | table               | <b>prop</b> <b>erti</b> <b>ess</b> <b>ize</b> | number |
| N size                                           |                     | <b>save</b> <b>size</b>                       | table  |
| N step                                           |                     | N size                                        |        |
| <b>init</b>                                      | overloaded function | N step                                        |        |
| <b>input</b> <b>size</b>                         | table               | <b>shell_</b> <b>escape</b>                   | string |
| N size                                           |                     | <b>string</b> <b>size</b>                     | table  |
| N step                                           |                     | N size                                        |        |
| <b>insert</b> <b>size</b>                        | table               | N step                                        |        |
| N size                                           |                     | <b>token</b> <b>size</b>                      | table  |
| N step                                           |                     | N size                                        |        |
| <b>kpse_</b> <b>init</b>                         | boolean             | N step                                        |        |

## texio (basic tex)

|                          |          |
|--------------------------|----------|
| <b>closeinput</b>        | function |
| <b>forceendoffile</b>    | function |
| <b>getinputindex</b>     | function |
| <b>getselectorvalues</b> | function |
| <b>getsourcefilename</b> | function |
| <b>setlogfile</b>        | function |
| <b>terminal</b>          | function |

|                        |          |
|------------------------|----------|
| <b>write</b>           | function |
| <b>write_nl</b>        | function |
| <b>writeln</b>         | function |
| <b>writeselector</b>   | function |
| <b>writeselectorlf</b> | function |
| <b>writeselectornl</b> | function |

## token (basic tex)

|                         |          |                          |          |
|-------------------------|----------|--------------------------|----------|
| <b>create</b>           | function | <b>getexpandable</b>     | function |
| <b>expand</b>           | function | <b>getexpansion</b>      | function |
| <b>expand_macro</b>     | function | <b>getfields</b>         | function |
| <b>expandmacro</b>      | function | <b>getflags</b>          | function |
| <b>future_expand</b>    | function | <b>getfrozen</b>         | function |
| <b>futureexpand</b>     | function | <b>getfunctionvalues</b> | function |
| <b>get_active</b>       | function | <b>getid</b>             | function |
| <b>get_cmdchrscs</b>    | function | <b>getimmutable</b>      | function |
| <b>get_cmdname</b>      | function | <b>getindex</b>          | function |
| <b>get_command</b>      | function | <b>getinstance</b>       | function |
| <b>get_csname</b>       | function | <b>getinteger</b>        | function |
| <b>get_cstoken</b>      | function | <b>getlength</b>         | function |
| <b>get_dimension</b>    | function | <b>getmacro</b>          | function |
| <b>get_expandable</b>   | function | <b>getmeaning</b>        | function |
| <b>get_fields</b>       | function | <b>getnoaligned</b>      | function |
| <b>get_flags</b>        | function | <b>getparameters</b>     | function |
| <b>get_frozen</b>       | function | <b>getpermanent</b>      | function |
| <b>get_id</b>           | function | <b>getprimitive</b>      | function |
| <b>get_immutable</b>    | function | <b>getprimitives</b>     | function |
| <b>get_index</b>        | function | <b>getprotected</b>      | function |
| <b>get_instance</b>     | function | <b>getrange</b>          | function |
| <b>get_integer</b>      | function | <b>gettok</b>            | function |
| <b>get_macro</b>        | function | <b>gettolerant</b>       | function |
| <b>get_meaning</b>      | function | <b>gobble</b>            | function |
| <b>get_noaligned</b>    | function | <b>gobble_dimension</b>  | function |
| <b>get_parameters</b>   | function | <b>gobble_integer</b>    | function |
| <b>get_permanent</b>    | function | <b>gobbledimension</b>   | function |
| <b>get_primitive</b>    | function | <b>gobbleinteger</b>     | function |
| <b>get_protected</b>    | function | <b>grab</b>              | function |
| <b>get_range</b>        | function | <b>grabstring</b>        | function |
| <b>get_tok</b>          | function | <b>inhibit</b>           | function |
| <b>get_tolerant</b>     | function | <b>is_defined</b>        | function |
| <b>getactive</b>        | function | <b>is_next_char</b>      | function |
| <b>getcmdchrscs</b>     | function | <b>is_token</b>          | function |
| <b>getcmdname</b>       | function | <b>isactivestring</b>    | function |
| <b>getcommand</b>       | function | <b>isdefined</b>         | function |
| <b>getcommandvalues</b> | function | <b>isnextchar</b>        | function |
| <b>getconstant</b>      | function | <b>istoken</b>           | function |
| <b>getcsname</b>        | function | <b>locatemacro</b>       | function |
| <b>getcstoken</b>       | function | <b>new</b>               | function |
| <b>getdimension</b>     | function | <b>peek_next</b>         | function |

|                                   |          |                                    |          |
|-----------------------------------|----------|------------------------------------|----------|
| <code>peek_next_char</code>       | function | <code>scan_tokenlist</code>        | function |
| <code>peek_next_expanded</code>   | function | <code>scan_toks</code>             | function |
| <code>peeknext</code>             | function | <code>scan_value</code>            | function |
| <code>peeknextchar</code>         | function | <code>scan_word</code>             | function |
| <code>peeknextexpanded</code>     | function | <code>scanargument</code>          | function |
| <code>pop_macro</code>            | function | <code>scanbox</code>               | function |
| <code>popmacro</code>             | function | <code>scancardinal</code>          | function |
| <code>push_macro</code>           | function | <code>scanchar</code>              | function |
| <code>pushmacro</code>            | function | <code>scancmdchr</code>            | function |
| <code>put_back</code>             | function | <code>scancmdchrexplanded</code>   | function |
| <code>put_next</code>             | function | <code>scancode</code>              | function |
| <code>putback</code>              | function | <code>scancsname</code>            | function |
| <code>putnext</code>              | function | <code>scancstoken</code>           | function |
| <code>save_lua</code>             | function | <code>scandelimited</code>         | function |
| <code>savelua</code>              | function | <code>scandetokened</code>         | function |
| <code>scan_argument</code>        | function | <code>scandimen</code>             | function |
| <code>scan_box</code>             | function | <code>scandimenargument</code>     | function |
| <code>scan_cardinal</code>        | function | <code>scandimension</code>         | function |
| <code>scan_char</code>            | function | <code>scandimensionargument</code> | function |
| <code>scan_cmdchr</code>          | function | <code>scanfloat</code>             | function |
| <code>scan_cmdchr_expanded</code> | function | <code>scanglue</code>              | function |
| <code>scan_code</code>            | function | <code>scanint</code>               | function |
| <code>scan_csname</code>          | function | <code>scaninteger</code>           | function |
| <code>scan_delimited</code>       | function | <code>scanintegerargument</code>   | function |
| <code>scan_dimen</code>           | function | <code>scankey</code>               | function |
| <code>scan_float</code>           | function | <code>scankeyword</code>           | function |
| <code>scan_glue</code>            | function | <code>scankeywordcs</code>         | function |
| <code>scan_integer</code>         | function | <code>scanletters</code>           | function |
| <code>scan_key</code>             | function | <code>scanluacardinal</code>       | function |
| <code>scan_keyword</code>         | function | <code>scanluainteger</code>        | function |
| <code>scan_keyword_cs</code>      | function | <code>scanluanumber</code>         | function |
| <code>scan_letters</code>         | function | <code>scannext</code>              | function |
| <code>scan_luacardinal</code>     | function | <code>scannextchar</code>          | function |
| <code>scan_luainteger</code>      | function | <code>scannextexpanded</code>      | function |
| <code>scan_luanumber</code>       | function | <code>scanposit</code>             | function |
| <code>scan_next</code>            | function | <code>scanreal</code>              | function |
| <code>scan_next_char</code>       | function | <code>scanscale</code>             | function |
| <code>scan_next_expanded</code>   | function | <code>scanskip</code>              | function |
| <code>scan_real</code>            | function | <code>scanstring</code>            | function |
| <code>scan_scale</code>           | function | <code>scantoken</code>             | function |
| <code>scan_skip</code>            | function | <code>scantokencode</code>         | function |
| <code>scan_string</code>          | function | <code>scantokenlist</code>         | function |
| <code>scan_token</code>           | function | <code>scantokenstring</code>       | function |
| <code>scan_token_code</code>      | function | <code>scantoks</code>              | function |



|                      |          |                           |          |
|----------------------|----------|---------------------------|----------|
| <b>scanvalue</b>     | function | <b>setmacro</b>           | function |
| <b>scanword</b>      | function | <b>setmacrofrommark</b>   | function |
| <b>serialize</b>     | function | <b>setmacrotonothing</b>  | function |
| <b>set_dimension</b> | function | <b>skip_next</b>          | function |
| <b>set_integer</b>   | function | <b>skip_next_expanded</b> | function |
| <b>set_lua</b>       | function | <b>skipnext</b>           | function |
| <b>set_macro</b>     | function | <b>skipnextexpanded</b>   | function |
| <b>setdimension</b>  | function | <b>type</b>               | function |
| <b>setinteger</b>    | function | <b>undefine_macro</b>     | function |
| <b>setlua</b>        | function | <b>undefinemacro</b>      | function |

# tokens (context)

**accessors** table

F active  
F cmd  
F cmdchrsc  
F cmdname  
F command  
F csname  
F expandable  
F frozen  
F index  
F name  
F protected  
F range

**astable** function

**bits** table

N active  
N alignment  
N begingroup  
N character  
N close  
N comment  
N endgroup  
N endofline  
N escape  
N ignore  
N invalid  
N letter  
N mathshift  
N open  
N other  
N parameter  
N space  
N subscript  
N superscript  
N whitespace

**cache** data table

**commands** data table

**compile** function

**constant** function

**converters** table

S toboolean

S todimen

S toglue

S tonumber

S tostring

**create** function

**defined** function

**flags** function

**getinteger** function

**getters** table

F box

F count

F cstoken

F dimen

F glue

F macro

F meaning

F skip

F token

**gobble** function

**isactivestring** function

**isdefined** function

**istoken** function

**new** function

**pickup** function

**popmacro** function

**presets** table

**pushmacro** function

**scanners** table

F argument

F argumentasis

F array

F boolean

F box

F bracketed

F bracketedasis

F cardinal

F char

F cmdchr

F cmdchrexplanded

F code

F conditional

|                     |                         |          |
|---------------------|-------------------------|----------|
| F count             | F tokencode             |          |
| F csname            | F tokenlist             |          |
| F csnameunchecked   | F tokens                |          |
| F delimited         | F tokenstring           |          |
| F detokened         | F toks                  |          |
| F dimen             | F value                 |          |
| F dimension         | F vbox                  |          |
| F dimensionargument | F verbatim              |          |
| F float             | F vtop                  |          |
| F glue              | F whd                   |          |
| F gluespec          | F word                  |          |
| F gluevalues        | <b>setinteger</b>       | function |
| F hash              | <b>setters</b>          | table    |
| F hbox              | F box                   |          |
| F integer           | F char                  |          |
| F integerargument   | F count                 |          |
| F ischar            | F dimen                 |          |
| F key               | F glue                  |          |
| F keyword           | F lua                   |          |
| F keywordcs         | F macro                 |          |
| F letters           | F skip                  |          |
| F list              | <b>shortcuts</b>        | table    |
| F lua               | N close                 |          |
| F luacardinal       | F inspect               |          |
| F luainteger        | N open                  |          |
| F luanumber         | F report                |          |
| F lxmlid            | F scanargument          |          |
| F next              | F scanargumentasis      |          |
| F nextchar          | F scanarray             |          |
| F nextexpanded      | F scanboolean           |          |
| F number            | F scanbox               |          |
| F optional          | F scanbracketed         |          |
| F peek              | F scanbracketedasis     |          |
| F peekchar          | F scancardinal          |          |
| F peekexpanded      | F scanclose             |          |
| F posit             | F scancode              |          |
| F real              | F scanconditional       |          |
| F scanclose         | F scanscsname           |          |
| F scanopen          | F scanscsnameunchecked  |          |
| F skip              | F scandelimited         |          |
| F skipexpanded      | F scandetokened         |          |
| F string            | F scandimen             |          |
| F table             | F scandimension         |          |
| F token             | F scandimensionargument |          |

|                       |                   |                   |
|-----------------------|-------------------|-------------------|
| F scanfloat           | F scanposit       |                   |
| F scan glue           | F scanstring      |                   |
| F scan glue spec      | F scantable       |                   |
| F scan glue values    | F scantokenlist   |                   |
| F scanhash            | F scantokenstring |                   |
| F scanhbox            | F scantoks        |                   |
| F scaninteger         | F scanvbox        |                   |
| F scanintegerargument | F scanverbatim    |                   |
| F scankeyword         | F scanvtop        |                   |
| F scankeywordcs       | F scanword        |                   |
| F scanletters         | F toboolean       |                   |
| F scanlist            | F todimen         |                   |
| F scanlua             | F tonumber        |                   |
| F scannumber          | F tostring        |                   |
| F scanopen            | <b>values</b>     | <b>data table</b> |
| F scanoptional        |                   |                   |

## tracers (context)

**printerror**    function  
**printwarning**    function

**showlines**    function

## trackers (context)

|                       |            |                 |          |
|-----------------------|------------|-----------------|----------|
| <b>data</b>           | data table | <b>register</b> | function |
| <b>default</b>        | function   | <b>report</b>   | function |
| <b>disable</b>        | function   | <b>reset</b>    | function |
| <b>done</b>           | table      | <b>savehash</b> | function |
| <b>dumphashtofile</b> | function   | <b>show</b>     | function |
| <b>enable</b>         | function   | <b>value</b>    | function |
| <b>list</b>           | function   | <b>values</b>   | function |
| <b>name</b>           | string     |                 |          |

# typesetters (context)

## **alignments**      table

F handler

F set

## **breakpoints**      table

F define

F handler

D mapping

F set

F setreplacement

## **cases**      table

F handler

F register

F set

## **characteralign**   table

F handler

F resetcharacteralign

F setcharacteralign

## **characters**      table

F handler

F replacensbp

F replacenspaces

## **checkers**      table

F handler

F reset

F set

## **cleaners**      table

F handler

F set

## **compare**      table

F handler

F reset

F set

F setpair

F setup

## **corrections**      table

F handler

## **digits**      table

F handler

F set

## **directions**      table

F getfences

F getglobal

F getmethod

F handler

F installhandler

F set

F setcolor

F tomode

## **firstlines**      table

F handler

F set

## **fontkerns**      table

F handler

## **initials**      table

F handler

F set

## **injectors**      table

F check

F mark

F reset

F set

F show

## **italics**      table

F enablemath

F enabletext

F handler

## **kerns**      table

D factors

F handler

F keepeligature

D mapping

F set

## **localboxes**      table

F handler

## **manipulators**      table

F apply

F applyspecification

F splitspecification

## **margins**      table

F finalhandler

F globalhandler

F localhandler

F save

**marksuspects**      function

**paragraphs**        table

  F addanchortoline

  F addtoline

  F calculatedelta

  F checkline

  F moveinline

**periodkerns**        table

  D factors

  F handler

  D mapping

  F set

**processors**         table

  F apply

  F register

  F reset

  F split

  F startapply

  F stopapply

  F stripped

  F tostring

**rubies**             table

  F attach

  F check

**showinvisibles**    function

**showsuspects**     function

**spacings**          table

  F define

  F handler

  D mapping

  F reset

  F set

  F setup

**stacking**           table

  F getindex

  F getselection

  F handler

**swapping**          table

  F handler

**synchronize**       table

  F handler

**tighten**            table

  F handler

**wrappers**          table

**xtables**            table

  F cleanup

  F construct

  F create

  F finish\_row

  F flush

  F initialize\_construct

  F initialize\_reflow\_height

  F initialize\_reflow\_width

  F next\_row

  F reflow\_height

  F reflow\_width

  F set\_construct

  F set\_reflow\_height

  F set\_reflow\_width



## updaters (context)

**apply**      function

**register**   function

## url (context)

|                  |          |
|------------------|----------|
| <b>addscheme</b> | function |
| <b>barepath</b>  | function |
| <b>construct</b> | function |
| <b>decode</b>    | function |
| <b>encode</b>    | function |
| <b>escape</b>    | function |
| <b>filename</b>  | function |

|                    |          |
|--------------------|----------|
| <b>hashed</b>      | function |
| <b>hasscheme</b>   | function |
| <b>query</b>       | function |
| <b>split</b>       | function |
| <b>toquery</b>     | function |
| <b>unescape</b>    | function |
| <b>unescapeget</b> | function |

## utf8 (basic lua)

**char**           function  
**charpattern** string  
**codepoint**     function

**codes**           function  
**len**            function  
**offset**         function

# utilities (context)

## **debugger**            table

F disable  
F enable  
F savestats  
F showstats  
F showtraceback

## **files**                table

F close  
F getposition  
F getsize  
F look  
F open  
F read2dot14  
F readbyte  
F readbytes  
F readbytetable  
F readcardinal  
F readcardinal1  
F readcardinal1le  
F readcardinal2  
F readcardinal2le  
F readcardinal3  
F readcardinal3le  
F readcardinal4  
F readcardinal4le  
F readcardinaltable  
F readchar  
F readcline  
F readcstring  
F readfixed2  
F readfixed4  
F readinteger  
F readinteger1  
F readinteger1le  
F readinteger2  
F readinteger2le  
F readinteger3  
F readinteger3le  
F readinteger4  
F readinteger4le  
F readintegertable

F readsignedbyte  
F readstring  
F readtag  
F setposition  
F size  
F skip  
F skiplong  
F skipposition  
F skipshort  
F writebyte  
F writecardinal1  
F writecardinal1le  
F writecardinal2  
F writecardinal2le  
F writecardinal3  
F writecardinal3le  
F writecardinal4  
F writecardinal4le  
F writeinteger1  
F writeinteger1le  
F writeinteger2  
F writeinteger2le  
F writeinteger3  
F writeinteger3le  
F writeinteger4  
F writeinteger4le  
F writestring

## **formatters**           table

F formatcolumns  
F list  
F stripzeros

## **ghostscript**           table

F convert  
F statistics

## **graphicmagick**        table

F convert  
F statistics

## **graphicsmagick**      table

F convert  
F statistics

## **json**                  table

|                                 |                                 |
|---------------------------------|---------------------------------|
| F load                          | F settings_to_array_with_repeat |
| F tojson                        | F settings_to_hash              |
| F tolua                         | F settings_to_hash_colon_too    |
| F tostring                      | F settings_to_hash_strict       |
| <b>lua</b>                      | F settings_to_hash_tolerant     |
|                                 | F settings_to_numbers           |
| B alwaysstripcode               | F settings_to_set               |
| F checkmemory                   | F simple_hash_to_string         |
| F compile                       | F splitmethod                   |
| F loadedluacode                 | F splitthousands                |
| F loadstring                    | F stepper                       |
| F loadstripped                  | F stripped_settings_to_array    |
| N nofstrippedbytes              | F totime                        |
| N nofstrippedchunks             | F unittotex                     |
| F registerfinalizer             | F unittoxml                     |
| B stripcode                     |                                 |
| F strippedloadstring            | <b>randomizer</b>               |
| <b>packers</b>                  | table                           |
|                                 | F get                           |
| F hashed                        | F getnumber                     |
| F new                           | F getrepeatable                 |
| F pack                          | F getrepeatableseed             |
| F simplehashed                  | F getseed                       |
| F strip                         | F newrepeatable                 |
| F unpack                        | F popseed                       |
| N version                       | F pushseed                      |
| <b>parsers</b>                  | F repeatable                    |
|                                 | F reuseseed                     |
| F add_settings_to_array         | F setseed                       |
| F arguments_to_table            | F setseedi                      |
| F array_to_string               | <b>randomizers</b>              |
| F csvsplitter                   | table                           |
| F getparameters                 | F poisson                       |
| F groupedsplitat                | <b>sequencers</b>               |
| F hash_to_string                | table                           |
| F iterator                      | F appendaction                  |
| F keq_to_hash                   | F appendgroup                   |
| F listitem                      | F compile                       |
| F make_settings_to_hash_pattern | F disableaction                 |
| F mergehashes                   | F disablegroup                  |
| F options_to_array              | F enableaction                  |
| F options_to_hash               | F enablegroup                   |
| F process_stripped_settings     | D functions                     |
| F rfc4180splitter               | F localize                      |
| F runtime                       | F new                           |
| F settings_to_array             | F nodeprocessor                 |
| F settings_to_array_obey_fences | F prependaction                 |
|                                 | F prependgroup                  |

|                  |              |                     |
|------------------|--------------|---------------------|
| F removeaction   |              | F read2dot14        |
| F replaceaction  |              | F readbyte          |
| F setkind        |              | F readbytes         |
| F toString       |              | F readbytetable     |
| <b>setters</b>   | <b>table</b> | F readcardinal      |
| F disable        |              | F readcardinal1     |
| F enable         |              | F readcardinal1le   |
| F initialize     |              | F readcardinal2     |
| F list           |              | F readcardinal2le   |
| F new            |              | F readcardinal3     |
| F register       |              | F readcardinal3le   |
| F report         |              | F readcardinal4     |
| F reset          |              | F readcardinal4le   |
| F show           |              | F readcardinaltable |
| <b>signals</b>   | <b>table</b> | F readchar          |
| S configuration  |              | F readcstring       |
| F initialize     |              | F readdouble        |
| F loadstate      |              | F readdoublele      |
| F report         |              | F readfixed2        |
| F savestate      |              | F readfixed4        |
| F serialclose    |              | F readfloat         |
| F serialfast     |              | F readfloatle       |
| F serialopen     |              | F readfloatle12     |
| S serialprefix   |              | F readfloatle6      |
| F serialsend     |              | F readinteger       |
| F serialwrite    |              | F readinteger1      |
| N version        |              | F readinteger1le    |
| <b>stacker</b>   | <b>table</b> | F readinteger2      |
| F new            |              | F readinteger2le    |
| <b>storage</b>   | <b>table</b> | F readinteger3      |
| F allocate       |              | F readinteger3le    |
| F checked        |              | F readinteger4      |
| F mark           |              | F readinteger4le    |
| F marked         |              | F readintegertable  |
| F setinitializer |              | F readline          |
| F sparse         |              | F readsignedbyte    |
| <b>streams</b>   | <b>table</b> | F readstring        |
| F close          |              | F setposition       |
| F getposition    |              | F size              |
| F getsize        |              | F skip              |
| F getstring      |              | F skipbytes         |
| F look           |              | F skiplong          |
| F open           |              | F skipshort         |
| F openstring     |              | F tocardinal1       |

|                      |                        |
|----------------------|------------------------|
| F tocardinal1le      | F migratetable         |
| F tocardinal2        | F removevalue          |
| F tocardinal2le      | F replacevalue         |
| F tocardinal3        | <b>tar</b> table       |
| F tocardinal3le      | F closearchive         |
| F tocardinal4        | F openarchive          |
| F tocardinal4le      | <b>templates</b> table |
| <b>strings</b> table | F load                 |
| F collapse           | F replace              |
| F collapsecrlf       | F replacer             |
| F newcollector       | F resolve              |
| F newrepeater        | <b>zipfiles</b> table  |
| F striplines         | F close                |
| F tabtospace         | F found                |
| <b>tables</b> table  | F hash                 |
| F accesstable        | F list                 |
| F definedtable       | F open                 |
| F definetable        | F unzip                |
| F encapsulate        | F unzipdir             |
| F insertaftervalue   | F zipdir               |
| F insertbeforevalue  |                        |

## vector (extra lua)

|                       |          |                   |          |
|-----------------------|----------|-------------------|----------|
| <b>add</b>            | function | F xrotation       |          |
| <b>append</b>         | function | F xyzscale        |          |
| <b>average</b>        | function | F yrotation       |          |
| <b>ceiling</b>        | function | F zrotation       |          |
| <b>checkoverlap</b>   | function | <b>homogenize</b> | function |
| <b>collectoverlap</b> | function | <b>identity</b>   | function |
| <b>concat</b>         | function | <b>inner</b>      | function |
| <b>contour</b>        | table    | <b>insert</b>     | function |
| F average             |          | <b>inverse</b>    | function |
| F bounds              |          | <b>ismesh</b>     | function |
| F checkoverlap        |          | <b>ispoints</b>   | function |
| F collectoverlap      |          | <b>issingular</b> | function |
| F getarea             |          | <b>isvector</b>   | function |
| F getmesh             |          | <b>iszero</b>     | function |
| F makemesh            |          | <b>mesh</b>       | table    |
| F overlapping         |          | F copy            |          |
| F show                |          | F get             |          |
| F sort                |          | F getaverage      |          |
| <b>copy</b>           | function | F getdimensions   |          |
| <b>crossproduct</b>   | function | F gettypevalues   |          |
| <b>delete</b>         | function | F ismesh          |          |
| <b>determinant</b>    | function | F new             |          |
| <b>div</b>            | function | F set             |          |
| <b>equal</b>          | function | F setaverage      |          |
| <b>exchange</b>       | function | F tostring        |          |
| <b>floor</b>          | function | F totable         |          |
| <b>get</b>            | function | F type            |          |
| <b>getdimensions</b>  | function | <b>meshname</b>   | function |
| <b>getepsilon</b>     | function | <b>mul</b>        | function |
| <b>getmemory</b>      | function | <b>negate</b>     | function |
| <b>getmesh</b>        | function | <b>new</b>        | function |
| <b>getmeshtypes</b>   | function | <b>normalize</b>  | function |
| <b>getpointtypes</b>  | function | <b>normals</b>    | table    |
| <b>getrc</b>          | function | F flip            |          |
| <b>getrow</b>         | function | F get             |          |
| <b>getsize</b>        | function | F isnormals       |          |
| <b>getstacking</b>    | function | F new             |          |
| <b>gettype</b>        | function | F set             |          |
| <b>helpers</b>        | table    | F tostring        |          |
| F points              |          | F totable         |          |
| F transform           |          | F type            |          |



|                  |          |                    |          |
|------------------|----------|--------------------|----------|
| <b>onecolumn</b> | function | F getxy            |          |
| <b>onerow</b>    | function | F getxytable       |          |
| <b>point</b>     | table    | F getxyz           |          |
| F add            |          | F getxyztable      |          |
| F copy           |          | F grow             |          |
| F cross          |          | F index            |          |
| F crossproduct   |          | F ispoints         |          |
| F dist           |          | F new              |          |
| F distance       |          | F prune            |          |
| F div            |          | F set              |          |
| F dot            |          | F setindex         |          |
| F dotproduct     |          | F setnormal        |          |
| F equal          |          | F setxy            |          |
| F get            |          | F setxyz           |          |
| F ispoint        |          | F size             |          |
| F iszero         |          | F tostring         |          |
| F len            |          | F totable          |          |
| F length         |          | F type             |          |
| F mul            |          | <b>product</b>     | function |
| F neg            |          | <b>remove</b>      | function |
| F negate         |          | <b>replace</b>     | function |
| F new            |          | <b>round</b>       | function |
| F norm           |          | <b>rowechelon</b>  | function |
| F normalize      |          | <b>set</b>         | function |
| F perp           |          | <b>setepsilon</b>  | function |
| F perpendicular  |          | <b>setnext</b>     | function |
| F set            |          | <b>setrc</b>       | function |
| F sub            |          | <b>setstacking</b> | function |
| F tostring       |          | <b>slice</b>       | function |
| F totable        |          | <b>sort</b>        | function |
| F type           |          | <b>sub</b>         | function |
| <b>points</b>    | table    | <b>swap</b>        | function |
| F copy           |          | <b>tostring</b>    | function |
| F get            |          | <b>totable</b>     | function |
| F getbounds      |          | <b>transpose</b>   | function |
| F getindex       |          | <b>triangulate</b> | function |
| F getnormal      |          | <b>truncate</b>    | function |
| F getpool        |          | <b>type</b>        | function |
| F gettriangle    |          |                    |          |

## visualizers (context)

|                              |            |                        |            |
|------------------------------|------------|------------------------|------------|
| <b>escapepatterns</b>        | data table | <b>signal</b>          | string     |
| <b>getspecification</b>      | function   | <b>signalpattern</b>   | userdata   |
| <b>getvisualizer</b>         | function   | <b>specifications</b>  | data table |
| <b>load</b>                  | function   | <b>visualize</b>       | function   |
| <b>makenested</b>            | function   | <b>visualizebuffer</b> | function   |
| <b>makepattern</b>           | function   | <b>visualizefile</b>   | function   |
| <b>newgrammar</b>            | function   | <b>visualizestring</b> | function   |
| <b>newhandler</b>            | function   | <b>write</b>           | function   |
| <b>parsedxml</b>             | function   | <b>writeargument</b>   | function   |
| <b>pattern</b>               | function   | <b>writedefault</b>    | function   |
| <b>register</b>              | function   | <b>writeemptyline</b>  | function   |
| <b>registerescapecommand</b> | function   | <b>writenewline</b>    | function   |
| <b>registerescapepline</b>   | function   | <b>writespace</b>      | function   |
| <b>registerescapepattern</b> | function   |                        |            |

## xcomplex (extra lua)

|                   |          |                   |          |
|-------------------|----------|-------------------|----------|
| <b>I</b>          | function | <b>erf</b>        | function |
| <b>__add</b>      | function | <b>erfc</b>       | function |
| <b>__div</b>      | function | <b>erfcx</b>      | function |
| <b>__eq</b>       | function | <b>erfi</b>       | function |
| <b>__mul</b>      | function | <b>exp</b>        | function |
| <b>__name</b>     | string   | <b>i</b>          | function |
| <b>__pow</b>      | function | <b>imag</b>       | function |
| <b>__sub</b>      | function | <b>log</b>        | function |
| <b>__tostring</b> | function | <b>new</b>        | function |
| <b>__unm</b>      | function | <b>pow</b>        | function |
| <b>abs</b>        | function | <b>proj</b>       | function |
| <b>acos</b>       | function | <b>real</b>       | function |
| <b>acosh</b>      | function | <b>sin</b>        | function |
| <b>arg</b>        | function | <b>sinh</b>       | function |
| <b>asin</b>       | function | <b>sqrt</b>       | function |
| <b>asinh</b>      | function | <b>tan</b>        | function |
| <b>atan</b>       | function | <b>tanh</b>       | function |
| <b>atanh</b>      | function | <b>topair</b>     | function |
| <b>conj</b>       | function | <b>tostring</b>   | function |
| <b>cos</b>        | function | <b>totable</b>    | function |
| <b>cosh</b>       | function | <b>voigt</b>      | function |
| <b>dawson</b>     | function | <b>voigt_hwhm</b> | function |

## xdecimal (extra lua)

|                   |          |                     |          |
|-------------------|----------|---------------------|----------|
| <b>__add</b>      | function | <b>bxor</b>         | function |
| <b>__band</b>     | function | <b>conj</b>         | function |
| <b>__bor</b>      | function | <b>copy</b>         | function |
| <b>__bxor</b>     | function | <b>exp</b>          | function |
| <b>__div</b>      | function | <b>getprecision</b> | function |
| <b>__eq</b>       | function | <b>ln</b>           | function |
| <b>__idiv</b>     | function | <b>log</b>          | function |
| <b>__le</b>       | function | <b>max</b>          | function |
| <b>__lt</b>       | function | <b>min</b>          | function |
| <b>__mod</b>      | function | <b>minus</b>        | function |
| <b>__mul</b>      | function | <b>new</b>          | function |
| <b>__name</b>     | string   | <b>plus</b>         | function |
| <b>__pow</b>      | function | <b>pow</b>          | function |
| <b>__shl</b>      | function | <b>rotate</b>       | function |
| <b>__shr</b>      | function | <b>setprecision</b> | function |
| <b>__sub</b>      | function | <b>shift</b>        | function |
| <b>__tostring</b> | function | <b>sqrt</b>         | function |
| <b>__unm</b>      | function | <b>toengstring</b>  | function |
| <b>abs</b>        | function | <b>tonumber</b>     | function |
| <b>band</b>       | function | <b>tostring</b>     | function |
| <b>bor</b>        | function | <b>trim</b>         | function |

## xinterval (context)

|                   |          |                     |          |
|-------------------|----------|---------------------|----------|
| <b>__add</b>      | function | <b>greaterequal</b> | function |
| <b>__div</b>      | function | <b>hull</b>         | function |
| <b>__eq</b>       | function | <b>intersect</b>    | function |
| <b>__le</b>       | function | <b>less</b>         | function |
| <b>__lt</b>       | function | <b>lessequal</b>    | function |
| <b>__mod</b>      | function | <b>log</b>          | function |
| <b>__mul</b>      | function | <b>log10</b>        | function |
| <b>__name</b>     | string   | <b>log1p</b>        | function |
| <b>__pow</b>      | function | <b>log2</b>         | function |
| <b>__sub</b>      | function | <b>mid</b>          | function |
| <b>__tostring</b> | function | <b>mod</b>          | function |
| <b>__unm</b>      | function | <b>new</b>          | function |
| <b>abs</b>        | function | <b>neweps</b>       | function |
| <b>acos</b>       | function | <b>pow</b>          | function |
| <b>acosh</b>      | function | <b>seteq</b>        | function |
| <b>asin</b>       | function | <b>setge</b>        | function |
| <b>asinh</b>      | function | <b>setgt</b>        | function |
| <b>asstring</b>   | function | <b>setle</b>        | function |
| <b>atan</b>       | function | <b>setlt</b>        | function |
| <b>atanh</b>      | function | <b>sin</b>          | function |
| <b>cos</b>        | function | <b>sinh</b>         | function |
| <b>cosh</b>       | function | <b>sqr</b>          | function |
| <b>diameter</b>   | function | <b>sqrt</b>         | function |
| <b>equal</b>      | function | <b>tan</b>          | function |
| <b>erf</b>        | function | <b>tanh</b>         | function |
| <b>erfc</b>       | function | <b>topair</b>       | function |
| <b>exp</b>        | function | <b>tostring</b>     | function |
| <b>exp10</b>      | function | <b>totable</b>      | function |
| <b>exp2</b>       | function | <b>usedendian</b>   | function |
| <b>expm1</b>      | function | <b>within</b>       | function |
| <b>greater</b>    | function |                     |          |

## xmath (extra lua)

|                     |          |                  |          |
|---------------------|----------|------------------|----------|
| <b>acos</b>         | function | <b>isnormal</b>  | function |
| <b>acosh</b>        | function | <b>j0</b>        | function |
| <b>asin</b>         | function | <b>j1</b>        | function |
| <b>asinh</b>        | function | <b>jn</b>        | function |
| <b>atan</b>         | function | <b>ldexp</b>     | function |
| <b>atan2</b>        | function | <b>lgamma</b>    | function |
| <b>atanh</b>        | function | <b>log</b>       | function |
| <b>cbrt</b>         | function | <b>log10</b>     | function |
| <b>ceil</b>         | function | <b>log1p</b>     | function |
| <b>copysign</b>     | function | <b>log2</b>      | function |
| <b>cos</b>          | function | <b>logb</b>      | function |
| <b>cosh</b>         | function | <b>modf</b>      | function |
| <b>crossproduct</b> | function | <b>nan</b>       | number   |
| <b>deg</b>          | function | <b>nearbyint</b> | function |
| <b>dotproduct</b>   | function | <b>nextafter</b> | function |
| <b>erf</b>          | function | <b>normalize</b> | function |
| <b>erfc</b>         | function | <b>pi</b>        | number   |
| <b>exp</b>          | function | <b>pow</b>       | function |
| <b>exp2</b>         | function | <b>rad</b>       | function |
| <b>expm1</b>        | function | <b>remainder</b> | function |
| <b>fabs</b>         | function | <b>remquo</b>    | function |
| <b>fdim</b>         | function | <b>round</b>     | function |
| <b>floor</b>        | function | <b>scalbn</b>    | function |
| <b>fma</b>          | function | <b>sin</b>       | function |
| <b>fmax</b>         | function | <b>sinh</b>      | function |
| <b>fmin</b>         | function | <b>sqrt</b>      | function |
| <b>fmod</b>         | function | <b>tan</b>       | function |
| <b>frexp</b>        | function | <b>tanh</b>      | function |
| <b>gamma</b>        | function | <b>tgamma</b>    | function |
| <b>hypot</b>        | function | <b>trunc</b>     | function |
| <b>inf</b>          | number   | <b>y0</b>        | function |
| <b>isfinite</b>     | function | <b>y1</b>        | function |
| <b>isinf</b>        | function | <b>yn</b>        | function |
| <b>isnan</b>        | function |                  |          |

# xml (context)

|                           |          |                        |            |
|---------------------------|----------|------------------------|------------|
| <b>addentitiesdoctype</b> | function | F family               |            |
| <b>addspecialcode</b>     | function | F fontspecification    |            |
| <b>after</b>              | function | F mappedstylevalue     |            |
| <b>all</b>                | function | F padding              |            |
| <b>applylpath</b>         | function | F size                 |            |
| <b>applyselector</b>      | function | F style                |            |
| <b>assign</b>             | function | F stylevalue           |            |
| <b>attribute</b>          | function | F todimension          |            |
| <b>badinclusions</b>      | function | F variant              |            |
| <b>before</b>             | function | F weight               |            |
| <b>body</b>               | function | <b>defaulthandlers</b> | table      |
| <b>builtinentities</b>    | table    | D file                 |            |
| B amp                     |          | D parsedxml            |            |
| B apos                    |          | D string               |            |
| B gt                      |          | D tex                  |            |
| B lt                      |          | D texline              |            |
| B quot                    |          | D texspace             |            |
| <b>CDATA</b>              | function | D verbose              |            |
| <b>CDATATOTEXT</b>        | function | <b>defaultprotocol</b> | string     |
| <b>checkbom</b>           | function | <b>delete</b>          | function   |
| <b>checkedempty</b>       | function | <b>delete_element</b>  | function   |
| <b>checkerror</b>         | function | <b>despecialized</b>   | function   |
| <b>checknamespace</b>     | function | <b>each</b>            | function   |
| <b>checkns</b>            | function | <b>each_element</b>    | function   |
| <b>cleansed</b>           | function | <b>elements</b>        | function   |
| <b>collect</b>            | function | <b>elementtocdata</b>  | function   |
| <b>collect_elements</b>   | function | <b>empty</b>           | function   |
| <b>collect_tags</b>       | function | <b>entities</b>        | data table |
| <b>collect_texts</b>      | function | <b>erase</b>           | function   |
| <b>collected</b>          | function | <b>errorhandler</b>    | function   |
| <b>collecttexts</b>       | function | <b>escaped</b>         | function   |
| <b>compactelement</b>     | function | <b>expand</b>          | function   |
| <b>content</b>            | function | <b>expressions</b>     | table      |
| <b>convert</b>            | function | F attribute            |            |
| <b>copy</b>               | function | F boolean              |            |
| <b>count</b>              | function | F child                |            |
| <b>cprint</b>             | function | F contains             |            |
| <b>CSS</b>                | table    | F count                |            |
| F applyselector           |          | F error                |            |
| F colorspecification      |          | F find                 |            |
| F dimension               |          | F hasclass             |            |

|                              |          |                           |            |
|------------------------------|----------|---------------------------|------------|
| F idstring                   |          | <b>insertcomment</b>      | function   |
| F lower                      |          | <b>inspect</b>            | function   |
| F name                       |          | <b>is_valid</b>           | function   |
| F number                     |          | <b>kind</b>               | function   |
| F oneof                      |          | <b>last</b>               | function   |
| F print                      |          | <b>lastmatch</b>          | function   |
| F quit                       |          | <b>load</b>               | function   |
| F tag                        |          | <b>lpath</b>              | function   |
| F text                       |          | <b>lpathcached</b>        | function   |
| F undefined                  |          | <b>lpathcalls</b>         | function   |
| F upper                      |          | <b>lshow</b>              | function   |
| <b>fillin</b>                | function | <b>makestandalone</b>     | function   |
| <b>filter</b>                | function | <b>match</b>              | function   |
| <b>finalizers</b>            | table    | <b>mml</b>                | table      |
| F paragraphs                 |          | F stripped                |            |
| <b>first</b>                 | function | F verbose                 |            |
| <b>found</b>                 | function | <b>name</b>               | function   |
| <b>functions</b>             | table    | <b>newhandlers</b>        | function   |
| F classes                    |          | <b>nodesettostring</b>    | function   |
| F hasclass                   |          | <b>originalload</b>       | function   |
| <b>getbuffer</b>             | function | <b>package</b>            | function   |
| <b>gethandlers</b>           | function | <b>parent</b>             | function   |
| <b>gethandlersfunction</b>   | function | <b>parsedentitylpeg</b>   | userdata   |
| <b>helpers</b>               | table    | <b>pihandlers</b>         | table      |
| F recursetext                |          | F injector                |            |
| <b>include</b>               | function | F tex                     |            |
| <b>inclusion</b>             | function | <b>placeholders</b>       | table      |
| <b>inclusions</b>            | function | F unknown_any_entity      |            |
| <b>index</b>                 | function | F unknown_dec_entity      |            |
| <b>inheritattributes</b>     | function | F unknown_hex_entity      |            |
| <b>inheritedconvert</b>      | function | <b>popmatch</b>           | function   |
| <b>inject</b>                | function | <b>position</b>           | function   |
| <b>inject_element</b>        | function | <b>privatecodes</b>       | data table |
| <b>inject_element_after</b>  | function | <b>privatetoken</b>       | function   |
| <b>inject_element_before</b> | function | <b>process</b>            | function   |
| <b>injectafter</b>           | function | <b>process_attributes</b> | function   |
| <b>injectbefore</b>          | function | <b>process_elements</b>   | function   |
| <b>insert</b>                | function | <b>processattributes</b>  | function   |
| <b>insert_element</b>        | function | <b>profiled</b>           | table      |
| <b>insert_element_after</b>  | function | <b>pure</b>               | function   |
| <b>insert_element_before</b> | function | <b>pushmatch</b>          | function   |
| <b>insertafter</b>           | function | <b>raw</b>                | function   |
| <b>insertbefore</b>          | function | <b>registerentity</b>     | function   |
| <b>insertcdata</b>           | function | <b>registerns</b>         | function   |



|                            |            |                            |          |
|----------------------------|------------|----------------------------|----------|
| <b>reindex</b>             | function   | <b>specialhandler</b>      | table    |
| <b>remap_name</b>          | function   | <b>sprint</b>              | function |
| <b>remap_namespace</b>     | function   | <b>string</b>              | function |
| <b>remap_tag</b>           | function   | <b>strip</b>               | function |
| <b>remapname</b>           | function   | <b>strip_whitespace</b>    | function |
| <b>remapnamespace</b>      | function   | <b>stripelement</b>        | function |
| <b>remaptag</b>            | function   | <b>stripleadingspaces</b>  | function |
| <b>rename</b>              | function   | <b>table</b>               | function |
| <b>renamespace</b>         | function   | <b>text</b>                | function |
| <b>reparsedentitylpeg</b>  | userdata   | <b>textonly</b>            | function |
| <b>replace</b>             | function   | <b>texttocdata</b>         | function |
| <b>replace_element</b>     | function   | <b>tocdata</b>             | function |
| <b>resetinjectors</b>      | function   | <b>toelement</b>           | function |
| <b>resolvens</b>           | function   | <b>tostring</b>            | function |
| <b>root</b>                | function   | <b>totable</b>             | function |
| <b>save</b>                | function   | <b>toxml</b>               | function |
| <b>selection</b>           | function   | <b>tprint</b>              | function |
| <b>separate</b>            | function   | <b>traverse</b>            | function |
| <b>serialize</b>           | function   | <b>unescaped</b>           | function |
| <b>serializetotext</b>     | function   | <b>unescapedentitylpeg</b> | userdata |
| <b>setcdata</b>            | function   | <b>unprivatized</b>        | function |
| <b>setcomment</b>          | function   | <b>unspecialized</b>       | function |
| <b>sethandlersfunction</b> | function   | <b>wipe</b>                | function |
| <b>setinjectors</b>        | function   | <b>withelement</b>         | function |
| <b>setproperty</b>         | function   | <b>withelements</b>        | function |
| <b>settings</b>            | function   | <b>wrap</b>                | function |
| <b>specialcodes</b>        | data table | <b>xmlns</b>               | table    |

## xzip (extra lua)

**adler32**           function  
**compress**        function  
**compresssize**   function

**crc32**            function  
**decompress**      function  
**decompresssize** function

## zbuffer (context)

|                       |          |                       |          |
|-----------------------|----------|-----------------------|----------|
| <b>bounds</b>         | function | <b>set</b>            | function |
| <b>composite</b>      | function | <b>setcolor</b>       | function |
| <b>crop</b>           | function | <b>setdepth</b>       | function |
| <b>edgepoint</b>      | function | <b>setnormal</b>      | function |
| <b>flatten</b>        | function | <b>setopacity</b>     | function |
| <b>get</b>            | function | <b>setup</b>          | function |
| <b>getcolor</b>       | function | <b>smoothen</b>       | function |
| <b>getconstants</b>   | function | <b>stipple_0</b>      | function |
| <b>getdepth</b>       | function | <b>stipple_1</b>      | function |
| <b>getnormal</b>      | function | <b>stipple_2</b>      | function |
| <b>getnormalmodes</b> | function | <b>stipple_3</b>      | function |
| <b>getopacity</b>     | function | <b>stipple_4</b>      | function |
| <b>getsize</b>        | function | <b>stipple_setup</b>  | function |
| <b>implicit</b>       | function | <b>tobytes</b>        | function |
| <b>iszbuffer</b>      | function | <b>tostring</b>       | function |
| <b>meshmesh</b>       | function | <b>totable</b>        | function |
| <b>new</b>            | function | <b>transform</b>      | function |
| <b>points</b>         | function | <b>trianglebounds</b> | function |
| <b>process</b>        | function | <b>triangled</b>      | function |
| <b>project</b>        | function | <b>triangles</b>      | function |
| <b>resolve</b>        | function | <b>type</b>           | function |

# zip (context)

|                     |          |                        |          |
|---------------------|----------|------------------------|----------|
| <b>archives</b>     | table    | <b>openarchive</b>     | function |
| <b>closearchive</b> | function | <b>registeredfiles</b> | table    |

## zlib (context)

|                     |          |
|---------------------|----------|
| <b>adler32</b>      | function |
| <b>compress</b>     | function |
| <b>compresssize</b> | function |

|                       |          |
|-----------------------|----------|
| <b>crc32</b>          | function |
| <b>decompress</b>     | function |
| <b>decompresssize</b> | function |